

THE STORY OF THE UNIVERSE



Volume 1 of 4

THE STARRY SKIES

By ESTHER SINGLETON

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The Zodiacal Light

THE STORY OF THE UNIVERSE

*Told by Great Scientists
and Popular Authors*

COLLECTED AND EDITED

By ESTHER SINGLETON

Author of "Turrets, Towers and Temples," "Wonders of Nature,"
"The World's Great Events," "Famous Paintings," Translator
of Lavignac's "Music Dramas of Richard Wagner"

FULLY ILLUSTRATED



VOLUME I
THE
STARRY
SKIES

P. F. COLLIER AND SON
NEW YORK

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P R E F A C E

In the following pages I have endeavored to present a comprehensive and general view of the material side of the universe. Instead of trying myself to tell the story of the universe, I have gone to the works of acknowledged weight and authority in this line of research and selected from them extracts of a popular character, especially those that are entertaining as well as merely instructive. The average reader is frequently repelled from the study of the sciences by the dry treatment adopted by those who try to instruct him. He cares little for laws, theories, or affinities; and he can not help being bored by attempts to make him understand classifications with their long lists of words manufactured from the names of modern celebrities or non-entities and roots from dead languages. I have therefore kept constantly in mind the person who seeks entertaining knowledge, and not the scientific specialist. I have tried to avoid all technicalities wherever possible.

Of late years, in fact ever since the foundation of the British Association, there has been a constantly increasing interest in the wonders of nature; and the specialist has responded to this popular interest in his scientific labors by speaking in language that an intelligent child can comprehend. People as a rule prefer to read of the habits, instincts, intelligence, and movements of animals and plants, rather than of their organs and structure. Thus the study of Natural History has received a great impetus from the writings of such men as Darwin and Lubbock; and Astronomy has been rendered more attractive to the lay reader by Flammarion, Gore, Proctor, and Ball. Every traveler who returns from remote or hitherto unknown Arctic or Torrid Zones has something fresh to tell us of the phenomena and life of our universe, which adds fresh stimulus to the popular interest in the Natural Sciences.

The Story of the Universe naturally falls under the following four heads:

First, the bodies moving in infinite space, including stars, dark and lucid, planets, nebulae, comets, and meteors.

Second, the Earth, considered as a separate world and the only one of which we have precise detailed knowledge. In this chapter we learn of the past of our globe from the evidence afforded by the rocks of which its crust is composed. The varying conformations of its present surface are described, as is its atmospheric envelope and attendant phenomena. The ocean and its movements and depths are likewise fully considered.

Third, the Earth's Garment—its flora. In this chapter we are told of the wonders and beauties of plant-life, its development and distribution.

Fourth, the Earth's Creatures. Here we have a general view of animal life, from the mighty mammoth to the fairy fly: even the beings visible only to the microscope are not forgotten. Special attention is also paid to man, from his origin to the present day.

I have made the selections from authentic editions of the writings of the scientists; and have taken no liberties with the text, with the exception of occasional cutting.

In the Introduction I have given a short sketch of the development of the Natural Sciences, from the dawn of written history to the present day.

E. S.

NEW YORK, *March, 1905.*

INTRODUCTION

The knowledge of the Natural Sciences among the Greeks and Romans was derived principally from the Egyptians and Babylonians. The Phœnicians in their voyages, also, necessarily paid considerable attention to Astronomy. Their Cynosura consisted of the tail of the Little Bear, by which they steered. The great names in Greek Astronomy are Aratus, Hipparchus, and Ptolemy.

From the fancies of Astrology, in which the early Arabs largely indulged, and which, though discountenanced by Mahomet himself, have never been wholly abandoned by their descendants, a not unnatural transition, led to the study of Astronomy. Under the patronage of the Abbaside Caliph Al-Mamun (813-833 A. D.) this science made rapid progress.

Astronomy was zealously studied in the famous schools of Bagdad and Cordova.

The *Almagest*, or System of Astronomy, by Ptolemy, was translated into Arabic by Alhazi and Sergius as early as 812. In the Tenth Century, Albaten observed the advance of the line of the apsides in the earth's orbit; Mohammed-ben-Jeber-al-Batani, the obliquity of the ecliptic; Alpetragius wrote a theory of the planets; and Abul-Hassan-Ali, on astronomical instruments. The obliquity of the ecliptic, the diameter of the earth, and even the precession of the equinoxes, were then calculated with commendable accuracy; and shortly after, Abul-Mezar's *Introduction to Astronomy* and his *Treatise on the Conjunction of the Planets*, with the *Elements* of Al-Furjanee (though this last author was largely indebted to the Egyptian labors of Ptolemy), proved that the caliph's liberality had been well bestowed. But Al-Batinee, a native of Syria (879-920 A. D.), surpassed all his predecessors in the nicety alike of his observations and computations. Geber, at Seville, constructed (1196 A. D.) the first astronomical observatory on record; and Ebn-Korrah in Egypt proved by

his example that the Arabs could be even better astronomers than the Greeks.

Ulug Bekh, grandson of the great Tamerlane, was a diligent observer. He established an academy of astronomers at Samarcand, the capital of his dominions, and constructed magnificent instruments. Ulug Bekh, too, made a catalogue of the fixed stars—the only one that had been compiled since that of Hipparchus, sixteen centuries previously.

Gradually, by their intercourse with civilized nations, the Arabian conquerors were themselves subjected to the humanizing influence of letters, and, after 749 A. D., or during the reign of the Abassides, literature, arts, and sciences appeared, and were generously fostered under the splendid sway, first of Almansor (754-775), and afterward of the celebrated Harun-al-Raschid (786-808). Learned men were now invited from many countries and remunerated for their labors with princely munificence; the works of the best Greek, Syriac, and old Persian writers were translated into Arabic, and spread abroad in numerous copies. The Caliph Al-Mamun, who reigned from 813 to 833, offered to the Greek emperor five tons of gold and a perpetual treaty of peace on condition that the philosopher Leo should be allowed to give instruction to the former. Under the same Caliph the famous schools of Bagdad, Basra, Bokhara, and Kufa were founded, and large libraries were collected in Alexandria, Cairo, and Bagdad. The school of Cordova in Spain soon rivaled that of Bagdad, and in the Tenth Century the Arabs were everywhere the preservers and distributors of knowledge.

Pupils from France and other European countries repaired to Spain in great numbers, to study mathematics and medicine under the Arabs. There were fourteen academies, with many preparatory and upper schools, in Spain, and five very considerable public libraries; that of the Caliph Hakem containing, as is said, more than 600,000 volumes.

In Geography, History, Philosophy, Medicine, Physics, and Mathematics the Arabians rendered important services to science; and the Arabic words still employed in science—such as algebra, alcohol, azimuth, zenith, nadir, with many names of stars, etc. (see *The Arabian Heavens*, pages 106-120 of Vol. I)—remain as indications of their influence on the early intellectual culture of Europe. But Geography owes most to them during the Middle Ages. In Africa and Asia, the boundaries of geographical science were

extended, and the old Arab treatises on geography and works of travels in several countries by Abulfeda, Edrisi, Leo Africanus, Ibn Batuta, Ibn Foslan, Ibn Jobair, Albiruni the astronomer, and others, are still interesting.

The structure of the earth received little attention from the ancients; the extent of its surface known was limited, and the changes upon it were neither so speedy nor violent as to excite special attention. The only opinions deserving to be noticed are those of Pythagoras and Strabo, both of whom observed the phenomena which were then altering the surface of the earth, and proposed theories for explaining the changes that had taken place in geological time. The first held that, in addition to volcanic action, the change in the level of sea and land was owing to the retiring of the sea; while the other maintained that the land changed its level, and not the sea, and that such changes happened more easily to the land below the sea because of its humidity.

From the fall of the Roman empire, during the Dark Ages, the physical sciences were neglected. In the Tenth Century, Avicenna, Omar, and other Arabian writers commented on the works of the Romans, but added little of their own.

Geological phenomena attracted attention in Italy in the Sixteenth Century, the absorbing question then being as to the nature of fossils; only a few maintained that they were the remains of animals. Two centuries elapsed before the opinion was generally adopted.

Aristotle was the first who collected, in his work *On Meteors*, the current prognostics of the weather. Some of these were derived from the Egyptians, who had studied the science as a branch of Astronomy, while a considerable number were the result of his own observation. The next writer upon this subject was Theophrastus, one of Aristotle's pupils, who classified the opinions commonly received regarding the weather under four heads, viz., the prognostics of rain, of wind, of storm, and of fine weather. The subject was discussed purely in its popular and practical bearings, and no attempt was made to explain phenomena whose occurrence appeared so irregular and capricious. Cicero, Virgil, and a few other writers also wrote on the subject; but the treatise of Theophrastus contains nearly all that was known down to comparatively recent times. Partial explanations were attempted by

Aristotle and Lucretius, but their explanations were vague, and often absurd.

In this dormant condition meteorology remained for ages, and no progress was made till proper instruments were invented for making real observations with regard to the temperature, the pressure, the humidity, and the electricity of the air.

Solomon spoke of “trees, from the cedar in Lebanon even to the hyssop that springeth out of the wall.” There is reason also to believe that Zoroaster devoted some attention to plants, and that this study early engaged some of the philosophers of Greece. The oldest botanical work which has come down to us is that of Theophrastus, the pupil of Aristotle, who flourished in the fourth century B. C. His descriptions of plants are very unsatisfactory, but his knowledge of their organs and of vegetable physiology may well be deemed wonderful. It was not, indeed, till after the revival of letters in Western Europe, that it was ever again studied as it had been by him. About four hundred years after Theophrastus, in the First Century of the Christian era, Dioscorides of Anazarbus, in Asia Minor—a herbalist, however, rather than a botanist—described more than 600 plants in a work which continued in great repute throughout the Middle Ages.

About the same time, the elder Pliny devoted a share of his attention to Botany, and his writings contain some account of more than 1,000 species, compiled from various sources and mingled with many errors. Centuries elapsed without producing another name worthy to be mentioned. It was among the Arabians that the science next began to be cultivated, about the close of the Eighth Century. The greatest name of this period is Avicenna. Among the Arabs, Botany, like Chemistry, was chiefly studied as subsidiary to medicine; but as an adjunct to the old herbal pharmacopœia, it received close attention. The principal mercurial and arsenical preparations of the *materia medica*, the sulphates of several metals, the properties of acids and alkalies, the distillation of alcohol—in fine, whatever resources chemistry availed itself of up to a very recent date—were, with their practical application, known to Er-Razi and Geber. In fact, the numerous terms borrowed from the Arabic language—for instance, alcohol, alkali, alembic, and others—with the signs of drugs and the like, still in use among modern apothecaries, remain to show how deeply this science is indebted to Arab research.

Aristotle seems to have been the first to study Zoology. Some of the groups he established still retain their place in the most modern classifications. His two great sections of the Animal Kingdom consisted of Enanima (red blood) and Anima (having a circulation of colorless fluid). Ælian and Pliny wrote on the subject, but they indulged largely in fables. There was little advance in the science during the Dark and Middle Ages. The *Bestiaries* were written for the sake of moral teaching, and the animals had to behave with that end in view. Albertus Magnus is the only famous name in this department before the revival of learning.

The shining light of the Thirteenth Century was Roger Bacon. His *Opus Majus* is “at once the Encyclopædia and the Novum Organum of the Thirteenth Century.” In this, besides other branches of scientific research, he devotes a rapid examination to questions of Climate, Hydrography, Geography, and Astrology. Scientific research, however, was out of date, and from the educated world Roger Bacon received small recognition. His writings earned only a prison from his own Order, and he died, in his own words, “unheard, forgotten, buried.”

The Revival of Learning, commonly known as the Period of the Renaissance, naturally entailed renewed interest in the sciences as well as the arts. Green gives a comprehensive view of it:

“The last royalist had only just laid down his arms when the little company who were at a later time to be known as the Royal Society gathered round Wilkins at Oxford. It is in this group of scientific observers that we catch the secret of the coming generation. From the vexed problems, political and religious, with which it had so long wrestled in vain, England turned at last to the physical world around it, to the observation of its phenomena, to the discovery of the laws which govern them. The pursuit of physical science became a passion; and its method of research, by observation, comparison, and experiment, transformed the older methods of inquiry in matters without its pale. In religion, in politics, in the study of man and of nature, not faith but reason, not tradition but inquiry, were to be the watchwords of the coming time. The dead-weight of the past was suddenly rolled away, and the new England heard at last and understood the call of Francis Bacon.

“Bacon had already called men with a trumpet-voice to such studies; but in England at least Bacon stood before his age. The beginnings of physical science were more slow and timid there than in any country of Europe. Only two discoveries of any real value came from English research before the Restoration; the first, Gilbert’s discovery of terrestrial magnetism in the close of Elizabeth’s reign; the next, the great discovery of the circulation of the blood, which was taught by Harvey in the reign of James. Apart from these illustrious names England took little share in

the scientific movement of the continent; and her whole energies seemed to be whirled into the vortex of theology and politics by the Civil War. But the war had not reached its end when a little group of students were to be seen in London, men 'inquisitive,' says one of them, 'into natural philosophy and other parts of human learning, and particularly of what hath been called the New Philosophy,... which from the times of Galileo at Florence, and Sir Francis Bacon (Lord Verulam) in England, hath been much cultivated in Italy, France, Germany, and other parts abroad, as well as with us in England.' The strife of the time indeed aided in directing the minds of men to natural inquiries. 'To have been always tossing about some theological question,' says the first historian of the Royal Society, Bishop Sprat, 'would have been to have made that their private diversion, the excess of which they disliked in the public. To have been eternally musing on civil business and the distresses of the country was too melancholy a reflection. It was nature alone which could pleasantly entertain them in that estate.' Foremost in the group stood Doctors Wallis and Wilkins, whose removal to Oxford, which had just been reorganized by the Puritan Visitors, divided the little company into two societies. The Oxford society, which was the more important of the two, held its meetings at the lodgings of Dr. Wilkins, who had become Warden of Wadham College, and added to the names of its members that of the eminent mathematician Dr. Ward, and that of the first of English economists, Sir William Petty. 'Our business,' Wallis tells us, 'was (precluding matters of theology and state affairs) to discourse and consider of philosophical inquiries and such as related thereunto, as Physick, Anatomy, Geometry, Astronomy, Navigation, Statics, Magneticks, Chymicks, Mechanicks, and Natural Experiments: with the state of these studies, as then cultivated at home and abroad. We then discoursed of the circulation of the blood, the valves in the *venæ lacteæ*, the lymphatic vessels, the Copernican hypothesis, the nature of comets and new stars, the satellites of Jupiter, the oval shape of Saturn, the spots in the sun and its turning on its own axis, the inequalities and selenography of the moon, the several phases of Venus and Mercury, the improvement of telescopes, the grinding of glasses for that purpose, the weight of air, the possibility or impossibility of vacuities, and Nature's abhorrence thereof, the Torricellian experiment in quicksilver, the descent of heavy bodies and the degree of acceleration therein, and divers other things of like nature.'

"The other little company of inquirers, who remained in London, was at last broken up by the troubles of the Second Protectorate; but it was revived at the Restoration by the return to London of the more eminent members of the Oxford group. Science suddenly became the fashion of the day. Charles was himself a fair chymist, and took a keen interest in the problems of navigation. The Duke of Buckingham varied his freaks of riming, drinking, and fiddling by fits of devotion to his laboratory. Poets like Dryden and Cowley, courtiers like Sir Robert Murray and Sir Kenelm Digby joined the scientific company to which in token of his sympathy with it the King gave the title of 'The Royal Society.' The curious glass toys called Prince Rupert's drops recall the scientific inquiries which, with the study of etching, amused the old age of the great cavalry leader of the Civil War. Wits and fops crowded to the meetings of the new society. Statesmen like Lord Somers felt honored at being chosen its presidents. Its definite establishment marks the opening of a great age of scientific discovery in England. Almost every year of the half century which

followed saw some step made to a wider and truer knowledge. Our first national observatory rose at Greenwich, and modern astronomy began with the long series of astronomical observations which immortalized the name of Flamsteed. His successor, Halley, undertook the investigation of the tides, of comets, and of terrestrial magnetism. Hooke improved the microscope, and gave a fresh impulse to microscopical research. Boyle made the air-pump a means of advancing the science of pneumatics, and became the founder of experimental chymistry. Wilkins pointed forward to the science of philology in his scheme of a universal language. Sydenham introduced a careful observation of nature and facts which changed the whole face of medicine. The physiological researches of Willis first threw light upon the structure of the brain. Woodward was the founder of mineralogy. In his edition of Willoughby's *Ornithology*, and in his own *History of Fishes*, John Ray was the first to raise zoology to the rank of a science; and the first scientific classification of animals was attempted in his *Synopsis of Quadrupeds*. Modern botany began with his *History of Plants*, and the researches of an Oxford professor, Robert Morison; while Grew divided with Malpighi the credit of founding the study of vegetable physiology. But great as some of these names undoubtedly are, they are lost in the lustre of Isaac Newton. Newton was born at Woolsthorpe in Lincolnshire, on Christmas Day, in the memorable year which saw the outbreak of the Civil War. In the year of the Restoration he entered Cambridge, where the teaching of Isaac Barrow quickened his genius for mathematics, and where the method of Descartes had superseded the older modes of study. From the close of his Cambridge career his life became a series of great physical discoveries. At twenty-three he facilitated the calculation of planetary movements by his theory of Fluxions. The optical discoveries to which he was led by his experiments with the prism, and which he partly disclosed in the lectures which he delivered as mathematical professor at Cambridge, were embodied in the theory of light which he laid before the Royal Society on becoming a Fellow of it. His discovery of the law of gravitation had been made as early as 1666; but the erroneous estimate which was then generally received of the earth's diameter prevented him from disclosing it for sixteen years; and it was not till the eve of the Revolution that the *Principia* revealed to the world his new theory of the Universe."

Ever since the Fifteenth Century, when Copernicus revived the ancient theory of Pythagoras that the planets revolved around the sun (a theory left in an imperfect state and demonstrated later by Kepler, Galileo, Newton, and others) astronomical research has progressed steadily. It must be remembered, however, that *De Revolutionibus Orbium*, which met with great opposition, contained nothing regarding the laws of motion, for these had not been as yet discovered, and Saturn marked the boundaries of the Solar System. Copernicus assigned the "fixed stars" to a sphere, as in Ptolemy's heavens (see [page 331](#)).

The great Danish astronomer, Tycho Brahe, whose idea of the Solar System is represented on [page 343](#), was his opponent. Brahe, however, a

devoted student, a man of wealth, the favorite of kings and princes, and the proud possessor of the Castle of Uraniberg (City of the Heavens), an observatory equipped with fine instruments and built for him by Frederick II, King of Denmark, on the island of Huen, and after his death the protégé of Rudolph II at Benatek, near Prague, contributed greatly to the advancement of the science by means of his discoveries, computations, solar and lunar tables, and catalogue of stars. He, like Copernicus, placed the “fixed stars” in an outer sphere. His observations on the planets were made to prove the truth of his system. This mass of observations was used instead by Johann Kepler, who had been his assistant at the Benatek Observatory, to prove Copernicus’s theory. Of Kepler, the discoverer of the three famous laws, who gave a complete theory of solar eclipses, calculated the transits of Mercury and Venus, and made numerous discoveries in optics and general physics, Proctor says:

“Kepler was not merely an observer and calculator; he inquired with great diligence into the physical causes of every phenomenon, and made a near approach to the discovery of that great principle which maintains and regulates the planetary motions. He possessed some very sound and accurate notions of the nature of gravity, but unfortunately conceived it to diminish simply in proportion to the distance, although he had demonstrated that the intensity of light is reciprocally proportional to the surface over which it is spread, or inversely as the square of the distance from the luminous body.”

Great names follow in rapid succession. One of Kepler’s contemporaries was Galileo Galilei, the discoverer of the “three laws of motion” and the relation of time and space in falling bodies, the first to apply the newly invented telescope to the observation of the heavens and the discoverer of four satellites of Jupiter (named by him the “Medeiran Stars” in honor of his patron). He also detected spots on the sun’s disk, the phases of Venus, and irregularities on the moon’s surface, and declared the Milky Way to be composed of a countless tract of separate stars.

When we remember the limited power of the telescope of the age, we can but marvel, not at how little, but how much was known regarding the starry skies.

During this period, numerous observers rendered great service to Astronomy, and other scientists were engaged in making useful drawings, charts, maps, tables, and catalogues of stars.

To this period also belongs John Bayer of Augsburg, who published a description of the constellations with maps upon which the stars were marked with the letters of the Greek Alphabet—a convenient method that was universally adopted and is still in use. Other names include Gassendi, Riccioli, Grimaldi, and Hevelius—the latter a rich citizen of Dantzic, who had a fine observatory of his own, where he worked for forty years. His drawings and descriptions of the moon, his researches on comets, which he still believed moved in parabolas, and his celestial charts engaged most of his attention.

The Dutch astronomer Huygens (born in 1629) is famous for his improvements in the telescope use of the pendulum clock and developments in the machinery of astronomical instruments. He discovered the ring of Saturn and four of his satellites. Edmund Halley, an English astronomer (born in 1656), also took a great interest in the telescope, and went to Dantzic to settle a controversy between Robert Hooke and Hevelius regarding the best glasses for use in astronomical observations; for Hevelius still worked with the ancient instruments, while Hooke believed in the lens.

Halley revived the ancient idea that comets belonged to the Solar System, and predicted that the comet of 1681 would return to its perihelion in 1759. This was the first prediction of its kind verified.

During the last quarter of the Seventeenth Century, the telescope assumes importance and two great observatories begin their work. In 1670 the Paris Observatory, of which Cassini was made director, was finished, and five years later the Greenwich Observatory, where Flamsteed was installed as royal astronomer.

Of Cassini, Lalande remarks that under him Astronomy underwent revolutions, and in France he was regarded as the “creator of the science.” Cassini discovered that Saturn’s ring was double and found four satellites of Jupiter.

Flamsteed’s observations on planets, satellites, comets, “fixed stars,” and his catalogue of 2,884 stars were valuable contributions to science; and his *Historia Cœlestis* is said to have “formed a new era in sidereal astronomy.”

Flamsteed was succeeded by Halley, particularly famed for his investigations of comets. The next great astronomical event was the

discovery of Uranus by Sir William Herschel in 1781. Sir William Herschel also discovered two more of Saturn's satellites, and began the great work of resolving the Milky Way and other clusters into swarms of suns, single stars into double and triple stars, inquiries into the mysteries of the nebulae, and in every way enlarging the general conception of the sidereal universe.

To the end of the Eighteenth and beginning of the Nineteenth Centuries belongs the brilliant French astronomer and mathematician Laplace, who published in 1799-1808 his *Mécanique Céleste*, in which he announced his Nebular Hypothesis (described on page 433 of Vol. II. The discoveries of the Planetoids are described on [pages 396-403](#), and that of Neptune in 1846 on [pages 430-432](#)). The latest important additions to the Solar System are the discovery by Prof. Barnard of Jupiter's Fifth Satellite in 1892 and Saturn's Ninth by Prof. W. H. Pickering in 1904. The discovery even of a Seventh Satellite of Jupiter has just been announced from the Lick Observatory.

It would be impossible to mention the names of the astronomers whose work from the middle of the last century to its closing years has been distinguished in various fields. Space only permits brief mention of the new methods of research by means of the spectroscope and celestial photography. With the first the name of the English astronomer, William Huggins, is identified and has yielded most important and startling information regarding the composition of heavenly bodies, and with the application of the photographic telescope these new methods have created a revolution in astronomical observation.

It may be interesting to gain a slight idea of the numbers of stars revealed by the camera by referring to Sir Robert Ball:

"If we take a position on the equator, from whence, of course, all the heavens can be completely seen in the lapse of six months, the number of stars that can be reckoned with the unaided eye will, according to Houzeau, amount to about six thousand. If we augment our unaided vision by a telescope of even small dimensions, such as three inches in diameter, the number of stars in the Northern Hemisphere alone is upward of three hundred thousand. We may assume that the Southern Hemisphere has an equally numerous star-population, so that the entire multitude visible with this optical aid is about six hundred thousand. Thus we see that the use of a telescope small enough to be carried in the hands suffices to multiply the lucid stars one-hundredfold. Great telescopes no doubt soon show us that the hundreds of thousands are only the brighter members of a host of millions,

and now we receive the assurance of photography that the telescopic stars are only the more conspicuous members of that vast universe. Mr. Roberts indeed declares that the multitudes of stars on the photographic plate grow with each increase of exposure to such a degree that it would almost seem as if the plate would be a wellnigh continuous mass of stars if the operations could be sufficiently protracted."

Naturally the past years have witnessed the making of new catalogues and maps of stars, and many valuable computations of parallaxes, etc. Some of the results obtained by these new methods are described in the chapters on the Nebulæ and Swarms of Suns, The Great Nebula of Orion, and The Colored, Double, Multiple, Binary, Variable, and Temporary Stars in Vol. I. From this brief survey of the progress of Astronomy the fact will be appreciated, therefore, that all the discoveries and researches have resulted in a larger conception of the universe, and the Solar System sinks into insignificance in the vast ocean of stars and suns.

The study of the Earth's crust and its contents divested of superstition dates from the end of the Seventeenth Century. Nicolaus Steno (1638-1687), a Dane, devoted himself to geology, and in 1669 observed successive layers of strata. He is called "the father of Palæontology." In 1680 Leibnitz proposed the theory that the Earth was originally in a molten state. The classification of strata was begun about the middle of the Eighteenth Century. The views of James Hutton (1788), who returned to the theories advanced by Ray (a return to the views of Pythagoras), were continued by Sir Charles Lyell.

Geology and Palæontology have progressed side by side. Among the most famous investigators are Cuvier, Dawson, Marsh, Owen, Huxley, Agassiz, De Blainville, Kaup, Sir Roderick Murchison, Boyd Dawkins, Sir William Flower, R. Lydekker, and E. D. Cope.

To the review of the new developments of meteorology and the science of probabilities by Sir Ralph Abercromby, on pages 784-792 of Vol. II, it is only necessary to add that the interest in meteorological research developed greatly after Torricelli's discovery in 1643 of weight and pressure in the atmosphere led to the perfection of the barometer and the development of the thermometer and hygrometer, both in the Seventeenth Century. The theory of trade-winds George Hadley announced in the *Philosophical Transactions* for 1735. Dalton's *Meteorological Essays*, published in 1793, and Dr. William Charles Wells's *Theory of Dew*, published in 1814,

attracted much attention. Regarding the inquiries into the laws of light by Snell, Newton, Descartes, Thomas Young, and Sir George Airy, the reader is referred to the chapter on The Rainbow in Vol. II, by John Tyndall, with whose researches in the latter half of the Nineteenth Century every one is more or less acquainted.

Little need be said here regarding the history of Botany, which is reviewed on pages 984-1000 of Vol. II. We may add, however, that one of the first to revive this study was Otto Brunfels, whose *Historia Plantarum Argentorati* was published in two folio volumes with cuts in Strasburg in 1530. He had many followers on the Continent and in England. During the revival of learning, chairs of Botany were founded in the universities; botanic gardens were established in many places (the Jardin des Plantes was founded in 1626); and botanists began to travel to remote countries to search for unknown flora.

To the Seventeenth Century belong the names of Dr. Turner, “the father of English Botany”; Robert Morison, professor of Botany at Oxford; John Ray, Nehemiah Grew, Malpighi, Henshaw, and Robert Hooke. The two latter were among the first to employ the newly invented microscope to the study of this science. It may be mentioned in passing, that Huygens is said to have taken from Holland to England microscopes about the size of a grain of sand, and that the first microscope consisting of a combination of lenses is attributed to Jansen, a spectacle-maker of Holland. Hooke, whom Herschel calls “the great contemporary and almost the rival of Newton,” gave a tremendous impetus to Microscopy, and practically laid the foundation of Histology or the Inner Morphology of Plants, due to Grew and Malpighi. Schleiden undertook to explain the mysteries of cell formation in 1838, further investigated by Schwann, and is now known as the Schleiden-Schwann theory. Nägeli and Von Mohl continued researches on this line. To the contents of the cell Von Mohl gave the name *protoplasm*.

In 1849, Hofmeister began investigations into the life-histories of plants, since when the study of Vegetable Physiology has progressed side by side with Chemistry. To Darwin great subjects are due: the cross-fertilization of plants, their reproduction, and their relations to insects and their movements. It may be mentioned, however, that in 1693 Ray attempted to

explain the movements of leaves, tendrils, and petals by physical and mechanical laws.

Since the middle of the Nineteenth Century, the branches of Botany that have been particularly studied are Vegetable Physiology and Pathology, Inner Morphology, and Fossil Botany—and the discoveries made have naturally had an effect upon the classification of vegetable life.

According to Agassiz:

“We must come down to the last century, to Linnæus, before we find the history taken up where Aristotle had left it, and some of his suggestions carried out with new freshness and vigor. Aristotle had already distinguished between genera and species; Linnæus took hold of this idea, and gave special names to other groups, of different weight and value. Besides species and genera, he gives us orders and classes—considering classes the most comprehensive, then orders, then genera, then species. He did not, however, represent these groups as distinguished by their nature, but only by their range; they were still to him, as genera and species had been to Aristotle, only larger or smaller groups, not founded upon and limited by different categories of structure. He divided the animal kingdom into six classes: Mammalia, Birds, Reptiles, Fishes, Insects, and Worms.”

Linnæus's classification was, therefore, the first attempt to group animals; but until Cuvier there was no great principle of classification. In 1707 Buffon succeeded in making Zoology, which had been regarded as a most uninteresting study, popular and respected. He also had the idea of collecting all the known facts of scientific investigation and arranging them systematically. Buffon was ridiculed as a scientist by his contemporaries, Hevelius, Diderot, D'Alembert, and Condillac, who opposed his explanations of natural phenomena. Buffon's *Histoire Naturelle Générale et Particulière* is his most important work. A complete edition in thirty-six volumes appeared in Paris in 1749-1788. Although it is said to “have made an epoch in the study of the natural sciences” in Buffon's day, it now possesses little scientific value.

Cuvier's classification has never been overthrown. His original investigations in various departments of science, and particularly that of fossil vertebrate animals, opened up new fields of study. His talents with both pen and pencil contributed largely to making that branch of science popular.

Lamarck, Cuvier's contemporary, divided the animal kingdom into Vertebrates and Invertebrates. Lamarck, like Geoffroy Saint-Hilaire, was a believer in the theory of evolution, which was opposed by Cuvier.

Lamarck turned from the study of Meteorology to that of Botany, and later again to that of Zoology. In 1793 he became professor of the natural history of the lower classes of animals in the Jardin des Plantes. His theories have greatly influenced modern science, particularly that of the "Variation of Species," which was set forth in his *Philosophie Zoologique* (two vols., Paris, 1809) and other works. Lamarck's *Histoire des Animaux sans Vertèbres* (seven vols., Paris, 1815-22) is his greatest work.

Karl Ernst von Baer, the Russian naturalist, a pupil of Döllinger in Würzburg, devoted himself chiefly to the study of embryology and made valuable discoveries.

Passing by many illustrious names, we come to that of Sir Richard Owen, of whom it has been said that "from the sponge to man, he has thrown light over every subject he has touched." His work in the Hunter Museum, his descriptions and restorations of extinct birds and animals, and his original works on every branch of animal life, form an enormous contribution to the progress of science. He promulgated the advanced views of John Hunter, the great physiologist and surgeon, of whose famous museum of more than ten thousand specimens, illustrative of anatomy and natural history, he became curator.

Three names shine with especial lustre upon the Nineteenth Century—Darwin, Huxley, and Spencer. The theory of evolution first appeared in De Maillet's work, *Telliamed*, published in 1758, but written in 1735. More than thirty writers before Darwin treated this theory, among whom were Erasmus Darwin, Goethe, Lamarck, and Geoffroy Saint-Hilaire. Largely owing to the opposition of Cuvier, it never succeeded until it was revived by Charles Darwin, who, after twenty-one years of work, published his results in 1858 in the *Journal of the Linnæan Society*, followed in the next year by *The Origin of Species by Means of Natural Selection* (see pages 1482-1512 of Vol. IV).

"The lifeless earth," says Sir Robert Ball, "is the canvas on which has been drawn the noblest picture that modern science has produced. It is Darwin who has drawn this picture. He has shown that the evolution of the lifeless earth from the nebula is

but the prelude to an organic evolution of still greater interest and complexity. He has taken up the history of the earth at the point where the astronomer left it, and he has made discoveries which have influenced thought and opinion more than any other discoveries that have been made for centuries.”

The neglected department of Marine Zoology the Nineteenth Century has made particularly its mission to investigate, but space only permits mention of four names: Edward Forbes, Lord Kelvin (Sir Wyville Thomson), Ernst Heinrich Haeckel, and the Prince of Monaco.

The first, whom Lord Kelvin considers “the most accomplished and original naturalist of his time,” was a pupil of Geoffroy Saint-Hilaire, Jussieu, and De Blainville. He is regarded as the originator of the use of the dredge for collecting specimens and the first who undertook the systematic study of Marine Zoology with reference to the distribution of fauna. In 1859 his *Natural History of the European Seas* appeared after his death.

One of the most important investigators in this line is Prof. Haeckel, famous for his studies of the lower class of marine animals. He is also distinguished for his researches in other branches of Zoology and Palæontology, and was one of the first followers of Darwin in Germany.

Entomology has also made enormous progress during the Nineteenth Century. At the end of the Seventeenth Century, Ray estimated the number of insects throughout the world at 10,000 species! The great entomologists of the Eighteenth Century include Linnæus, De Geer, and Fabricius. Next follow Latreille, Kirby and Spence, and a host of distinguished scientists in Europe and the United States, of whom Sir John Lubbock (Lord Avebury) heads the list. A comparatively new line of investigation is that of the Chalcididæ (see Fairy Flies, pages 1449-1458, in Vol. IV).

ESTHER SINGLETON.

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THE STORY OF THE UNIVERSE

THE HEAVENS.—AMÉDÉE GUILLEMIN

What are the heavens? Where the shores of that limitless ocean; where the bottom of that unfathomable abyss?

What are those brilliant points—those innumerable stars, which, never dim, shine out unceasingly from the dark profound? Are they sown broadcast—orderless, with no other bond save that which perspective lends to them? Or, if not immovable, as we have so long imagined, if not golden nails fixed to a crystal vault, whither are they bound? And, finally, what are the parts assigned to the sun, our earth, and all the earths attendant on the glorious orb of day in this tremendous concert of celestial spheres—this sublime harmony of the universe?

These are magnificent problems of which the most fertile imagination would have in vain attempted the solution, if, for the greater glory of the human mind, astronomy—first born of the sciences—had not at length come to our aid.

How wonderful is the power of man! Chained down to the surface of the earth, an intelligent atom on a grain of sand lost in the immensity of a space, he invents instruments which multiply a thousand-fold his vision, he sounds the depths of the ether, gauges the visible universe, and counts the myriads of stars which people it; next, studying their most complicated movements, he measures exactly their dimensions and the distances of the nearest of them from the earth, and next deduces their masses; then,

discovering in the seeming disorder of the stellar groupings real bonds of union, he at last evolves order from apparent confusion.

Nor is this all. Rising by a supreme flight of thought to the most abstract speculations, he discovers the laws which regulate all celestial movements, and defines the nature of the universal force which sustains the worlds.

Such are the fruits of the unceasing labors of twenty generations of astronomers. Such the result of the genius and of the patient perseverance of men who have devoted themselves for two thousand years to the study of the phenomena of the heavens. The Chaldean shepherds were, they say, the first astronomers. We can well believe it. Dwelling in the midst of vast plains, where the mildness of the seasons permitted them to pass the night in the open air, where the clear sky unfolded before them perpetually the most glorious scenes, they ought to have been, and they were, contemplative astronomers. And all of us would be what they were did not the rigor of our climate and our variable atmosphere so often prevent us observing the heavens; and did not, moreover, the turmoil and cares of civilized life deprive us of the necessary leisure.

Nothing is more fitted to elevate the mind toward the infinite than the pensive contemplation of the starry vault in the silent calm of night. A thousand fires sparkle in all parts of the sombre azure of the sky. Varied in color and brilliancy, some shine with a vivid light, perpetually changing and twinkling; others, again, with a more constant one—more tranquil and soft; while very many only send us their rays intermittently, as if they could scarce pierce the profundity of space.

To enjoy this spectacle in all its magnificence, a night must be chosen when the atmosphere is perfectly pure and transparent—one neither illuminated by the moon, nor by the glimmer of twilight or of dawn. The heavens then resemble an immense sea, the broad expanse of which glitters with gold dust or diamonds.

In presence of such splendor, the senses, mind and imagination are alike enthralled. The impression gathered is an emotion at once profound and religious, an indefinable mixture of admiration, and of calm and tender melancholy. It seems as if these distant worlds, in shining earthward, put themselves in close communication with our thoughts.

At a first glance at the starry firmament the stars seem pretty regularly distributed; nevertheless, look at that whitish, undecided, vapory glimmer which girdles the heavens as with a belt. It is the Milky Way.^[1] As we approach the borders of this star-cloud in our inspection, the stars appear more and more crowded together, and most of them so small that the eye can scarcely distinguish them. The accumulation of stars in the direction of the Milky Way is more especially visible when we examine the heavens with the aid of a powerful telescope.

The Milky Way itself is nothing more than an immensely extended zone of stars, that is, of suns, since each star, from the most brilliant to the faintest, is a sun.

Here, then, is an immense group, a gigantic assemblage of worlds, which seems to embrace all the universe, if it be true that the greater number of the scattered stars situated out of the Milky Way nevertheless form part of it. In reality, this multitude of millions of suns is divided into numerous and distinct groups, and those into others still more restricted in number, each composed of two or three suns.

What breadth of space does each of these groups occupy? What is the measure of the space which holds them all? The most powerful imagination in vain attempts to answer these questions intelligibly; here numbers fail us.

Let us add—a fact well proved, and one which will seem strange to many —

Our sun himself is a star of the Milky Way.

In examining attentively every part of the starry vault, a keen eye perceives here and there whitish spots resembling little clouds. One would say they were so many patches detached from the Milky Way, from which, however, they are often very distinct and very distant. The telescope discovers by thousands those cloud-patches, these—to give them their astronomical name—*Nebulæ*.

It was formerly imagined that each of these star-clouds was nothing more than an accumulation of stars, very close together, and very numerous—so many Milky Ways lying outside our own, and for the most part so distant that the most powerful instruments were able only to distinguish a confused glimmering. One of the most important observations of modern times,

however, has shown that many of these nebulae, including the most glorious one in our northern hemisphere—that in the sword-handle of Orion—are but masses of glowing gases.

Others, again, of these cloud-like masses—cloud-like by reason of their distance—show us, faintly shining on a background of apparent nebulae, brilliant stars, larger no doubt, or more brilliant, than their fellows, and some of these objects called “Star-Clusters,” which are nearest to us, are among the most glorious objects revealed to us by our telescopes.

Let us attempt now to conceive what fearful distances separate these archipelagoes of worlds from our own!

Unfathomable abysses whose unspeakable depths the most powerful telescopes increase indefinitely! Profound, endless, bottomless, but lighted up by millions of suns!

Such appears to us the universe from the natural observatory where we are placed. But to obtain a more complete idea of its constitution, of the infinite variety of its members, we must descend from those regions, where the sight and mind are lost, to a group, nearer to us, and therefore more accessible to the investigations of man—to that group, or system, of which the earth forms part.

Of this the sun is the centre.

Round this focus of light and heat, but at various distances, revolve more than a hundred secondary bodies—Planets, some of which are accompanied by smaller ones—Satellites. Not self-luminous, they would be invisible to us, if the light, which they receive from the sun, were not reflected toward the earth, making them also appear as luminous points spread over the celestial vault like so many stars. Such would be the appearance of the earth seen in space, at a distance sufficiently great.

A common character distinguishes all the celestial bodies that form part of this group—the Solar System—from the multitude of other stars. For while the suns, composing what is called the Sidereal Universe, are situated at distances seemingly infinite, the bodies composing the group of which we speak are relatively much nearer the earth, are, in fact, our neighbors.

What results from this double fact? Two very simple consequences, easily understood.

The first is, that the stars do not undergo any sensible change of position in the starry vault. Their distance is such that they appear actually at rest in the depths of space; hence the term *Fixed Stars*—now abandoned, because a minute and elaborate study of their relative positions has established the fact that the stars really do move in the remote regions of the heavens. The apparent immobility of which we have spoken, and which is one of their characteristics, is evidenced by the uniformity of appearance preserved for centuries by the artificial groups of stars, to which the name of Constellations has been given.

Now, it is otherwise with the bodies that revolve round our sun: they are near enough to the earth to allow of their displacements in space being perceived in short intervals of time. Traveling, by virtue of their proper motions along the starry vault, distances which appear greater as their own distance from us is less, these bodies received at the outset the name they have since retained—*Planets*, or Wandering Stars.

It is thus that, when we stand in the middle of an extensive plain, we judge distant objects—those that border the horizon—to be immovable; while we instantly perceive the slightest change of place in the near ones. It is true that when we ourselves move, the real movements become complicated with the apparent movements, but the former must be distinguished, if we wish to have an exact idea of the actual course traveled. This complication of the apparent movements of the planets—a necessary consequence of the movement of the earth—is one of the most striking testimonials to the reality of the latter; but it must also be added, that this was precisely the stone of stumbling of ancient astronomy until the time—and that not long ago—when the real movements were made known. Movements of rotation, movements of revolution, around the common centre, the duration of these movements, distances, forms and dimensions, distribution of light and heat, all change in passing from one planet to another. And yet, marvelous thing, the same laws govern, all in such a way that the unity of plan is not less marked than the astonishing variety of the phenomena.

One circumstance common to all the bodies of the solar system forcibly strikes the imagination. It is, that these enormous masses—these globes, many of which are much heavier than the earth, and lastly, the earth itself—

are not only suspended in space, but move through the ether with velocities truly stupendous.

Imagine yourself a spectator, standing immovable in space. A luminous body appears in the distance, little by little you see it approach and increase in size; its immense circumference, which exceeds a hundred thousand leagues, is in rapid rotation, which makes each point on its periphery travel through nine miles a second. The globe itself passes before you, carried through space with a velocity twenty-four times greater than that of a cannon-ball. In such a way Jupiter would appear to you traveling in its orbit. This headlong course would banish it forever to the most remote regions of the visible universe, if it were not subdued and held by the powerful attraction of a globe a thousand times larger than its own—by the sun himself. Not only does astronomy show, by undeniable proofs, the reality of these marvelous movements—not only has she arrived at the knowledge of their invariable constancy, at least during thousands of centuries; but she has found in their very rapidity the cause of the stability of all the celestial bodies.

If there is difficulty in imagining such masses freely circulating in the ether, how much more are we impressed when we consider that these rapid movements are not confined to the planets; and when we look upon the sun with all his retinue as moving in an orbit yet unknown, himself attracted no doubt by a more powerful sun, or by a group of suns! All the stars which by reason of their infinite distances appear immovable, move in different directions; and we shall see later, that if these movements are performed with extreme slowness, the slowness is apparent only. In reality, these are the most rapid celestial movements that we know of.

Thousands of centuries will be necessary before these immense sidereal voyages are accomplished. Their vast periods are to the length of our year what the dimensions of the earth are to the distances of the stars; and, according to the happy expression of Humboldt, they make of the universe an eternal timekeeper. Thus, in the contemplation of celestial phenomena, the idea of infinite duration impresses itself on the mind with the same irresistible power as the idea of the infinity of space.

FOOTNOTES:

[\[1\]](#) Via Lactea. It is also called the Galaxy, from the Greek word for the same thing.

SPACE.—RICHARD A. PROCTOR

Although astronomy tells us in the clearest words of the vast depths of space which surround our earth on all sides, we are not thereby enabled to realize their enormous extension. It is not merely that the unknown depths beyond the range of our most powerful telescopes are inconceivable, but that the parts of space which we can examine are on too large a scale for us to conceive their real dimensions. It is hardly going too far to say that our powers of actual conception are limited to the extent of space over which the eye *seems* to range in the daytime. Of course, in the daytime, at least in clear weather, there is one direction in which the eyesight ranges over a distance of many millions of miles—namely, where we see the sun. But the sense of sight is not cognizant of that enormous distance, and simply presents the sun to us as a bright disk in the sky, or perhaps rather nearer to us than the sky. Even the distance of the sky itself is underestimated. A portion of the light we receive from the sky on a clear day comes from parts of the atmosphere distant more than thirty or forty miles from us; but the eye does not recognize the fact. The blue sky seems a little further off than the clouds, but not much; the light clouds of summer seem a little, but not much, further off than the heavier clouds of a winter sky; a cloud-covered winter sky seems a little further off than heavy rain-clouds. The actual varieties of distance among clouds of various kinds are not much more clearly discerned than the actual varieties of distance among the heavenly bodies. The estimate formed of the distance of a cloud-covered sky overhead probably amounts to little more than a mile, and it is very doubtful whether the mind presents the remotest depths of a blue sky overhead at more than two miles. Toward the horizon the distance seems greater, and probably on a cloudy day the sky near the horizon is unconsciously regarded as at a distance of about five miles, while blue sky near the horizon may be regarded as lying at a distance of six or seven miles, the arch of a blue sky seeming to be far more deeply curved than that of a cloud-covered sky.

It is to distances such as these that the mind unconsciously refers the celestial bodies. We know that the moon is about 2,000 miles in diameter,

but the mind refuses to present her to us as other than a round disk much smaller than those other objects in sight which occupy a much larger portion of the field of vision. The sun can not be conceived to exceed the moon enormously in size, seeing that he appears no larger; and all the multitude of stars are judged by the sight to be mere bright points of light in reality as they appear to be.

How, then, can we hope to appreciate the vastness of space whereof astronomy tells us? To the student of science attempting to conceive the immensities of whose existence he is assured, the same lesson might be taught in parable which the child of St. Augustine's vision taught the Numidian theologian. As reasonably might an infant hope to pour the waters of ocean into a hollow, scooped with his tiny fingers in the sand, as man to picture in his narrow mind the length and breadth and depth of the abysses of space in which our earth is lost.

Yet, as a picture of a great mansion may be so drawn on a small scrap of paper as to convey just ideas of its proportions, so may the great truths which astronomy has taught us about the depths of space be so presented that just conceptions may be formed of the proportions of at least those parts of the universe which lie within the range of scientific vision, though it would be hopeless to attempt to conceive their real dimensions.

When we learn that a globe as large as our earth, suspended beside the moon, would seem to have a diameter exceeding hers nearly four times, so that the globe would cover a space in the heavens about thirteen times as large as the moon covers, we form a just conception of the size of the moon as compared with the earth, though the mind can not conceive such a body as the moon or the earth really is. When, in turn, we are told that if a globe as large as the earth, but glowing as brightly as the sun, were set beside the sun, it would look a mere point of light, we not only learn to picture rightly to ourselves how largely the sun exceeds the earth, but also how enormous must be the real distance of the sun.

Another step leads us to a standpoint whence we can form a correct estimate of the vast distance of the fixed stars; for we can learn that so enormous is the distance of even the nearest fixed star, that the tremendous space separating the earth from that star sinks in turn into the merest point, insomuch that if a globe as bright as the sun had the earth's orbit as a close-

fitting girdle, then this glorious orb (with a diameter of some 184,000,000 of miles) would look very much smaller than such a globe as our earth would look at the sun's distance—would, in fact, occupy but about one-fortieth part of the space in the sky which she, though she would then look a mere point, would occupy if viewed from that distance.

But there is a way of viewing the immensities of space which, though not aiding us indeed to conceive them, enables the mind to picture their proportions better than any other. The dimensions of the earth's path around the sun sink into insignificance beside those of the outermost planets; but these in their turn dwindle into nothingness beside those of some among the comets. From the path of these comets, if only sentient and reasoning beings could trace out in a comet's company those mighty orbits, and could have for the duration of their existence not the brief span of time which measures the longest human life, but many circuits of their comet home around the same ruling orb (as we live during many circuits of our globe around the sun), the dimensions of the star-depths, which even to scientific insight are all but immeasurable, would be directly discernible. Not only would the proportions of that mighty system be perceived, whose fruits and blossoms are suns and worlds, but even the gradually changing arrangement of its parts could be discerned.

Some comets, indeed, do not travel around the sun, but flit from sun to sun on journeys lasting millions of years, paying each sun but a single visit. A being inhabiting such a comet, and having these interstellar journeys as the years of his existence, so that he could live through many of them, would have a wonderful insight into the economy of the stellar system. If his powers of conception as far exceeded ours as the range of his travels and the duration of his existence, he would be able to recognize the proportions of a large part of the stellar universe as clearly as we recognize the proportions of the solar system.

But leaving these wonderful wanderers, whose journeys are as far beyond our powers of conception as the immensity of the regions of star-strewn space, we may find, among the comets belonging to the sun's domain, bodies whose range of travel would give their inhabitants far clearer views of the architecture of the heavens than even the profoundest terrestrial astronomer can possibly obtain.

Such a comet as Halley's, for instance, though one of comparatively limited range in space, yet travels so far from the sun that, from the extreme part of its path, it sees the stars displaced nearly twenty times as much (owing to its own change of position) as they are from the earth on opposite sides of her comparatively narrow orbit. And the length of this comet's year, if it indicated the lives of all creatures traveling along with it, would suggest a power of patiently watching the progress of changes lasting not a few of our years only, but for centuries. Seventy-five or seventy-six years elapse between each return of this comet to the sun's neighborhood, and one who should have lived during sixty or seventy circuits of this body around its mighty orbit would have been able to watch the rush of stars, with their velocities of many miles per second, until visible displacements had taken place in their positions.

This, however, is as nothing compared with the mighty range in space and the enormous period of the orbit of the great comet of the year 1811. This comet is, on the whole, the most remarkable ever known. It was visible for nearly seventeen months, and though it did not approach the sun within 100,000,000 miles, and was therefore not subject to that violence of action which has caused enormous tails to be thrown out from comets which have come within a few million miles of him, or even within less than a quarter of his own diameter, it flourished forth a tail 120,000,000 of miles in length. Its orbit has, according to the calculations of the astronomer Argelander, a space exceeding the earth's distance from the sun 211 times, and thus surpassing even the mighty distance of Neptune fully seven times. It occupies in circuiting this mighty path no less than 3,065 of our years (with a possible error either way of about forty-three years). So that, according to Bible chronology, this comet's last appearance probably occurred during the rule of the Judge Tola, son of Puah, son of Dodo, over the children of Israel, though it may have occurred during the rule of his predecessor Abimelech, or during that of his successor Jair.^[2] During one-half of the enormous interval between that time and 1811 the comet was rushing outward into space, reaching the remotest part of its path somewhere about the year 278 (A. D.), and from that time to 1811 it was on its return journey. It is strange to think, however, that though the remotest part of its path lay 211 times further from the sun than the earth's orbit, yet even this mighty path, requiring more than 3,000 years for a single circuit, can not be said to have carried the comet into the star-depths. If the earth were to shift its position

by the some enormous amount, the nearest fixed star would have its apparent position changed only by about an eighth part of the apparent diameter of the sun or moon, or by about one-quarter of the distance separating the middle star of the Bear's tail from its close companion.

But this fact of itself is most strikingly suggestive of the vast distance of the stars. For consider what it means. Imagine the middle star of the Bear's tail to be the really nearest of all the stars instead of lying probably twenty or thirty times further away. Conceive a comet belonging to that sun after making its nearest approach to it to travel away upon an orbit requiring 3,000 years for each circuit. *Then* (supposing that star equal to our sun in mass) the comet, though rushing away from its sun with inconceivable velocity during 1,500 years, would, at the end of that vast period, seem to be no further away than one-fourth of the distance separating the sun from its near companion. Look at the middle star of the Bear's tail on any clear night, and on its small satellite, remembering this fact, and the awful immensity of the star-depths are strongly impressed upon the mind. But the observer must not fail to remember that the star really is many times more remote than we have here for a moment supposed, and that such a comet's range of travel would be proportionately reduced. Moreover, many among the stars are doubtless hundreds, even thousands, of times still further away.

Let us turn lastly to the amazing comet of the year 1744. We find that though it had the longest period of any which has ever been assigned to a comet as the result of actual mathematical calculation, yet its range in space would scarcely suffice to change the position of the stars in such sort that the aspect of the familiar constellations would be materially altered. Euler, the eminent mathematician, calculated for this comet a period of 122,683 years, which would correspond, I find, to a distance of recession equal to 2,469 times the distance of the earth from the sun, or about eighty times the distance of Neptune. Yet this is but little more than twelve times the greatest distance of the comet of 1811. Probably the actual range of such an orbit from the middle star of the Bear's tail would be equal in appearance to the range described above on the supposition that the star is no further from us than the nearest known star (Alpha Centauri). That is, such a comet, if it could be seen and watched during a period of about 122,000 years, would seem to recede from the star to a distance equal to about one-fourth the

space separating it from its close companion, and then to return to the point of nearest approach to its ruling sun.

Such are the immensities of star-strewn space! The journey of a comet receding from the sun with inconceivable velocity during hundreds of thousands of years carries it but so small a distance from him compared with the distance of the nearest star as scarcely to change the appearance of the celestial landscape; and yet the distances separating the sun from the nearest of his fellow suns are but as hairbreadths to leagues when compared with the proportions of the scheme of suns to which he belongs. These distances, though so mighty that by comparison with them the inconceivable dimensions of our own earth sink into utter nothingness, do not bring us even to the threshold of the outermost court of that region of space to which the scrutiny of our telescopes extends. Yet the whole of that region is but an atom in the infinity of space.

FOOTNOTES:

[2] It might be suggested that the appearance of this blazing comet among the stars drove the more superstitious of the Israelites at that time to the worship of star-gods, as we read how, during the Judgeship of Jair, they “served Baalim and Ashtaroth, and the gods of Syria and the gods of Moab, and the gods of the Philistines, and forsook the Lord and served not Him.” To a people like the Jews, who seem to have been in continual danger of returning to the Sabaistic worship of their Chaldean ancestors, the appearance of a blazing comet may have been a frequent occasion of backsliding.

EXTENT OF THE SIDEREAL HEAVENS.—

SIR ROBERT S. BALL

Of all the discoveries that have ever been made in science there are two which especially baffle our powers of comprehension. They lie at the opposite extremes of nature. One relates to objects which are infinitely small, the other relates to objects which are almost infinitely great. The microscope teaches us that there are animals so minute that if a thousand of them were ranged abreast they would easily swim without being thrown out of line through the eye of the finest cambric needle. Each of those minute creatures is a highly organized number of particles, capable of moving about, of finding and devouring its food, and of behaving in all other respects as becomes an animal as distinguished from an unorganized piece of matter. The mind is capable of realizing the structure of these little creatures, and of fully appreciating their marvelous adaptation to the life they are destined to lead. If these animals excite our astonishment by reason of their extreme minuteness, there is an appeal made to conceptions of an entirely different character when we learn the lessons which the telescope teaches. As the microscope reveals the excessively minute, so does the telescope disclose the sublimely great. In each case myriads of objects are submitted to our astonished view, but while the microscope brings before us creatures of which countless millions could swim about freely in a thimbleful of water, the telescope conducts our vision to uncounted legions of stars, many of them millions of times larger than the earth.

The grandest truth in the whole of nature is conveyed in that first lesson in astronomy which answers the question: What are the stars? This is a question that a child will ask, and I have heard of a child's pretty idea that the stars were little holes in the sky to let the glory of heaven shine through. The philosopher will replace this explanation by another hardly less poetical, which will enable us to form some more adequate notion of the real magnificence of the universe. Each star that we see is, it is true, only a glittering little point of light, but that is merely because we are a long way from it. An electric light which will dazzle your eye when quite close will be reduced to an agreeable illumination if it is at a little distance, will

become a faint light a mile away, and at no great distance will become altogether invisible. We must remember that out in space there is plenty of room—there are no bounds; and therefore when we see light glistening in the far distant depths we can not at once conclude that the light is a faint one because it appears to us to be faint. It may be that the light is only faint because it comes from such a tremendous distance. In fact, the brightest light conceivable could be reduced to the insignificance of a small star if only it were removed sufficiently far.

The most intense light we know of comes, of course, from the light which rules by day, from our sun himself. The sun pours his unrivaled beams around us in all directions with prodigal abundance, notwithstanding his enormous distance of ninety-three millions of miles. Let me describe an experiment with respect to our sun, an experiment, it is needless to say, which could never be performed, but the results to which it leads us are none the less certain. Astronomers have demonstrated them in many other ways.

Suppose that the sun were gradually to be moved away further and further into space; suppose that by this time to-morrow the great luminary should be twice as far as it is now, and the next day should be three times as far, and the day after that four times, and so on until in a year's time we should find that the sun was 365 times the distance from us that it is at present. Let us now trace the changes which we should see in the brilliancy of our orb of day. When he had reached double his distance from us, we should find that the light had decreased to a quarter of its present amount, and the heat which we derived from his beams would have decreased in the same proportion. In ten days we should find that the light had become so feeble as to be only one-hundredth part of that which we enjoy now. The apparent size of the sun would also be steadily decreasing, for as the distance of a body increases its apparent dimensions diminish. Sometimes the diminution of apparent size with distance is well illustrated on a clock tower. You would hardly believe that the hands and face of a clock like that at Westminster were so large until you happen to see a man cleaning or repairing it, when he appears a mere pigmy in comparison with the mighty dial which points out the hours. In a similar way with every increase of distance, the apparent size of the sun would decline, and in the lapse of a year the sunlight would be reduced to a feeble twilight. The sun itself would

remain visible for many years, even if it were steadily moving away, though its lustre would continually decline, and its size would continually diminish, until at last it would have shrunk to the insignificance of a small point of light, still visible as a glittering object, but too minute to enable any definite form to be perceived.

Further still, the sun might recede until it passed beyond the reach of vision of the unaided eye; the telescope would, however, be able to pursue the retreating luminary until at last it sank into the depths of space beyond the reach of any instrument whatever.

This little argument will prepare us for an explanation of the stars. They merely appear to us to be points of light of varying degrees of brightness, but we have seen that our own sun might be reduced in lustre to that of the very dimmest of the stars if only it were removed sufficiently far. If, therefore, the stars are at a great enough distance from our system, it may indeed be that they also are suns, possibly equaling, or possibly even surpassing, our own sun in magnificence.

Here is indeed an imposing suggestion. Can it be that the host of stars which adorn our midnight sky are actually suns themselves of an importance comparable with that of our own? This is a great thought, and we desire to test it by every means in our power. You will see from the reasoning I have given that the whole question turns simply on one point, and that is: How far off are the stars?

The tiniest point of light that is just seen as a glimmer in the mightiest of telescopes may be indeed a sun as great, or indeed a million times greater, than our sun, if only that star be sufficiently far off. To find the distance of a star is a problem which taxes the utmost powers of the painstaking astronomer; every refinement of skill in making his measurements and of care in the calculation of his observations have to be lavished on the operation. Alas! it but too often happens that the astronomer's labors prove to be futile. The surveying navigator often has to mark on his chart that no bottom could be found in the depths of the sea. His appliances would not work, or work reliably, in those ocean abysses; so, too, the astronomer, when he tries to sound the depths of space to the distances of the stars, has also to mark, generally speaking, "No bottom here," as the result of most of his investigations. When this is the case we know for certain that the star on

which his calculations have been made must be a gorgeous sun, because we are assured of the greatness of its distance, even though we have not been able to find out what that distance was. There are, however, some few places through the sky where the astronomer's sounding line can, so to speak, touch bottom; there are a few stars of which we do know the distance, and the result is not a little significant. Were our sun to be withdrawn from us to a distance so great as that of the very nearest of the stars, our magnificent ruler and benefactor would certainly have lost all his splendor; he would, in fact, have shrunk to the similitude of a little star not nearly so bright as many of those which we see over our heads every night. Imagine the sun's light subdivided into two hundred thousand parts, each of which would give us only a feeble illumination, and then imagine that each of these parts was again divided into two hundred thousand parts more, and it is one of these last fragments that would represent the miserable lustre which the sun would then display.

From these considerations we can enunciate the magnificent truth which astronomy discloses to us. I do not think that in the whole range of nature there is any thought so magnificent or so imposing as that which teaches us to regard every star of every constellation as a sun. We can not indeed assert that they are all so great as our sun, but we can affirm with certainty that many of them are far greater and far more splendid. Considering that our sun presides over a system of worlds of which the earth is one, that it gives light and heat to those worlds, and guides them in their movements, it would greatly enlarge our conceptions of the universe if we were assured that there was even one more sun as large and as splendidly attended as is our own. But now we find that not only is there one additional sun, but that they teem in uncounted thousands through space. Look, for example, on the next fine night at the Great Bear, the best known of all our northern constellations, and there you see seven stars forming the well-known feature. Figure in your mind's eye each one of those stars in the likeness of a majestic sun, as big, warm, and bright as our sun, and look at other parts of the sky and repeat the process with the other constellations, and your conception of the magnificence of the starry system will begin to assume proper proportions. But this is only the first step, you must next look at the smaller stars, and reflect that they, too, are also suns, only much further off as a general rule than the brighter stars, though this is by no means invariably the case. Thus your estimate of the number of suns in the

universe will rise to thousands, but you will not stop there, you will get a telescope to help you, and, to your extreme delight and wonder, you will find that there are hosts of stars—too faint to be visible to the eye, but which the telescope will immediately disclose. You will get a more powerful instrument, and then you will perceive that the stars are to be numbered by tens of thousands, and even by millions, and with every fresh accession of power in your telescope fresh troops and myriads of suns are revealed. Suns in clusters, suns strewn thickly here and sparsely there, so as to give us the notion that the only limit to the number we can see is the power of the telescopes we are using. Attempts at actual numeration are futile, for who can tell the number of the stars?

We can, however, form an estimate, and by taking samples, so to speak, of the sky here and other samples there, we have been enabled to learn the overwhelming fact that our universe does contain at the very least one hundred millions of suns.

In discussing the extent of the visible universe, it must always be borne in mind that the further a source of light is from us the fainter is the illumination which we receive from it. Suppose that a star which just lies on the limits of naked-eye visibility were somehow to be transported to a distance which is twice as great, then the lustre of that star would be diminished to one-fourth of its original amount. It would, therefore, be of course invisible to the unaided eye, but could still be easily perceived by a telescope. Indeed, the very word *telescope* means an instrument for looking at objects a long way off, and the effect of the telescope is to reduce the apparent distance of the object.

The bulk of a grain of sand as compared with the bulk of a football may illustrate the space accessible to our eyes when compared with the space accessible to one of the great telescopes. The larger of these spaces has a thousand times the diameter of the others; therefore, the relative quantities of these spaces are to be obtained by multiplying 1,000 by 1,000 and by 1,000 again. Thus we finally learn that the amplitude of our vision is augmented to one thousand million times its original extent by the use of our greatest telescopes. It need, therefore, be no matter for surprise that the number of stars visible through our great telescopes or recorded on the sensitive films of photographic plates should number scores of millions. In fact, it would sometimes seem surprising that the number of telescopic stars

is not even greater than it actually appears to be. If we are able to explore one thousand million times as much space, we might expect that the number of objects disclosed would be also increased about a thousand million-fold, but this is certainly not the case. The truth seems to be that our sun is but one star of a mighty cluster of stars; we happen to lie near the middle of the cluster, and the rest of the stars belonging to it form what we know as the Milky Way. There are, of course, other clusters scattered through the heavens, some of them, perhaps, as great as that body of stars which forms the Milky Way. Owing to our residence in this cluster we see the neighboring suns in multitudes, and thus we receive the impression that the solar system lies in an exceptionally rich part of the universe in as far as the distribution of stars is concerned.

On the outskirts of the universe lie those faintest and dimmest of objects which we can just perceive through our greatest telescopes. We know that many of the stars around us would still remain visible in great instruments, even though they were removed a thousand times as far off. Among the myriads of faint stars which we see from our observatories, there may be many, indeed there must be many, which are fully a thousand times as distant as the bright stars which twinkle in our comparative neighborhood. We thus obtain some conception of the stupendous distance at which the outskirts of the universe are situated.

There are different ways of illustrating this point, but I think the simplest, as well as the most striking, is that which is founded on the velocity of light. It is a remarkable fact that the beautiful star known as Vega^[3] has a distance from us so tremendous that its light must have taken somewhere about eighteen years to travel hither from thence. Notwithstanding that the light dashes along with such inconceivable speed that it will cover 185,000 miles in every second, notwithstanding that a journey at this pace will complete the entire circuit of this globe seven or eight times between two successive ticks of the clock, the light will, nevertheless, take eighteen years to reach our eye from the time it leaves Vega. We do not, therefore, see the star as it is at present; we see it as it was eighteen years ago. For the light which this evening enters our eyes has been all that time on its journey. Indeed, if Vega were actually to be blotted out from existence it would still continue to shine out as vividly as ever for eighteen years before all the light on its way had reached us.

We have been led to the belief that among the more distant stars in the universe there must be many which are fully a thousand times as far from us as is Vega, hence we arrive at the startling conception that the light they emit has been on its journey for 18,000 years before it reached us. When we look at those lights to-night we are actually viewing them as they were 18,000 years ago. In fact, those stars might have totally vanished 17,000 years ago, though we and our descendants may still see them glittering for yet another thousand years.

We shall realize a little more fully what this reasoning involves if we suppose that astronomers dwelt on such a star, and that they had eyes and telescopes sufficiently keen not only to discern our little earth, but even to scrutinize its surface with attention. Let us suppose that the stellar astronomers looked at England: do you think they would see a network of railways joining mighty and populous cities, furnished with immense manufactories and with countless institutions? Such would be the England of to-day. But from the distance at which these astronomers are situated light takes 18,000 years for its journey, and, therefore, what they would see would be England as it was 18,000 years ago. To them England would even now appear as a country mainly covered with forests inhabited by bears and wolves, and totally void of any trace of civilization. This illustration will, at all events, serve to convey some conception of the distance at which the outskirts of our visible universe are plunged in the depths of space.

FOOTNOTES:

[3] Vega is the brightest star in the Lyre and is nearly always at night directly overhead in our latitude.—E. S.

THE STARS.—AMÉDÉE GUILLEMIN

No sight is at once so awe-inspiring and so grand as that of the heavens on a beautiful night. If care be taken to choose as a standpoint for observation an open place, such as a plain or the summit of a hill on land, or, again, the open sea, and if the atmosphere, somewhat charged with dew, possesses all its transparency and purity, we shall see thousands of luminous points twinkling in all directions, accomplishing slowly and together their silent march. The contrast of the obscurity which reigns on the surface of the earth with the brightness of that resplendent vault gives an indefinite depth to the celestial ocean that deepens over our heads. But let us here leave the magnificence of the spectacle to study it in its most minute details.

Let us commence with the appearances. A characteristic common to all the stars is an incessant and very rapid change of brightness, which has received the name of *scintillation*. This is accompanied by variations of color equally rapid, due to the same cause as the successive disappearances and reappearances. All stars scintillate, whatever may be their brilliancy, at least in our temperate regions. But the intensity of this luminous movement is not the same in all, and it varies, moreover, both with the degree of purity of the sky, the elevation of the stars above the horizon, and the temperature of the night.

According to Arago, scintillation is due to the difference of velocity of the various colored rays traversing the unequally warm, unequally dense, unequally humid atmospheric strata. Thus, in tropical regions, where the atmospheric strata are more homogeneous, scintillation is rarely observed in stars the elevation of which above the horizon is more than 15° , or the sixth of the distance of the horizon from the zenith. "This circumstance," says Humboldt, "gives to the celestial vault of these countries a particularly calm and soft character."

Another specific character of the stars is that their diameters are without appreciable dimensions. To the naked eye, this distinction would be insufficient, since, the moon and the sun excepted, the most considerable planets have not sensible diameters. But, while the magnifying power of

optical instruments shows us the principal planets under the form of clearly defined disks, the most powerful glasses only show a star as a luminous point. The distance which separates us from these bodies is so great that there is nothing to astonish us in such a result.

Wollaston affirms that the apparent diameter of the most brilliant star in the heavens, Sirius, is not more than the fiftieth part of a second of an arc. But let us hasten to say that this result still leaves a good margin as to the real dimensions of the star, since, at the distance of Sirius, an apparent diameter would represent a real diameter of 11,000,000 miles; that is, twelve times the diameter of our sun.

Let us add, lastly, that the absence of appreciable dimensions does not suffice to distinguish absolutely the stars from the planets, since a certain number of the latter, as we have before seen, appear in telescopes only as simple luminous points. Let us come, then, to a permanent specific characteristic, the knowledge of which will always prevent us from confounding a star with one of the known or unknown bodies which form part of our solar group. This characteristic is as follows:

The stars, properly so called, preserve among themselves—nearly enough for our present purpose—the same relative distances. They form, then, on the celestial vault apparent groups, the configuration of which is nearly invariable. Centuries must elapse to show a change of form, unless we employ extremely delicate measures. A planet, on the contrary, moves rapidly across these groups, to such a degree that, in the interval of a night, or at most of a few nights, this displacement is very perceptible; hence the old denomination of *fixed stars*, in opposition to the *wandering* ones, or planets.

We must be careful, however, to guard against assigning to this word a rigidity which it does not possess, for the stars really move with a velocity not inferior to that which animates the members of our system. Their immense distance is the only cause of their apparent immobility, which vanishes when precise observations, embracing a sufficient interval of time—some years, for example—are made.

A fact which strikes every one is the great diversity of brightness in the stars which people the heavens. All degrees of intensity are remarked, from

the resplendent light of Sirius to the scarcely perceptible glimmer of those hardly visible to the naked eye.

Whence arises this difference of brightness? This question we can not answer for any star in particular, but it is easy to imagine that it may result from various circumstances, such as their less or greater distance, the real and various dimensions of the bodies, and, lastly, the intrinsic brightness of the light peculiar to each. However this may be, astronomers without regard to the unknown causes which may influence the intensity of the stellar light, have divided stars into *classes* or *magnitudes*; and when we speak of a star of the first, second, or fifth magnitude, it is understood that this way of speaking refers only to the apparent brightness, and that nothing is affirmed either as to the real dimensions or distance, or even intrinsic brightness.

Besides, as the stars, arranged in the order of their brightness, would form a progression decreasing by imperceptible degrees, the classes adopted are themselves conventional and arbitrary. The first six magnitudes comprise all stars visible to the naked eye. But the use of the most powerful telescopes brings to view stars of feebler light, descending to the sixteenth and seventeenth magnitudes. In truth, the progression has no inferior limit: it extends more and more in proportion as the progress of the optician's art increases the penetrating power of our instruments.

To gain an idea of the respective intensities of the light emitted by the stars of the first six magnitudes, following the scale adopted by astronomers, the accompanying illustration (Fig. 1), should be inspected; in it the stars are figured by disks, the surfaces of which are in proportion to their brilliancy.

But, we repeat, it must not be thought that the stars ranked in the same class are, on that account, of the same brightness. Thus the light of Sirius is estimated at four times the star Alpha Centauri; but both, nevertheless, are included by astronomers in the number of the stars of the first magnitude.

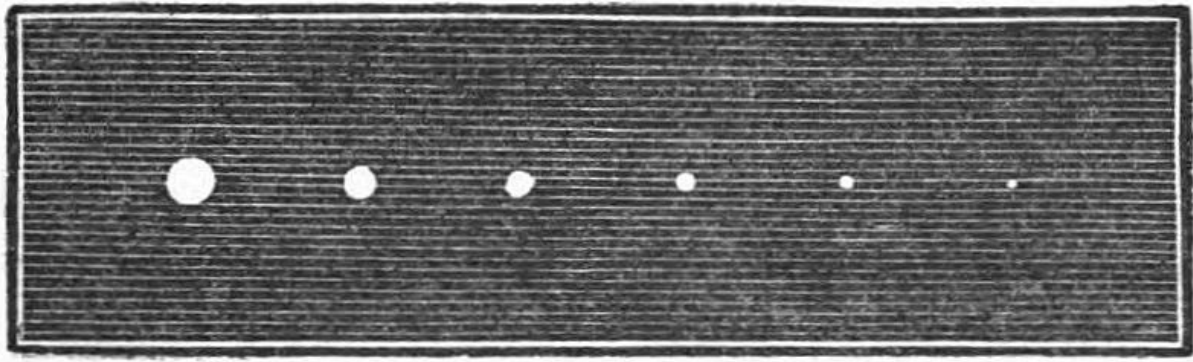


Fig. 1.—Relative Brilliancy of Stars of the first Six Magnitudes

We here give the names of the twenty most brilliant stars of the two hemispheres which it is usual to consider as forming the first class. They are here arranged in the order of their brightness:

- | | |
|-------------------|-------------------|
| 1. Sirius | 11. Achernar |
| 2. Eta Argus | 12. Aldebaran |
| 3. Canopus | 13. Beta Centauri |
| 4. Alpha Centauri | 14. Alpha Crucis |
| 5. Arcturus | 15. Antares |
| 6. Rigel | 16. Altair |
| 7. Capella | 17. Spica |
| 8. Vega | 18. Fomalhaut |
| 9. Procyon | 19. Beta Crucis |
| 10. Betelgeuse | 20. Pollux |

Lastly, Regulus, a bright star in the constellation of the Lion, is also ranked by some astronomers in the first magnitude, while others only admit in this class the first seventeen stars in the above list. These divergences are of no importance.

In proportion as the scale of brilliancy or magnitude is descended, the number of the stars contained in each class rapidly increases. The number of second magnitude stars in the heavens is about 65; of the third, about 200; of the fifth, 1,100; and of the sixth magnitude, 3,200. Adding these numbers together, we obtain a few over 5,000 stars of the first six magnitudes, and these comprise very nearly all those that can be seen with the naked eye.

The smallness of this number nearly always astonishes those who have not tried to form an exact estimate of the number of stars which shine in the celestial vault on the most favorable nights.

The aspect of the multitude of sparkling points which are scattered over the sky makes us disposed to believe that they are innumerable, and to be counted, if not by millions, at all events by hundreds of thousands. This is, nevertheless, an illusion. All observers who have taken the trouble to make an exact enumeration of the stars visible to the naked eye have arrived at a maximum of 3,000 as the mean number which can be observed in every part of the heavens, visible at the same time, at the same place; this, of course, is but half of the entire heavens.

Argelander has published an exact catalogue of the stars visible on the horizon of Berlin during the course of the year. This catalogue comprises 3,256 stars. According to Humboldt, there are 4,146 visible on the horizon of Paris in the whole course of the year; and as this number increases in proportion as we approach the Equator, that is to say, in proportion as the double movement of the earth unfolds to us during a year a more extensive portion of the heavens, 4,638 stars are already visible to the naked eye on the horizon of Alexandria.

We repeat, the maximum number is comprised between 5,000 and 6,000 stars for the entire heavens, including those seen by the most piercing and most accustomed eyes in the best nights for observation. When the atmosphere is lit up by the moon, or by twilight, or, as happens in the great centres of population, by the illumination of the houses and streets, the lowest magnitude stars are effaced altogether, and the number of those visible is consequently much more limited. We may add in conclusion, that the more the scintillation, the more easy it is to distinguish very faint stars.

A word now on the number of stars that can be seen with the help of the telescope. Here we shall find the numbers which our imagination had erroneously led us to believe are visible to the naked eye.

According to the illustrious director of the Observatory of Bonn—Argelander—the seventh magnitude comprises nearly 13,000 stars; the eighth, 40,000; and, lastly, the ninth, 142,000. The calculations of Struve give the total number of stars visible in the entire heavens by the aid of Sir William Herschel's 20-foot reflector as more than 20,000,000. But, without

doubt, these approximate numbers are much below the real ones. It will be seen, besides, that the richness of the heavens in stars is very unequal. The bright zone known under the name of the Milky Way alone contains, according to Herschel, 18,000,000.

THE LUCID STARS.—J. E. GORE

The term “lucid” has been applied to the stars visible to the naked eye, without optical aid of any kind.^[4] Many people think that the number of stars visible in this way is very large. But in reality the number visible to the naked eye is comparatively small. Some persons are, of course, gifted with very keen eyesight—“miraculous vision” it is sometimes called—and can see more stars than others; but to average eyesight the number visible in this way, and which can be individually counted, is very limited. The famous Hipparchus formed a catalogue of stars in the year 127 B. C. This presumably contained all the most conspicuous stars he could see in his latitude, and it includes only 1,025 stars. Al-Sûfi, the Persian astronomer, in his *Description of the Fixed Stars*, written in the Tenth Century, describes the positions of only 1,018 stars, although he refers to a number of other faint stars, of which he does not record the exact places. Pliny thought that about 1,600 stars were visible in the sky of Europe.

In modern times, however, a considerable number of fainter stars have been recorded as visible to the naked eye. The famous German astronomer, Heis, who had keen eyesight, records the positions of 3,903 stars north of the Equator, and 1,040 between the Equator and 20 degrees south declination, or a total of 4,943 stars between the North Pole and 20 degrees south of the Equator. This would, I find, give a total of about 7,366 stars for both hemispheres if the stars were equally distributed. Behrmann, in his *Atlas of Southern Stars*, between 20 degrees south declination and the South Pole, shows 2,344 stars as visible to the naked eye. This would give a total of 7,124 for both hemispheres. The actual number seen by Heis and Behrmann in both hemispheres is $4,943 + 2,344$, or 7,287 stars. The Belgian astronomer, Houzeau, published a catalogue and atlas of the stars in *both* hemispheres, made from his own observations in Jamaica and South America, and finds a total of 5,719 stars in the whole sky. As all these observers had good eyesight, we may take a mean of the above results as the total number visible to the naked eye in the whole star sphere. This gives 6,874 stars, or in round numbers we may say that there are about 7,000 stars visible to average eyesight in both hemispheres. This gives, of

course, about 3,500 stars to one observer at the same time at any point on the earth's surface.

As the whole star sphere contains an area of 41,253 square degrees, we have an average of one star to six square degrees. In other words there is, *on an average*, one lucid star in a space equal to about thirty times the area covered by the full moon! This result may seem rather surprising considering the apparently large number of stars visible to the naked eye on a clear night, but the fact can not be denied. The stars are not, of course, equally distributed over the surface of the sky, but are gathered together in some places, and sparsely scattered in others, and this may perhaps help to give the impression of a greater number than there really are.

That the stars are of various degrees of brightness was recognized by the ancient astronomers. Ptolemy divided them into six classes, the brightest being called first magnitude, those considerably fainter the second, those much fainter still the third, down to the sixth magnitude, which were supposed to be the faintest just visible to the naked eye on a clear moonless night. Ptolemy only recorded whole magnitudes, but Al-Sûfi, in the Tenth Century, divided these magnitudes, for the first time, into thirds. Thus a star slightly less than an average star of the second magnitude he called 2—3, that is nearer in brightness to 2 than to 3; one a little brighter than the third he recorded as 3—2, or nearer to 3 than to 2, and so on. This method has been followed by Argelander, Behrmann, Heis, and Houzeau, but in the photometric catalogues of Harvard, Oxford, and Potsdam the magnitudes are measured in decimals of a degree. This has been found necessary for greater accuracy, as the heavens contain stars of all degrees of brightness.

The term “magnitude” means the ratio between the light of a star of a given magnitude and that of another exactly one magnitude fainter. This ratio has been variously estimated by different astronomers, and ranges from 2.155, found by Johnson in 1851, to 3.06, assumed by Pierce in 1878. The value now universally adopted by astronomers is 2.512 (of which the logarithm is 0.4). This number is nearly a mean of all the estimates made, and agrees with the value found by Pogson in 1854 by means of an oil flame, and by Rosen with a Zöllner photometer in 1870. It simply means that an average star of the first magnitude is 2.512 times the brightness of a star of the second magnitude; a star of the second, 2.512 times brighter than

one of the third, and so on. This makes a star of the first magnitude just 100 times brighter than one of the sixth.

There are several stars brighter than an average star of the first magnitude, such as Aldebaran. These are Sirius, which is nearly 11 times brighter than Aldebaran (according to the revised measures at Harvard); Canopus, the second brightest star in the heavens, and about two magnitudes brighter than Aldebaran; Arcturus, Capella, Vega, Alpha Centauri, Rigel, Procyon, Alpha Eridani, Beta Centauri, and Alpha Orionis. Al-Sûfi rated 13 stars of the first magnitude, visible at his station in Persia, and Halley enumerates 16 in the whole sky. According to the Harvard photometric measures, there are 13 stars in both hemispheres brighter than Aldebaran, which is rated 1.07.

As average stars of the different magnitudes the following may be taken as examples, derived from the Harvard measures: First magnitude, Aldebaran and Spica; second magnitude, β Aurigæ and β Canis Majoris; third magnitude, ι Aurigæ and β Ophiuchi; fourth magnitude, θ Herculis and ϵ Draconis; and fifth magnitude, ρ Ursæ Majoris and ω Sagittarii. Stars of about the sixth magnitude are, of course, numerous, and lie near the limit of naked-eye vision for average eyesight, although on clear moonless nights still fainter stars may be “glimpsed” by keen-eyed observers.

The stars have been divided into groups and constellations, now chiefly used for the purpose of reference, but in ancient times they were associated with the imaginary figures of men and animals, etc. The origin of these constellation figures is doubtful, but they are certainly of great antiquity. Ptolemy's constellations were 48 in number, but different writers from the First Century B. C. give different numbers, ranging from 43 to 62. Bayer's *Uranometria*, published in 1603, contains 60, 12 new constellations in the Southern Hemisphere having been added by Theodorus to Ptolemy's original 48.

The figures representing the constellations were originally drawn on spheres, or celestial globes, as they are now called. The ancient astronomers attributed the invention of the sphere to Atlas. It seems certain that a celestial sphere was constructed by Eudoxus in the Fourth Century B. C. Strabo speaks of one made by Krates about the year 130 B. C., and according to Ovid, Archimedes had constructed one at a considerably

earlier period. None of these ancient spheres has been preserved. There is, however, in the Vatican a fragment in marble of a Græco-Egyptian planisphere, and a globe in the museum of Arolsen, but these are of much later date. Our knowledge of the original constellation figures is derived from the accounts given by Ptolemy and his successors, and from a few globes which only date back to the Arabian period of astronomy. Among the Arabian globes still existing the most famous is one made of copper, and preserved in the Borgia Museum at Velletri in Italy. It is supposed to have been made by a person called Caisar, who was executed by the Sultan of Egypt in A. D. 1225. The most ancient of all is one discovered some years ago at Florence. It is supposed to date back to A. D. 1081, and to have been made by Meucci. There is also one in the Farnese Museum at Naples, made in A. D. 1225. Of modern celestial globes the oldest is one made by Jansson Blaeu in 1603. This gives all the constellations of the Southern Hemisphere as well as the Northern.

Ptolemy's figures of the constellations were restored by the famous painter Albert Dürer of Nuremberg in 1515. The figures on modern globes and maps have been copied from this restoration. Dürer's maps are now very rare.

In 1603, an atlas was published by Bayer. This was the first atlas to show the southern sky, and the first to designate the brightest stars by the letters of the Greek alphabet.^[5] Flamsteed published an atlas in 1729. Maps and catalogues of the lucid stars have been published in recent times by Argelander, Behrmann, Heis, Houzeau, Proctor, and others. Of these Heis's is, perhaps, the most reliable, at least so far as accurate star magnitudes are concerned. Houzeau shows *both* hemispheres, all the stars had been observed by himself in Jamaica and South America. Behrmann's maps are confined to the Southern Hemisphere, between the South Pole and 20 degrees south of the Equator. The maps of the *Uranometria Argentina*, made at Cordoba in the Argentine Republic, show all the southern stars to the seventh magnitude, but many of these are beyond the reach of ordinary eyesight.

It is a well-known fact that the planets Venus and Jupiter are bright enough to form shadows of objects on a white background. It has also been found that the brightest stars, especially Sirius, are sufficiently brilliant to

cast shadows. Kepler stated that a shadow was formed by even Spica, but I am not aware that this has been confirmed by modern observations.

There are some remarkable collections or clusters of stars visible to the naked eye, of these the Pleiades are probably the best known. To ordinary eyesight 6 stars are visible, but Möstlin, Kepler's tutor, is said to have seen 14 with the naked eye, and some observers in modern times have seen 11 or 12. Other naked-eye clusters are the Hyades in Taurus, called *Palilicium* by Halley, and the Præsepe, or Bee-Hive in Cancer. Of larger groups, the Plow or Great Bear, Cassiopeia's Chair, and Orion are probably known to most people.

Many of the lucid stars are double, that is, consist of two components, but most of these are only visible in powerful telescopes. There are, however, a few objects visible to the naked eye as double, and these have been called "naked-eye doubles," although not strictly double in the correct sense of the term.

Ptolemy applied the term double to the star ν Sagittarii, which consists of two stars separated by a distance of fourteen minutes of arc, or about half the apparent diameter of the moon. According to Riccioli, Van der Hove saw two naked-eye doubles, one in Capricornus, 5 to $5\frac{1}{2}$ minutes distant, and the other in the Hyades, $4\frac{1}{2}$ or 5 minutes apart. The one in Capricornus was probably α , and the one in the Hyades θ Tauri. The middle star in the tail of the Great Bear, or handle of the Plow, has near it a small star, Alcor, which to many eyes is distinctly visible without optical aid. The famous Belgian astronomer, Houzeau, who seems to have had excellent sight, saw the star χ Tauri double, and 51 and 56 Tauri separated, also ι Orionis, and others.

Many of the stars are variable in their light, and several hundred of these curious and interesting objects are now known to astronomers. In a few of these the light changes may be followed with the naked eye. It is an interesting question whether any of the lucid stars have disappeared or changed in brightness since the early ages of astronomical observations. Al-Sûfi failed to find seven of Ptolemy's stars, and Ulug Bekh, comparing his observations with the catalogues of Ptolemy and Al-Sûfi, announced twelve cases of supposed disappearance. Some of these may, however, be due to errors of observation. Montanari, writing in 1672, mentions two stars as

having disappeared, namely β and γ of the constellation Argo, but these stars are now visible in the positions originally assigned to them.

In a careful examination of Al-Sûfi's description of the stars written in the Tenth Century, and a comparison with modern estimates and measures, I have found several very interesting cases of apparent change in the brightness of the lucid stars. Al-Sûfi was an excellent and careful observer, and as a rule his estimates agree well with modern observations. We can therefore place considerable reliance on his estimates of star magnitudes. *The Story of Theta Eridani* has been well told by Dr. Anderson, and there seems to be no doubt that this southern star, which is now only of the third magnitude, was a bright star of the first magnitude in Al-Sûfi's time! The following are other interesting cases of apparent change which I have met with in my examination of Al-Sûfi's work. The Pole Star was rated third magnitude by both Ptolemy and Al-Sûfi, but it is now of the second magnitude, or a little less. The star γ Geminorum was rated third magnitude by Ptolemy and Al-Sûfi, or equal to δ Geminorum, but γ is now of the second magnitude, and its great superiority in brightness over δ is noticeable at a glance. Another interesting case is that of ζ and $\omicron$ Persei, two stars which lie near each other, about seven degrees north of the Pleiades. Al-Sûfi distinctly describes these stars as *both* of the 3—4 magnitude; but Argelander, Heis, and the photometric measures at Harvard agree in making ζ about one magnitude brighter than $\omicron$. The stars being close are easily compared, and their present great difference in brightness is very noticeable. This is one of the most remarkable cases I have met with in Al-Sûfi's work, and strongly suggests variation in $\omicron$, as ζ is still about the same brightness as Al-Sûfi made it. The identity of the stars is beyond all doubt, as Al-Sûfi describes their positions very clearly, and says there is no star between them and the Pleiades, a remark which is quite correct for the naked eye. The remarkable decrease in brightness of β Leonis (Denebola) since Al-Sûfi's time has been considered in my paper on *Some Suspected Variable Stars*. That it was a bright star of the first magnitude is fully proved by the observations of Al-Sûfi and Tycho Brahe. These were careful and accurate observers, and they could not have been mistaken about a star of the first magnitude. β Leonis is now fainter than an average star of the second magnitude, and there can be no reasonable doubt that it has faded considerably since the Tenth Century.

There are some other discrepancies between Al-Sûfi's observations and modern estimates, but the above are perhaps the most remarkable. With reference to lucid stars not mentioned by Al-Sûfi, he has not, I think, omitted any star brighter than the fourth magnitude in that portion of the sky visible from his station. There are, however, a number of stars between the fourth and sixth magnitudes which he does not mention. Of these the brightest seem to be ϵ Aquilæ, ρ and μ Cygni, and ζ Coronæ Borealis.

With reference to the distribution of the lucid stars in the sky there seems to be a well-marked tendency to congregate on the Milky Way. It is a remarkable fact that of the 15 brightest stars in the heavens, no less than 11 lie on or near the Milky Way, although the space covered by the Galaxy does not exceed one-fifth or one-sixth of the whole sky. From a careful enumeration of the stars in or near the Milky Way which I made some years ago, I found that of stars brighter than the fourth magnitude there are 118 on the Milky Way out of a total of 392, or about 30 per cent. From the Southern catalogue known as the *Uranometria Argentina*, Colonel Markwick, F.R.A.S., found 121 out of 228 stars to fourth magnitude, or a percentage of 53 per cent. These results seem to show some intimate relation between the lucid stars and the Galaxy.

FOOTNOTES:

[4] Except concave spectacles used by short-sighted persons.

[5] This custom has since prevailed. The following are the letters and their names:

α	Alpha	η	Eta	ν	Nu	τ	Tau
β	Beta	θ	Theta	ξ	Xi	υ	Upsilon
γ	Gamma	ι	Iota	$\omicron$	Omicron	ϕ	Phi
δ	Delta	κ	Kappa	π	Pi	χ	Chi
ϵ	Epsilon	λ	Lambda	ρ	Rho	ψ	Psi
ζ	Zeta	μ	Mu	σ	Sigma	ω	Omega

THE CONSTELLATIONS.—CAMILLE FLAMMARION

The earth is forgotten, with its small and ephemeral history. The sun himself, with all his immense system, has sunk in the infinite night. On the wings of inter-sidereal comets we have taken our flight toward the stars, the suns of space. Have we exactly measured, have we worthily realized the road passed over by our thoughts? The nearest star to us reigns at a distance of 275,000 times 37 millions of leagues—that is to say, at ten trillions^[6] of leagues (about twenty-five billions of miles); out to that star an immense desert surrounds us, the most profound, the darkest, and the most silent of solitudes.

The solar system seems to us very vast, the abyss which separates our world from Mars, Jupiter, Saturn, and Neptune appears to us immense; relatively to the fixed stars, however, our whole system represents but an isolated family immediately surrounding us: a sphere as vast as the whole solar system would be reduced to the size of a simple point if it were transported to the distance of the nearest star. The space which extends between the solar system and the stars, and which separates the stars from each other, appears to be entirely void of visible matter, with the exception of nebulous fragments, cometary or meteoric, which circulate here and there in the immense voids. Nine thousand two hundred and fifty systems like ours (bounded by Neptune), would be contained in the space which isolates us from the nearest star!

If a terrible explosion occurred in this star, and if the sound could traverse the void which separates it from us, this sound would take more than three millions of years to reach us.

It is marvelous that we can perceive the stars at such a distance. What an admirable transparency in these immense spaces to permit the light to pass, without being wasted, to thousands of billions of miles! Around us, in the thick air which envelops us, the mountains are already darkened and difficult to see at seventy miles; the least fog hides from us objects on the horizon. What must be the tenuity, the rarefaction, the extreme transparency of the ethereal medium which fills the celestial spaces!

Let us suppose ourselves, then, on the sun nearest to ours. From there our dazzling furnace is already lost like a little star, hardly recognizable among the constellations: earth, planets, comets sail in the invisible. We are in a new system. If we thus approach each star we find a sun, while all the other suns of space are reduced to the rank of stars. Strange reality!—the normal state of the universe is night. What we call day only exists for us because we are near a star.

The immense distance which isolates us from all the stars reduces them to the state of motionless lights apparently fixed on the vault of the firmament. All human eyes, since humanity freed its wings from the animal chrysalis, all minds since the minds have been, have contemplated these distant stars lost in the ethereal depths; our ancestors of Central Asia, the Chaldeans of Babylon, the Egyptians of the Pyramids, the Argonauts of the Golden Fleece, the Hebrews sung by Job, the Greeks sung by Homer, the Romans sung by Virgil—all these earthly eyes, for so long dull and closed, have been fixed from age to age on these eyes of the sky, always open, animated, and living. Terrestrial generations, nations and their glories, thrones and altars have vanished: the sky of Homer is always there. Is it astonishing that the heavens were contemplated, loved, venerated, questioned, and admired even before anything was known of their true beauties and their unfathomable grandeur?

Better than the spectacle of the sea calm or agitated, grander than the spectacle of mountains adorned with forests or crowned with perpetual snow, the spectacle of the sky attracts us, envelops us, speaks to us of the infinite, gives us the dizziness of the abyss; for, more than any other, it seizes the contemplative mind and appeals to it, being the truth, the infinite, the eternal, the all. Writers who know nothing of the true poetry of modern science have supposed that the perception of the sublime is born of ignorance, and that to admire it is necessary not to know. This is assuredly a strange error, and the best proof of it is found in the captivating charm and the passionate admiration which divine science now inspires, not in some rare minds only, but in thousands of intellects, in a hundred thousand readers impassioned in the search for truth, surprised, almost ashamed at having lived in ignorance of and indifference to these splendid realities, anxious to incessantly enlarge their conception of things eternal, and feeling admiration increasing in their dazzled minds in proportion as they penetrate

further into Infinitude. What was the universe of Moses, of Job, of Hesiod, or of Cicero, compared to ours! Search through all the religious mysteries, in all the surprises of art, painting, music, the theatre, or romance, search for an intellectual contemplation which produces in the mind the impression of truth, of grandeur, of the sublime, like astronomical contemplation! The smallest shooting star puts to us a question which it is difficult not to hear; it seems to say to us, What are we in the universe? The comet opens its wings to carry us into the profundities of space: the star which shines in the depths of the heavens shows us a distant sun surrounded with unknown humanities who warm themselves in his rays. Wonderful, immense, fantastic spectacles, they charm by their captivating beauty and transport into the majesty of the unfathomable the man who permits himself to soar and wing his flight to Infinitude.

Nel ciel che più della sua luce prende
Fu' io, e vidi cose che ridire
Né sa, né può qual di lassù discende.

“I have ascended into the heavens, which receive most of His light, and I have seen things which he who descends from on high knows not, neither can repeat,” wrote Dante in the first canto of his poem on “Paradise.” Let us, like him, rise toward the celestial heights, no longer on the trembling wings of faith, but on the stronger wings of science. What the stars would teach us is incomparably more beautiful, more marvelous, and more splendid than anything we can dream of.

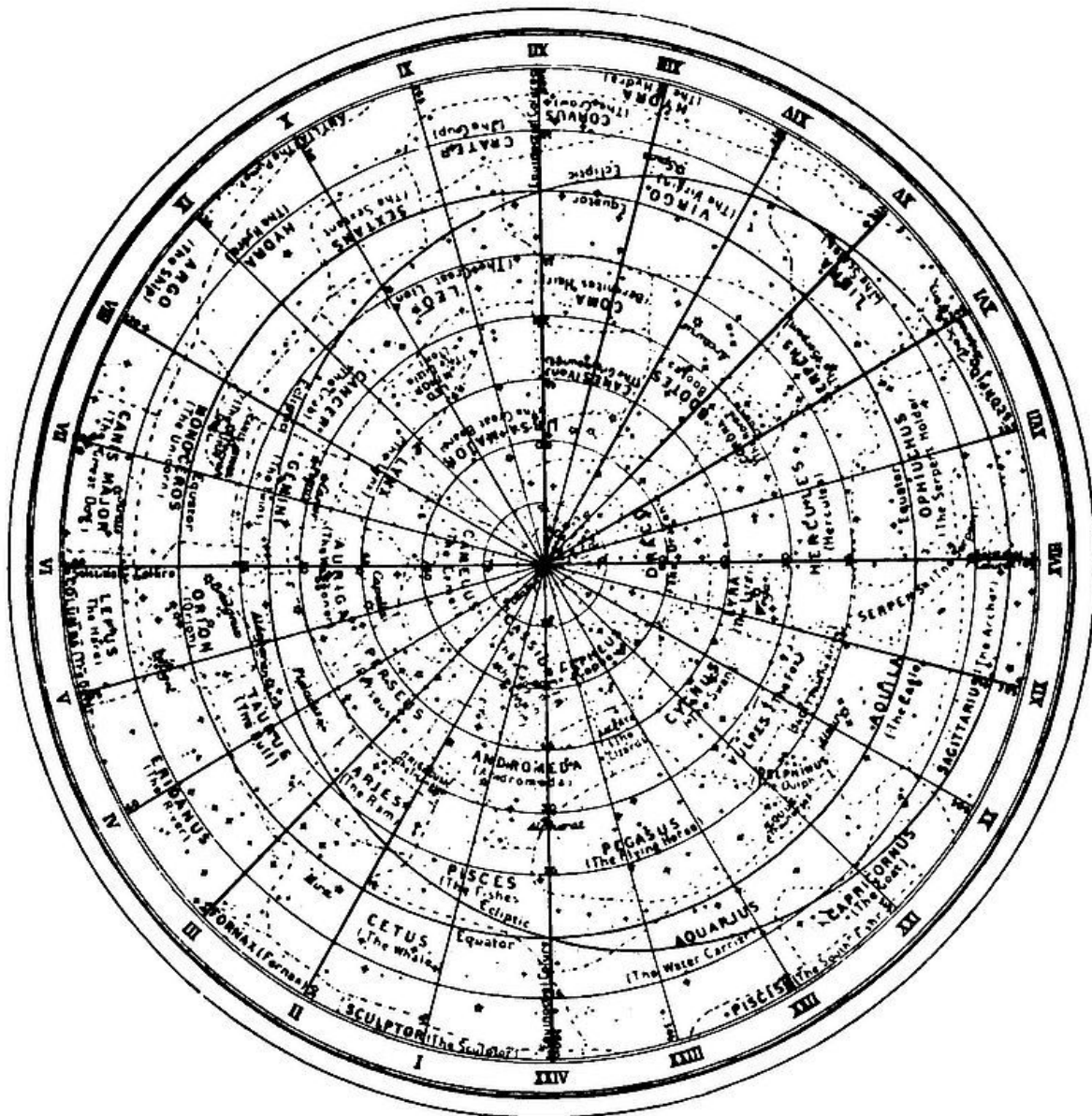


Chart of the Northern Constellations
Showing the principal Stars of the first five magnitudes visible to the naked eye

Among the innumerable army of stars which sparkle in the infinite night, the gaze is especially arrested by the most brilliant lights and by certain groups which vaguely present a mysterious bond between the worlds of space. These groups have been noticed at all epochs, even among the rudest races of men, and from the earliest ages of humanity they have received names, usually derived from the organic kingdom, which give a fantastic life to the solitude and the silence of the skies. Thus were early distinguished the seven stars of the North, or the Chariot, of which Homer

speaks; the *Pleiades*, or the “*Poussinière*”; the giant *Orion*; the Hyades in the head of Taurus; *Boötes*, near the Chariot or Great Bear. These five groups were already named more than 3,000 years ago, and so were the brightest stars of the sky, *Sirius* and *Arcturus*, etc.

The epoch of the formation of the constellations is unknown, but we know that they were established successively. The centaur Chiron, Jason’s tutor, has the reputation of having first divided the sky on the sphere of the Argonauts. But this is mythology; and, besides, Job lived before the epoch at which Chiron is supposed to have flourished, and Job had already spoken of Orion, the Pleiades, and the Hyades 3,000 years ago. Homer also speaks of these constellations in describing the famous shield of Vulcan. “On its surface,” says he, “Vulcan, with a divine intelligence traces a thousand varied pictures. He represents the earth, the heavens, the sea, the indefatigable sun, the moon at its full, and all the stars which wreath the sky: the Pleiades, the Hyades, the brilliant Orion, the Bear, which they also call the Chariot, and which revolves round the pole; this is the only constellation which does not dip into the ocean waves” (*Iliad*, chapter xviii.).

Several theologians have affirmed that it was Adam himself, in the terrestrial paradise, who gave their names to the stars; the historian Josephus assures us that it was not Adam, but his son Seth, and that in any case astronomy was cultivated long before the Deluge. This nobility is sufficient for us.

Attentive observation of the sky also noticed from the beginning the beautiful stars *Vega* of the Lyre, *Capella* of Auriga, *Procyon* of the Little Dog, *Antares* of the Scorpion, *Altair* of the Eagle, *Spica* of the Virgin, the *Twins*, the *Chair* of Cassiopeia, the Cross of the White Swan, stretched in the midst of the *Milky Way*. Although noticed at the epoch of Hesiod and Homer, these constellations and stars were probably not yet named, because doubtless men had not yet felt the necessity of registering them for any application to the calendar, to navigation, or to voyages.^[2]

At the epoch when the maritime power of the Phœnicians was at its apogee, about 3,000 years ago, or twelve centuries before our era, it was the star β of the Little Bear which was the nearest bright star to the pole, and the skilful navigators of Tyre and Sidon (O purpled kings of former times!

what remains of your pride?) had recognized the seven stars of the Little Bear, which they named the Tail of the Dog, *Cynosura*; they guided themselves by the pivot of the diurnal motion, and during several centuries they surpassed in precision all the mariners of the Mediterranean. The Dog has given place to a Bear, doubtless on account of the resemblance of the configuration of these seven stars to the seven of the Great Bear, but the tail remains long and curled up, in spite of the nature of the new animal.

Thus the stars of the North at first served as points of reference for the first men who dared to venture on the seas. But they served at the same time as guides on the mainland for the nomadic tribes who carried their tents from country to country. In the midst of savage nature, the first warriors themselves had nothing but the Little Bear to guide their steps.

Imperceptibly, successively, the constellations were formed. Some groups resemble the names which they still bear, and suggested their denomination to the men of ancient times, who lived in the midst of nature and sought everywhere for relations with their daily observations. The Chariot; the Chair; the Three Kings, also named the Rake; Jacob's Staff and the Belt of Orion; the Pleiades, or the Hen and Chickens; the Arrow (*Sagitta*); the Crown; the Triangle; the Twins; the Dragon; the Serpent; and even the Bull, the Swan, the Giant Orion, the Dolphin, the Fishes, the Lion, Water and Aquarius (the Water-bearer), etc., have given rise to the analogy. These resemblances are sometimes vague and far-fetched, like those we find in the clouds; but it appears much more natural to admit this origin than to suppose, with the classic authors, that these names were suggested by the concordance between the seasons or the labors of the fields and the presence of the stars above the horizon. That the name of the Balance (*Libra*) was given to the constellation of the equinox because then the days are equal, seems to us more than questionable; that Cancer (the Crab) signifies that the sun goes back to the solstice, and that the Lion has for its object to symbolize the heat of summer, and Aquarius the rain and inundations, appears to us no less imaginary. However, they have also had other origins. Thus, the Great Dog Sirius certainly announced the rising of the Nile and the dog-days (which remain in our calendar as a fine type of anachronism). Poetry, gratitude, the deification of heroes, mythology, afterward transferred to the sky the names of personages and sovereigns—Hercules, Perseus, Andromeda, Cepheus, Cassiopeia, Pegasus; later, in the

Roman epoch, they added the Hair of Berenice and Antinous; later still, in modern times, they added the Southern Cross, the Indian, the Sculptor's Workshop (Cœlum), the Lynx, the Giraffe (Camelopardus), the Greyhounds (Canes Venatici), the Shield of Sobieski, and the little Fox (Vulpecula). They even placed in the sky a Mountain, an Oak, a Peacock, a Swordfish, a Goose, a Cat, a Crane, a Lizard, and a Fly, for which there was no necessity.

This is not the place to describe and draw in detail all these constellations, with their more or less strange figures. The important point for us here is to form a general idea.

The sky remains divided into provinces, each of which continues to bear the name of the primitive constellation. But it is important to understand that the positions of the stars themselves, as we see them, are not absolute, and that the different configurations which they may show us are only a matter of perspective. We already know that the sky is not a concave sphere on which brilliant nails could be attached; that it is not a species of vault; that an immense infinite void envelops the earth on all sides, in all directions. We know also that the stars, the suns of space, are scattered at all distances in the vast immensity. When, therefore, we remark in the sky several stars near each other, that does not imply that these stars form the same constellation, that they are on the same plane, and at an equal distance from the earth. By no means; the arrangement which they assume to our eyes is but an appearance caused by the position of the earth relatively to them. This is a mere matter of perspective. If we could leave our world, and transport ourselves to a point in space sufficiently distant, we should see a variation in the apparent arrangement of the stars so much the greater as our station of observation were more distant from where we are at present. A moment's reflection is sufficient to convince us of this fact, and save us from insisting further on this point.

Once these illusions are appreciated at their true value, we can begin the description of the figures with which the ancient mythology has constellated the sphere. A knowledge of the constellations is necessary for the observation of the heavens and for the researches which a love of the sciences and curiosity may suggest; without it we find ourselves in an unknown country, of which the geography has not been made, and where it would be impossible to know our exact position. Let us make, then, this

celestial geography; let us see how to find our way, in order to read readily in the great book of the heavens.

There is a constellation which everybody knows; for greater simplicity we will begin with it. It will serve us well as a point of departure from which to go to the others, and as a point of reference to find its companions. This constellation is the *Great Bear*, which has also been named the *Chariot of David*.

It may well boast of being celebrated. If, notwithstanding its universal notoriety, some of our readers have not yet made its acquaintance, the following is a description by which they may recognize it.

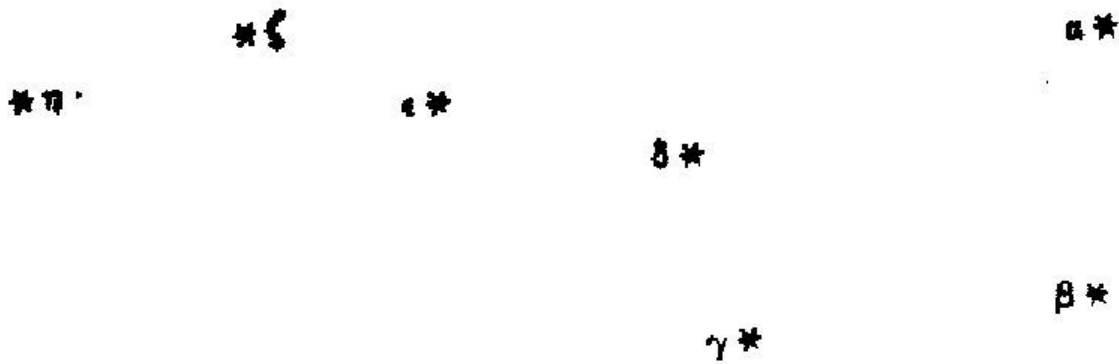


Fig. 2

Turn yourself toward the north—that is to say, opposite to the point where the sun is found at noon. Whatever may be the season of the year, the day of the month, or the hour of the night, you will always see there a large constellation formed of seven fine stars, of which four are in a quadrilateral, and three at an angle with one side; all are arranged as we see in Fig. 2.

You have all seen it, have you not? It never sets. Night and day it watches above the northern horizon, *turning slowly* in twenty-four hours round a star of which we shall speak directly. In the figure of the Great Bear, the three stars of the extremity form the tail, and the four in the quadrilateral lie in the body. In the Chariot, the four stars of the quadrilateral form the wheels, and the other three the pole, the horses, or the oxen. Above the second of these latter stars, ζ, good sight distinguishes quite a little star named Alcor, which is also called the Cavalier. It serves to test the power of the sight. Each star is designated by a letter of the Greek alphabet: α and β mark the first two stars of the quadrilateral, γ and δ the two following, ε, ζ, η, the

three of the pole. Arabic names have also been given to these stars, which we will pass in silence, because they are generally obsolete, with the exception, however, of that of the second horse—Mizar. With reference to the Greek letters, many persons think that it would be preferable to suppress them and to replace them by numbers. But this would be impossible in the practice of astronomy; and, moreover, inevitable confusion would result, on account of the numbers which the stars bear in the catalogues.

The Latins gave to plowing oxen the name of *triones*; instead of speaking of a chariot and three oxen, they came to call them the seven oxen (*septemtriones*). From this is derived the word septentrion, and there are now doubtless but few persons who, in writing this word, know that they are speaking of seven oxen. It is the same, however, with many other words. Who remembers, for example, in using the word *tragedy*, that he speaks of a song of a goat: *tragôs-ode*?

Let us go back to Fig. 2. If we draw a straight line through the two stars marked α and β which form the right side of the square, and produce it beyond α to a distance equal to five times that from β to α , or to a distance equaling that from α to the end of the tail, η , we find a star a little less brilliant than at the extremity of a figure similar to the Great Bear, but smaller and pointing in the opposite direction. This is the *Little Bear*, or the *Little Chariot*, also formed of seven stars. The star to which our line leads us—that which is at the tip of the tail of the Little Bear, or at the end of the pole of the Little Chariot—is the *polar star*.

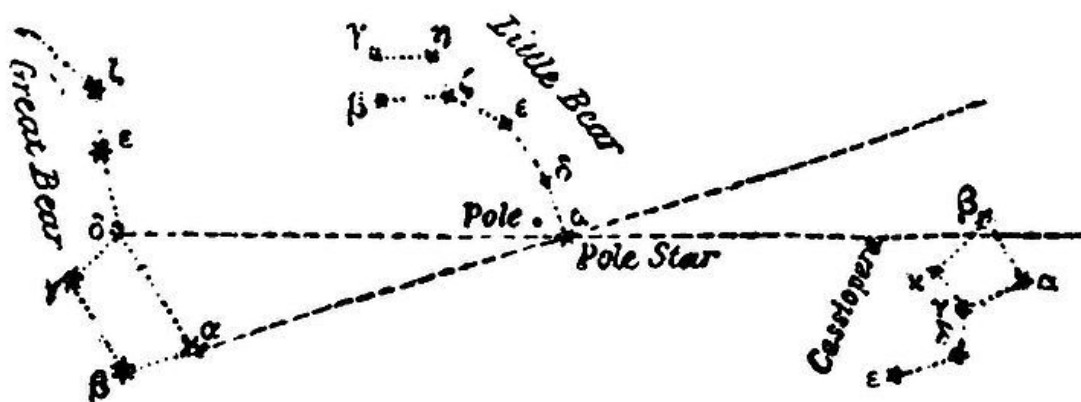


Fig. 3

The polar star enjoys a certain fame, like all persons who are distinguished from the common, because, among all the bodies which

scintillate in the starry night, it alone remains motionless in the heavens. At any moment of the year, by day or by night, when you observe the sky, you will always find it. All the other stars, on the contrary, turn in twenty-four hours round it, taken as the centre of this immense vortex. The pole star remains motionless at the pole of the world, from whence it serves as a fixed point to navigators on the trackless ocean, as well as to travelers in the unexplored desert.

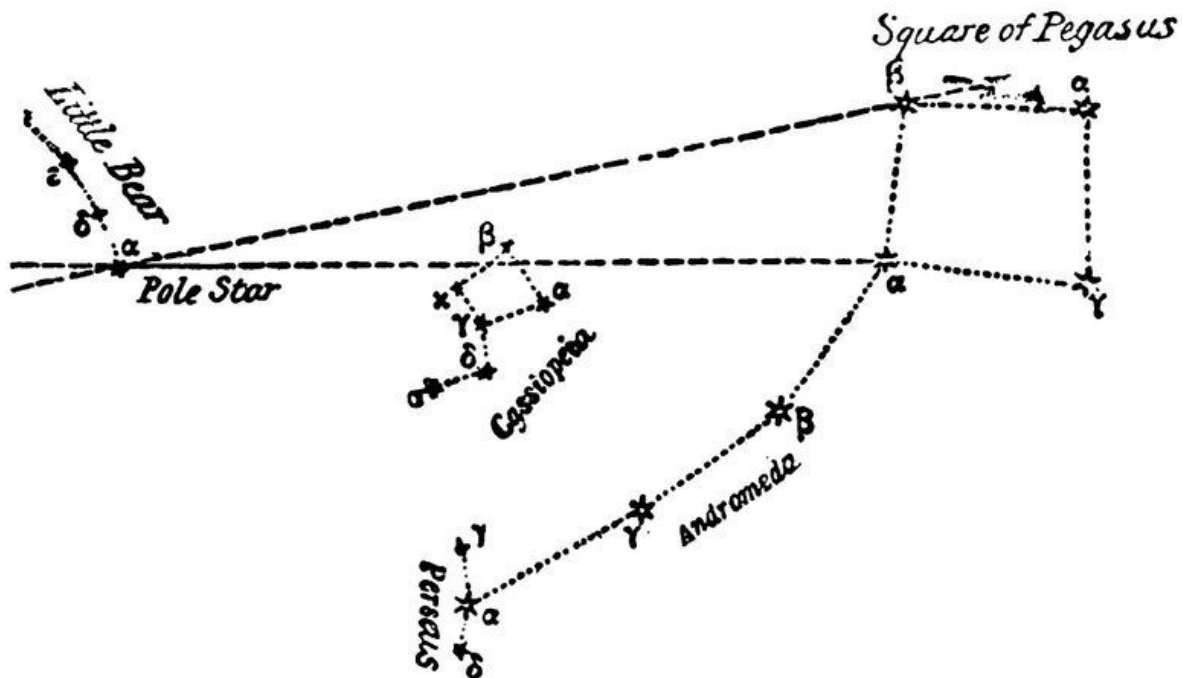


Fig. 4

In looking at the pole star, motionless in the midst of the northern region of the sky, we have the south behind us, the east to the right, the west to the left. All the stars turn round the pole star in a direction contrary to that of the hands of a watch; they should, then, be recognized according to their mutual relations rather than by reference to the cardinal points.

On the other side of the pole star, with reference to the Great Bear, is found another constellation which we can also recognize at once. If from the middle star, δ , we draw a line to the pole, and produce this line by the same distance (see [Fig. 3](#)), we arrive at *Cassiopeia*, formed of five principal stars arranged somewhat like the strokes of the letter M. The little star χ , which completes the square, gives the constellation the form of a *chair*. This group assumes all possible positions in turning round the pole; it is

found sometimes above, sometimes below, sometimes to the right, and sometimes to the left; but it is always easily recognized, for, like the preceding group, it never sets, and is always opposite to the Great Bear. The pole star is the axle round which both these constellations turn.



Fig. 5

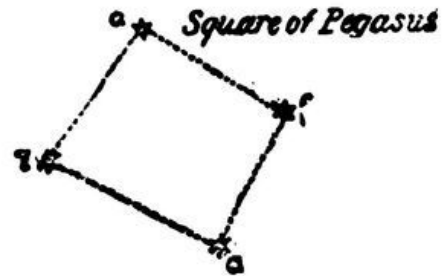


Fig. 6

If, now, we draw from the stars α and δ of the Great Bear two lines through the pole, and produce them beyond Cassiopeia, we come to the Square of Pegasus (see [Fig. 4](#)), which shows a line of three stars somewhat similar to the tail of the Great Bear. These three stars belong to *Andromeda*, and lead to another constellation, *Perseus*. The last star of the Square of Pegasus is, as we see, the first (α) of *Andromeda*; the three others are named γ , α , and β . To the north of β of *Andromeda* is found, near a little star, ν , an oblong nebula, which can be distinguished with the naked eye. In *Perseus*, α , the brightest—on the prolongation of the three principal stars of *Andromeda*—appears between two others less brilliant, which form with it a concave arc very easy to distinguish. This arc serves us for a new alignment. Producing it in the direction of δ , we find a very brilliant star of the first magnitude; this is *Capella* (the Goat). Forming a right angle with this prolongation toward the south we come to the *Pleiades* (Fig. 5). Not far from that is a variable star, *Algol*, or the *Head of Medusa*, which varies from the second to the fourth magnitude^[8] in 2 days, 20 hours, 48 minutes, 51 seconds. We may add, that in this region the star γ of *Andromeda* is one of the most beautiful double stars (it is even triple).

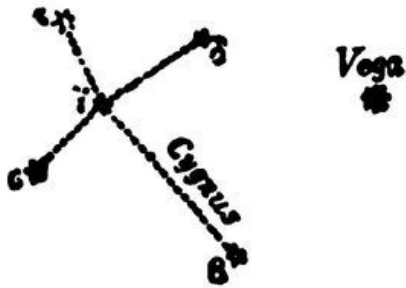


Fig. 7



Fig. 8

If, now, we produce beyond the Square of Pegasus (Fig. 6) the curved line of Andromeda, we reach the Milky Way, and we meet in these parts Cygnus, like a cross; the Lyre, where Vega shines (Fig. 7); the Eagle, and Altair (not Atair, as it is sometimes written) with two companions (Fig. 8).

Such are the principal constellations visible in the circumpolar regions on one side; we shall make a fuller acquaintance with them directly. While we are tracing the lines of reference let us still have a little patience and finish our summary review of this part of the sky.

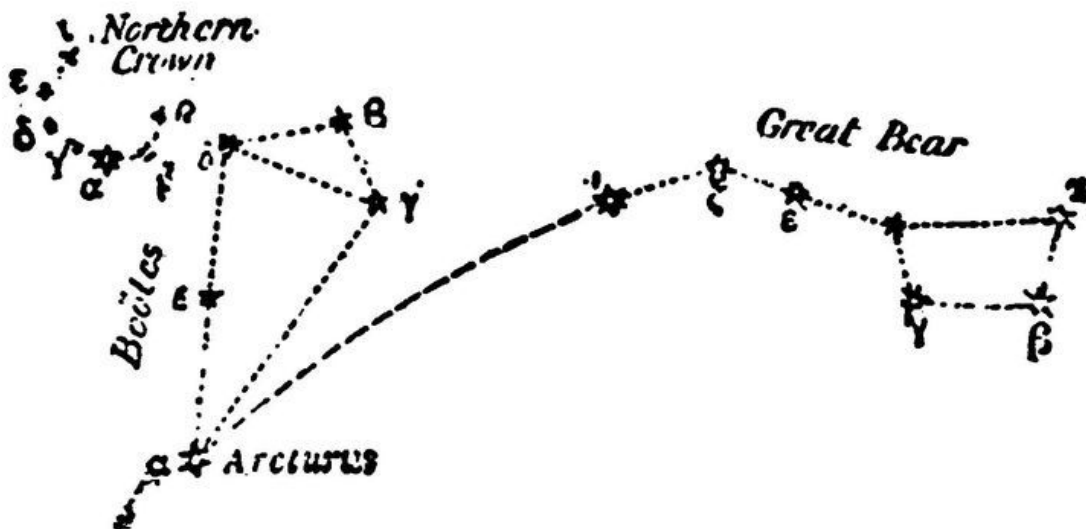


Fig. 9

Look now at the side opposite to that of which we have just spoken. Let us return to the Great Bear. Producing the tail along its curve, we find at some distance from that a star of the first magnitude, *Arcturus* (Fig. 9), or α of Boötes. A little circle of stars which we see to the left of Boötes constitutes the *Northern Crown* (Corona Borealis). In the month of May,

1866, there was seen shining there a fine star, the brightness of which lasted only fifteen days. The constellation of Boötes is traced in the form of a pentagon. The stars which compose it are of the third magnitude, with the exception of Arcturus, which is of the first. This is one of the nearest to the earth; at least, it is one of a small number whose distance has been measured. It shines with a beautiful golden yellow color. The star ϵ , which we see above it, is *double*—that is to say, the telescope resolves it into two distinct stars, one yellow, the other blue.

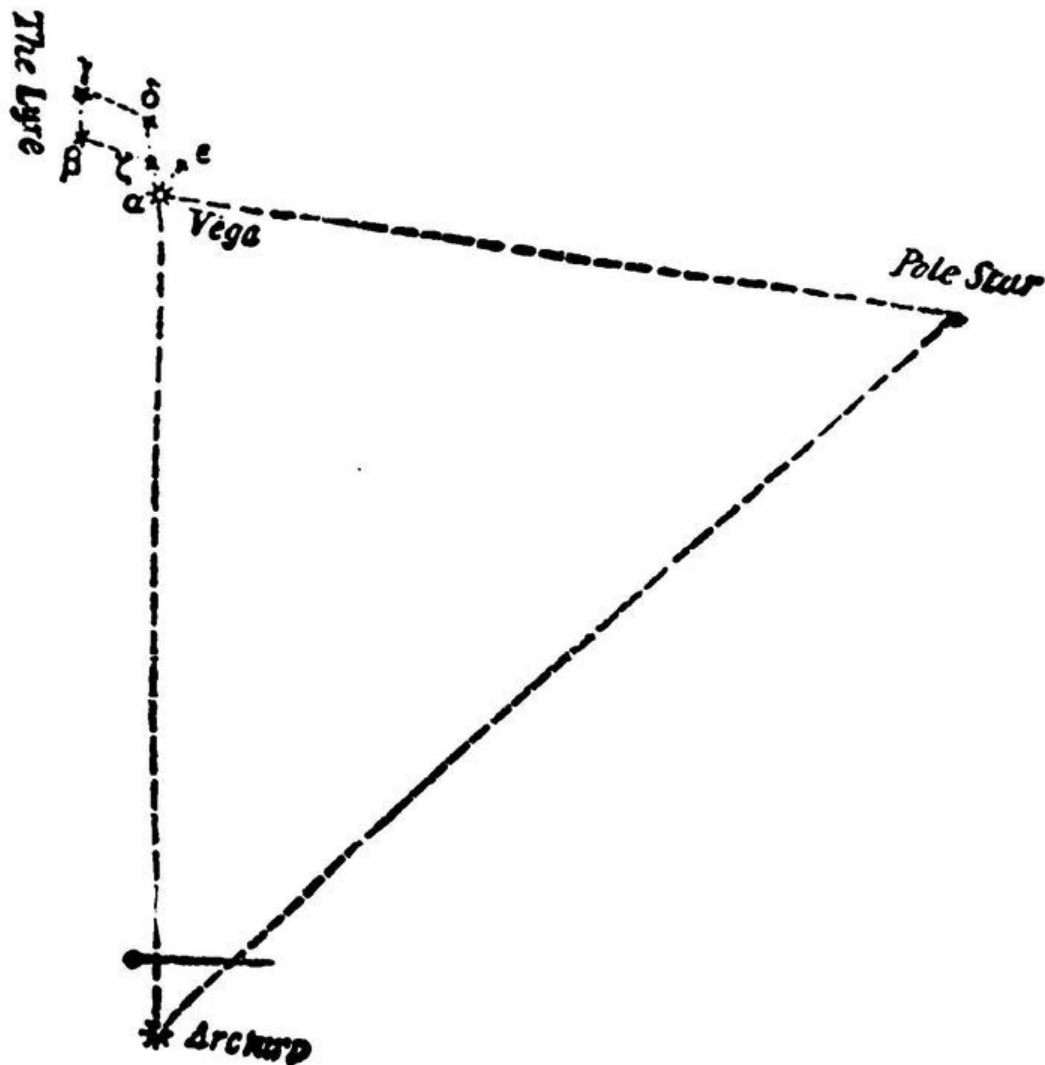


Fig. 10.

This technical description is far from the poetry of Nature; but it is especially important here to be clear and precise. Let us suppose ourselves,

however, under the starry vault on a beautiful summer's night, splendid and silent, and let us consider that each of these points which we seek to recognize is a world, or rather a system of worlds. Look at this equilateral triangle (Fig 10); it permits us to cast our eyes successively on three important suns: Vega of the Lyre, Arcturus of Boötes, and the pole star, which watches above the solitudes of our mysterious North Pole. Many martyrs of science have died looking at it! In twelve thousand years our descendants will see the Lyre at the pole, ruling the harmony of the heavens.

The stars which are near the pole, and which have for that reason received the name of circumpolar stars, are distributed in the groups which have just been indicated. I earnestly invite my readers to profit by fine evenings, and try to find for themselves these constellations in the sky.

We have here the principal stars and constellations of the Northern Hemisphere, the North Pole being at the centre of the circle. We come now in the order of our description to the twelve constellations of the zodiacal belt, which makes the circuit of the sky, inclined at 23° to the Equator, and of which the ecliptic, the apparent path of the sun, forms the centre line.

The name of zodiac, given to the zone of stars which the sun traverses during the course of the year, comes from ζῳδία, *animals*, an etymology which is due to the species of figures traced on this belt of stars. Animals, in fact, predominate in these figures. The entire circumference of the sky has been divided into twelve parts, which have been named the twelve signs of the zodiac; our ancestors called them the "houses of the sun," or "the monthly abodes of Apollo," because the day star visits them each month, and returns every spring to the beginning of the zodiacal city. Two memorable Latin verses of the poet Ausonius present to us these twelve signs in the order in which the sun travels through them, and this still appears the easiest method of learning them by heart.

*Sunt Aries, Taurus, Gemini, Cancer, Leo, Virgo,
Libraque, Scorpius, Arciteneus, Caper, Amphora, Pisces;*

or, in English, the Ram ♈, the Bull ♉, the Twins ♊, the Crab ♋, the Lion ♌, the Virgin ♍, the Balance ♎, the Scorpion ♏, the Archer ♐, Capricornus ♑, Aquarius ♒, and the Fishes ♓. The signs placed beside these names are a vestige of the primitive hieroglyphics which described

them: ♈ represents the horns of the Ram, ♉ the head of the Bull; ♊ is a stream of water, etc.

If we now know our northern sky, if its most important stars are sufficiently noted down in our mind, with the reciprocal relations which they preserve among themselves, we have no more confusion to fear, and it will be easy to recognize the zodiacal constellations. This zone may be of use to us as a line of division between the north and the south. Here is a description of it:

The Ram, which, moving in front of the herd, and regulating, so to say, the march, opens the series. This constellation has in itself nothing remarkable; the brightness of its stars indicates the base of one of the horns of the leader of the sheep; it is but of the second magnitude. After the *Ram* comes the Bull. Admire on a fine winter's night the charming Pleiades which scintillate in the ether; not far from them shines a fine red star—this is the *eye* of the Bull—Aldebaran, a star of the first magnitude and one of the finest of our sky. We now arrive at the Twins, whose heads are marked by two fine stars of the second magnitude, situated a little above a star of the first magnitude—*Procyon*, or the Little Dog; *Cancer*, or the Crab, a constellation very little conspicuous (its most visible stars are but of the fourth magnitude, and occupy the body of the animal); the *Lion*, a fine constellation, marked by a star of the first magnitude, *Regulus*, by one of the second, β , and by several others of the second and third magnitudes arranged in a trapezium; the *Virgin*, indicated by a very brilliant star of the first magnitude; *Spica*, situated in the neighborhood of a star, also of the first magnitude, Arcturus, which is found on the prolongation of the tail of the Great Bear; the *Balance* (Libra), indicated by two stars of the second magnitude, which would exactly resemble the Twins if they were nearer to each other; the *Scorpion*, a remarkable constellation; a star of the first magnitude, of a fine red color, marks the *Heart* (Antares), in the middle of two stars of the third magnitude, above which are three bright stars arranged in a diadem; *Sagittarius*, the Archer, of which the arrow, indicated by three stars of the second and third magnitudes, is pointed toward the tail of the Scorpion; *Capricornus*, a constellation not conspicuous, which is recognized by two stars of the third magnitude very near each other, and representing the base of the horns of the hieroglyphic animal; *Aquarius*, indicated by three stars of the third magnitude arranged in a triangle, of which the most northern occupies a point on the equator; *Pisces*, the *Fishes*, composed of stars, barely conspicuous, of the third to fourth magnitudes, situated to the south of a large and magnificent quadrilateral—the Square of Pegasus—of which we have already spoken.

We have now enumerated the zodiacal constellations in the order of the direct motion (from west to east) of the sun, moon, and planets which traverse them. They marked at the epoch of their formation, the monthly passage of the sun into each of them. The distribution of the stars in

figurative groups was the first truly hieroglyphical writing; it was engraved on the firmament in indelible characters.

The zodiac has played a great part in the ancient history of every nation—in the formation of the calendar, in the appointment of public festivals, and in the constitution of eras. The zodiac of Denderah, discovered by the French *savants* in Egypt at the end of the Eighteenth Century, was at first believed to have an antiquity of 15,000 years; but it is now proved that it is necessary to deduct from that number of years half the cycle of precession—that is to say, nearly 13,000 years—which brings down the date of this sculpture to 2,000 years before our epoch; and this in fact corresponds with the evidence of archæology. It is remarkable that all the ancient zodiacs and calendars which have been preserved to us begin the year with the constellation of the Bull, as we have already noticed. The zodiac of the Elephanta Pagoda (Salsette) has at the head of the procession the sign of the sacred Bull, the ox Apis, Mithra—of which the promenade of the fat ox, which is still performed in the environs of Paris, is a vestige. The ceiling of a sepulchral chamber at Thebes shows the Bull at the head of the procession. The zodiac of Esne, the astronomical picture discovered by Champollion in the Ramesseum of Thebes, carries us back to the same origin, between two and three thousand years before our era; Biot supposes the date of this to be the year 3285, the vernal equinox passing through the Hyades on the forehead of Taurus. Father Gaubil has proved that from ancient times the Chinese have referred the beginning of the apparent motion of the sun to the stars of Taurus; and we have a Chinese observation of the star η of the Pleiades as marking the vernal equinox in the year 2357 before our era. Hesiod sings of the Pleiades as ruling the labors of the year, and the name of Vergilia, which the ancient Romans gave them, associates them with the beginning of the year in spring.



Fig. 11

Without entering into any details of the different zodiacs which have been preserved to us from the most ancient and diverse nations, a glance at those which are reproduced here will lead us to appreciate the part which they have played in ancient religions. Several zodiacal signs have become veritable gods. The zodiac represented by Fig. 11 was engraved, in the Thirteenth Century, on an Arabic magic mirror, and dedicated to the sovereign prince Aboulfald, "Victorious Sultan, Light of the World," if we are to believe the bombastic inscription which encircles it. Fig. 12 shows an ancient Hindoo zodiac. Fig. 13 shows a Chinese zodiac stamped upon a

talisman, even now in use. The twelve signs differ from ours; they are: the Mouse, the Cow, the Tiger, the Rabbit, the Dragon, the Serpent, the Horse, the Ram, the Ape, the Hen, the Dog, and the Pig. Fig. 14 represents a Chinese medal, on which we see the constellation *Teou*, the Great Bear^[9] (which they call the Bushel), the Serpent, the Sword, and the Tortoise. This is a talisman intended to give courage; it appears that it is in great demand among the Chinese, and is as well circulated as the medals of the Immaculate Conception are in France.

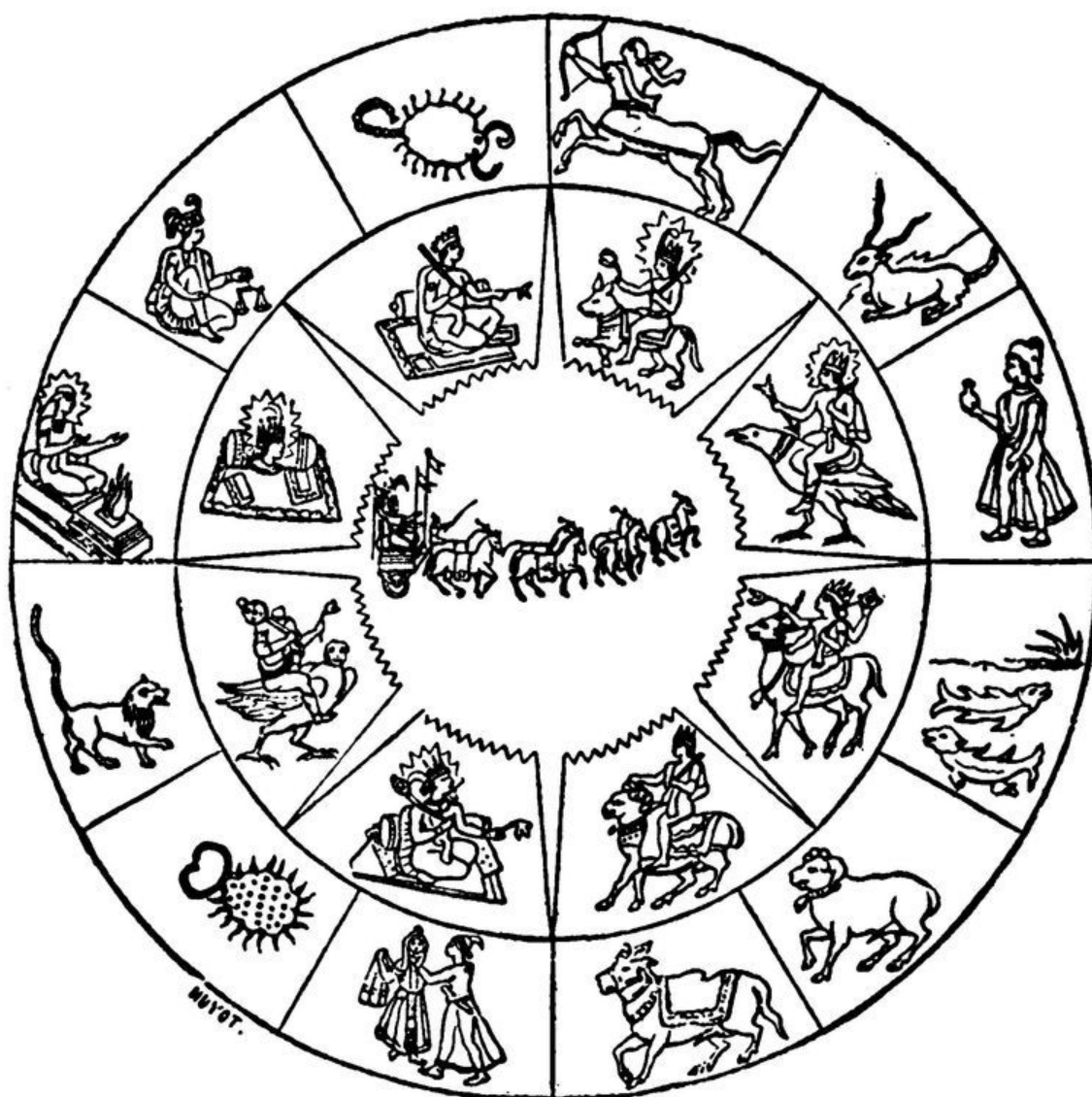


Fig. 12.—Ancient Hindoo Zodiac

Of all the zodiacal constellations, that of the Bull has played the principal rôle in ancient myths; and in this constellation it was the sparkling cluster of the Pleiades which appears to have regulated the year and the calendar among all the ancient nations. The Mosaic deluge itself, referred to 17 Athir (November), in commemoration of an important inundation, had its date coincident with the appearance of the Pleiades. ^[10]



Fig. 13.—Chinese Zodiac, from a Talisman



Fig. 14.—Chinese Medal, showing the Great Bear

But we forget the stars. If our descriptions have been carefully followed, the reader will now know the zodiacal constellations as well as those of the north. There remains but little to do to know the entire sky. But there is an indispensable addition to be made to what precedes. The circumpolar stars are perpetually visible above the London horizon; at any time of the year when we wish to observe them it is sufficient to turn to the north, and we shall always find them, either above the pole star or below it, to one side or the other, and always maintaining among themselves the relations which we have employed to find them. The stars of the zodiac do not resemble them from this point of view, for they are sometimes above the horizon, sometimes below. It is necessary, then, to know at what epoch they are visible. For this purpose it will be sufficient to remember the constellation which is found in the middle of the sky at *nine o'clock in the evening* on the first day of each month—that, for example, which crosses at that moment a line descending from the zenith to the south. This line is the *meridian*, of

which we have already spoken; all the stars cross it once a day, moving from east to west—that is to say, from left to right. In indicating each of the constellations which pass at the hour indicated, we also give the centre of the visible constellations.

On January 1 Taurus passes the meridian at 9 o'clock in the evening; notice Aldebaran, the Pleiades. On February 1 the Twins (Gemini) are not yet there; we see them a little to the left. March 1, Castor and Pollux have passed; Procyon to the south, the little stars of the Crab (Cancer) to the left. April 1, the Lion, Regulus. May 1, β of the Lion, Berenice's Hair. June 1, Spica of the Virgin, Arcturus. July 1, the Balance (Libra), the Scorpion. August 1, Antares, Ophiuchus. September 1, Sagittarius, Aquila. October 1, Capricornus, Aquarius. November 1, Pisces, Pegasus. December 1, Aries, the Ram.

Our general review of the starry sky must now be completed by the stars of the southern heavens.

Below Taurus and Gemini, to the south of the zodiac, you notice the giant Orion, who raises his club toward the forehead of the Bull. Seven brilliant stars are here distinguished; two of them, α and β , are of the first magnitude; the five others are of the second magnitude, α and γ mark the shoulders, κ the right knee, β the left knee; δ , ϵ , ζ mark the belt or girdle. Below this line is a luminous train of three stars, very near each other; this is the Sword. Between the western shoulder and Taurus is seen the Shield, composed of a row of small stars. The head is marked by a little star (λ) of the fourth magnitude.

On a fine winter's night turn toward the south, and you will immediately recognize this giant constellation. The four stars, α , γ , β , κ , occupy the angles of a great quadrilateral. The three others, δ , ϵ , ζ , are crowded in an oblique line in the middle of this quadrilateral; α , at the northeast angle, is named *Betelgeuse* (not Beteigeuse, as some books print it); β , at the southeast angle, is called *Rigel*.

The line of the Belt, produced both ways, passes to the northeast near *Aldebaran*, the Eye of the Bull, which we know already, and to the southeast near *Sirius*, the finest star of the sky, which we shall soon consider.

It is during the fine nights of winter that this constellation shines in the evening above our heads. No other season is so magnificently constellated

as the months of winter. While nature deprives us of certain enjoyments in one way, it offers us in exchange others no less precious. The marvels of the heavens present themselves from Taurus and Orion in the east to Virgo and Boötes on the west. Of eighteen stars of the first magnitude which are counted in the whole extent of the firmament, a dozen are visible from nine o'clock to midnight, not to mention some fine stars of the second magnitude, remarkable nebulae, and celestial objects well worthy of the attention of mortals. It is thus that nature establishes a harmonious compensation, and while it darkens our short and frosty days of winter, it gives us long nights enriched with the most opulent creations of the sky.

The constellation of Orion is not only the richest in brilliant stars, but it conceals for the initiated treasures which no other is known to afford. We might almost call it the California of the sky.

To the southeast of Orion, on the line of the Three Kings, shines the most magnificent of all the stars, *Sirius*, or α of the constellation of the Great Dog. This star of the first magnitude marks the upper eastern angle of a great quadrilateral, of which the base near the horizon of London, is adjacent to a triangle. This constellation rises in the evening at the end of November, passes the meridian at midnight at the end of January, and sets at the end of March. It played the greatest part in Egyptian astronomy, for it regulated the ancient calendar. It was the famous Dog Star; it predicted the inundation of the Nile, the summer solstice, great heats and fevers; but the precession of the equinoxes has in 3,000 years moved back the time of its appearance by a month and a half, and now this fine star announces nothing, either to the Egyptians who are dead or to their successors.

The *Little Dog*, or Procyon, is found above the Great Dog and below the Twins (Castor and Pollux), to the east of Orion. With the exception of α Procyon, no brilliant star distinguishes it.

Hydra is a long constellation, which occupies a quarter of the horizon, under Cancer, the Lion, and the Virgin. The head, formed of four stars of the fourth magnitude, is to the left of Procyon, on the prolongation of a line drawn from that star to Betelgeuse. The western side of the great trapezium of the Lion, like the line from Castor and Pollux, points to α , of the second magnitude. This is the Heart of Hydra; we remark the asterisms of the second class, Corvus the Crow, and Crater the Cup.

Eridanus, *Cetus*, *Piscis Australis*, and the *Centaur* are the only important constellations which remain to be described. We find them, in the order which we have indicated, to the right of Orion. Eridanus is a river composed of a train of stars winding from the left foot of Orion and losing itself below the horizon. After following long windings, it ends with a fine star of the first magnitude, α Eridani, or Achernar. This is the river into which Phaeton fell when he unskilfully directed the Chariot of the Sun. It was placed in the sky to console Apollo for the death of his son.

To find the Whale (*Cetus*), we may notice below the Ram a star of the second magnitude which forms an equilateral triangle with the Ram and the Pleiades; this is α of *Cetus*, or the Jaw; α , μ , ξ , and γ form a parallelogram which represents the head. The base, α , γ , may be produced to a star of the third magnitude, δ , and to a star of the neck marked σ . This star is one of the most curious in the heavens. It is named the Wonderful, *Mira Ceti*. It belongs to the class of variable stars. Sometimes it equals in brightness stars of the second magnitude, sometimes it becomes completely invisible.^[11] Its variations have been followed since the end of the Sixteenth Century, and it has been found that they are reproduced periodically every 331 days on the average. The study of these singular stars presents us with curious phenomena.

Lastly, the constellation of the Centaur is situated below Spica of the Virgin. The star θ , of the second magnitude, and the star ι , of the third, mark the head and the shoulder. This is the only part of this figure which rises above our horizon. The Centaur contains the *nearest star* to us (α) of the first magnitude, the distance of which is about twenty-five billions of miles. The feet of the Centaur touch the *Southern Cross*, formed of four stars of the second magnitude, always hidden below our horizon. It reigns in silence above the icy solitudes of the Southern Pole, where ships proceed only with difficulty. Further on, at the centre of the other hemisphere, is the southern celestial pole, which is not marked by any remarkable star.

It was from this region, Dante relates, that, having visited hell, inclosed in the centre of the earth, he went to the Mountain of Purgatory, and from there to the Heights of Paradise. These beautiful dreams have disappeared in the sunshine of modern astronomy.

We will complete these descriptions by a little astronomical chronology, which is not without interest. From a careful examination of the most ancient historical sources of classical astronomy, the following is the order in which the constellations appear to have been noticed, formed, and named, beginning with the most ancient:

	Most Ancient Reference
The Great Bear	<i>Job</i> (ch. xxxviii. ver. 32) (Seventeenth Century before our era), <i>Homer</i> (Ninth Century).
Orion	<i>Job</i> (ch. ix. ver. 9), <i>Homer</i> , <i>Hesiod</i> .
The Pleiades (the Hyades)	<i>Job</i> (ch. xxxviii. ver. 31), <i>Homer</i> , <i>Hesiod</i> .
Sirius and the Great Dog	<i>Hesiod</i> mentions it. <i>Homer</i> calls Sirius the Star of Autumn.
Aldebaran (Taurus)	<i>Homer</i> , <i>Hesiod</i> .
Boötes, Arcturus	<i>Job</i> (ch. xxxviii. ver. 32), <i>Homer</i> , <i>Hesiod</i> .
The Little Bear	<i>Thales</i> (Seventh Century), <i>Eudoxus</i> , <i>Aratus</i> .
Draco (the Dragon)	<i>Eudoxus</i> (Fourth Century), <i>Aratus</i> (Third Century).
The Man on his Knees, or Hercules	<i>Id.</i>
The Branch and Cerberus ^[12]	<i>Id.</i>
Corona Borealis	<i>Id.</i>
Ophiuchus or Serpentarius	<i>Id.</i>
The Scorpion	<i>Id.</i>
Virgo and Spica	<i>Eudoxus</i> (Fourth Century), <i>Aratus</i> (Third Century)
Gemini (the Twins)	<i>Id.</i>
Procyon	<i>Id.</i>
Cancer (the Crab)	<i>Id.</i>
Leo (the Lion)	<i>Id.</i>
Auriga (the Charioteer)	<i>Id.</i>
Capella (the Goat, the Kids)	<i>Id.</i>
Cepheus	<i>Id.</i>
Cassiopeia	<i>Id.</i>
Andromeda	<i>Id.</i>
Pegasus (the Horse)	<i>Id.</i>
Aries (the Ram)	<i>Id.</i>
The Triangle	<i>Id.</i>
Pisces (the Fishes)	<i>Id.</i>
Perseus	<i>Id.</i>
Lyra	<i>Id.</i>
The Bird, or Cygnus (the Swan)	<i>Id.</i>

Aquila (the Eagle)	<i>Id.</i>
Aquarius	<i>Id.</i>
Capricornus	<i>Id.</i>
Sagittarius	<i>Id.</i>
Sagitta (the Arrow)	<i>Id.</i>
Delphinus (the Dolphin)	<i>Id.</i>
Lepus (the Hare)	<i>Id.</i>
Argo (the Ship)	<i>Id.</i>
Canopus (afterward written Canopus)	<i>Id.</i>
Eridanus	<i>Id.</i>
Cetus (the Whale)	<i>Id.</i>
Piscis Australis (the Southern Fish)	<i>Id.</i>
Corona Australis	<i>Id.</i>
The Altar	<i>Id.</i>
The Centaur	<i>Id.</i>
The Wolf (Lupus)	<i>Id.</i>
Hydra	<i>Id.</i>
Crater (the Cup)	<i>Id.</i>
Corvus (the Crow)	<i>Id.</i>
Libra (the Balance)	<i>Manetho</i> (Third Century B. C.) <i>Geminus</i> (First Century) B. C.).
The Hair of Berenice ^[13]	<i>Callimachus, Eratosthenes</i> (Third Century).
Feet of the Centaur	<i>Hipparchus</i> (First Century B. C.).
Propus (η of Gemini)	<i>Hipparchus.</i>
The Manger and Donkeys	<i>Id.</i>
The Little Horse (Equuleus)	<i>Id.</i>
The Head of Medusa	<i>Id.</i>
Antinous ^[13]	Under the Emperor Adrian (130 A. D.).
The Peacock (Pavo)	<i>John Bayer</i> , 1603.
Toucan	<i>Id.</i>
Grus (the Crane)	<i>Id.</i>
Phoenix	<i>Id.</i>
Doradus	<i>Id.</i>
The Flying Fish	<i>Id.</i>
Hydrus	<i>Id.</i>
Chamæleon	<i>Id.</i>
The Bee (Musca)	<i>Id.</i>
The Bird of Paradise (Apus)	<i>Id.</i>

Triangulum Australis	<i>Id.</i>
The Indian (Indus)	<i>Id.</i>
The Giraffe (Camelopardus)	<i>Bartschius</i> , 1624.
The Fly (Musca)	<i>Id.</i>
The Unicorn (Monoceros)	<i>Id.</i>
Noah's Dove (Columba)	<i>Id.</i>
The Oak of Charles II	<i>Halley</i> , 1679.
The Southern Cross (already seen by the ancients)	<i>Augustine Royer</i> , 1677.
The Great and Little Cloud (Magellanic Clouds)	<i>Hevelius</i> , 1690.
The Fleur de Lys	<i>Id.</i>
The Greyhounds (Canes Venatici)	<i>Id.</i>
The Fox and Goose (Vulpecula et Anser)	<i>Id.</i>
The Lizard (Lacerta)	<i>Id.</i>
The Sextant of Urania (Sextans)	<i>Id.</i>
The Little Lion (Leo Minor)	<i>Hevelius</i> , 1690.
The Lynx	<i>Id.</i>
The Shield of Sobieski	<i>Id.</i>
The Little Triangle	<i>Id.</i>
Mount Mænalus	<i>Flamsteed</i> , 1725.
The Heart of Charles II (α Canum Venaticorum)	<i>Id.</i>
The Sculptor's Workshop (Sculptor)	<i>Lacaille</i> , 1752.
The Chemical Furnace (Fornax)	<i>Id.</i>
The Clock (Horologium)	<i>Id.</i>
The Rhomboid Reticule (Reticulum)	<i>Id.</i>
The Engraver's Pen	<i>Id.</i>
The Painter's Easel (Pictor)	<i>Id.</i>
The Compass (Circinus)	<i>Id.</i>
The Air Pump (Antlia)	<i>Id.</i>
The Octant (Octans)	<i>Id.</i>
The Compass and Square	<i>Id.</i>
The Telescope (Telescopium)	<i>Id.</i>
The Microscope (Microscopium)	<i>Id.</i>
The Table Mountain (Mensa)	<i>Id.</i>
The Reindeer	<i>Lemonnier</i> , 1774.
The Solitaire (Indian Bird)	<i>Id.</i>

Le Messier	<i>Lalande</i> , 1776.
The Bull of Poniatowski	<i>Poczobut</i> , 1877.
The Honors of Frederick	<i>Bode</i> , 1786.
The Harp of the Georges	<i>Hell</i> , 1789.
The Telescope of Herschel	<i>Bode</i> , 1787.
The Electrical Machine	<i>Id</i> , 1790.
The Printer's Workshop	<i>Id</i> .
The Mural Quadrant	<i>Lalande</i> , 1795.
The Air Balloon	<i>Id.</i> , 1798.
The Cat	<i>Id.</i> , 1799.

Such are the constellations, ancient and modern, venerable or recent, into which the celestial sphere has been divided. The ancient names are respectable and respected, on account of their relations, known or unknown, with the origins of history and religion; the new ones must be ephemeral. It is useful to know them, because several stars celebrated under different titles have for their principal designation their position in these asterisms; but what we should wish would be to see them disappear.^[14]

Many other substitutions have, however, been attempted. I have in my library a splendid folio of the year 1661, containing twenty-nine engraved plates, illuminated in gold and silver, among which are two which represent the sky delivered from the pagans and peopled with Christians. Instead of divinities more or less virtuous, in place of animals of forms more or less fantastic, we behold the elect—apostles, saints, popes, martyrs, sacred persons of the Old and New Testament—seated in the celestial vault, clothed in rich costumes of all colors, embroidered with gold, and carefully installed in the place of all the pagan heroes who for so many ages reigned in the sky.

The author of this metamorphosis was named Jules Schiller, and it was in the year 1627 that he introduced it, coupling his name with that of John Bayer. He began his dissertation by showing how the pagan constellations are opposed to Christian opinion and even to common-sense. He quoted the Fathers of the Church who expressly disapprove of them: Isodorus, who treats them as diabolical; Lactantius, who condemns the corruption of the human race; Augustine, who sends their heroes to hell, etc.

These constellations formed by chance, in the course of ages, without a fixed object; their inconvenient size, the uncertainty of their boundaries; the

complicated designations, for which it was sometimes necessary to exhaust whole alphabets; the bad taste with which observers have introduced into the southern sky the frigid nomenclature of instruments used in science alongside mythological allegories—all these accumulated defects have often suggested plans of reform for the stellar divisions, and even the banishing of all configuration. But ancient customs are difficult to overcome, and it is very probable that, except the recently named groups, which we may now suppress, the venerable constellations will always reign.

Such are the provinces of the sky. But these provinces are of no intrinsic value; the important point for us is to make acquaintance with the inhabitants.

FOOTNOTES:

[6] The French trillion is equivalent to the English billion, or a million times a million (1,000,000,000,000).—J. E. G.

[7] The Chinese had designated them all, it is true, at the same epoch, but their groups as well as their denominations are absolutely different from ours, and do not appear to have exercised any influence on the foundations of astronomical history. It was another world, other methods, other inspirations, as if Asia and Europe formed two distinct planets. A distinguished author, M. Schlegel, published in 1875 a Chinese Uranography, which is composed of 670 asterisms, and of which he believes he can trace back the origin to 17,000 years before our era. His argument is not convincing, and it seems to me that the origin of the astronomy of the Celestial Empire can not be very much anterior to the reign of the Emperor Hoang-Ti—that is to say, to the Twenty-seventh Century before our era—and would go back at furthest to the time of Fou-Hi that is to say, to the Twenty-ninth Century. It was about the same epoch—the Twenty-eighth Century before our era—that the Egyptians, observing Sirius, the early rising of which announced the inundation of the Nile, formed their canicular year of 365 days.

[8] More correctly, from 2.3 magnitude to 3.5 magnitude.—J. E. G.

[9] The author possesses in the Museum of the Observatory at Juvisy a Japanese executioner's sword, on the guard of which this constellation is engraved. Was it believed that the souls of executed criminals were sent there?

[10] See *Astronomical Myths, based on Flammarion's History of the Heavens*. By J. F. Blake. London, 1876.

[11] That is, to the naked eye; it never descends below the tenth magnitude, and always remains visible in a 3-inch telescope.—J. E. G.

[12] A constellation wrongly attributed by Arago and others to Hevelius. It is found on the sphere of Eudoxus.

[13] Constellations incorrectly attributed to Tycho Brahe. The first is given by Eratosthenes, the second dates from the Emperor Adrian.

[14] Especially those which are absolutely superfluous, and occupy places stolen from the ancient constellations, like the Heart of Charles II, the Fox and Goose, the Lizard, the Sextant, the Shield of Sobieski, Mount Mænalus, the Reindeer, the Solitaire, the Messier, the Bull of Poniatowski, the Honors of Frederick, the Harp, the Telescope, the Mural Circle, the Air Balloon, the Electrical Machine, the Printer's Workshop, and the Cat. I know, however, with reference to this last animal, that Lalande wrote: "I love cats! I adore cats! I may be pardoned for having placed one in the sky after my sixty years of assiduous labors." But the illustrious astronomer had no necessity for this plea in order that his name should remain inscribed in letters of gold on the tablets of Urania. The Heart of Charles II is but the flattery of a courtier; the Shield of Sobieski, the Bull of Poniatowski, should fall from the sky; the Messier is but a play on words which makes the celestial flocks guarded by a pastor whose name is the same as that of the prolific hunter of comets, Messier. As for the Honors of Frederick, they usurp an unmerited place, for, in order to make room for them, Andromeda has been obliged *to draw in her arm, which she had stretched out there for three thousand years.*

THE ARABIAN HEAVENS.—LUDWIG IDELER

The majority of Arabic star-names mentioned by Kazwini owe their origin to the astronomy of the Greeks. For instance, to the latter belong El-dschediain, the two Kids (Hædi); El-ma'lef, the Manger; El-hhimârain, the two Asses; Kalb el-ased, the Lion's Heart; El-sumbela, the Ears; El-zubênâ, the two Claws. Others indicate the positions of the stars in the Greek constellations as Râs el-tinnîn, Dragon's Head; Râs el-hhauwâ, Head of the Snake Man; Râs el-dschêthi, Head of the Kneeling (Hercules); Dseneb el-dedschâdsche, the Hen's Tail (Swan's); Dseneb el-dschedi, Goat's Tail (Wild-goat); Dseneb Kaitos, Whale's Tail; Fom el-hhût, Jaw of the (southern) Fish; Ridschl el-dschebbâr, Giant's Foot (Orion), etc. Still others, such as Khebd el-ased, Dafîra el-ased, El-dsirâ el-mebsûta, and el-mekbûda, El-nethra, El-dschebha, El-zubra, Sâk el-ased, Adschaz el-ased, refer to the Arabic Lion, which is a caricature of the Greek one.

Now if we separate these and many similar expressions from the astronomical nomenclature of the Arabs, there remains a class of star-names that present sufficient internal evidence to show plainly that they are indigenous to Arabia. It is worth while taking the trouble to collect and compare them. We shall in this way obtain a clearer idea of the sky that was altogether peculiar to this people.

In the first place, a large number of names of animals attracts our notice. In the vicinity of the North Pole, a shepherd (El-râï, Gamma in Cepheus), accompanied by his dog (Khelb el-râï, Zeta in Cepheus), is pasturing a herd of sheep (El-firk and El-agnâm, Alpha, Beta, Eta, and smaller stars in Cepheus), to which group also seem to belong two calves (El-ferkadain, Beta and Gamma in the Little Bear), a she-goat (El-anâk, Zeta in the Great Bear), a he-goat (El-tais in the Dragon), a young he-goat (El-dschedi, Alpha in the Little Bear), four mother-camels, a camel-foal, and a single camel pasturing by itself (El-awaîd, El-raba, and El-râfid, collectively on the head of the Dragon).

Various predatory animals are slinking around this herd, two Jackals (El-dsîbain, Zeta and Eta in the Dragon), which are specially stalking the

camel-foal; a male-hyena (El-dsîch, Iota in the Dragon) and many other she-hyenas (El-dibâ, Beta, Gamma, Delta and Mu in Boötes), and other she-hyenas with their young (Aulâd el-dibâ, Theta, Iota, Kappa, Lambda, and others in the same figure).

In the neighborhood of the two jackals (two stars in the Dragon) bear the name of their claws (Adhfâr el-dsîb).

Another shepherd (El-râï, Alpha in Ophiuchus) pastures his sheep (El-agnâm, small stars in the region of Hercules's Club) on a mead (El-rauda), which is defended on the side of the above-mentioned hyenas by two hurdles (Nasak schâmi and Nasak jemêni, rows of stars in Hercules and in the upper part of the Snake), and is open in the direction of the shepherd's two dogs (Khelb el-râï, Alpha in Hercules and Beta in Ophiuchus).

A third shepherd and a third herd are to be found further to the south in the Milky Way. The latter was represented as a river in which four animals (camels or sheep) are drinking, while four others (El-naâim el-sâdira, Zeta, Sigma, Tau, and Phi, in the Archer), are going away from it after having quenched their thirst. Lambda in the Archer was regarded as their shepherd (Râï el-naâim).

Yet another shepherd was signified by the star Beta in Orion (Rigel). He was called Râï el-dschauzâ, the shepherd in the Dschauzâ, or Nut-region, *i. e.*, in the region of Orion, which is splendid with many conspicuous stars. The herd which he was given to pasture are probably the "Thirst-quenched Camels" (El-nihâl), which were regarded as being the stars Alpha, Beta, Gamma, and Delta, in the Hare in the vicinity of the Milky Way.

Besides these groups of animals, there are several others scattered over the heavens. The three pairs of stars standing close together at the feet of the Great Bear were likened to the footmarks of a gazelle. They were called the Gazelle's Springs, or Hoofs (Kafzât el-dhibâ or Dhufra el-gizlân). Naturally the animal itself was regarded as being in the neighborhood of its tracks. On the one hand, Omicron, Pi, Rho, Sigma, A and d, on the head of the Great Bear, and on the other, as it appears, the stars of the Little Lion were included under the name Gazelle (El-dhibâ). The latter group also appears under the names the Gazelles and their Young (El-dhibâ w' aulâdhâ).

The five stars of the Virgin, Beta, Eta, Gamma, Delta, and Epsilon, were looked upon as so many yelping dogs (El-auwâ); Alpha and Beta in the Archer as a pair of birds peculiar to Arabia (El-suradain); Alpha (Fomalhaut) in the Southern Fish and Beta (Diphda) in the Whale as two Frogs (El-difda el-awel and El-difda el-thâni); four stars in the Great Dog and the Dove and as many Monkeys (El-kurûd), and the two bright stars of the latter constellation as a pair of Ravens (El-ag'riba).

All the creatures so far mentioned are familiar to the Arabs, the camel most of all. Just as their language is rich in words which refer to this useful animal, so also it plays the chief rôle in their astronomical nomenclature. We have already met with some camel-groups in the Arabian heavens. We find two more in the Bull and in the Crow. The brightest star in the Hyades has the name of "the Large Camel" (El-fenîk or El-fetîk), the others are called "the Small Camels" (El-kilâs or El-kalâjis). The four principal stars of the Crow were regarded as so many male-camels (El-adschmâl), analogous to the above-mentioned four female-camels in a similar figure at the head of the Dragon.

Just as frequently do we come across the ostrich in the Arabian heavens. The Southern Crown bears the name of the Ostrich Nest (Udha el-naâm), to which two pairs of ostriches (El-dhalîmain, Lambda and Mu in the Archer) appear to belong. A second ostrich-nest was formed from a number of stars in the upper part of Eridanus. In the neighborhood are five hen-ostriches (El-naâmât, Zeta, Eta, Theta, Tau, and Upsilon) in the belly of the Whale, and somewhat further away are two male birds (El-dhalîm, Alpha in the Southern Fish and Alpha in the River). The latter have a number of young ostriches (El-rijâl stars in the Phoenix) between them. Ostrich eggs (El-baid), or their shells (El-kaid), are represented by small stars in the vicinity of the nest.

Besides the groups, we also find various isolated animals in the starry heavens of the Arabs. Among these is the Black Horse (El-dschaun, Epsilon in the Great Bear), perhaps belonging to the neighboring Governor (El-kâid, Eta, in the same constellation); the beast of prey (Anâk el-ard, Gamma in Andromeda); the Male Camel (El-fahl), which was represented by Canopus and the Dog running in front of Sirius (El-khelb, Beta in Canis Major). This nomenclature, borrowed from the animal kingdom, to which must be added the Maidens (El-adsâra Omicron, Eta, Delta, and Epsilon in Canis Major);

the Outrider and the Man-riding-behind (El-fawâris and El-ridf, Delta, Gamma, Epsilon, Zeta, and Alpha in the Swan); this nomenclature, I say, is peculiar in that only one star was always used to distinguish one animal.

The Arabs with so lively an imagination saw in the skies sheep, camels, ostriches, but without being led to it by the resemblance of the contour of the entire star group, as was the case of the designers of the Greek heavens. They therefore had no animal figures proper, but only animal names, such as the She-goat, the two He-goats, and the two Asses of the Greeks. On two occasions, however, it happened that more than one star was given to one animal. When the eight stars of the Archer, which were represented under the figure of only four animals at pasture going to and returning from drinking, were regarded by some as two ostriches, this does not seem to be an exception to the rule, but a misunderstanding instead, caused by the resemblance of two words (Naâim and Naâm). The case is probably the same with the four stars, Delta, Pi, Rho, and Epsilon, in the Dragon which are called the He-goat by a very late Arab astronomer; for a star-name given by the lexicographer, Firuzabadi, would argue that analogy held true here also.

The two unmistakable cases to which I refer are those of the falling and flying Eagle (El-nesr el-wâki and El-nesr el-tâir), the former of which was made up of three stars in the form of an equilateral triangle, and the latter of three standing in a straight line (Alpha, Epsilon, and Zeta of the Lyre, Alpha, Beta and Gamma of the Greek Eagle).

We need not take into consideration in this connection either the Arabic Lion or the complete Horse, since most probably both owe their origin to false interpretations of later grammarians.

It is quite different with a second class of Arabic star-names which signify inanimate objects. These have to do with real forms throughout, which, however, for the most part consist of only a few stars after the manner of the Greek Arrow and Triangle. To these belong El-chibâ, the tent of the Arab nomads resting on three or four supports. One of these was represented by three stars of the Charioteer (Lambda, Mu, and Sigma), and another by the four chief stars of the Crow.

El-athâfi, the three stones which the nomadic Arab placed under his pot or kettle in the form of an equilateral triangle to form the hearth. Every triad

of stars standing in a similar figure might be called an Athâfi; for instance, Delta, Epsilon, and Rho in the Ram, and the three on the head of Orion, which were actually likened to one of these. In just so many words, however, the only stars that occur under this name are Alpha, Epsilon, and Zeta in the Lyre, and Sigma, Tau, and Upsilon in the Dragon.

El-kidr, the Pot, a ring of stars in the vicinity of the last Athâfi, which was formed from a number of small stars of Cepheus and the Swan.

El-midschdah, the wooden twirling-stick (spit). A kitchen utensil of similar triangular form was represented by the Hyades. The name in course of time came to be restricted to the chief star of this group.

El-fekka, the sounding plate with the broken rim, or Kas'a el-masâkhîm, the Beggar's-dish. This name was given to the stars of the Northern Crown, which stand in a circle open toward the northeast.

El-mîzân, the Scale-beam, an appropriate name for three stars in a straight line. The ancient Arabs used it for Theta, Eta, and Delta in the Eagle; the modern ones use it to distinguish the three stars on the Belt and the three on the Sword of Orion, the former of which, on account of their equal distance from each other, are called the true scale-beam, and the latter the false one, on account of the unequal intervals.

El-dsirâ, the Ell, a term which may fitly be applied to every pair of conspicuous stars standing a certain distance from one another. It was used for the two pairs of stars on the head of the Twins and in the Little Dog.

El-ma'lef, the Manger—the name of the stars of the Cup which stand in a circular form. The more familiar Manger in the Crab belongs to the Greek Heavens.

El-kubba, the Traveling-tent, drawn by camels of the Arab's female apartment. This name was given by some to the stars of the Southern Crown, while others, as has already been remarked, regard it as an Ostrich Nest.

El-zaurak, the Boat, was represented by the chief stars of the Phoenix. El-delv, the Well-Bucket, represented by the Square of Pegasus, occurred more frequently than any other, as is shown by the star-names relating to it—El-ferg, El-arkûwa, El-khereb, and Elnaâim. Elna'sch, the Bier, was applied to the well-known quadrangle in the Great and Little Bear. The term

particularly signifies the death-bier, and taken in this sense each of the two biers is accompanied by three mourning women—Benât—biers and mourners combined are called Benât na'sch, literally Daughters of the Bier, *i. e.*, belonging to the Bier.

El-salîb, the Cross: one of these was referred to under the four stars on the head of the Dragon, which others regarded as four mother camels. A second was found in the stars of the Dolphin.

El-serîr, El-khursi, El-arsch, various kinds of Thrones. One, named Serîr benât na'sch, was represented by seven stars standing in the form of a bow on the head of the Great Bear, which were also called El-hhûd, the Pond. Two other thrones under the names Khursi, or Arsch el-dschauzâ, were distinguished under four stars of Eridanus, and four in the Hare, and yet another, named Arsch el-simâkh el-a'zal, in the stars of the Crow.

El-nidâm and El-nedm, every set of things arranged in a row, especially the Pearl Necklace, which was the name given to the four stars 1, 2, 3, 4, and Phi of the Whale standing in a straight line, and the three on Orion's Belt. Synonymous with this, among words taken in their common acceptation is El-nasak, a name used for two rows of stars in the upper part of the Snake and Hercules, which also has a picturesqueness about it, since the two rows were regarded as hurdles around the meadow on which the above-mentioned shepherd pastures his flock.

El-fikrat, El-fekâr, and El-kelâda, the Brooch: the first of these appears as the name of the stars on the vertebra of the Scorpion's tail; the second, for Orion's Belt; the third, for stars of the Archer. El-dschauzâ, the Nuts, and El-lekat, the Golden-grains or Spangles. The former name was used for the stars of Orion and the neighboring Twins collectively, the latter merely for those on Orion's Sword. Finally, to this class belongs El-khaf el-chadîb and El-khaf el-dschedsmâ, the Dyed and the Mutilated Hand, which figures were represented by the five chief stars of Cassiopeia and the five better known on the head of the Whale. Several of these figures, as we have seen, appear at more than one place in the sky. Hence arose, for astronomers at least, the necessity for distinguishing epithets. Thus the Cross on the head of the Dragon was called "the falling," the Tent in the Crow "the southern," one of the Biers "the smaller," the other, "the greater"; one of the thrones in the vicinity of Orion, "the front"; the other, "the back."

When these distinctions are wanting, as in the case of the Athâfis, it is probably because the astronomers only made use of the one in the Dragon. Ulug Bekh does not name the other in the Lyre; Kazwini also states that it only occurred in the speech of the common people.

There is still a third and very numerous class of genuine Arabic star-names, which, borrowed neither from animate nor inanimate objects, are consequently names that do not represent any figures. They owe their origin to many circumstances, the majority of which are lost to us. I will content myself with mentioning only a few of them whose origin is not shrouded in doubt.

The small star over the middle of the Great Bear's tail is called El-suhâ, the Forgotten, the Lost, because it is only noticeable to a sharp eye; also El-saidak, the Touchstone (test-stone), because by it the eyesight was tested; Arcturus, Hâris el-semâ, the Warder of the Heavens, because it is never entirely lost in the rays of the sun; Capella, Rakîb el-thorejâ, the Watchman of the Pleiades, because it rises at the same time as they do; Alpha (Aldebaran), in the Bull, Hhâdi el-nedschm, the Driver of the Seven Stars; also El-tâbi and El-debarân, the Follower, because it rises immediately after that constellation; Beta (Denebola) in the Lion, El-serfa, the Breaker-up (Upsetter), because at its rising and setting in the morning twilight the hot and cold weather change; Alpha (Ras Alhague) in the Watersnake; El-ferd, the Isolated, because it is situated in a starless region, etc. Besides this, among this class we must include the Su'ûd, or fortunate stars, four of which are in Pegasus, two in the Wild Goat, and four in the Waterman.

It will already have been noticed that in this nomenclature single stars frequently appear under several names. Thus the stars of the Crow are sometimes called El-adschmâl, the Camels; sometimes El-chibâ el-jemêni, the Southern Tent; sometimes Arsch el-simâkh el-a'zal, the figure of the throne in the neighborhood of Spica—three quite different names which express so many various notions and have also so many separate authors.

Who were the originators of this nomenclature as a whole?

The Arabs, and particularly the nomad Arabs. To prove this we have only to cast a glance at the names in the first two classes.

The inhabitants of the northern part of the Arabian peninsula, the so-called “desert” and “stony” Arabia, for the most part, lead a nomadic kind of life.

The country is a treeless and waterless plain covered with naked rocks and sand-drifted hills, on which lie scattered single oases watered by springs and glorified with a luxuriant vegetation. On these the Arabs camp with their herds, and do not leave them until the provender is consumed, or until more powerful tribes force them to depart. They call themselves *Bedâvi* (Bedouins), that is, *Scenitæ*, Nomads, as they were called by the Greeks. These nomads, cut off from all intercourse with the world around them, who have never been subjugated by a foreign power, have preserved their character and their customs unchanged for several thousand years. Their most important occupation is cattle-breeding. Besides this, they follow the chase, or war upon their enemies, regarding as such all those not belonging to their race or who are not under their protection. They dwell in tents. Several families are under a *Schech* and several *Schechs* generally under an *Emîr*, who rules over the whole tribe.

The majority of these nomadic Arabs were Sabians, or Star-worshippers, before the adoption of Islam. History has preserved for us the names of several tribes who paid divine honors to single planets, or conspicuous fixed stars. No wonder that they should have fallen into such idolatry! The dust raised by the desert wind, which, as a rule, only blows during the day, and the heat of the sun compel them to pasture their herds and to undertake their hostile expeditions during the night. Leisure and necessity bid them gain information by directing their gaze at the starry sky, which is presented to them in a splendor of which we in our northern regions can scarcely form any idea. Since, therefore, the aborigines must have noticed at an early period that the nearly regular succession of changes in their climate took place in conformity with the annually recurring phenomena of the fixed stars, they ascribed to the latter a divine power. Thus originated the worship of the stars; and this once established, no other motives were needed to induce them to devote their constant attention to the starry skies. One result of this was that they applied proper names to the most conspicuous stars and groups of stars which were borrowed partly from the animal world around them, partly from their simple household effects, partly from various qualities and circumstances which they noticed in the stars. One tribe

selected one name; another, another; and so it came to pass that one star, or group of stars, frequently bears more than one name. When, on the other hand, stars no less bright bear no names at all, the probable reason is that only fragments of the astronomical nomenclature of the Arab nomads have come down to us.

After this terminology had been transmitted by oral tradition, and especially by folk-songs, for hundreds, perhaps thousands, of years in its original condition, it was combined into an entirely heterogeneous mass—that variegated mixture which we find in the works of Kazwini, Ulug Bekh, and others.

When the Arabs in their fanatic zeal for the spread of Mohammed's doctrines had conquered a great part of Asia, Africa, and Europe, and established in the heart of the ancient world a mighty empire, they adopted from the Greeks, with whom they had now come in contact, their astronomy among other sciences, and with it the Greek constellations and their method of distinguishing the stars according to their position in the figures. [\[15\]](#)

Their astronomers now generally discriminated between the two classes of names in attributing the one to the Arabs, the other to the astronomers.

Abdelrahman Sufi, in the preface to his work on the constellations, says there are two kinds of heavens to become acquainted with—that of the astronomers and that of the Arabs. In the work itself he first describes the constellations used by the astronomers, *i. e.*, the Greek ones, and then the old constellations of the Arabs. Kazwini in every case mentions a genuine Arabic star-name when he speaks of the Arabic, which is the case with almost every constellation.

Our early astronomers had very false notions of this relation of the nomadic heavens of the ancient Arabs to the mythological one of the Greeks adopted by their descendants. Schickard, in his *Astroscopium*, says: "Instead of the Dragon the Arabs depict two wolves and five dromedaries." He means the two jackals and the family of camels which the nomads represented under the five stars on the head of the Dragon. The Arab astronomers drew the Greek dragon on their charts and globes just as we do. They only looked on the old jackals and camels as *names* for some of its stars. In Golius and Hyde we find a more correct view of the case.

FOOTNOTES:

[\[15\]](#) Already in the ancient book of *Job*, whose hero has quite the characteristics of a Nomadic Emir, we find some astronomical terms whose analogy with the true Arabic star-names is unmistakable. See *Job*, ix. 9; xxxviii. 31, 32.

ASTRONOMY WITHOUT A TELESCOPE.—J. E. GORE

It must be remembered that astronomy was studied ages before the invention of the telescope, and that the ancient astronomers gained, without any optical assistance, a considerable amount of knowledge respecting the heavenly bodies.

Let us first consider the stars visible to the naked eye. The number of these down to the sixth magnitude—about the faintest that average eyesight can see—is, for both hemispheres, about 6,000. The number, therefore, visible at *one* time from any given place is about 3,000. Possibly double this number might be seen by those gifted with exceptionally keen eyesight; but even this is a comparatively small number, scattered as it is over so large an area. Those who do not possess the power of effective enumeration estimate the number visible to the naked eye as considerably greater than is really the case. This is partly due to the irregular distribution of the lucid stars over the celestial vault, and partly to the effect which the aspect of the starry sky produces on the imagination; the fact of the stars increasing in number as they diminish in brightness inducing us to suspect the presence of points of light which we do not actually see. An attempt to count those visible with *certainly* in any selected portion of the sky will, however, convince any intelligent person that the number, far from being large, is really very small, and that the idea, which some entertain, of a countless multitude is merely an optical illusion, and a popular fallacy which has no foundation in fact. Of course, the number visible in telescopes is very considerable. Perhaps with the largest telescopes 100,000,000 could be seen; but even this large number is very far from being “countless.” The present population of the earth is about 1,400,000,000, or about fourteen times the number of the *visible* stars!

The first thing to be done in studying the heavens with the naked eye is to learn the positions and names of the brighter stars; and from these the fainter ones may easily be identified by means of a star atlas. Those who study the stars in this way have probably a more intimate knowledge of the starry heavens than professional astronomers, who generally find the stars

—at least the fainter ones—by referring to a catalogue of stars, and then setting their telescope to the place indicated by the figures given in the catalogue. Although the famous astronomer Sir William Herschel possessed several large telescopes, he also studied the stars with the naked eye, and it is related of this great observer that he could without hesitation identify any star he could see in this way by its name, letter, or number! Such an exhaustive knowledge of the heavens is, of course, very rare; but an acquaintance with all the brighter stars can easily be acquired by any person of ordinary intelligence.

The “Plow,” or Great Bear,^[16] is familiar to most people. This remarkable group of seven stars will be found very useful in identifying some of the brighter stars. The two stars furthest from the “tail” are called “pointers,” as they point nearly to the Pole Star, or star to which the axis of the earth nearly points. I say “nearly,” for the Pole Star is not *exactly* at the pole, but distant from it about three diameters of the moon. The northern of these stars is known to astronomers by the Greek letter Alpha and the southern as Beta. The others, following the order of the figure, are known by the letters Gamma, Delta (the faintest of the seven), Epsilon, Zeta, and Eta.^[17] Now, if the curve formed by the three stars in the tail, Epsilon, Zeta, and Eta, is continued on, it will pass near a very bright star. This is Arcturus (Alpha of the constellation Boötes), one of the brightest stars visible. Again, if we draw an imaginary line from Gamma to Beta, and produce it, it will pass near another bright star. This is Capella (Alpha of Auriga, “the Charioteer” referred to by Tennyson).

Again, if we draw a line from Delta to Beta, and produce it, it will pass near the tolerably bright stars, Castor and Pollux (Alpha and Beta of the constellation Gemini, or the Twins), the northern of the two being Castor. Another line from Delta to Gamma produced will pass near a bright star called Regulus (Alpha of Leo, the Lion). Another line from Beta to Eta will pass near a group called Corona Borealis, or the Northern Crown.



Constellation of Orion, showing the Belt and Sword

On the opposite side of the Pole Star from the Plow, a group of five conspicuous stars will be found, forming a figure shaped somewhat like a W. This is Cassiopeia's Chair. Commencing with the most westerly of the five, these stars are known as Beta, Alpha, Gamma, Delta, and Eta. Like the stars of the Plow, those of Cassiopeia's Chair may be used to find other stars. For instance, a line drawn from Beta to Alpha passes close to a star known as Gamma in Andromeda; and the same line produced in the opposite direction will pass a little north of the bright star Vega (Alpha Lyræ), one of the brightest stars in the northern heavens. A line from Gamma to Alpha produced will pass through the well-known "Square of Pegasus."

To the east of Vega lies Cygnus, or the Swan, a well-known northern constellation. It may be recognized by the long cross formed by its principal stars, Alpha, Beta, Gamma, Delta, and Epsilon; Alpha, or Deneb, being the most northern and brightest, and Beta the most southern and faintest of the five.

To the southeast of Cassiopeia's Chair lies the constellation Perseus, distinguished by its well-known festoon, or curve, of stars. South of this lies the constellation Taurus or the Bull, which contains the well-known groups or clusters, the Pleiades and the Hyades. The Pleiades form perhaps the most remarkable group of stars in the heavens, and are easily found, when above the horizon. To ordinary eyesight the cluster consists of six stars. Some persons gifted with exceptionally keen eyesight have, however, seen eleven or twelve. A map of the Pleiades made in the sixteenth century shows eleven stars very correctly. This was drawn, of course, from observations made with a measuring instrument, but without the aid of a telescope. The observer (I think it was Möstlin, Kepler's tutor) must have possessed wonderfully sharp eyesight. The Hyades form a V-shaped figure, and contain the bright reddish star Aldebaran.

South of Taurus and Gemini will be found the splendid constellation of Orion, perhaps the most brilliant group of stars visible in either hemisphere. A remarkable quadrilateral figure is formed by its four stars, Betelgeuse (Alpha) and Gamma^[18] on the north, and Rigel (Beta) and Kappa on the south. Of these Betelgeuse and Rigel are bright stars of the first magnitude.

Betelgeuse is distinctly reddish and also slightly variable in its light. Rigel is a beautiful white star. In the middle of the quadrilateral are three stars of the second magnitude, nearly in a straight line, known as Delta, Epsilon, and Zeta, Delta being the northern of the three. These form Orion's "belt." South of these are three faint stars, also in a straight line, forming the "sword" of Orion. Surrounding the central star of the "sword" is "the great nebula of Orion," one of the finest objects in the heavens. It is barely visible to the naked eye, but may be seen with a good opera-glass.

To the southeast of Orion will be found Sirius, the brightest star in the heavens. It is the chief star of the constellation Canis Major, or the Great Dog, and has been well termed "the monarch of the skies," from its great brilliancy.

The bright star Regulus, referred to above, is situated in a remarkable group of stars shaped like a sickle, and known as "the Sickle in Leo." Regulus lies at the extremity of the handle. Leo is well placed for observation in April and May.

The famous group called the Southern Cross forms a conspicuous object in the southern heavens. It has formed a subject of interest since the earliest ages of antiquity. Its component stars, are, however, not so brilliant as some suppose, the two brightest being between the first and second magnitudes, the next of the second, and one between the third and fourth magnitudes. Near the Southern Cross are two bright stars known as Alpha^[19] and Beta of the Centaur.

Among the stars are many objects known as "double stars." These consist of two stars very close together, but which appear to the naked eye only as single stars. Some are triple, and even quadruple. Of these double stars there are now about 10,000 known to astronomers, but they are only visible with a telescope. Some, indeed, are so close that the highest powers of the very largest telescopes are necessary to see them as anything but single stars. Of the naked-eye stars there are, however, some apparently so close that they present very much the appearance of real double stars as seen in a telescope. These, although not recognized by astronomers as double stars, have been termed "naked-eye doubles." Houzeau found that the brighter the stars are the easier it is to separate them; and that for small stars, about 15'

of arc, or half the moon's apparent diameter, is about the limit below which the naked eye can not see a faint star double.

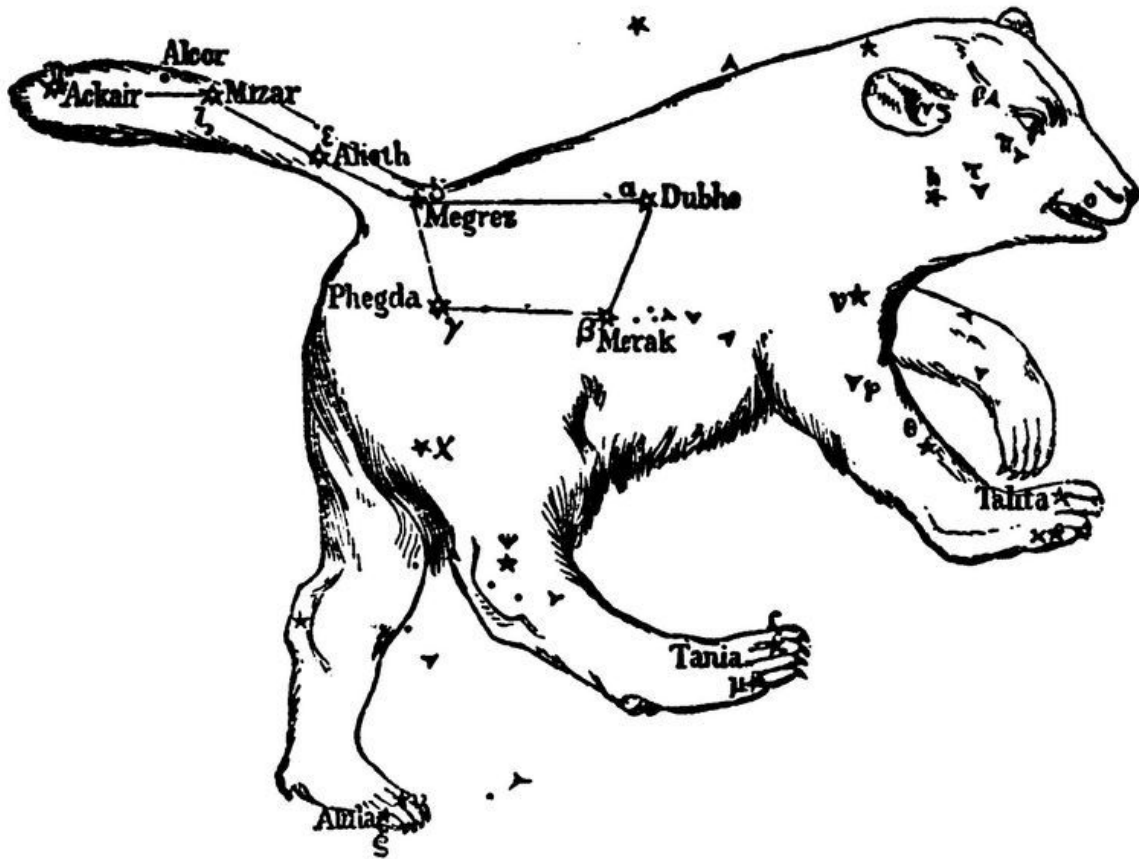


Fig. 15.—Constellation of the Great Bear

Of the “naked-eye doubles,” perhaps the most remarkable is Mizar, the middle star in the “tail” of the Great Bear. Close to it is a small star, sometimes called “Jack on the Middle Horse.” It was known to the ancient astronomers as Alcor, or “the test,” as it was *then* considered a test of excellent eyesight. Whether it has really brightened seems doubtful, but at present it is perhaps visible to *ordinary* eyesight. Some, however, fail to see it, while to others with keener vision it seems as plain as the proverbial “pike-staff.” The star Alpha Capricorni consists of two stars which, although closer than Mizar and Alcor, are more equal in brightness, and may be easily seen with the naked eye on a clear night. Nu Sagittarii may also be seen double in this way. Theta Tauri, in the Hyades, is another object which some eyes can see distinctly double; also Kappa Tauri, a little to the north of the Hyades; Omicron Cygni, a little to the west of Alpha

Cygni (Deneb), is another example. On a very fine night two stars may be seen in Iota Orionis, the most southern star in the “sword.” Near Gamma Leonis, one of the brightest stars in the “sickle,” is a star of the sixth magnitude, which some can see without optical aid.

The most severe test is, however, Epsilon Lyræ, the northern of two small stars which form a little triangle with the brilliant Vega. This, to some eyes, appears double. The famous German astronomer Bessel is said to have seen it at thirteen years of age. To most people, however, it will perhaps appear only elongated. This is a very remarkable star, as each of the components is seen to be a close double when examined with a good telescope; and between the pairs are several fainter stars.

Among those interesting objects, the variable stars, are several which may be well observed without optical assistance. Of these may be mentioned Algol, of which all the fluctuations of light may be easily observed with the naked eye; Mira Ceti, which may be well observed when at its brightest; Lambda Tauri, a variable star of the Algol type; Betelgeuse (Alpha Orionis), which is slightly variable; Zeta Geminorum, a fourth magnitude star, which varies about three-quarters of a magnitude in a period of about ten days; R. Hydræ, which is visible to the naked eye at maximum; Beta Lyræ, period about thirteen days; Eta Aquilæ, period about seven days; and Delta Cephei, which varies about one magnitude in a period of a little over five days. Of all these stars useful observations may be made without optical assistance of any sort.

Observations, and even discoveries, of new or “temporary” stars may also be made with the naked eye. This occurred in the case of the “temporary” stars of 1572, 1604, 1670, 1866, and 1870, but, of course, these were bright objects at the time of their discovery. Hind’s “new star” of 1848 in Ophiuchus was, however, only of the fifth magnitude when it appeared, and it might have escaped detection with the naked eye. A star of this magnitude might, however, be easily detected by an observer who is familiar with the principal stars of a constellation.

The Milky Way may, perhaps, be better seen with the naked eye than with any instrument, although an opera-glass brings out well, in some places, its more delicate details. A mere passing glance might lead a casual observer to suppose that the Galaxy stretched as a band of nearly uniform

brightness across the heavens. But good eyesight, careful attention, and a clear sky will soon disclose numerous details previously unsuspected; streams and rays of different brightness, intersected by rifts of darkness, and interspersed with spots and channels of comparatively starless spaces. An excellent drawing of the Milky Way—the result of five years' observations with the naked eye alone—has recently been completed by Dr. Otto Boeddicker at Lord Rosse's observatory in Ireland. This beautiful picture is exquisitely drawn, and shows a wonderful amount of detail. A writer in the *Saturday Review* of November 30, 1889, says: "His maps are in many respects a completely new disclosure. Features barely suspected before come out in them as evident and persistent; every previous representation appears, by comparison, *structureless*." This shows what can be done with the naked eye in the study of this wonderful zone.

Among the nebulae and clusters there are not many objects visible to the naked eye. A hazy appearance about the middle star in Orion's "sword" indicates the presence of the "great Nebula," one of the finest objects in the heavens. The "great Nebula in Andromeda," aptly termed "the Queen of the Nebulae," is distinctly visible to the naked eye on a very clear night. It lies near the four and a half magnitude star, Nu Andromedæ (a few degrees north of Beta Andromedæ), and may be well seen in the early evening hours in the month of January, when it is high in the sky. It somewhat resembles a small comet. This nebula was known long before the invention of the telescope, and it was described by one of the earlier astronomers as resembling "a candle shining through horn," a not inapt description.

Of star clusters visible without optical aid may be mentioned the double cluster Chi Persei, which appears to the eye as a luminous spot in the Milky Way; the cluster known as 35 Messier, a little north of Eta Geminorum, just visible to the naked eye on a very clear night; and there are others in the Southern Hemisphere, notably the globular cluster known as Omega in the Centaur, which shines as a hazy star of the fourth magnitude. Among the clusters may perhaps be included the Præsepe, or the "Beehive," in Cancer, which has a nebulous appearance to the naked eye.

Coming now to the Solar System, the sun and moon, of course, first attract attention. Cases of sun-spots visible to the naked eye are recorded, but, of course, spots of such enormous size are of rare occurrence. Of lunar detail little can be seen without a telescope of some sort, but the larger

markings are sufficiently distinct to good eyesight to convince the observer that they do not alter perceptibly, thus showing clearly that the moon always turns the same side to the earth.

Of the planets, nothing of their appearance in the telescope can, of course, be seen with the naked eye, but it is easy to identify the brighter planets. Mercury, owing to its proximity to the sun, is rarely visible in Europe and North America, but when favorably situated, it may sometimes be detected near the sun shortly after sunset or a little before sunrise. Notwithstanding the difficulty of seeing it, it was well known to the ancients, an observation of the planet dating back to 264 B. C. It is easier, however, to see in more southern latitudes, and I have frequently observed it as bright as a star of the first magnitude in the clear air of the Punjab sky. I have also seen it on several occasions in Ireland, and the Rev. S. S. Johnson, F.R.A.S., tells me he has seen it with the naked eye no less than one hundred times in the south of England. The brilliant planet Venus can hardly be mistaken when seen in the morning or evening sky. When at its brightest it considerably exceeds Jupiter and Mars, and far surpasses Sirius, the brightest star in the heavens.

If a very bright planet is seen rising at sunset, it can not be Venus, which is never seen beyond a limited distance from the sun. The observer may, therefore, conclude with certainty that the planet is either Jupiter or Mars. The latter, which occasionally rivals Jupiter in brilliancy, may be easily distinguished from the “giant planet” by its distinctly reddish color. Saturn shines with a yellowish light, and is never so bright as Mars or Jupiter when at their brightest. The planet Uranus is just visible to the naked eye, and may be found without optical assistance when its position is accurately known.

Some observers think that they can see the crescent of Venus with the naked eye when the planet is in that phase, but this seems very doubtful. Cases have been recorded of one or two satellites of Jupiter having been seen with the unaided eyesight, but few are gifted with such keen vision.

Occultations of bright stars may be well seen with the naked eye, especially when they pass behind the moon’s *dark* limb, and as the disappearance of a star is practically instantaneous, really valuable

observations may be made without a telescope, by merely noting the exact time at which the star vanishes.

Most of the comets discovered by astronomers are small and faint, and only visible in good telescopes. At intervals, however, a brilliant visitor appears on the scene, and its path among the stars may be watched from night to night with the naked eye. Before the invention of the telescope, bright comets were watched in this way, and their course recorded so carefully that it has been found possible to calculate their orbits with some approach to accuracy. In these days of large telescopes and instruments of almost mathematical precision, such a method of observation is, of course, superseded; but we may still watch the movements of a bright comet with interest, and note its apparent path across the sky with pleasure and profit. Shooting stars and fire-balls may be best observed with the naked eye, and the excellent work done in this way by Mr. W. F. Denning, F.R.A.S., should encourage others to take up this interesting branch of astronomy.

Another object which may be well seen with the naked eye—indeed, it may best be observed in this way—is the Zodiacal Light. This is a lenticular or cone-shaped beam of light, which makes its appearance at certain times of the year, above the eastern horizon before the dawn, and above the western horizon after sunset, when the sky is clear and the moon absent. In the tropics it is much more easily seen, the twilight being shorter, and I have often observed it in India shining with great brilliancy.

From the above sketch my readers will see how much may be learned of astronomy without optical assistance of any kind, and I hope that those who do not possess a telescope will use their eyes instead, and thus gain some knowledge of the wonders and beauties of the starry heavens. The knowledge thus gained will stimulate their curiosity and will give them keener interest in reading books which describe the still greater wonders revealed by the telescope.

FOOTNOTES:

[16] Also known as the Dipper and Charles's Wain.—E. S.

[17] The Arabian names Dubhe (Alpha), Merak (Beta), Phecda (Gamma), Megrez (Delta), Alioth (Epsilon), Mizar (Zeta), and Alkaid (Eta).—E. S.

[\[18\]](#) Bellatrix.

[\[19\]](#) This is the nearest star to the earth.—E. S.

THE MILKY WAY.—RICHARD A. PROCTOR

To those who rightly appreciate its meaning the Milky Way is the most magnificent of all astronomical phenomena. However opinions may vary as to the configuration of the star-streams composing this object, no doubt now exists among astronomers that the Milky Way consists really of suns, some doubtless falling short of our own sun in brilliancy, but many probably surpassing it. Around these suns, we may fairly conceive, there revolve systems of dependent orbs, each supporting its myriads of living creatures. We have afforded to us a noble theme for contemplation in the consideration of the endless diversities of structure, and of arrangement, which must prevail throughout this immensity of systems.

The Galaxy traverses the constellation Cassiopeia. Thence it throws off a branch toward Alpha Persei (Mirfak), prolonged faintly toward the Pleiades. The main stream, here faint, passes on through Auriga, between the feet of Gemini and the Bull's horns, over Orion's club to the neck of Monoceros. Thence, growing gradually brighter, the stream passes over the head of Canis Major, in a uniform stream, until it enters the prow of Argo, where it subdivides. One stream continues to Gamma Argus, the other diffuses itself broadly, forming a fan-like expanse of interlacing branches, which terminate abruptly on a line through Lambda and Gamma Argus. Here there is a gap beyond which the Milky Way commences in a similar fan-shaped grouping, converging on the brilliant (and in other respects remarkable) star Eta Argus. Thence, it enters the Cross by a narrow neck, and then directly expands into a broad, bright mass, extending almost to Alpha Centauri. Within this mass is a singular cavity known as the Coal-Sack. At Alpha Centauri the Milky Way again subdivides, a branch running off at an angle of 20° , and losing itself in a narrow streamlet. The main stream increases in breadth, until, "making an abrupt elbow," it subdivides into one continuous but irregular stream, and a complicated system of interlacing streams covering the region around the tail and following claw of Scorpio. A wide interval separates this part of the Galaxy from the great branch on the northern side, terminating close on Beta Ophiuchi.

The main stream, after exhibiting several very remarkable condensations, passes through Aquila, Sagitta, and Vulpecula to Cygnus. In Cygnus there is a “confused and patchy” region marked by a broad vacancy, not unlike the Coal-Sack. From this region there is thrown off the offset to Beta Ophiuchi, already mentioned; the main stream is continued to Cassiopeia.

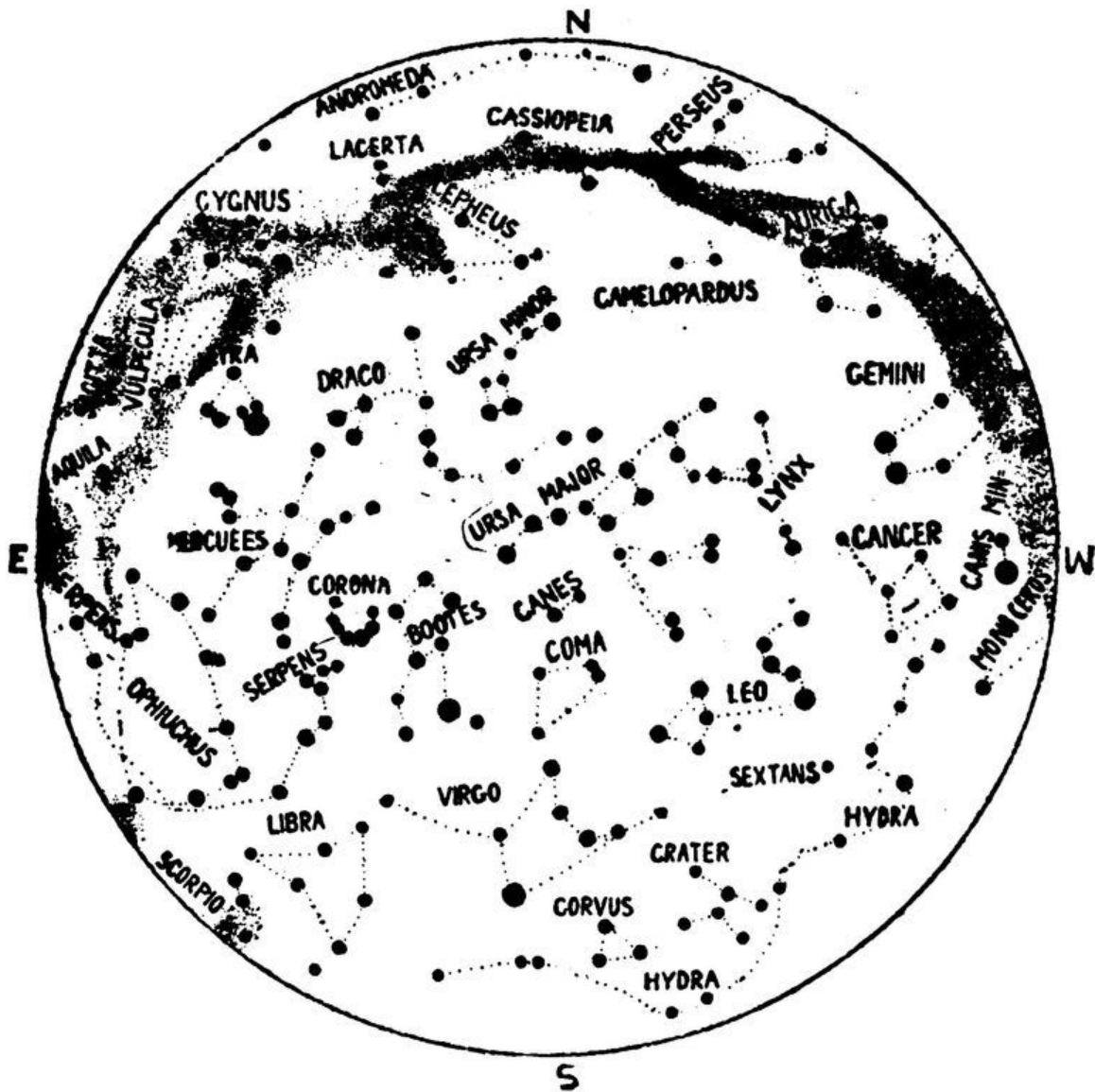


Fig. 16.—The Midnight Sky, with Milky Way

There only remains to be noticed “a considerable offset or protuberant appendage,” thrown from the head of Cepheus directly toward the pole. Galileo was the first to prove, though earlier astronomers had entertained the notion, that the Milky Way was composed of a vast number of stars

crowded closely together. But no attempt was made to offer a theory of its structure until, in 1754 Thomas Wright, in his *Theory of the Universe*, propounded views closely according with those entertained later by Sir W. Herschel. Wright, having examined a portion of the Galaxy with a reflecting telescope, only one foot in focal length, came to the conclusion that our sun is in the midst of a vast stratum of stars; that it is when we look along the direction in which this stratum extends that we see the zone of light constituting the Milky Way; and that as the line of sight is inclined at a greater and greater angle to the mean plane of the stratum, the apparent density of the star-grouping gradually diminishes.

But it is to Sir W. Herschel, and the supplementary labors of Sir J. Herschel, that we owe the more definite views now commonly entertained respecting the Via Lactea. The elder Herschel, whose nobly speculative views of nature were accompanied by practical common-sense, and a wonderful power of patient observation, applied to the heavens his celebrated method of gauging. He assumed as a first principle, to be modified by the results of observation, that there is a tolerable uniformity in the distribution of stars through space. Directing his twenty-foot reflector successively toward different parts of the heavens, he counted the number of stars which were visible at any single view. The field of view of this reflector was fifteen minutes in diameter, so that the portion of the sky included in any one view was less than one-fourth of that covered by the moon. He found the number of stars visible in different parts of the heavens in a field of view of this size to be very variable. Sometimes there were but two or three stars in the field;^[20] indeed, on one occasion he counted only three stars in four fields. In other parts of the heavens the whole field was crowded with stars. In the richer parts of the Galaxy as many as four hundred or five hundred stars would be visible at once, and on one occasion he saw as many as five hundred and eighty-eight. He calculated that in one-quarter of an hour 116,000 stars traversed the field of his telescope, when the richest part of the Galaxy was under observation. Now, on the assumption above named, the number of stars visible when the telescope was pointed in any given direction was a criterion of the depth of the bed of stars in that direction. Thus, by combining a large number of observations, a conception—rough, indeed, but instructive—might be formed of the figure of that stratum of stars within which our sun is situated.

Sir J. Herschel, during his residence at the Cape of Good Hope, carried out an extensive series of observations of the southern heavens. Applying his father's methods of gauging with a telescope of equal power, he obtained a result agreeing, in a most remarkable manner, with those obtained by Sir William Herschel. It appeared, however, that the Southern Hemisphere is somewhat richer in stars than the Northern—a result which has been accepted as indicating that our system is probably somewhat nearer the southern than the northern part of the galactic nebula. Moreover, Sir J. Herschel was led to believe that the sidereal system forms a cloven flat ring rather than a disk.

I think no one who has attentively examined the glories of Orion, the richly jeweled Taurus, the singular festoon of stars in Perseus, and the closely set stars of Cassiopeia, but must have felt that the association of splendor along this streak of the heavens is not wholly accidental. The stars here seem to form a system, and a system which one can hardly conceive to be wholly unconnected with the neighboring stream of the Milky Way. But in the southern portion the arrangement is yet more remarkable and significant. From Scorpio, over the feet of the Centaur, over the keel of Argo, to Canis Major, there is a clustering of brilliant stars, which it seems wholly impossible not to connect with the background of nebulous light. It is noteworthy, also, that this stream of stars merges into the stream commencing with the group of Orion, already noticed. Nor is this all. It is impossible not to be struck by the marked absence of bright stars in the region of the heavens between Algol, Crux, and Corvus. One has the impression that the stars have been attracted toward the region of the stream indicated, so as to leave this space comparatively bare.

Now, this last circumstance would appear less remarkable if the paucity of stars here noticed were common also in parts of the heavens far removed from the Milky Way. But this is not the case. Beyond this very region, which we find so bare of stars, we come to a region in which stars are clustered in considerable density, a region including Crater, Corvus, and Virgo, with the conspicuous stars Algore, Alkes, and Spica. But what is very remarkable, while we can trace a connection between the stream of bright stars over the Milky Way and the stream of nebulous light in the background, it is obvious that the two streams are not absolutely coincident in direction.

The stream lies on one side of the Milky Way near Scorpio, crosses it in the neighborhood of Crux, and passes to the other side along Canis Major, Orion, and Taurus. Does the stream return to the Milky Way? It seems to me that there is clear evidence of a separation near Aldebaran, one branch curving through Auriga, Perseus, and Cassiopeia, the other proceeding (more nearly in the direction originally observed) through Aries (throwing out an outlier along the band of Pisces), over the Square of Pegasus, and along the streams which the ancients compared to water from the urn of Aquarius (but which in our modern maps are divided between Aquarius and Grus). The stream-formation here is very marked, as is evident from the phenomenon having attracted the notice of astronomers so long ago. But modern travels have brought within our ken the continuation of the stream over Toucan, Hydrus, and Reticulum (the two latter names being doubtless suggested by the convolutions of the stream in this neighborhood). Here the stream seems to end in a sort of double loop, and it is not a little remarkable that the Nubecula Major lies within one loop, the Nubecula Minor within the other. It is also noteworthy that from the foot of Orion there is another remarkable stream of stars, recognized by the ancients under the name of the River Eridanus, which proceeds in a sinuous course toward this same region of the Nubeculæ.

Having thus met with evidence—striking at least, if not decisive—of a tendency to aggregation into streams, let us consider if, in any other parts of the heavens, similar traces may not be observable. We traced a stream from Scorpio toward Orion, and so round in a spiral to the Nubeculæ. Let us now return to Scorpio, and trace the stream (if any appear) in the contrary direction. Now, although over the Northern Hemisphere star-streams are not nearly so marked as over the Southern, yet there appears a decided indication of stream-formation along Serpens and Corona over the group on the left hand of Boötes to the Great Bear. A branch of this stream, starting from Corona, traverses the body of Boötes, Berenice's Hair, the Sickle in Leo, the Beehive in Cancer, passing over Castor and Pollux in Gemini, toward Capella. A branch from the feet of Gemini passes over Canis Minor, along Hydra (so named doubtless from the obvious tendency to stream-formation along the length of this constellation), and so to the right claw of Scorpio.

One other remarkable congeries of stars is to be mentioned. From the northern part of the Milky Way there will be noticed a projection toward the North Pole from the head of Cepheus. This projection seems to merge itself in a complex convolution of stars, forming the ancient constellation Draco, which doubtless included the ancient (but probably less ancient) constellation Ursa Minor. After following the convolutions of Draco, we reach the bright stars Alwaid and Etanin (Beta and Gamma) of this constellation, and thence the stream passes to Lyra, where it seems to divide into two, one passing through Hercules, the other along Aquila, curving into the remarkable group Delphinus.

The streams here considered include every conspicuous star in the heavens. But the question will at once suggest itself, whether we have not been following a merely fanciful scheme, whether all these apparent streams might not very well be supposed to result from mere accident. Now, from experiments I have made, I am inclined to believe that in any chance distribution of points over a surface, the chance against the occurrence of a single stream as marked as that which lies (in part) along the back of Grus, or as the curved stream of bright stars along Scorpio, is very great indeed; I am certain that the occurrence of *many* such streams is altogether improbable. And wherever one observes a tendency to stream-formation in objects apparently distributed wholly by chance, one is led to suspect, and thence often to detect, the operation of law. I will take an illustration, very homely perhaps, but which will serve admirably to explain my meaning. In soapy water, left in a basin after washing, there will often be noticed a tendency to the formation of spiral whorls on the surface. In other cases there may be no definite spirality, but still a tendency to stream-formation. Now, in this case, it is easy to see that the curved bottom of the basin has assisted to generate streams in the water, either circulating in one direction or opposing and modifying each other's effects, according to the accidental character of the disturbance given to the water in the process of washing.^[21] Here, of course, there can be no doubt of the cause of the observed phenomena; and I believe that in every case in which even a single marked stream is seen in any congeries of spots or points, a little consideration will suggest a regulating cause to which the peculiarity may be referred.

It is hardly necessary to say that, if the stream-formation I have indicated is considered to be really referable to systematic distribution, the theory of a

stratum of stars distributed with any approach to uniformity, either as respects magnitude or distance, must be abandoned. It seems to me to be also quite clear that the immense extent of the Galaxy, as compared with the distances of the lucid stars from us, could no longer be maintained. On this last point we have other evidence, which I will briefly consider.

First, there is the evidence afforded by clusterings in the Milky Way. I will select one which is well known to every telescopist, namely, the magnificent cluster on the sword-hand of Perseus. No doubt can be entertained that this cluster belongs to the galactic system, that is, that it is not an *external* cluster: the evidence from the configuration of the spot and from the position it occupies is conclusive on this point. Now, within this spot, which shows no stars to the naked eye, a telescope of moderate power reveals a multitude of brilliant stars, the brightest of which are of about the seventh magnitude. Around these there still appears a milky unresolved light. If a telescope of higher power be applied, more stars are seen, and around these there still remains a nebulous light. Increase power until the whole field blazes with almost unbearable light, yet still there remains an unresolved background. "The illustrious Herschel," says Professor Nichol, "penetrated, on one occasion, into this spot, until he found himself among depths whose light could not have reached him in much less than 4,000 years; no marvel that he withdrew from the pursuit, conceiving that such abysses must be endless." It is precisely this view that I wish to controvert. And I think it is no difficult matter to show at least a probability against the supposition that the milky light in the spot is removed at a vast distance behind the stars of the seventh magnitude seen in the same field.

The supposition amounts, in fact, to the highly improbable view that we are looking here at a range of stars extending in a cylindrical stratum directly from the eye—a stratum whose section is so very minute in comparison with its breadth that, whereas the whole field within which the spot is included is but small, the distance separating the nearest parts of the group from the furthest is equivalent to the immense distance supposed to separate the sphere of seventh magnitude stars from the extreme limits of our Galaxy. And the great improbability of this view is yet further increased when it is observed that within this spot there is to be seen a very marked tendency to the formation of minor streams, around which the milky light seems to cling. It seems, therefore, wholly improbable that the cluster really

has that indefinite longitudinal extension suggested by Professor Nichol. In fact, it becomes practically certain that the milky light comes from orbs really smaller than the seventh magnitude stars in the same field, and clustering round these stars in reality as well as in appearance.

The observations applied to this spot may be extended to all clusters of globular form; and where a cluster is not globular in form, but exhibits, on examination, either (1) any tendency within its bounds to stream-formation, or (2) a uniform increase in density as we proceed from any part of the circumference toward the centre, it appears wholly inconceivable that the apparent cluster is not really a cluster, but a long range of stars extending to an enormous distance directly from the eye of the observer. When, in such a case, many stars of the higher magnitudes appear within the cluster, we seem compelled to admit the probability that they belong to it; and, in any case, we can not assign to the furthest parts of the cluster a distance greatly exceeding (*proportionally*) that of the nearest parts.

Of a like character is the evidence afforded by narrow streams and necks within the Galaxy itself. If we consider the convolutions over Scorpio, it will seem highly improbable that in each of these we see, not a real convolution or stream, but the edge of a *roll* of stars. For instance, if a spiral roll of paper be viewed from any point taken at random, the chances are thousands to one against its appearing as a spiral *curve*, and, of course, the chance against several such rolls so appearing is very much greater. The fact that we are assumed to be not very far from the supposed mean plane of the Milky Way would partly remove the difficulty here considered, if it were not that the thickness and extent of the stratum, as compared with the distances of the lucid stars, must necessarily be supposed very great, on the assumption of any approach to uniformity of distribution.

Evidence pointing the same way is afforded by circular apertures in the Galaxy, or indeed by apertures of other forms. Another peculiarity of these cavities is also noticeable; whereas on the borders of every one there are many lucid stars, or in some cases two or three very bright stars, *within* the cavity there is a marked paucity of stars. This phenomenon seems to indicate a much closer connection between the brighter stars and the milky light beyond than is supposed on the stratum theory. One can hardly conceive the phenomenon to be wholly accidental.

There are some other points on which I fain would dwell, but space will not permit me. I will merely note that there are peculiarities in the distribution of red double and multiple stars, in the position in which temporary stars have made their appearance, and in the distribution of nebulæ, which seem very worthy of notice.

One point, however, immediately connected with my subject remains to be mentioned. I have traced streams of stars *more* conspicuous than those forming the Milky Way. We have also evidence of streams of light yet more delicate and evanescent than the light of our own Galaxy. In Sir John Herschel's great work on the southern skies, he notes the frequent recurrence of "an exceedingly delicate and uniform dotting, or *stippling*, of the field of view by points of light too small to admit of any one being steadily or fully examined, and too numerous for counting, were it possible so to view them." In thirty-seven places he detected this remarkable and significant phenomenon; a phenomenon so faint that he says, "The idea of illusion has continually arisen subsequently"; an idea well befitting the modesty of the philosophic observer, but which those who appreciate Sir John Herschel's skill as an observer will be very unwilling to accept. As Professor Nichol remarks, "It is enough to read from Herschel's notebook —'I feel satisfied the stippling is no illusion, for its dark mottling moves with the stars as I move the tube to and fro'—to feel convinced that the phenomenon is real." Now a remarkable fact connected with those observations is, that when Sir J. Herschel marked down in a star-chart the places in which he had detected this nebulous appearance, he found that, "with the exception of *three* which appeared outlying and disconnected, they formed several *distinct but continuous streams*."

FOOTNOTES:

[20] Field means the actual space covered by the lens.—E. S.

[21] Sometimes a singular regularity of curvature is noticed, and a spiral is formed closely resembling in configuration some of the great spiral nebulæ, as drawn by Lord Rosse, so that one is tempted to see in the centrifugal tendency of the disturbed water, and the centripetal effects caused by reflection from the

basin's surface, causes which may in some sense illustrate the laws operating in wider domains of space.

THE MAGELLANIC CLOUDS—ZODIACAL LIGHT—STAR GROUPS.—AMÉDÉE GUILLEMIN

When we look on the region of the celestial vault which surrounds the South Pole, we can not help being struck with the contrast presented by the small quantity of stars which it contains, with the brilliant zone which borders the Milky Way, from Orion and Argo to the Centaur, passing by the Southern Cross. One solitary star of the first magnitude, Achernar, more distant from the pole than are the beautiful stars of the Centaur and of the Cross, shines in this part of the sky.

But even this circumstance renders the singular aspect of the two nebulous spots, which seem two detached pieces of the great galactic zone, still more striking. These half-stellar, half-nebulous systems, unequal in magnitude and brightness, but easily seen with the naked eye on a clear, moonless night, are situated, one, the larger and more brilliant, between the pole and Canopus, in the constellation of Doradus; the other, the smaller and less brilliant, ordinarily visible during the full moon, in Hydrus, between Achernar and the pole.

Both are known by astronomers and navigators under the name of “Cape Clouds,” or again, “Magellanic Clouds.” And, to distinguish them, we have again the Great Cloud (*Nebecula Major*) and the Small Cloud (*Nebecula Minor*).

The Clouds of Magellan are distinguished from all other nebulae by their great apparent dimensions, and by their physical structure; this last character distinguishes them from most of the branches and offshoots of the Milky Way, with which, we may also add, they do not appear connected in any way.

The Great Cloud extends over a space which embraces not less than forty-two square degrees—about two hundred times the apparent surface of the lunar disk. The Small Cloud occupies in extent four times less than the other; according to Humboldt, it is surrounded “with a kind of desert,” where, it is true, shines the magnificent stellar cluster of Toucan. If the

exterior aspect of these two remarkable nebulae, and their situation in a celestial region poor in stars, give to the southern sky a peculiar appearance, their real structure makes them one of the wonders of the heavens.

In the Great Cloud, Herschel has counted 582 single stars, among which one only is of the fifth magnitude; six others are of the order immediately inferior, and would doubtless be visible to the naked eye if their light were not effaced by the general glare.

In the Small Cloud, the single stars are proportionally more numerous, since 200 have been counted, among which three are of the sixth magnitude, while it only includes thirty-seven of the nebulae and seven star-clusters. These immense aggregations, the elements of which are themselves swarms of suns, remind us of the largest, in appearance at least, of all the clusters which the eye contemplates in the depths of the sky—the Milky Way.

In the evenings, about the time of the vernal equinox—in March and April, when in our climate the twilight is of short duration—if we examine the horizon toward the west, a little after sunset, we may perceive a faint light that rises in the form of a cone among the starry constellations.

This is what astronomers call the Zodiacal Light. Those unfamiliar with it, or little accustomed to the ordinary aspect of the sky, might confuse the glimmering either with the Milky Way or with the ordinary twilight, or even with an aurora. But, with a little attention, it is impossible to mistake it.

The triangular form of this luminous cone, its elevation and its inclined position to the horizon, make it a thing apart, and one eminently deserving particular mention.

As the days lengthen, and with them the duration of twilight, the Zodiacal Light disappears; it becomes invisible, at least in our climate. But it may again be seen in the morning, in the east, about the time of the autumnal equinox, in September and October, when the dawn has an equally short duration—again, however, to disappear during the period of long nights and long twilights.

It is needless to add that the sky must be clear and the night moonless for observations of the Zodiacal Light to be possible.

Among the explanations that have been given, the most probable one is that which likens the Zodiacal Light to a flattened nebulous ring surrounding the sun at some distance. It is to be remarked that the direction of the axis of the cone, or of the pyramid, prolonged below the horizon, always passes through the sun.

It was believed at first that this direction precisely coincided with the solar equator; but it seems more certain that it coincides with the plane of the earth's orbit, or the ecliptic.

Now, what is the nature of this luminous mass? Must it be considered as a zone of vapors thrown off by the sun, when in the process of consolidation, when our central star passed from a nebulous state to that of a condensed fluid sphere? This was the opinion of Laplace.

Another hypothesis, also connected with the first, is that the Zodiacal Light is formed of myriads of solid particles, analogous to the aerolites, possessing a general movement, but traveling separately around the focus of our solar world. The light of the ring would be thus produced by the accumulation of this multitude of brilliant points, reflecting toward us the light borrowed by each of them from the sun.

This explanation accounts for the intensity of the Zodiacal Light at different epochs; it would suffice to admit that the condensation of the particles or the density of the ring is not the same throughout its extent, and that its movement of circulation round the sun presents successively different parts to the earth. In this case, it becomes a question whether this lenticular ring of matter is distinct from the zone of aerolites.

Lastly, some astronomers regard the Zodiacal Light as a vaporous ring which belongs to the earth, surrounding it at some distance. But this is an opinion which appears somewhat wild, and is utterly at variance with observation.

Are the stars that are visible to the naked eye spread orderless on the celestial vault? or is there not between those apparently most closely connected some real or physical connection which requires us to rank them in natural groups?

These questions have been already partly solved by what is known of the double and multiple star systems. Soon, exploring the regions of the sky

visible by means of the telescope, we shall have to pass in review a multitude of stellar associations, in which suns are found so compact and so numerous, and the form of the groups so regular, that it is impossible to deny their reciprocal dependence.

But long before the discovery of these islands, these archipelagos as worlds, scattered with such astonishing profusion over the infinite, the naked eye had already distinguished a certain number of groups, the stars composing which were so near together that it was impossible to doubt their physical connection.

Such, for example, is the group of the Pleiades. Such, again, are the groups known under the names of the Hyades, of Præsepe, and of Berenice's Hair. All are visible to the naked eye, and good eyes distinguish without difficulty the principal stars of the first-named groups. The Pleiades are situated in the constellation of the Bull, which we can distinguish so easily to the northwest of Orion and Aldebaran.

Of about eighty stars which form the group of the Pleiades, six are visible without the help of telescopes. Formerly, the Latin poet tells us, seven were counted, which may be held to prove that one of them is variable, and has diminished in brightness, or else has disappeared.

The most brilliant, Alcyone, is of the third magnitude; Electra and Atlas are of the fourth; Merope, Maia, and Taygete of the fifth. Three others again have received particular names, although they are below the limit of ordinary vision; these are Pleione, Celeno, and Asterope, from the sixth to the eighth magnitude. All the others are only visible by the aid of a telescope; but with an ordinary glass it is possible to distinguish a large number. The Pleiades are known under the name of the Hen-coop, doubtless because Alcyone appears in the group as a hen surrounded with her chickens.

The Hyades, which are near the Pleiades, form a less numerous and more scattered group. The bright light of Aldebaran, which is, as is known, of the first magnitude, renders them more difficult to distinguish with the naked eye.

They appear in the rainy season. Hence their name of Hyades, from the Greek word which signifies to rain.

The connection of the stars which compose this group is not so striking as in the case of the Pleiades. Nevertheless, it seems difficult to admit that they are quite independent of each other's attraction. In examining the position of these two groups in the vicinity of the Milky Way, and observing that both are situated in the prolongation of a branch of the great zone, we are almost entitled to consider them as two clusters of stars, belonging to the immense stellar stratum which surrounds us, and in the midst of which the sun himself is placed.

In Berenice's Hair, most of the stars are visible to the naked eye, and are perfectly distinguished in the sky, a little to the east of the Lion. No very brilliant star in the vicinity inconveniences the eye by effacing their light.

The next group is situated in the Crab, and is known under the name of *Præsepe*: it is visible to the unassisted sight; but it is impossible to distinguish the separate stars without the help of a telescope. Nevertheless, an instrument of moderate power easily separates them.

The groups which we have just described form a transition between the stars scattered over the celestial vault and the more condensed clusters, the undefined aspect of which caused them formerly to be designated under the general name of *nebulæ*.

Doubtless, if we could place ourselves in space, and contemplate from a sufficiently distant standpoint the whole of the stars which appear to us isolated, we should see them condensed into one or several distinct groups, analogous to those of the Pleiades; while, were we to penetrate into the midst of one of those compact clusters, we should see the stars of which it is formed separated and scattered over the celestial vault in such a way as to give it the aspect of our own heavens.

THE NEBULÆ AND SWARMS OF SUNS.—J. E. GORE

We will now consider the nebulæ, properly so called, that is to say, objects which the spectroscope shows to consist of glowing gas. These are sometimes large and irregular in form, like the great nebula in the “Sword” of Orion, sometimes with spiral convolutions, and sometimes of a definite shape, like the planetary and annular nebulæ.

Of the large and irregular nebulæ, one of the most remarkable is that known as “the great nebula in Orion.” It surrounds the multiple star, Theta Orionis. It is a curious fact that it escaped the searching eye of Galileo, although he gave special attention to the constellation of Orion, for even with a good opera-glass a nebulous gleam is distinctly visible round the central star of the “Sword.” The nebula seems to have been discovered by Cysat, a Swiss astronomer, in the year 1618, and it was sketched by Huygens in 1656. It has been called the “fish-mouth” nebula, from the fancied resemblance of the centre portion to the mouth of a fish. A number of small stars are visible over the surface of the nebula, and at one time Lord Rosse thought it showed indications of resolution into stars when examined with his giant telescope; but this is now known to have been a mistake, for Dr. Huggins finds, with the spectroscope, that it consists of nothing but glowing gas.

The brightest line in the nebular spectrum—the “chief nebular line,” as it is called—has not yet been identified with that of any terrestrial substance.

Mr. W. H. Pickering and Dr. Max Wolf have photographed another nebula surrounding the star Zeta Orionis—the southern star of the “Belt,” which seems to be connected with the nebula in the “Sword”; and Professor Barnard, using the “lens of a cheap oil lantern” of 1½ inches aperture and 3½ inches focal length, has photographed “an enormous curved nebulosity” stretching over nearly the whole of the constellation of Orion, and involving the “great nebula.”

Professor Keeler found, with the spectroscope, that the Orion nebula is apparently receding from the earth at the rate of nearly eleven miles a second, but this motion may be, in part at least, due to the sun’s motion in

space in the opposite direction. Professor Pickering considers that the parallax of the nebula is probably not more than 0."003, which corresponds to a thousand years' journey for light!

In the southern constellation, Argo is a magnificent nebula, somewhat similar in appearance to the great nebula in Orion. It surrounds the famous variable star Eta Argûs. It is sometimes spoken of as the "keyhole" nebula, owing to a curious opening of that shape near its centre. It was carefully drawn by Sir John Herschel at the Cape of Good Hope in the years 1834-38. It lies in a very brilliant portion of the Milky Way, and Sir John Herschel thus describes it: "It is not easy for language to convey a full impression of the beauty and sublimity of the spectacle which the nebula offers as it enters the field of view of a telescope, fixed in right ascension, by the diurnal motion, ushered in as it is by so glorious and innumerable a procession of stars, to which it forms a sort of climax, and in a part of the heavens otherwise full of interest," and he adds: "In no part of its extent does this nebula show any appearance of resolvability into stars, being, in this respect, analogous to the nebula of Orion. It has, therefore, nothing in common with the Milky Way, on the ground of which we see it projected, and may therefore be, and not improbably is, placed at an immeasurable distance behind that stratum." Sir John Herschel's conclusion as to its physical constitution has been fully confirmed by the spectroscope, which shows it to consist of luminous gas. As in the Orion nebula, there are numerous stars scattered over it. Some of these may possibly have a physical connection with the nebula, while others may belong to the Milky Way. The nebula is of great extent, covering an apparent space about five times the area of the full moon, and its real dimensions must be enormous. It was photographed by Mr. Russell, director of the Sydney Observatory, in July, 1890, and the photograph shows that "one of the brightest and most conspicuous parts of the nebula"—the swan-shaped form near the centre of Herschel's drawing—has "wholly disappeared," and its place is now occupied by "a great, dark oval." Mr. Russell first missed the vanished portion of the nebula in the year 1871, while examining it with a telescope of 11½ inches aperture, and the photograph now confirms the disappearance, which is very remarkable, and shows that changes are actually in progress in these wonderful nebulae, changes which may be detected after a comparatively short interval of time.

Smaller than the nebula in Argo, but somewhat similar in general appearance, is that known as 30 Doradus, which forms one of the numerous and diverse objects which together constitute the greater Magellanic Cloud. Sir John Herschel drew it carefully at the Cape of Good Hope, and describes it as “one of the most singular and extraordinary objects which the heavens present,” and he says “it is unique even in the system to which it belongs, there being no other object in either nubecula to which it bears the least resemblance.” It is sometimes called the “looped nebula,” from the curious openings it contains. One of these is somewhat similar to the “key-hole” opening in the Argo nebula. Near its centre is a small cluster of stars, and scattered over the nebula are many faint stars, of which Sir John Herschel gives a catalogue of 105, ranging from the ninth to the seventeenth magnitude. I do not know whether this nebula has been examined with the spectroscope, but its appearance would suggest that it is gaseous. It is remarkable as being the only object of its class which is found outside the zone of the Milky Way.

Among the nebulae of irregular shape, although its spectrum is said to be not gaseous, may be mentioned that known as the “trifid nebula,” or 20 Messier. It lies closely north of the star 4 Sagittarii in a magnificent region of the heavens. In the drawing made by Sir John Herschel at the Cape of Good Hope, the principal portion consists of three masses of nebulous matter separated by dark “lanes” or “rifts.” Near the junction of the three “rifts” is a triple star. A beautiful drawing of this nebula has also been made by Trouvelot. It agrees fairly well with that of Sir John Herschel, but shows more detail.

Among other gaseous nebulae may be mentioned that called by Sir John Herschel the “dumb-bell” nebula. It lies a little south of the sixth magnitude star 14 Vulpeculae, and was discovered by Messier in 1779, while observing Bode’s comet of that year. In small telescopes it has the appearance of a dumb-bell, or hour-glass, but in larger telescopes the outline is filled in with fainter nebulous light, giving to the whole an elliptical form. Several faint stars have been seen in it, but these probably belong to the Milky Way, as Dr. Huggins finds the spectrum gaseous. Dr. Roberts has photographed it, and he thinks that “the nebula is probably a globular mass of nebular matter which is undergoing the process of condensation into stars, and the faint protrusions of nebulosity in the *south following* and *north preceding* ends

are the projections of a broad ring of nebulosity which surrounds the globular mass. This ring, not being sufficiently dense to obscure the light of the central region of the globular mass, is dense enough to obscure those parts of it that are hidden by the increased thickness of the nebulosity, thus producing the 'dumb-bell' appearance. If these inferences are true, we may proceed yet a step, or a series of steps, further, and predict that the consummation of the life-history of this nebula will be its reduction to a globular cluster of stars."

Among the gaseous nebulae may also be included those known as "annular nebulae." These are very rare objects, only a few being known in the whole heavens. The most remarkable is that known as 57 Messier, which lies between the stars Beta and Gamma Lyræ, south of the bright star Vega. It was discovered by Darquier, at Toulouse, in 1779, while following Bode's comet of that year. Lord Rosse thought it resolvable into stars, and so did Chacornac and Secchi, but no stars are perceptible with the great American telescopes, and Dr. Huggins finds it to be gaseous. The central portion is not absolutely dark, but contains some faint nebulous light. Examined with the great telescope of the Lick Observatory, Professor Barnard finds that the opening of the ring is filled in with fainter light "about midway in brightness between the brightness of the ring and the darkness of the adjacent sky. The aperture was more nearly circular than the outer boundary of the nebula, so that the ends of the ring were thicker than the sides." The entire nebula was of a milky color. A central star, noticed by some observers, was usually seen by Professor Barnard, but was never a conspicuous object. He found the extreme dimensions of the nebula about 81" in length by about 59" in width, or more than double the apparent area of Jupiter's disk. It has been beautifully photographed by Dr. Roberts, and he says "the photograph shows the nebula and the interior of the ring more elliptical than the drawings and descriptions indicate; and the star of the *following* side is nearer to the ring than the distance given. The nebulosity on the *preceding* and *following* ends of the ring protrudes a little, and is less dense than on the *north* and *south* sides. This probably suggested the filamentous appearance which Lord Rosse shows. Some photographs of the nebula have been taken between 1887 and 1891, and the central star is strongly shown on some of them, but on others it is scarcely visible, which points to the star being variable." On a photograph taken by MM. Andoyer and Montaugerand of the Toulouse Observatory, with an exposure of nine

hours (in multiple exposures), about 4,800 stars are visible on and near the nebula in an area of three square degrees.

Another object of the annular class will be found a little to the southwest of the star Lambda Scorpii. It is thus described by Sir John Herschel: "A delicate, extremely faint, but perfectly well defined, annulus. The field crowded with stars, two of which are on the nebula. A beautiful, delicate ring of a faint, ghost-like appearance, about 40" in diameter in a field of about 150 stars, eleven and twelve magnitude and under."

Near the stars 44 and 51 Ophiuchi is another object of the annular class, which Sir John Herschel describes as "exactly round, pretty faint, 12" diameter, well terminated, but a little cottony at the edge, and with a decided darkness in the middle, equal to a tenth magnitude star at the most. Few stars in the field, a beautiful specimen of the planetary annular class of nebula."

The Planetary Nebulae form an interesting class. They were so named by Sir William Herschel from their resemblance to the disks of the planets, but, of course, much fainter. They are generally of uniform brightness, without any nucleus or brighter part in the centre. There are numerous examples of this class, one of the most remarkable being that known as 97 Messier, which is situated about two degrees southeast of Beta Ursae Majoris—the southern of the two "pointers" in the Plow. It is of considerable apparent size, and even supposing its distance to be not greater than that of 61 Cygni, its real dimensions must be enormous. Lord Rosse observed two openings in the centre with a star in each opening, and from this appearance he called it the "owl nebula." One of the stars seems to have disappeared since 1850, and a photograph recently taken by Dr. Roberts confirms the disappearance.

Another fine object of the planetary class is one which lies close to the pole of the ecliptic. Webb saw it "like a considerable star out of focus." Smyth found it pale blue in color. Dr. Huggins finds a gaseous spectrum, the first discovery of the kind made. Professor Holden, observing it with the great Lick telescope, finds its structure extraordinary. He says it "is apparently composed of rings overlying each other, and it is difficult to resist the conviction that these are arranged in space in the form of a true helix," and he ranks it in a new class which he calls "helical nebulae."

A somewhat similar nebula lies a little to the west of the star Nu Aquarii. Secchi believed it to be in reality a cluster of small stars, but Dr. Huggins finds its spectrum gaseous. A small nebula on each side gives it an appearance somewhat similar to the planet Saturn, with the rings seen edgewise. The great Lick telescope shows it as a wonderful object—"a central ring lies upon an oval of much fainter nebulosity." Professor Holden says "the color is a pale blue," and he compares the appearance of the central ring "to that of a footprint left in the wet sand on a sea beach."

About two degrees south of the star Mu Hydræ is another planetary nebula, which Smyth describes as resembling the planet Jupiter in "size, equable light and color." Webb saw it of "a steady, pale blue light," and Sir John Herschel, at the Cape of Good Hope, speaks of its color as "a decided blue—at all events, a good sky-blue," a color which seems characteristic of these curious objects. Although Sir William Herschel, with his large telescopes, failed to resolve it into stars, Secchi thought he saw it breaking up into stars with a "sparkling ring." Dr. Huggins, however, finds the spectrum to be gaseous, so that the luminous points seen by Secchi could not have been stellar.

Sir John Herschel, in his *Cape Observations*, describes a planetary nebula which lies between the stars Pi Centauri and Delta Crucis. He says it is "perfectly round, very planetary, color fine blue ... very like Uranus, only about half as large again, and blue.... It is of the most decided independent blue color when in the field by itself, and with no lamplight and no bright star. About 10' north of it is an orange-colored star, eighth magnitude. When this is brought into view, the blue color of the nebula becomes intense ... color, a beautiful rich blue, between Prussian blue and verditer green."

There are some rare objects called "nebulous stars." The star Epsilon Orionis—the centre star of Orion's Belt—is involved in a great nebulous atmosphere. The triple star Iota Orionis is surrounded by a nebulous haze. The star Beta in Canes Venatici is a 4½ magnitude star surrounded by a nebulous atmosphere.

The term elliptical nebulae has been applied to those of an elliptical or elongated shape. This form is probably due in many cases to the effect of perspective, their real shape being circular, or nearly so. Perhaps the most remarkable object of this class is the well-known "nebula in Andromeda,"

known to astronomers as 31 Messier. It can be just seen with the naked eye, on a clear moonless night, as a hazy spot of light near the star Nu Andromedæ, and it is curious that it is not mentioned by the ancients, although it must have been very visible to their keen eyesight in the clear Eastern skies. It was, however, certainly seen so far back as 905 A. D., and it is referred to as a familiar object by the Persian astronomer, Al-Sûfi, who wrote a description of the heavens about the middle of the Tenth Century. Tycho Brahe and Bayer failed to notice it, but Simon Marius saw it in December, 1612, and described it "as a light seen from a great distance through half-transparent horn plates." It was also observed by Bullialdus, in 1664, while following the comet of that year. It has frequently been mistaken for a comet by amateur observers in recent years. Closely northwest of the great nebula is a smaller one discovered by Le Gentil in 1749, and another to the south, detected by Miss Caroline Herschel in 1783. The great nebula is of an elliptical shape and considerable apparent size. The American astronomer, Bond, using a telescope of 15 inches aperture, traced it to a length of about four degrees, and a width of two and a half degrees. A beautiful photograph taken by Dr. Roberts in December, 1888, shows an extension of nearly two degrees in length, and about half a degree in width, or considerably larger than the apparent size of the full moon. Bond could not see any symptom of resolution into stars, but noticed two dark rifts or channels running nearly parallel to the length of the nebula. In Dr. Roberts's photograph these rifts are seen to be really dark intervals between consecutive nebulous rings into which the nebula is divided. Dr. Roberts says: "A photograph which I took with the 20-inch reflector on October 10, 1887, revealed for the first time the true character of the great nebula, and one of the features exhibited was that the dark bands, referred to by Bond, formed parts of divisions between symmetrical rings of nebulous matter surrounding the large diffuse centre of the nebula. Other photographs were taken in 1887, November 15; 1888, October 1; 1888, October 2; 1888, December 29; besides several others taken since, upon all of which the rings of nebulosity are identically shown, and thus the photographs confirm the accuracy of each other, and the objective reality of the details shown of the structure of the nebula." Dr. Roberts adds: "These photographs throw a strong light on the probable truth of the *Nebular Hypothesis*, for they show what appears to be the progressive evolution of a gigantic stellar system."

The largest telescopes have hitherto completely failed to resolve this wonderful object into stars. Dr. Huggins, however, finds that the spectrum is *not* gaseous, so that if the nebula really consists of stellar points, they must be of very small dimensions.

The question may be asked, What is the probable size and distance of this wonderful nebula? and could it be an external universe?

The temporary star which appeared near the nucleus of the nebula in August, 1885, was of the seventh magnitude. I find that our sun, if placed at the distance indicated by a parallax of $\frac{1}{200}$ th of a second, would be reduced to a star of about the eleventh magnitude, or four magnitudes fainter than the temporary star appeared to us. That is to say, the star would have been—with the assumed distance—about forty times brighter than the sun. With any greater distance, the star would have been proportionately brighter, compared with the sun. This seems improbable, and tends to the conclusion that the nebula is *not* an external galaxy, but a member of our own sidereal system, a system which probably includes all the stars and nebulae visible in our largest telescopes. Dr. Common, indeed, suggests that it may be comparatively near our system. He says: “It is difficult to imagine that such an enormous object, as the Andromeda nebula must be, is not very near to us; perhaps it may be found to be the nearest celestial object of all beyond the Solar System. It is one that offers the best chance of the detection of parallax, as it seems to be projected on a crowd of stars, and there are well-defined points that might be taken as fiducial points for measurement,” and he adds: “Apart from the great promise this nebula seems to give of determining parallax, there is a fair presumption that in the course of time the rotation of the outer portion may perhaps be detected by observation of the positions of the two outer detached portions in relation to the neighboring stars.”

The spiral nebulae are wonderful objects, and were discovered by the late Lord Rosse with his great six-foot telescope. Their character has been fully confirmed by photographs taken by Dr. Roberts. One of the most remarkable of these extraordinary objects is that known as 51 Messier. It lies about three degrees southwest of the bright star Eta Ursae Majoris—the star at the end of the Great Bear’s tail. It was discovered by Messier while comet-hunting on October 13, 1773. Telescopes of moderate power merely show two nebulae nearly in contact, but Lord Rosse saw it as a wonderful

spiral, and his drawing agrees fairly well with a photograph taken by Dr. Roberts in April, 1889. The nebula has also been photographed by Dr. Common. Dr. Roberts says: "The photograph shows both nuclei of the nebula to be stellar, surrounded by dense nebulosity, and the convolutions of the spiral in this as in other spiral nebulae are broken up into star-like condensations with nebulosity around them. Those stars that do not conform to the trends of the spiral have nebulous trails attached to them, and seem as if they had broken away from the spirals." A tendency to a spiral structure in the smaller nebula is also visible on the original negative. Dr. Huggins finds that the spectrum is *not* gaseous.

The nebula known as 99 Messier is of the spiral form. It lies on the borders of Virgo and Coma Berenices, near the star 6 Comæ. In large telescopes it somewhat resembles a "Catherine wheel." D'Arrest and Key thought it resolvable into stars. It has been photographed by M. Von Gothard.

Among the clusters and nebulae, we may class the Magellanic Clouds, or Nubeculae in the Southern Hemisphere, as they consist of stars, clusters, and nebulae.

Among the so-called nebulae are many objects which, when examined with telescopes of adequate power, are seen to be resolved into myriads of small stars; their comparative isolation from surrounding objects impresses us forcibly with the idea that they form, as it were, families of stars connected by some physical bond of union. Of these clusters, as they are called, we have naked-eye examples in the Pleiades and the "Bee-Hive" in Cancer. Others may be partially seen with a good opera-glass or binocular, but most of them require telescopes of considerable power to view them to advantage. They are of various forms and of all degrees of condensation. Some are comparatively large and irregular, others small and compressed, with the component stars densely crowded. Many are of such uniform shape as to have received the name of globular clusters. These have been aptly termed "balls of stars," and are among the most interesting objects in the stellar heavens.

The most remarkable object of this class visible in the Northern Hemisphere is that known as 13 Messier. It lies between the tolerably bright stars Zeta and Eta Herculis, nearer the latter star. It may be seen with an

opera-glass as a hazy-looking star of about the sixth magnitude, with a star on each side of it. Examined with a powerful telescope, it is resolved into numerous small stars. Sir William Herschel estimated them at 14,000, but the real number is probably much less. Assuming the average magnitude of the components at twelve and a half, I find that an aggregation of 14,000 stars of this brightness would shine as a star of about the second magnitude, or a little fainter.

Another object of the globular class, but less resolvable, is that known as 92 Messier, which lies between the stars Eta and Iota in Hercules, nearer the latter. Sir William Herschel's telescopes showed it as seven or eight minutes of arc in diameter. It is considerably brighter at the centre. The larger components are easily visible in moderate-sized telescopes, but even Lord Rosse's giant instrument failed to resolve the central blaze. There is no doubt, however, that it consists wholly of small stars, as the unerring eye of the spectroscope shows a stellar spectrum, similar to that of the neighboring 13 Messier.

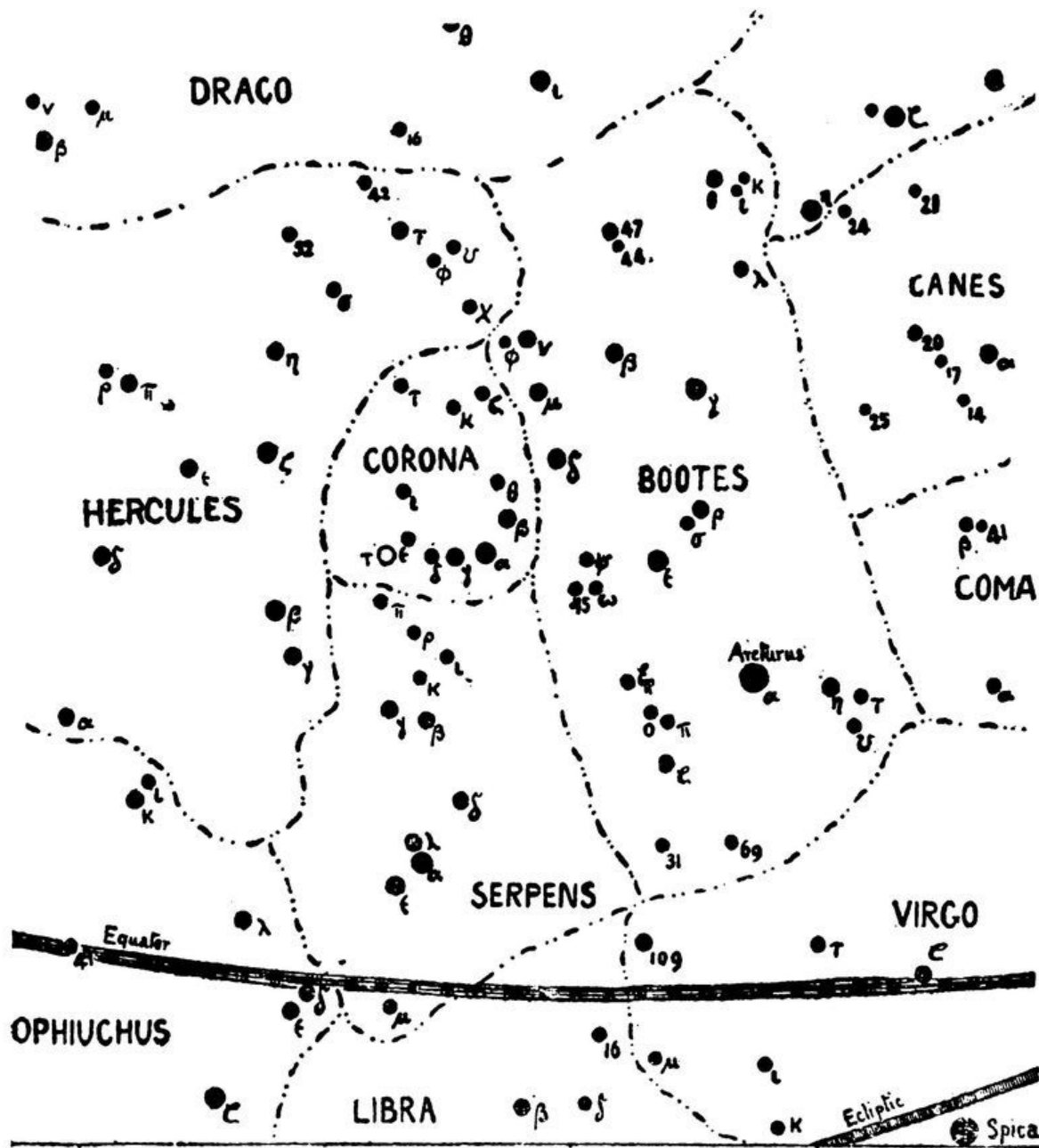


Fig. 17.—The Region of Boötes and Hercules

Another fine example of the globular class is 5 Messier, which lies closely north, preceding the fifth magnitude star, 5 Serpentis. It is considerably compressed at the centre. Sir William Herschel counted 200 stars, but failed to resolve the central nebulosity. Messier, its discoverer, found it visible with a telescope only one foot long.

Another fine object is 3 Messier, in Boötes. Admiral Smyth describes it as “a brilliant and beautiful globular aggregation of not less than 1,000 small stars.” It is beyond the power of small telescopes, but it was resolved by Buffham, even in the centre, with a 9-inch reflector.

Numerous fine examples of the globular class are found in the Southern Hemisphere, which indeed seems to be richer in these marvelous objects than the northern sky. Of these the most interesting are those known as Omega Centauri and 47 Tucani. Omega Centauri, from its great apparent size—about two-thirds of the moon’s diameter—and its visibility to the naked eye, may perhaps be considered as the most remarkable object of its kind in the heavens. It shines as a hazy star of the fourth magnitude, and I have often so seen it in the Punjab sky. Its large size and globular form are clearly visible in a binocular field-glass, but, of course, its component stars are far beyond the reach of such an instrument. Sir John Herschel, observing it with his large telescope at the Cape of Good Hope, found it a “truly astonishing object. All clearly resolved into stars of two magnitudes, viz., thirteen and fifteen, the larger lying in lines and ridges over the smaller;... the larger form rings like lace-work on it.” If we take the average magnitude of the components at thirteen and a half, the apparent brightness of the cluster would imply that it contains about 15,000 stars.



The Great Nebula in the Constellation Cygnus

The other wonderful cluster is that known as 47 Tucani, which lies close to the smaller Magellanic Cloud. It is smaller in apparent size than Omega Centauri, but Dr. Gould, observing it at Cordoba, speaks of it as “one of the most impressive and perhaps the grandest of its kind in either hemisphere,” and he estimates its magnitude at four and a half, as seen with the naked eye. It is thus described by Sir John Herschel: “A most magnificent globular cluster. It fills the field with its outskirts, but within its more compressed part I can insulate a tolerably defined, circular space, of 90" diameter, wherein the compression is much more decided, and the stars seem to run together, and this part, I think, has a pale pinkish or rose color, which contrasts, evidently, with the white light of the rest. The stars are equal, fourteen magnitude, immensely numerous and compressed.... Condensation in three distinct stages.... A stupendous object.” Sir John Herschel’s drawing of this cluster reminds one of a swarm of bees, and perhaps suggested to Tennyson the lines:

“Clusters and beds of worlds, and bee-like swarms
Of suns and starry streams.”

There are other interesting specimens of the globular class in the Southern Hemisphere, but not of such large apparent dimensions as those already described. Of these may be mentioned 22 Messier, which lies about midway between the stars Mu and Sigma Sagittarii. It is described by Sir John Herschel as a fine globular cluster, with stars of two magnitudes, namely eleven or twelve, and fifteen or sixteen, the larger being visibly reddish, and he suggested that it consists of “two layers, or one shell over another.” Owing to the comparative brightness of the larger components, this cluster forms a good object for small telescopes. I saw the brighter stars well with a 3-inch refractor in the Punjab sky, but, of course, the greater portion of the cluster has a nebulous appearance in a telescope of this size.

Between Alpha and Beta Scorpii there is a condensed globular cluster. With small telescopes it very much resembles a telescopic comet, but with larger instruments its true character is revealed. Sir William Herschel considered it “the richest and most condensed mass of stars in the firmament.” In May, 1860, a “temporary star” of the seventh magnitude suddenly appeared in the centre, almost blotting out the cluster by its superior light. The star faded away before the end of June of the same year,

and has not been seen with any certainty since. It has been suggested that this temporary star lay *between* the cluster and the earth, but it seems to me much more probable that the outburst took place *in* the cluster itself, and that it was possibly caused by a collision between two of the component stars, or by a swarm of meteors rushing with a high velocity through the cluster.

The beauty and sublimity of the spectacle presented by these globular clusters, when viewed with a powerful telescope, is such as can not be adequately described, and it has been said that when seen for the first time, “few can refrain from a shout of rapture.” The component stars, although distinctly visible as points of light, defy all attempts at counting them, and seem literally innumerable. Placed like a mass of glittering diamond-dust on the dark background of the heavens, they impress us forcibly with the idea that if each of these lucid points is a sun, the thousands which seem massed together in so small a space must be in reality either relatively close and individually small, or else the system of suns must be placed at a distance almost approaching the infinite.

The distance of these globular clusters from the earth is, however, certainly very great. Attempts to accurately determine their position in space have not been attended with success. As the component stars are at practically the same distance from the eye, we have no comparison stars to measure from, and their exact distance, therefore, remains unknown. We may, however, estimate their probable distance with some show of plausibility. We may assume that the stars of the Hercules cluster would, if concentrated in a point, shine as a star of about the fourth magnitude. As the components are of about the twelfth and thirteenth magnitudes, this would imply that the cluster consists of about 2,500 stars. With the data assumed, we may therefore conclude that the components of the Hercules cluster are suns of comparatively small size, separated by considerable distances, but apparently massed together by the effect of distance.

Among less condensed star clusters there are many interesting objects. The Pleiades have been already referred to. On a photograph of this remarkable group, taken at the Paris Observatory, over 2,000 stars can be counted of all degrees of brilliancy, from those visible without optical aid down to points of light so faint as to be invisible to the eye in the telescope with which they were photographed. Here we have a cluster of probably

larger size than that in Hercules, probably at a greater distance from the earth, and with its larger components of considerably greater mass than our sun.

Near the bright star Pollux, I see a small cluster of stars of about the seventh and eight magnitudes, which, with a binocular field-glass, very much resembles the Pleiades as seen with the naked eye. A smaller cluster (known as 39 Messier) may be seen near the star Pi Cygni.

The well-known Chi Persei may be also seen with an opera-glass, but a telescope is necessary to show the component stars to advantage, and the larger the telescope the greater the number of faint stars in these wonderful objects.

The cluster known as 35 Messier, a little north of the star Eta Geminorum, is visible in an opera-glass, but a small telescope is required to see the component stars. A well-marked clustering tendency is visible among the brighter stars of the group, two, three, four, and sometimes five stars being grouped together in subordinate collections. Admiral Smyth says: "It presents a gorgeous field of stars from the ninth to the sixteenth magnitude, but with the centre of the mass less rich than the rest. From the small stars being inclined to form curves of three or four, and often with a large one at the root of the curve, it somewhat reminds one of the bursting of a sky-rocket." This tendency to "stream" formation in the components of star clusters is also well marked in a photograph of the cluster 38 Messier (kindly sent to me by MM. Henry of the Paris Observatory). It was described by Webb as "a noble cluster arranged in an oblique cross," and Smyth says: "The very unusual shape of this cluster recalls the sagacity of Sir William Herschel's speculations upon the subject, and very much favors the idea of an attractive power lodged in the brightest part. For although the form is not globular, it is plainly to be seen that there is a tendency toward sphericity, by the swell of the dimensions as they draw near the most luminous part, denoting, as it were, a stream or tide of stars, setting toward the centre."

Sir W. Herschel, speaking of a compressed cluster in Perseus, says "the large stars are arranged in lines like interwoven letters," and Webb says "it is beautifully bordered by a brighter foreshortened pentagon."

Observing with a 3-inch telescope in India, I noticed a beautiful cluster of stars, about 4° north of Gamma and Upsilon Scorpii, resembling in shape a bird's foot, with remarkable streams of stars. This cluster is visible to the naked eye as a star of about the fifth magnitude.

Although these loosely associated star clusters do not show such evidence in favor of family connection as the more closely compacted globular clusters, still we can hardly escape from the conviction that their apparent aggregation is really due to some physical bond of union, and not merely the result of a fortuitous scattering of stars at different distances in the line of sight.

THE GREAT NEBULA OF ORION.—SIR ROBERT S. BALL

The telescope, ever an ally in the study of the heavens, is in this part of the science absolutely indispensable. In other branches of astronomy we can learn something without its aid. Indeed, many great astronomical discoveries were made long before the telescope was invented. But ere this memorable event in the history of science it was impossible for us to know anything of the existence of the nebulae. It is indeed true that there is one of these objects which can be just detected by the naked eye. It lies in the constellation of Andromeda, where, on a clear and dark night, a faint spot of light can just be discerned by a good eye. But a mere glimpse gives us really no adequate notion of the true character of the object. It might only, so far as the naked eye discloses its nature, be a cluster of stars like that we have already discerned in Perseus, or like the similar group that, under the name of the Beehive, is comparatively familiar in the constellation of Cancer. With the single exception of the nebula in Andromeda, all the objects so called are entirely telescopic, yet how important a constituent the nebulae form in the contents of the heavens will be shown by a look at some of the lists of these objects. There are now several thousands of nebulae known, and their positions in the sky, as well as the details of their appearances, are set forth in the catalogues.

The most glorious constellation of stars in the firmament is undoubtedly that of Orion. This splendid group is seen in the south during the winter months, and toward the close of January it is situated in a very convenient position for observing early in the evening. The group is specially characterized by the number of unusually bright stars which it includes, and the three stars in the centre, forming the so-called Belt of Orion, is as well known a celestial figure as the sky contains. Directly under the belt are three much smaller stars nearly in a line, which points straight upward to the middle star of the belt. These three lower stars are usually known as the sword handle of Orion, this being the position which they occupied in the fanciful old sketches of the constellation. The three stars of the sword handle of Orion are plunged in the Great Nebula. This object can not be seen by the unassisted eye, though doubtless around the central star a little

haziness is perceptible, and even the slightest telescopic aid will suffice to indicate that the central star of the sword handle is attended by a surrounding glow of light, which renders it quite unlike other stars. This can indeed be sufficiently shown with an ordinary opera-glass, one glance through which will awaken in the beholder a keen desire to study the object under more favorable conditions. But to do justice to the object, telescopes of large power are desirable.

To realize fully the magnificence of the Great Nebula, the observer who is being introduced to the object for the first time should not, strange to say, direct the telescope at the nebula; the instrument should rather be pointed at the heavens, just a little to the west of the nebula. The clock driving the equatorial should not be started, and the observer should take his seat and look through the eye-piece before the nebula has entered the field. He will see, no doubt, a few stars on the black background, which gradually pass in procession across his field of view. This is merely the ordinary diurnal journey of the heavens, by which all the objects move slowly from east to west; I ought rather to say *appear* to move, for, of course, the motion on the heavens is only apparent, the fact being that it is the earth which is turning round.

After the observer's eye for a minute or so has become familiarized with the dark aspect of the heavens under ordinary circumstances, he will begin to perceive on the eastern side (it will appear in the telescope no doubt as on the western side) a faint dawn of light. Gradually there will steal across his field of view a sort of ghostlike luminosity that is in marked contrast to the darkness in the rest of the field; as the seconds move on, this object will disclose itself until the full splendor of the Great Nebula comes into view; then the entire field will be filled with the light, and then it will gradually advance and gradually pass away again to emphasize the contrast between the brilliance of the nebula and the darkness of the sky. Unless this method is adopted, the full interest of a telescopic view of the Great Nebula is not attained, for when the entire field is full of the glow the beginner will hardly recognize the nebula. He will be apt to think that the fainter part of the field he sees is the ordinary groundwork of the sky, and this illusion can only be dispelled by enabling him to witness the actual contrast in the way I have described. The central portions of the nebula are, however, so brilliant and

so wonderfully marked with interesting detail, that even a small instrument will suffice to reveal much of its beauties.

In the centre of the nebula is the star known to astronomers as Theta Orionis, the most prominent star of the sword handle. To the eye this looks like an ordinary star, but the telescope speedily dispels that notion. Theta Orionis is found to consist of four, or rather six, stars all so close together that the unaided eye fails to distinguish them separately. A structure so complex gives to this star quite a special, indeed a unique, interest, wholly apart from the marvelous nebula of which it is the focus. We must dwell a little on the peculiarities of this star. We are familiar with stars which are called double; there are indeed some ten thousand objects so designated known to astronomers and duly registered in catalogues.

Many of these double stars are objects of extreme telescopic beauty; sometimes they offer to our admiration a delightful contrast of colors; perhaps one will be topaz color and the other bluish, or on rare occasions a pair of emerald gems will be seen with an invisible band of mutual connection. Sometimes triple stars are found, in which three stars are obviously in alliance; but multiple stars of greater complexity are comparatively rare; and so marvelous a spectacle as Theta Orionis, in which no fewer than six stars are obviously an allied group, is almost unique. It is not a little remarkable that we find the most exquisite multiple star which the sky can show, beautifully framed or set in the centre of the grandest of the nebulæ. Of course it might conceivably happen that the apparent concurrence of these objects was fortuitous. The actual phenomenon could be accounted for by the belief that the Great Nebula was either very much nearer or very much further than the multiple star, and that they chanced to lie in the same line of sight, and had no other connection. But to me it appears that this view is quite at variance with every reasonable probability; that the most wondrous multiple star should have happened to lie in line with the very centre of the most wondrous nebula would have been a coincidence against the occurrence of which the probabilities were almost infinite. There can scarcely be any doubt that the multiple star and the Great Nebula are part of the same system, and that the star is, in truth, placed in the middle of the nebula, as it actually appears to be.

And now as to the composition of this mysterious object.

The word nebula means, of course, a little cloud, but the expression is apt to be a misleading one. In a sense no doubt they are little, inasmuch as the patch of the sky which a nebula covers would be small compared with one of our ordinary clouds. Indeed, a nebula which covered as large an apparent part of the sky as the size of the moon would be ranked as a large object of its class, while even the greatest of them is perhaps not more than ten or twelve times as great. Nor is the word cloud, as applied to nebula, an appropriate one. What we mean by a cloud is only a vast mass of watery vapor raised by the sun from the sea, and poised aloft until such time as it shall be again dispersed into invisible water, or until it shall descend to the earth as rain. Such clouds are, of course, within the limits of our atmosphere, and are rarely more than a few miles above the earth's surface. The light which renders clouds visible only comes from reflected sunbeams, and consequently at night clouds become invisible, though the astronomer is often only too unpleasantly made acquainted with their presence by the opacity with which they shut out the stars from his view.

Utterly different in all respects are the *nebulæ*. They are not masses of watery vapor. It may no doubt possibly be that water in some form is there, but it is not water which we see. We are looking at some gaseous material of a bluish hue. The light with which it glows is no reflected sunlight. The nebula is indeed indebted to no foreign source for that weird—I had almost said ghostlike—radiance which it gives forth. The light comes from the nebula itself. But how, it may well be asked, should a purely gaseous substance be able to radiate forth light? It is easy for us to comprehend how stars or suns or comparatively solid bodies can, in virtue of their tremendous temperature, glow with heat like red-hot or white-hot iron. It is true that flame is gas in an incandescent state, but in flame a vehement chemical union of oxygen with some other substance is in progress, and this is the source of the heat and the light that flame gives forth. We can not regard the Great Nebula in Orion as originating in anything resembling flame.

We can, however, in our physical laboratories arrange an experiment which seems to throw some light on the composition of the nebula. Into a glass tube a small quantity of hydrogen gas is admitted, the air having been previously extracted. Then, by means of two wires, one at each end of the tube, an electric current is transmitted through the gas. Here there is no

combustion; the gas is merely the vehicle by which the electricity flows from one pole to the other. In doing so the gas instantly begins to glow with an intense bluish light, and a very beautiful effect is produced, which can be renewed or terminated at will by simply making or breaking the electric current. It would seem as if the gas we see in the nebula were in a condition somewhat analogous to the gas in the tube. I do not mean that the passage of electricity through the nebula is the source of its luminosity. There is, indeed, no ground for such a supposition. It is the property of electricity when passing through a conductor to warm that conductor; thus we know that if a powerful current be transmitted through a wire of the most infusible of all metals, platinum, the wire will not only get warm, but it may become red hot, white hot, and even melt under the influence of the heat which is generated. In those beautiful incandescent electric lamps which are now happily coming into extensive use a current of electricity flows through a filament of carbon, and kindles that exquisite incandescence which is maintained while the current flows. It would appear that so long as the electricity is flowing through the glass tube its action on the gas is to impart a very high temperature. It is in consequence of this temperature that the gas glows. Now we can offer a reasonable account of the luminosity of the Great Nebula in Orion. The particles of gaseous or vaporous material of which it is formed are of an extremely high temperature, sufficient to enable them to glow with the brilliancy which renders them visible.

It is now almost twenty years since a marvelous accession to our knowledge of such objects as the Great Nebula in Orion was made by Dr. Huggins. I have used our gas hydrogen as an illustration in describing the character of the nebula, but I have now to add that the presence of hydrogen is no mere fiction but a substantial verity. Truly we here open up one of the most marvelous chapters which science has to disclose. The chemist can analyze the different substances on the earth with his test tubes, and he can tell the elements of which they are composed. But in this old-fashioned chemistry it was at least reasonable for the chemist to demand a portion of the substance he was expected to analyze. Unless he were provided with a sample, how could it be possible for him to grind it up or submit it to the various operations of his laboratory? In these modern days the chemist can perform operations of which his predecessors never even dreamed. No doubt the old method is still used—nay, is indeed at this moment cultivated with greater skill and means than in any previous age—but side by side

with the old method, and as an invaluable supplement thereto, the new method of chemical research, called spectrum analysis, has been created, and has already conducted to many profoundly interesting discoveries in the most varied branches of science.

In the application of the spectroscopic method it is not indispensably necessary that we actually have a fragment of the substance; all we require is a beam of light which that substance can be made to yield when heated to a sufficiently high temperature.

When a beam of the nebular light is transmitted through the prisms, it declares at once that the object from which that light has come is totally different from a star like the sun. Instead of the beautifully colored band, decked in all the glowing hues of the rainbow, the nebular beam is seen to be composed simply of six or seven widely separated strips. It is important to test the character of the light in these strips. Fortunately this can be done in a way that is completely satisfactory. We can produce artificial lights from known sources, and observe them through the spectroscope simultaneously with the light of the nebula.

There are in the composition of this globe some sixty or seventy different elementary substances, and under suitable conditions each one of these substances can afford a perfectly characteristic spectrum. Thus the way of making the comparison with the nebula is to try the different elements one after another, until one can be discovered which pours forth a light that behaves under the prism as does the light from the nebula. Pursuing this inquiry, Dr. Huggins found that when hydrogen gas was ignited to incandescence by the passage of electricity, it emitted light which, after passage through the prisms, came to coincidence with one of the lines in the spectrum of nebula; and the hydrogen character of two of the other lines has been since demonstrated. It was thus established that hydrogen is one of the constituents of the Great Nebula in Orion. Further confirmation of this important discovery was forthcoming when the photographs of the spectrum of the Great Nebula were subsequently obtained. On these photographs lines were present which are constituted by light of such a nature as to be wholly invisible to the eye, though perceptible on the photographic plate. It is of the greatest interest to discover that these invisible rays from the nebula are also indicative of the presence of

hydrogen. Thus we obtain a beautiful confirmation of the fact that the nebula is partly composed of glowing hydrogen.

There are, however, some remaining lines, the character of which has not yet been ascertained.

It would be a little premature to assert that there must be some substance in the Great Nebula not at present known to us on the earth. This would be, no doubt, one interpretation of the facts. We must, however, admit the possibility of another explanation. It is frequently found that the lines yielded by an incandescent material vary to some extent when the physical conditions of temperature and of pressure are modified. It is, therefore, not impossible that the unknown lines in the spectrum of the Great Nebula may be due to some element known to us, but which has not yet been tested under the conditions which would make it yield the particular rays we are speaking of.

The composition of a nebula as disclosed to us by these researches is very instructive. Here we are looking at an object which seems to lie at the very limits of the visible universe—an object so remote that our attempts to fathom its distance are quite unsuccessful; yet in this inconceivably distant part of our system we find at least one ingredient which we know well on the earth. Previous to actual trial no one would have expected, I think, to find the Great Nebula largely constituted from such a familiar element as hydrogen. This gas enters into the composition of water, and is thus an element of extreme abundance on the earth. That an element so common with us here should also be abundant in these awfully distant regions of the universe is one of the most astonishing facts that modern science has revealed.

As the eye follows these ramifications of the Great Nebula, ever fading away in brightness until it dissolves in the blackness of the sky; as we look at the multitudes of bright stars which sparkle out from the depths of the great glowing gas; as we ponder on the marvelous outlines of a portion of the nebula, we are tempted to ask what the true magnitude of this object must really be. Here, again, we have to confess that science is unable to satisfy this very legitimate curiosity. The only means of learning the true length and breadth of a celestial object depends upon our first having discovered the distance from us at which the object is situated. Unhappily

we are, as I have said, entirely ignorant of what this distance may be in the case of the Great Nebula in Orion. Our ordinary methods of conducting such an inquiry are hardly applicable to such an object, and its position so near the Equator introduces fresh difficulties into the problem. We shall, however, certainly not err on the side of exaggeration if we assert that the Great Nebula must be many millions of times larger than that group of bodies which we call the Solar System.

COLORED, DOUBLE, MULTIPLE, BINARY, VARIABLE AND TEMPORARY STARS.—J. E. GORE

On a clear night a careful observer will notice a marked difference in the colors of the brighter stars. The brilliant white or bluish-white light of Sirius, Rigel, and Vega contrasts strongly with the yellowish color of Capella, the deeper yellow, or orange, of Arcturus, and the ruddy light of Aldebaran and Betelgeuse. These colors are, however, limited to various shades of yellow and red. No star of a *decided* blue or green color is known, at least among those visible to the naked eye in the Northern Hemisphere. The third magnitude star Beta Libræ is described by Webb as of a “beautiful pale green hue,” but probably such a tint in the light of this star will to most people prove quite imperceptible. Dr. Gould, observing it in the Southern Hemisphere—under, of course, more favorable conditions—says: “There is a decidedly greenish tinge to the light of Beta Libræ, although its color can not properly be called conspicuous.”

Among the ruddy stars visible to the naked eye, Mu Cephei, Herschel’s “garnet star,” is generally admitted to be the reddest, but it is not sufficiently bright to enable its color to be well distinguished without the aid of an opera-glass. With such an instrument, however, its reddish hue is striking and beautiful, and very remarkable when compared with other stars in its vicinity. Like so many of the red stars, Mu Cephei is variable in its light, but seems to have no regular period, and often remains for many weeks without perceptible change. It may be seen near the zenith in the early evening hours toward the end of October, and when in this position its ruddy color is very conspicuous.

Among the brightest stars, Betelgeuse is perhaps the reddest, and the contrast between its ruddy tint and the white color of Rigel in the same constellation (Orion) is very noticeable. Like Mu Cephei, Betelgeuse is irregularly variable in its light, but not to such an extent, and, like the “garnet star,” it frequently remains for protracted periods nearly constant in brightness. There are other cases of reddish color among the naked-eye stars. Among these may be mentioned Antares (Alpha Scorpii), Alphard

(Alpha Hydræ), noted as red by the Persian astronomer Al-Sûfi, in the Tenth Century, and called by the Chinese “The Red Bird”; Eta and Mu Geminorum; Mu and Nu Ursæ Majoris; Delta and Lambda Draconis; Beta Ophiuchi; Gamma Aquilæ, and others in the Southern Hemisphere.

But it is among the stars below the limit of naked-eye vision that we meet with the finest examples of the red stars. Some of these are truly wonderful objects. The small star, No. 592 of Birmingham’s Catalogue of Red Stars (No. 713 of Espin’s edition), which lies a little south of the 5½ magnitude star 79 Cygni, was described as “splendid red” by Birmingham, “very deep red” by Copeland and Dreyer, and “orange vermilion” by Franks. The star 248 Birmingham, which lies about 5° south of Gamma Hydræ, is another fine specimen. Birmingham described it as “fine red” and “ruby”; Copeland as “brown red”; Dreyer as “copper red”; and Espin as “magnificent blood red.” This star is variable in light, as the estimates of magnitude range from 6.7 to below 9. About 3° to the northeast of this remarkable object is another highly-colored star, known as R Crateris. It is easily found, as it lies in the same telescopic field of view with Alpha Crateris, a 4½ magnitude star. Sir John Herschel described it as “scarlet, almost blood-color; a most intense and curious color.” Birmingham called it “crimson”; and Webb “very intense ruby.” Observing it with a 3-inch refractor in India in 1875, I noted it as “full scarlet.” It varies in light from above the eighth magnitude to below the ninth, and has near it a star of the ninth magnitude of a paler blue tint.

Another very red star is No. 4 of Birmingham’s Catalogue, which will be found about 5° north, preceding the great nebula in Andromeda. It is of about the eighth magnitude, and may be well seen with a 3-inch refractor. Krüger describes it as “*intensiv roth*”; Birmingham as “fine red” and “crimson”; Franks as “fine color, almost vermilion”; and Espin as “intense red color, most wonderful.”

Another fine object is R Leporis, which forms roughly an equilateral triangle with Kappa and Mu Leporis. This is also variable from 6½ to 8½ magnitude. It was discovered by Hind in 1845, and described by him as “of the most intense crimson, resembling a blood-drop on the background of the sky; as regards depth of color, no other star visible in these latitudes could be compared with it.” Schönfeld called it “*intensiv blutroth*,” but

Dunér, observing its spectrum in 1880, gives its color as a less intense red than that of other stars. Possibly it may vary in color as well as in light.

The variable star U Cygni, which lies between Omicron and Omega Cygni, is also very red. Webb described it as showing “one of the loveliest hues in the sky.” It varies from about the seventh to $11\frac{1}{2}$ magnitude, with a period of about 461 days.

Another deeply colored star is the well-known variable R Leonis. Hind says: “It is one of the most fiery-looking variables on our list—fiery in every stage from maximum to minimum, and is really a fine telescopic object in a dark sky about the time of greatest brilliancy, when its color forms a striking contrast with the steady white light of the sixth magnitude a little to the north.” This latter star is 19 Leonis.

In the Southern Hemisphere there are some fine examples of red stars. Epsilon Crucis, one of the stars in the Southern Cross, is very red. Mu Muscæ is described by Dr. Gould as of “an intense orange red.” Delta² Gruis is a very reddish star of about the fourth magnitude. Pi¹ Gruis was observed by Gould as “deep crimson,” and forming a striking contrast with its white neighbor Pi² Gruis, which he notes as “conspicuously white.” The variable L₂ Puppis is described as “red in all its stages, and remarkably so when faint.” Miss Clerke, observing—at the Cape of Good Hope—R Doradûs, another southern variable, says: “This extraordinary object strikes the eye with the glare of a stormy sunset,” and with reference to the variable R Sculptoris, described by Gould as “an intense scarlet,” she says: “The star glows like a live coal in the field,” a description I have found myself very applicable to other small red stars.

An eighth magnitude star about 5° north of Beta Pictoris is noted by Sir John Herschel, in his *Cape Observations*, as “vivid sanguine red, like a blood-drop. A superb specimen of its class.” With reference to a star of about $8\frac{1}{2}$ magnitude in the field with Beta Crucis, Herschel says: “The fullest and deepest maroon red; the most intense blood-red of any star I have seen. It is like a drop of blood when contrasted with the whiteness of Beta Crucis.”

Of stars of other colors, the asserted green tint of Beta Libræ has already been referred to. Among the brighter stars of the Southern Hemisphere, Theta Eridani, Epsilon Pavonis, Upsilon Puppis, and Gamma Tucanæ are

said to be decidedly blue. The wonderful cluster surrounding the star Kappa Crucis contains several bluish, greenish and red stars, and is described by Sir John Herschel as resembling “a superb piece of fancy jewelry.”

Among the double stars we find many examples of colored suns. Of these may be mentioned Epsilon Boötis, of which the colors are “most beautiful yellow” and “superb blue,” according to Secchi; Beta Cephei, “yellow and violet”; Beta Cygni, “golden yellow and smalt blue”; Gamma Delphini, of which I noted the colors in 1874 as “reddish yellow and grayish lilac”; Alpha Herculis, “orange and emerald or bluish green,” and described by Admiral Smyth as “a lovely object, one of the finest in the heavens”; Zeta Lyrae, “pale yellow and lilac” (Franks); and Beta Piscis Australis, of which I observed the colors in India as white and reddish lilac.

Some distant telescopic companions to red stars have been described as blue. This may be in some case due, partly at least, to the effect of contrast. In others the blue color seems to be real. This has been shown spectroscopically to be the case with the bluish companions of Beta Cygni.

The physical cause of the difference of color is still more or less a matter of mystery. Although we can not consider it proved that the red stars are cooling and “dying out” suns, as has been suggested, we may, I think, conclude that their temperature, although doubtless very high, must be lower than that of the white stars. We know that a bar of iron when heated to redness is not so hot as when raised to “white heat,” and although the analogy between hot iron and stellar photospheres may not be a perfect one, it seems probable that the higher the temperature of a star, the whiter its color will be. Most of the white stars, as Sirius, Vega, and those only yellow or slightly colored, show spectra of Secchi’s first and second types, while the great majority of the red stars exhibit banded spectra of the third and fourth types.

To this rule there are, however, like other rules, some notable exceptions. For instance, Aldebaran, Alpha Hydræ, Xi Cygni, and 31 Orionis, although distinctly reddish stars, show well-marked spectra of the second or solar type. On the other hand, Rho Ursæ Majoris and Omega Virginis, which, according to Dunér, are only slightly yellow, have well-marked spectra of the third type.

An apparent change of color seems in some cases to be well established. The supposed red color of Sirius in ancient times is well known. A certain established change is found in the case of the famous variable star Algol, which is distinctly described as red by Al-Sûfi in the Tenth Century. It is now pure white, or nearly so, and this is probably the best attested instance on record of change of color in a bright star.

Schmidt's Nova Cygni of 1876 was noted as "golden yellow" on the night of its discovery. When it had faded to the eighth magnitude, Dr. Copeland called it "decided red," but when examined at Lord Crawford's observatory in September, 1877, its color was recorded as "faint blue"! The new star in the Andromeda nebula was considered to be yellowish or reddish by most observers when near its maximum, but about a month later its color was noted as "bluish."

Among the red and variable stars, there are many suspected cases of color variation. Espin and other observers have noted that the wonderful variable Mira Ceti is much less red at maximum than at minimum. My own observations confirm this. When at its maximum brightness, Mira does not seem to me a very highly-colored star, while at one of its minima I noted it as "fiery red." Possibly, however, the great difference between its maximum and minimum brilliancy may have an influence on estimations of its color. The remarkable variable Chi Cygni is said to be "strikingly variable in color." Espin's observations in different years show it "sometimes quite red, at others only pale orange red." The star Birmingham 118 was described by Schjellerup in 1863 as "decided red," but it was found yellow by Secchi in 1868; "bluish" by Birmingham, 1873-76; "no longer red" by Schjellerup in March, 1876; and "white" by Franks in 1885. Espin omits it from his revised edition of Birmingham's Catalogue.

Birmingham 169 was found red by Struve, blue or bluish-white by Birmingham in 1874, and white at Greenwich in the same year. Espin also saw it white in March, 1888. The star Birmingham 30, which lies close to Phi Persei (54 Andromedæ), was described by Schweizer as a "red star with a little disk" in January, 1843; Birmingham noted it as "light red" in December, 1875; Copeland "deep red" in January, 1876; and Dreyer "reddish" in September, 1878; but Espin, in November and December, 1887, found it "certainly not red, and nothing peculiar in the star's appearance." It might be expected that these curious changes of color, if

real, would be accompanied by corresponding changes in the star's spectrum. Such may be the case, and observations in this direction would probably lead to some interesting results.

There seems to be some law governing the distribution of the colored stars. The white stars appear to be most numerous, as a rule, in those constellations where bright stars are most abundant, for instance, in Orion, Cassiopeia, and Lyra; yellow and orange stars in large and ill-defined constellations, such as Cetus, Pisces, Hydra, Virgo, etc. The very reddish stars are most numerous in or near the Milky Way, and one portion of the Galaxy—between Aquila, Lyra, and Cygnus—was termed by Birmingham “the red region in Cygnus.”

Many of the stars when examined with a good telescope are seen to be double, some triple, and a few quadruple, and even multiple. These when viewed with the naked eye, or even a powerful binocular, seem to be single, and show no sign of consisting of two components. These telescopic double stars should be carefully distinguished from those which appear very close together with the naked eye, and which in opera-glasses or telescopes of small power might be mistaken for wide double stars by the inexperienced observer. These latter stars, such as Mizar—the middle star in the tail of the Great Bear—and its small companion, Alcor, have been called “naked-eye doubles,” but they are not, properly speaking, double stars at all. Telescopic double stars are far closer, and even the widest of them could not possibly be seen double without optical aid, even by those who are gifted with the keenest vision. Of these so-called “naked-eye doubles,” we may mention Alpha Capricorni, which on a very clear night may be seen with the naked eye to consist of two stars. On a very fine night two stars may be seen in Iota Orionis, the most southern star in Orion's Sword. The star Zeta Ceti has near it a fifth magnitude star, Chi, which may be easily seen with the unaided vision. The star Epsilon Lyrae (near Vega) is a severe test for naked-eye vision. Bessel, the famous German astronomer, is said to have seen it when thirteen years of age. Omicron Cygni (north of Alpha and Delta Cygni) forms another naked-eye double, and other objects of this class may be noticed by a sharp-eyed observer.

The star Mizar, already referred to, is itself a wide telescopic double, and it seems to have been the first double star discovered with the telescope (by Riccioli in 1650). It consists of two components, of which one is

considerably brighter than the other. It will give an idea of the closeness of even a “wide” telescopic double when we say that the apparent distance between Mizar and Alcor is nearly forty times the distance which separates the close components of the bright star. From this it will be seen that even a powerful binocular field-glass would fail to show Mizar as anything but a single star. The components may, however, be well seen with a 3-inch telescope, or even with a good 2-inch. The colors of the two stars are pale green and white. Between Mizar and Alcor is a star of the eighth magnitude, and others fainter. Mizar was the first double star photographed by Bond.

The Pole Star has a small companion at a little greater distance than that which separates the components of Mizar, but owing to the faintness of this small star, the object is not so easy as Mizar.

The star Beta Cygni is composed of a large and small star, of which the colors are described as “golden yellow and smalt blue.” This is a very wide double, and may be seen with quite a small telescope. Another fine double star is that known to astronomers as Gamma Andromedæ. The magnitudes of the components are about the same as those of Mizar, but a little closer. Their colors are beautiful (“gold and blue”). This is one of the prettiest double stars in the heavens. It is really a triple star, the fainter of the pair being a very close double star; but this is beyond the reach of all but the largest telescopes. The star Gamma Delphini is another beautiful object, the components being a little more unequal in magnitude, but the distance between them about the same as in Gamma Andromedæ. I have noted the colors with a 3-inch telescope as “reddish yellow and grayish lilac.” Gamma Arietis, the faintest of the three well-known stars in the head of Aries, is another fine double star, a little closer than Gamma Delphini. This is an interesting object, from the fact that it was one of the first double stars discovered with the telescope—by Hooke, in 1664, when following the comet of that year.

Another beautiful double star is Eta Cassiopeiæ, the components being about equal in brightness to those of Gamma Delphini, but the distance less than one-half. The colors are, according to Webb, yellow and purple; but other observers have found the smaller star garnet or red. This is a very interesting object, the components revolving round each other, and forming what is called a binary star.

Another fine double star is Castor, which is composed of two nearly equal stars separated by a distance about half that between the components of Gamma Andromedæ. This is also a binary, or revolving double star, but the period is long. Gamma Virginis is another fine double star, with components at about the same distance as those of Castor, and the colors very similar. It is also a remarkable binary star.

Among double stars of which the components are closer than those mentioned above, but which are within the reach of a good 3-inch telescope—a common size with amateur observers—the following may be noticed: Alpha Herculis, colors, orange or emerald green; the light of this star is slightly variable. Gamma Leonis, another binary star with a long period; colors, pale yellow and purple. Epsilon Boötis, a lovely double star, the colors of which Secchi described as “most beautiful yellow, superb blue.”

For observers in the Southern Hemisphere, the following fine double stars may be seen with a 3-inch telescope: Alpha Centauri; this famous star, the nearest of all the fixed stars to the earth, is also a remarkable binary; its period, as recently computed by Dr. See, is eighty-one years. Theta Eridani is a splendid pair, but closer than Alpha Centauri. It is, however, an easy object with a 3-inch telescope, and with a telescope of this size I noted the colors in India as light yellow and dusky yellow. The star known as *f* Eridani is a very similar double to Theta, but the components are fainter. I noted the colors in India as yellowish-white and very light green.

Of triple, quadruple, and multiple stars, there are several which may be well seen with a small telescope. Of these may be mentioned Iota Orionis, the lowest star in the Sword of Orion, which consists of a bright star accompanied by two small companions. In Theta Orionis, the middle star of the Sword, four stars may be seen forming a quadrilateral figure, known to observers as the “trapezium.” There are two fainter stars in this curious object, which lie in the midst of the Orion nebula, but a somewhat larger telescope is required to see them. Within the trapezium are two very faint stars, which are only visible in the largest telescopes. In Sigma Orionis—a star closely south of Zeta, the lowest star in Orion’s Belt—six stars may be seen with a 3-inch telescope.

Double and multiple stars may be either optical or real. Optical double stars are those in which the component stars are merely apparently close

together, owing to their being seen in nearly the same direction in space. Two stars may *seem* to be close together, while, in reality, one of them may be placed at an immense distance behind the other. Just as two lighthouses at sea may, on a dark night, appear close together when viewed from a certain point, whereas they may be really miles apart. In the case of double stars it is, of course, always difficult to determine whether the apparent closeness of the stars is real or merely optical. But when, from a long series of observations of their relative position, we find that one is apparently moving round the other, we know that the stars must be comparatively close, and linked together by some physical bond of union. These most interesting objects are known to astronomers as binary, or revolving double stars. The probable existence of such objects was predicted from abstract reasoning by Mitchell in the Eighteenth Century; but the discovery of their actual existence was made by Sir William Herschel, while engaged on an attempt to determine the distance of some of the double stars from the earth. Unlike the planetary orbits, which are nearly circular, at least those of the larger planets of the Solar System, it is found that the orbits of these double stars differ, in many cases, widely from the circular form, in some cases, indeed, approaching in shape more the orbit of a comet than a planet.

The binary stars are among the most interesting objects in the heavens. The number now known probably amounts to nearly one thousand. In most of them, however, the motion is very slow, and in only about seventy cases has the change of position, since their discovery, been sufficient to enable an orbit to be computed.

Savary, in 1830, was the first astronomer who attempted to compute the orbit of a binary star, namely, the star Xi Ursæ Majoris. This remarkable pair was discovered by Sir William Herschel in 1780, and as the period of revolution is about sixty-one years, a considerable portion of the ellipse had been described in 1830, when it was attacked by Savary.

The binary star with the shortest period known at present seems to be the fourth magnitude star Kappa Pegasi. It was discovered as a wide double star by Sir William Herschel in 1786, the companion star being of the ninth magnitude. In August, 1880, Mr. Burnham, the famous American double star observer, examining the star with the 18½-inch refractor of the Dearborn Observatory, found the brighter star to be a very close double, with a distance between the components of only a quarter of a second of

arc. A few years' observations showed that this pair were in rapid motion round each other (about eleven years).

Another binary star, with a period of about the same length, is Delta Equulei, which was discovered to be a close double by Otto Struve in 1851. Next in order of shortness of period comes the southern binary star Zeta Sagittarii, for which an orbit was first computed in the year 1886 by the present writer. The orbit of this star will, I think, require still further revision, but the period of about eighteen years is probably not far from the truth.

Another remarkably rapid binary star is 85 Pegasi. Next in order of rapidity of motion we have the southern binary star 9 Argûs.

The star 42 Comæ Berenices has a period of about $25\frac{3}{4}$ years, according to Otto Struve. The orbit is remarkable from the fact that its plane passes through, or nearly through, the earth, and is, therefore, projected into a straight line, the companion star oscillating backward and forward on each side of its primary.

The star Beta Delphini—the most southern of the four stars in the “Dolphin's Rhomb”—is also a fast-moving binary, discovered by Burnham in 1873. Burnham thinks the period will prove to be about twenty-eight years. The spectrum of the light of Beta Delphini is similar to that of our sun, so that the two bodies should be comparable in intrinsic brilliancy.

Another remarkable binary star with a comparatively short period is Zeta Herculis. This pair have now performed three complete revolutions since their discovery in 1782 by Sir William Herschel. Several orbits have been computed, but Dr. See's period of thirty-five years is probably the best. The companion is, however, rather faint, being only $6\frac{1}{2}$ magnitude, while the primary star is of the third.

In the case of the binary star, Eta Coronæ Borealis, it was, some forty years ago, uncertain whether its period was forty-three or sixty-six years, but now that two complete revolutions have been performed since its discovery by Sir William Herschel in 1781, the question has been finally decided in favor of the shorter period.

The brilliant star Sirius is also an interesting binary star. The companion, which is relatively very faint—about tenth magnitude—was discovered by

Alvan Clark in 1862. The existence of some such disturbing body was previously suspected by astronomers, owing to observed irregularities in the proper motion of Sirius. Several orbits, giving periods of about fifty years, have been computed. The great brilliancy of Sirius, the brightest star in the heavens, naturally suggests a sun of great size. Recent investigations do not favor this idea. Its spectrum is, however, of the first type, and the star is therefore not comparable with the sun in brilliancy. The above result would indicate that stars of the first, or Sirian type, are intrinsically brighter than our sun.

Sirius is about eleven magnitudes brighter than its faint companion. This makes the light of Sirius about 25,000 times the light of the small star. The two bodies must, therefore, be differently constituted, and, indeed, the companion must be nearly a dark body. If Sirius has any planets revolving round it—like those of our solar system—they must forever remain invisible in our largest telescopes. This remark, of course, applies to all the fixed stars, single and double. They may possibly have attendant families of planets, like our sun, but if so, the fact can never be ascertained by direct observation.

The star Zeta Cancri is a well-known triple star, the close pair revolving in a period of about sixty years. Nearly two revolutions have now been completed since its discovery by Sir William Herschel in 1781. All three stars probably form a connected system, but the motion of the third star round the binary pair is very slow and irregular.

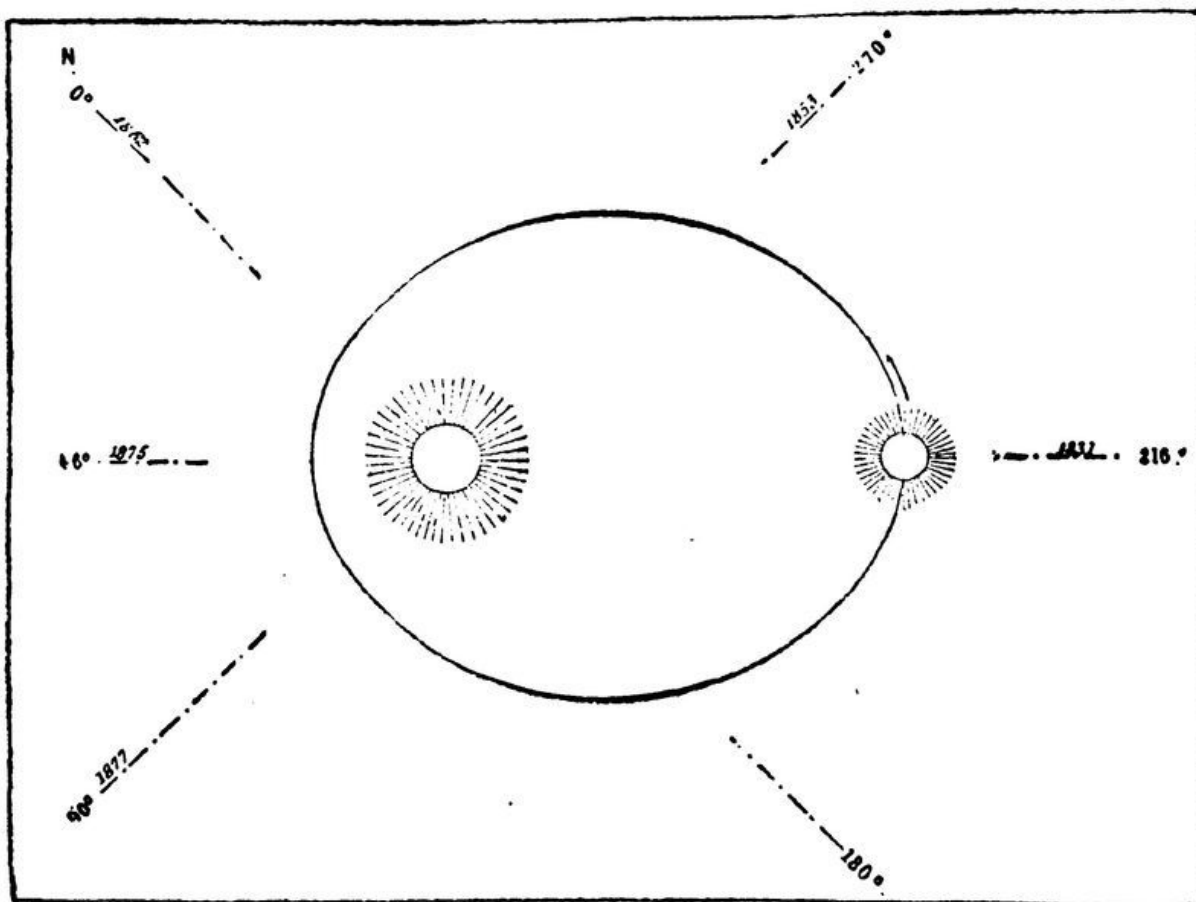


Fig. 18.—System of the Double Sun Alpha Centauri

Another interesting binary star is Xi Ursæ Majoris. As already stated, this was the first pair for which an orbit was computed. More than a complete revolution has now been performed since its discovery by Sir William Herschel in 1780. The period has, therefore, been well determined, and seems to be about sixty years.

The bright southern star, Alpha Centauri, the nearest of all the fixed stars to the earth, so far as is known at present, is also a remarkable binary star. It seems to have been first noticed as a double star by Richaud in 1690.

Assuming my value of the sun's stellar magnitude (about 27), I find that the sun, if placed at the distance of Alpha Centauri, would appear of about the same brightness as the star does to us. As, according to Professor Pickering, the spectrum of Alpha Centauri is of the second or solar type, it would seem that in mass, brightness, and physical condition the star closely resembles our sun.

We next come to another very interesting binary star, known to astronomers as 70 Ophiuchi. It is a very fine double star, the magnitudes of the components being about four and six, and the colors yellow and orange. More than a complete revolution has now been described by the components since its discovery by Sir William Herschel in 1779. Placed at the distance indicated by Krüger's parallax, I find that our sun would be reduced to a star of about magnitude $3\frac{1}{2}$, which shows that the sun and star are of about equal brightness. The spectrum is of the solar type, according to Vogel.

A very famous binary star is that known to astronomers as Gamma Virginis. Its history is a very interesting one. It lies close to the celestial equator, about one degree to the south and about fifteen degrees to the northwest of the bright star Spica (Alpha of the same constellation), with which it forms the stem of a Y-shaped figure formed by the brightest stars of the constellation Virgo, or the Virgin, Gamma being at the junction of the two upper branches. The brightness of Gamma Virginis is a little greater than an average star of the third magnitude. Variation of light has, however, been suspected in one or both components. The Persian astronomer, Al-Sûfi, in his description of the heavens, written in the Tenth Century, rates it of the third magnitude, and describes it as "the third of the stars of *al-auvâ*, which is a mansion of the moon," the first and second stars of this "mansion" being Beta and Eta Virginis, the fourth star Delta, and the fifth Epsilon, these five stars forming the two upper branches of the Y-shaped figure above referred to. Gamma was called *Zawiyah-al-auvâ*, "the corner of the barkers!" perhaps from its position in the figure, which formed the thirteenth Lunar Mansion of the old astrologers. It was also called *Porrina* and *Postvarta* in the old calendars. The fact that Gamma Virginis really consists of two stars very close together seems to have been discovered by the famous astronomer, Bradley, in 1718. The rapid decrease in the apparent distance from 1780-1834 indicated that the apparent orbit is very elongated, and that possibly the two stars might "close up" altogether, and appear as a single star even in telescopes of considerable power. This actually occurred in the year 1836, or, at least, the stars were then so close together that the most powerful telescopes of that day failed to show Gamma Virginis as anything but a single star. Of course, it would not have been beyond the reach of the giant telescopes of our day. From the year 1836 the pair began to open out again.

Another interesting binary star is Eta Cassiopeiae. Periods ranging from 149 to 222½ years have been found by different computers. The most recent computation makes it about 196 years.

The bright star Gamma Leonis, situated in the well-known “Sickle in Leo,” is also a binary star, but only a small portion of the orbit has been described since its discovery by Sir William Herschel in 1782. Dr. Doberck finds a period of 407 years. It is remarkable for its very high “relative brightness.” This pair forms a fine object for a small telescope.

The star known as 12 Lyncis is a triple star, the components being 5, 6, and 7½ magnitude. The close pair forms a binary system, for which an orbit has been computed by the present writer, who finds a period of about 486 years. Sir John Herschel predicted in 1823 that the angular motion of the pair would “bring the three stars into a straight line in 57 years.” This prediction was fulfilled in 1887, when measures by Tarrant showed that the stars were then exactly in a straight line.

The bright star Castor is a famous double star, and has been known since the year 1718, when it was observed by Bradley and Pond. It was also observed by Maskelyne in 1759, and frequently by Sir William Herschel from 1799 to 1803. Numerous orbits have been computed, with periods ranging from 199 years by Mädler and 1,001 years by Doberck. I find that the mass of the system of Castor is only $\frac{1}{19}$ th of the sun’s mass, a result which would imply that the components are masses of glowing gas! Dr. B  lopolsky has found, with the spectroscope, that the brighter component is a close binary star with a dark companion, like Algol. The period of revolution is about three days, and the relative orbital velocity about 20¾ miles a second. Dr. B  lopolsky’s observations show that the system is receding from the earth at the rate of about 4½ miles per second.

With reference to the colors of the components of binary stars, the following relation between color and relative brightness has been established:

(1) When the magnitudes of the components are equal, or approaching equality, the colors are generally the same, or similar.

(2) When the magnitudes of the components differ considerably, there is also a considerable difference in color.

A new class of binary stars has been discovered within the last few years by means of the spectroscope. These have been called “spectroscopic binaries,” and the brighter component of Castor, referred to above, is an example of the class. They are supposed to consist of two component stars, so close together that the highest powers of the largest telescopes fail to show them as anything but single stars. Indeed, the velocities indicated by the spectroscope show that they must be so close that the components must forever remain invisible by the most powerful telescopes which could ever be constructed by man. In some of these remarkable objects, the doubling of the spectral lines indicates that the components are both bright bodies, but in others, as in Algol, the lines are merely shifted from their normal position, not doubled, thus denoting that one of the components is a dark body. In either case, the motion in the line of sight can be measured by the spectroscope, and we can, therefore, calculate the actual dimensions of the system in miles, and thence its mass in terms of the sun’s mass, although the star’s distance from the earth remains unknown. Judging, however, from the brightness of the star, and the character of its spectrum, we can make an estimate of its probable distance from the earth.

The bright star Spica has also been found by the spectroscope to be a close binary star. Vogel finds a period of four days with a distance between the components of about $6\frac{1}{4}$ millions of miles, and assuming that the components have equal mass and are moving in a circular orbit, he finds the mass of the system about 2.6 times the mass of our sun. In addition to its orbital motion, Vogel finds that Spica is approaching the sun at the rate of over nine miles per second.

To ordinary observers, the light of the stars seems to be constant. Even to those who are familiar with the constellations, the stars appear to maintain their relative brilliancy unchanged. To a great extent this is, of course, true; the great majority of the stars remaining of the same brightness from day to day, and from year to year. There are, however, numerous exceptions to this rule. Many of the stars, when carefully watched, are found to fluctuate in their light, being sometimes brighter and sometimes fainter. These are known as “variable stars”—one of the most interesting class of objects in the heavens. Some of these have been known for a great number of years, and their variations having been carefully watched, the laws governing their light changes have been well determined.

We will first consider the variable stars with long periods of variation, as these generally show the largest fluctuations of light. Among these, the first star in which variation of light seems to have been noticed is the extraordinary object, Omicron Ceti, popularly known as Mira, or the “wonderful” star. It appears to have been first noticed by David Fabricius in the year 1596. He observed that the star now called Omicron, in the constellation Cetus, was of the third magnitude on April 13 of that year, and that in the following year it had disappeared. Bayer saw it again in 1603, when forming his maps of the constellations, and assigned to it the Greek letter Omicron, but does not seem to have noticed the fact that it was the same star which had been observed by Fabricius seven years previously. No further attention seems to have been paid to it until 1638 and 1639, when it was observed at Francker by Professor Phocylides Holwarda to be of the third magnitude in December, 1638, invisible in the following summer, and again visible in October, 1639. From 1648 to 1662 it was carefully observed by Hevelius, and in subsequent years by several observers. Its variations are now regularly followed from year to year, and it forms one of the most interesting objects of its kind in the heavens. Its light varies from about the second magnitude to the ninth, but its brightness at maximum is variable to a considerable extent.

Perhaps the long period variable star next in order of interest—at least to observers in the Northern Hemisphere—is that known as Chi Cygni. It was discovered by Kirch in 1686. The star varies at maximum from 4 to $6\frac{1}{2}$ magnitude, and at the minimum it sinks to below the thirteenth magnitude. At some maxima, therefore, it is easily visible to the naked eye, and at others it is just below the limit of ordinary vision. At the maximum of 1847, it was visible to the naked eye for a period of 97 days. The average period is about 406 days; but according to Schönfeld—a well-known authority on the variables—observations indicate a small lengthening of the period. Chi Cygni is said to be “strikingly variable in color.” Espin’s observations in different years show it “sometimes quite red, at others only pale orange-red.” In the spectroscope, its light shows a splendid spectrum of the third type (or banded spectrum, very characteristic of these long period variables), in which bright lines were observed by Espin in May, 1889.

R Leonis is another remarkable variable star, which is sometimes visible to the naked eye at maximum. It lies closely south of the star known as 19

Leonis. It was discovered by Koch in 1782. At the maximum, its brightness varies from 5.2 to 7 magnitude, and at minimum it fades to about the tenth magnitude. The mean period is about 313 days. The star is red in all phases of its light, and forms a fine telescopic object. Close to it are two small stars, which form, with the variable, an isosceles triangle.

There is a very remarkable variable star in the Southern Hemisphere known as Eta Argûs. It lies in the midst of the great nebula in Argo, and the history of its fluctuations in light is very interesting. Observed by Halley in 1677 as a star of the fourth magnitude, it was seen of the second magnitude by Lacaille in 1751. After this, it must have again faded, for Burchell found it of only the fourth magnitude from 1811 to 1815. From 1822 to 1826 it was again of the second magnitude, as observed by Fallows and Brisbane; but on February 1, 1827, it was estimated of the first magnitude by Burchell. It then faded again, for on February 29, 1828, Burchell found it of the second magnitude. From 1829 to 1833 Johnson and Taylor rated it of the second magnitude; and it was still of this magnitude, or a little brighter, when Sir John Herschel commenced his observations at the Cape of Good Hope in 1834. It does not seem to have varied much in brightness from that time until December, 1837, when Herschel was astonished to find its light “nearly tripled.” He says: “It very decidedly surpassed Procyon, which was about the same altitude, and was far superior to Aldebaran. It exceeded Alpha Orionis, and the only star (Sirius and Canopus excepted) which could at all be compared with it was Rigel.”

From this time its light continued to increase. On the 28th December it was far superior to Rigel, and could only be compared with Alpha Centauri, which it equaled, having the advantage of altitude, but fell somewhat short of it as the altitudes approached equality. The maximum of brightness seems to have been obtained about the 2d of January, 1838, on which night, both stars being high and the sky clear and pure, it was judged to be very nearly matched, indeed, with Alpha Centauri. In 1843 it again increased in brightness, and in April of that year it was observed by Maclear to be brighter than Canopus, and nearly equal to Sirius! It then faded slightly, but seems to have remained nearly as bright as Canopus until February, 1850, since which time its brilliancy gradually decreased. It was still of the first magnitude in 1856, according to Abbott, but was rated a little below the second magnitude by Powell in 1858. Tebbutt found it of the third

magnitude in 1860; Abbott a little below the fourth in 1861. Ellery rated it fifth magnitude in 1863, and Tebbutt sixth magnitude in 1867. In 1874 it was estimated 6.8 magnitude at Cordoba, and only 7.4 in November, 1878. Tebbutt's observations from 1877-86 show that it did not rise above the seventh magnitude in those years, and in March, 1886, it was rated 7.6 magnitude by Finlay at the Cape of Good Hope. This seems to have been the minimum of light, for in May, 1888, Tebbutt found that it "had increased fully half a magnitude" since April, 1887. The star is very reddish in color.

We will now consider the variables of short period, which are particularly interesting objects, owing to the comparative rapidity of their light changes. The periods vary in length from about $17\frac{1}{4}$ days down to a few hours. Perhaps the most interesting of these short period variables, at least to the amateur observer, is the star Beta Lyræ, which is easily visible to the naked eye in all phases of its light. It can be readily identified, as it is the nearest bright star to the south of the brilliant Vega, and one of two stars of nearly the same magnitude, the second being Gamma Lyræ. The variability of Beta Lyræ was discovered by Goodricke in the year 1784. The period is about 12 days, 21 hours, 46 minutes, 58 seconds. Recent observations with the spectroscope indicate that the star is a very close double or "spectroscopic binary," although it does not seem certain that an actual eclipse of one component by the other takes place, as in the case of Algol. Bright lines were detected in the star's spectrum by Secchi so far back as 1866. In 1883 M. Von Gothard noticed that the appearance of these bright lines varied in appearance, and from an examination of photographs taken at Harvard Observatory in 1891, Mrs. Fleming found displacements of bright and dark lines in a double spectrum, the period of which agreed fairly well with that of the star's light changes.

Another interesting star of short period is Delta Cephei, which is one of three stars forming an isosceles triangle a little to the west of Cassiopeia's Chair, the variable being at the vertex of the triangle, and the nearest of the three to Cassiopeia. Its variability was also discovered by Goodricke in 1784. It varies from 3.7 to 4.9 magnitude, with a period of 5 days, 8 hours, 47 minutes, 40 seconds. The amount of the variation is, therefore, the same as in the case of Algol, the star's light at maximum being about three times its light at minimum. The observations also show that Delta Cephei is

approaching the earth at the rate of about $8\frac{3}{4}$ miles a second. The color of the star is yellow, and it has a distant bluish companion of about the fifth magnitude, which may possibly have some physical connection with the brighter star, as both stars have a common proper motion through space.

Another remarkable star of short period is Eta Aquilæ, the variability of which was discovered by Pigott in 1784. It varies from magnitude 3.5 to 4.7, with a period of 7 days, 4 hours, 14 minutes, but Schönfeld found marked deviations from a uniform period. Its color is yellow, and its spectrum, like that of Delta Cephei, of the second or solar type.

A remarkable variable star of short period was discovered in 1888 by Mr. Paul in the southern constellation Antlia. It varies from magnitude 6.7 to 7.3, with the wonderfully short period of 7 hours, 46 minutes, 48 seconds, all the light changes being gone through no less than three times in twenty-four hours! It was for some years believed that the variation was of the Algol type, but recent measures made at the Harvard College Observatory show that it belongs to the same class as Delta Cephei and Eta Aquilæ.

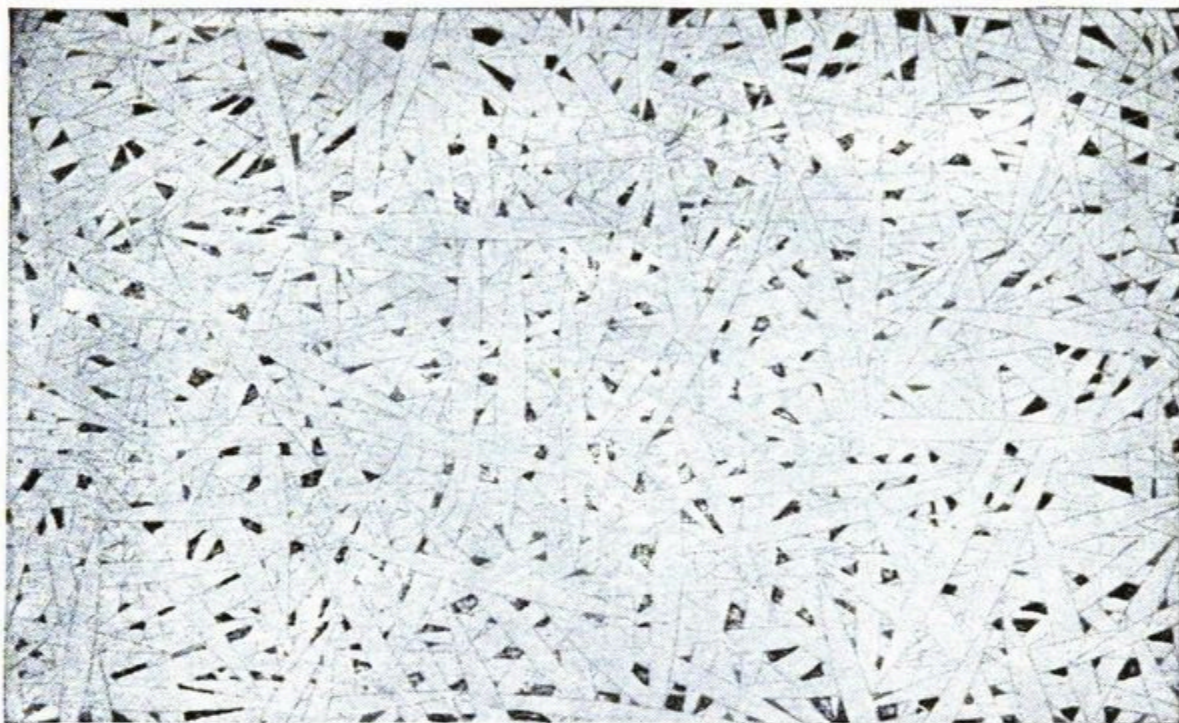
A telescopic variable with a wonderfully short period was discovered by Chandler in 1894. It lies a little to the west of the star Gamma Pegasi, and has been designated U Pegasi. It varies from magnitude 8.9 to 9.7, and was first supposed to be of the Algol type with a period of about two days, but further observations showed that the period was much shorter, and only 5 hours, 31 minutes, 9 seconds. The remarkable rapidity of its light changes, which are gone through four times in less than twenty-four hours, make this remarkable star a most interesting object. Possibly there may be other stars in the heavens with a similar rapidity of variation which have hitherto escaped detection.

Unlike the variable stars of long period which seemed scattered indifferently over the surface of the heavens, the great majority of the short period variables are found in a zone which nearly coincides with the course of the Milky Way. The most notable exceptions to this rule are W Virginis with the comparatively long period of $17\frac{1}{4}$ days, and U Pegasi, above described, which has the shortest known period of all the variable stars. Another peculiarity is that most of them are situated in what may be called the following hemisphere, that is between 12 hours and 24 hours of right ascension. The most remarkable exception to this rule is Zeta Geminorum.

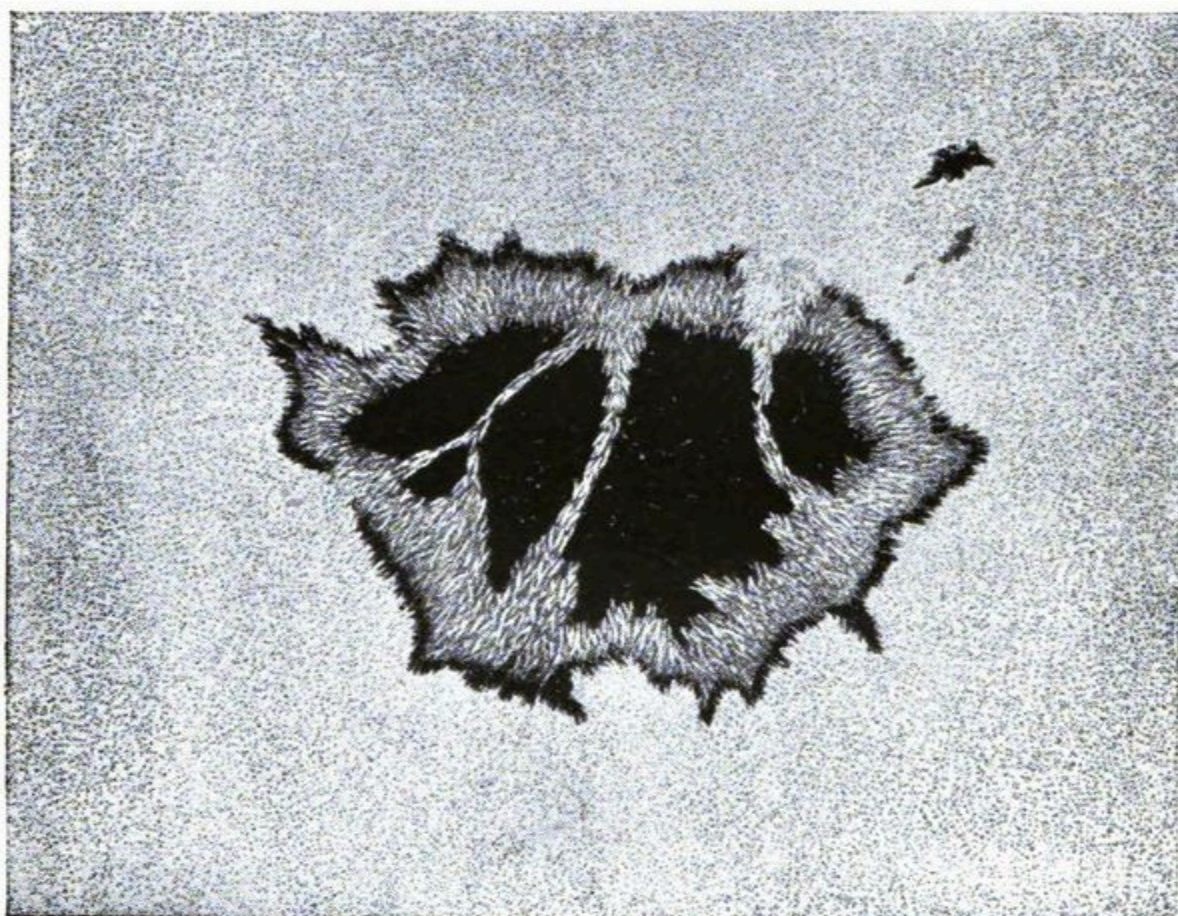
Algol, or Beta Persei, is a famous variable star, and the typical star of the class to which it belongs. Its name, Algol, is derived from a Persian word, meaning the “demon,” which suggests that the ancient astronomers may have detected some peculiarity in its behavior. The real discovery of its variation was, however, made by Montanari in 1667, and his observations were confirmed by Maraldi in 1692. Its fluctuations of light were also noticed by Kirch and Palitzsch, but the true character of its variations was first determined by the English astronomer, Goodricke, in 1782. Its fluctuations of light are very curious and interesting. Shining with a constant, or nearly constant, brightness for a period of about 59 hours as a star of a little less than the second magnitude, it suddenly begins to diminish in brightness, and in about $4\frac{1}{2}$ hours it is reduced to a star of about magnitude $3\frac{1}{2}$. In other words, its light is reduced to about one-third of its normal brightness. If we suppose three candles placed side by side at such a distance that their combined light is merged into one, and equal to the usual brightness of Algol, then, if two of these candles are extinguished, the remaining candle will represent the light of Algol at its minimum brilliancy. The star remains at its minimum, or faintest, for only about 15 minutes. It then begins to increase, and in about 5 hours recovers its normal brightness, all the light changes being gone through in a period of about 10 hours out of nearly 69 hours, which elapse between successive minima. These curious changes take place with great regularity, and the exact hour at which a minimum of light may be expected can be predicted with as much certainty as an eclipse of the sun.

Goodricke, comparing his own observations with one made by Flamsteed in the year 1696, found the period from minimum to minimum to be 2 days, 20 hours, 48 minutes, $59\frac{1}{2}$ seconds, and he came to the conclusion that the diminution in the light of the star is probably due to a partial eclipse by “a large body revolving round Algol.” This hypothesis was fully confirmed in the years 1888-89 by Professor Vogel with the spectroscope. As no close companion to Algol is visible in the largest telescopes, we must conclude that either the satellite is a dark body, or else so close to the primary that no telescope could show it. Now, if the diminution in Algol’s light is due to a dark body revolving round it, and periodically coming between us and the bright star, it follows that both components will be in motion, and both will revolve round the common centre of gravity of the pair. A little before a minimum of light takes place, the dark companion should therefore be

approaching the eye, and, consequently, the bright companion will be receding. During the minimum there will be no apparent motion in the line of sight, as the motion of both bodies will be at right angles to the visual ray. After the minimum is over, the motion of the two bodies will be reversed, the bright one approaching the eye, and the dark one receding. Now, this is exactly what Vogel found. Before the diminution in the light of Algol begins, the spectroscope showed that the star is receding from the earth and after the minimum that it is approaching the eye. That the companion is dark and not bright, like the primary, is evident from the fact that the spectral lines are merely shifted from their normal position and not doubled, as would be the case were both components bright, as in the case of some of the “spectroscopic binaries”—for example, Beta Aurigæ. Vogel found that before the minimum of light, Algol is receding from the earth with the velocity of $24\frac{1}{2}$ miles a second, and after the minimum it is approaching at the rate of $28\frac{1}{2}$ miles a second. The difference between the observed velocities indicates that the system is approaching the earth with a velocity of about 2 miles a second. Knowing, then, the orbital velocity, which is evidently about $26\frac{1}{2}$ miles a second, and assuming the orbit to be circular, it is easy, with the observed period of revolution, or the period of light variation, to calculate the diameter of the orbit in miles, although the star’s distance from the earth remains unknown. Further, comparing its period of revolution and the dimensions of the orbit with that of the earth round the sun, it is easy to calculate, by Kepler’s third law of motion, the mass of the system in terms of the sun’s mass, and the probable size of the component bodies. Calculating in this way, Vogel computes that the diameter of Algol is about 1,061,000 miles, and that of the dark companion 830,300 miles, with a distance between their centres of 3,230,000 miles, and a combined mass equal to two-thirds of the sun’s mass, the mass of Algol being four-ninths, and that of the companion two-ninths, of the mass of the sun. Taking the diameter of the sun as 866,000 miles, and its density as 1.44 (water being unity), I find that the above dimensions give a mean density for the components of Algol of about one-third that of water, so that the components are probably gaseous bodies, as Hall has already concluded.



10 0 10 20 30
SCALE OF 60000 MILES



It is a curious fact that Al-Sûfi, the Persian astronomer, in his *Description of the Heavens*, written in the Tenth Century, speaks distinctly of Algol as a red star (*étoile, brillant; d'un éclat, rouge*), while at present it is white or at the most of a yellow color. A similar change of color is supposed to have taken place in the case of Sirius, but the change in Algol seems more certain, as Al-Sûfi's descriptions are generally most accurate and reliable.

Stars of the Algol type of variable are very rare objects, only a dozen or so having been hitherto discovered in the whole heavens. Those visible to the naked eye, when at their normal brightness, are: Algol, Lambda Tauri, Delta Libræ, R Canis Majoris, and U Ophiuchi.

A remarkable peculiarity about the variable stars in general is that none of them has any considerable proper motion. As a large proper motion is generally considered to indicate proximity to the earth, we may conclude, with great probability, that the variable stars, as a rule, lie at a great distance from our system. In other words, it appears that the sun does not lie in a region of variable stars, and, with the exception of Alpha Cassiopeiæ and Alpha Herculis, a measurable parallax has not yet been found, so far as I know, for any known variable star.

We now come to the interesting and mysterious class of objects known as "new" or "temporary" stars. These phenomena are of very rare occurrence, and but few undoubted examples of the class are recorded in the annals of astronomy. Possibly in some cases they have been merely variable stars, of irregular period and fitful variability; but others may have been due to a real catastrophe, such as the collision of two dark bodies in space, or, possibly, the passage of a bright or dark body through a gaseous nebula.

The earliest temporary star of which we have any reliable information seems to be one which is recorded in the Chinese annals of Ma-tuan-lin, as having appeared in the year 134 B. C. in the constellation Scorpio. Its position seems to have been somewhere between the stars Beta and Rho of Scorpio. Pliny informs us that it was the sudden appearance of a new star which induced the famous astronomer Hipparchus to form his catalogue of stars, the first ever constructed. As the date of Hipparchus's catalogue is 125 B. C., it seems highly probable that the new star referred to by Pliny was

the same as that recorded by the Chinese astronomer as having appeared nine years previously.

A new star is said to have appeared in the year 76 B. C. between the stars Alpha and Delta in the Plow, but the accounts are vague.

In 101 A. D., a small “yellowish-blue” star is said to have appeared in the “sickle” in Leo, but its exact position is not known. In 107 A. D., a new star is mentioned near Delta, Epsilon and Eta in Canis Major, three bright stars southeast of Sirius. In 123 A. D., another new star is recorded by Ma-tuan-lin to have appeared between Alpha Herculis and Alpha Ophiuchi.

The Chinese annals record that on December 10, 173 A. D., a brilliant star appeared between Alpha and Beta Centauri in the Southern Hemisphere. It remained visible for eight months, and is described as resembling “a large bamboo mat!”—a curious description. There is at present, close to the spot indicated, a known variable star—R Centauri—of which the period seems to be long and the variation of light irregular. Possibly an unusually bright maximum of this variable star formed the star of the Chinese annals, or perhaps the variable star is the remnant of the outburst which took place in the First Century. The variable is a very reddish star, and at present varies from about the sixth to the tenth magnitude.

A new star is recorded in the year 386 A. D. as having appeared between Lambda and Phi Sagittarii. Near the position indicated, Flamsteed observed a star, No. 65 of his catalogue, which is now missing; and it has been conjectured that the star seen by Flamsteed may possibly have been a return of the star mentioned in the Chinese annals.

Cuspius relates that a star as bright as Venus appeared near Altair in 389 A. D., during the reign of the Emperor Honorius, and that he had himself seen it. There is some doubt, however, about the exact date, as other accounts give the year 388 or 398. The star seems to have disappeared in about three weeks.

In the year 393 A. D., another strange star is recorded in the tail of Scorpio. An extraordinary star is said to have been seen near Alpha Crateris in 561 A. D. Here again a known variable and red star—R Crateris—is close to the position indicated by the ancient records.

The Chinese annals record a new star in 829 A. D., somewhere in the vicinity of the bright star Procyon, and in this locality there are several known variable stars.

The Bohemian astronomer, Cyprianus Leoviticus, mentions the appearance of new stars in Cassiopeia in the years 945 A. D. and 1264, and it has been conjectured that perhaps these were apparitions of Tycho Brahe's famous star of 1572 (to be presently described), forming a variable star with a period of over 300 years. Lynn and Sadler, however, have shown that the supposed stars of 945 and 1264 were, in all probability, comets.

Extraordinary stars are recorded near Zeta Sagittarii in 1011 A. D., near Mu Scorpii in 1203, and near Pi Scorpii on July 1, 1584. It is remarkable how many of these objects seem to have appeared in this portion of the heavens.

A very brilliant star is mentioned by Hepidannus as having appeared in Aries in May, 1012. He describes it as "dazzling the eye." Other temporary stars are mentioned in 1054 A. D., near Zeta Tauri, and in 1139 near Kappa Virginis; but the accounts of these are very vague, and it seems by no means certain that they were really new stars.

No possible doubt, however, can be entertained with reference to the appearance of the object which suddenly blazed out in Cassiopeia's Chair in November, 1572. It was called the "Pilgrim Star," and was observed by the famous astronomer, Tycho Brahe, who has left us a very elaborate account of its appearance, position, etc. Although usually spoken of as Tycho Brahe's star, it seems to have been really discovered by Cornelius Gemma on the evening of November 9. That its appearance was very sudden may be inferred from Cornelius Gemma's statement that it was not visible on the preceding night in a clear sky. Tycho Brahe's attention was first attracted to it on November 11. His description of the new star is as follows—as quoted by Humboldt: "On my return to the Danish islands from my travels in Germany, I resided for some time with my uncle, Steno Bille, in the old and pleasantly situated monastery of Herritzwadt, and here I made it a practice not to leave my chemical laboratory until the evening. Raising my eyes, as usual, during one of my walks, to the well-known vault of heaven, I observed with indescribable astonishment, near the zenith in Cassiopeia, a radiant fixed star of a magnitude never before seen. In my amazement, I

doubted the evidence of my senses. However, to convince myself that it was no illusion, and to have the testimony of others, I summoned my assistants from the laboratory, and inquired of them, and of all the country people that passed by, if they also observed the star that had thus suddenly burst forth. I subsequently heard that in Germany, wagoners and other common people first called the attention of astronomers to this great phenomenon in the heavens—a circumstance which, as in the case of non-predicted comets, furnished fresh occasion for the usual raillery at the expense of the learned. This new star I found to be without a tail, not surrounded by any nebula, and perfectly like all other fixed stars, with the exception that it scintillated more strongly than stars of the first magnitude. Its brightness was greater than that of Sirius, Alpha Lyræ, or Jupiter. For splendor, it was only comparable to Venus when nearest to earth (that is, when only a quarter of her disk is illuminated). Those gifted with keen sight could, when the air was clear, discern the new star in the daytime, and even at noon. At night, when the sky was overcast, so that all other stars were hidden, it was often visible through the clouds, if they were not very dense (*nubes non admodum densas*). Its distances from the nearest stars of Cassiopeia, which throughout the whole of the following year I measured with great care, convinced me of its perfect immobility. Already in December, 1572, its brilliancy began to diminish, and the star gradually resembled Jupiter, but by January, 1573, it had become less bright than that planet. Toward the month of November the new star was not brighter than the eleventh in the lower part of Cassiopeia's Chair. The transition to the fifth and sixth magnitudes took place between December, 1573, and February, 1574. In the following month the new star disappeared, and, after having shone seventeen months, was no longer discernible to the naked eye." (The telescope was not invented until thirty-seven years afterward.) Humboldt adds: "At its first appearance, as long as it had the brilliancy of Venus and Jupiter, it was for two months white, and then passed through yellow into red. In the spring of 1573, Tycho Brahe compared it to Mars; afterward he thought it nearly resembled Betelgeuse, the star in the right shoulder of Orion. The color for the most part was like the red tint of Aldebaran. In the spring of 1573, and especially in May, its white color returned (*albedinam quandam sublividam induebat, qualis Saturni stellæ subesse videtur*). So it remained in January, 1574; being, up to the time of its entire disappearance in the month of March, 1574, of the fifth magnitude, and white, but of a

duller whiteness, and exhibiting a remarkably strong scintillation in proportion to its faintness.”

Ma-tuan-lin speaks of a star in 1578 “as large as the sun” (!) but does not state its position.

The star known as P (34) Cygni is sometimes spoken of as a “Nova,” or new star; but it is still visible to the naked eye as a star of the fifth magnitude. It was observed of the third magnitude by Jansen in 1600 and by Kepler in 1602. After the year 1619 it appears to have diminished in brightness, and is said to have vanished in 1621; but it may merely have become too faint to be seen with the naked eye. It was again observed of the third magnitude by Dominique Cassini in 1655, and it afterward disappeared. It was again seen by Hevelius in November, 1655. In 1667, 1682, and 1715 it is recorded as of the sixth magnitude, and there is no further record of any marked increase in its light. A period of about 18 years was assumed by Pigott; but this is now disproved, and it seems probable that the star is a variable of irregular period and fitful variability, and not, properly speaking, a temporary star. Its present color is yellow, and bright lines have been seen in its spectrum.

A new star of the third magnitude was observed near Beta Cygni by the Carthusian monk Anthelmus in 1670. It remained visible for about two years, and is said to have increased and diminished several times before its final disappearance. Schönfeld computed its exact position from observations made by Hevelius and Picard. Quite close to the spot indicated, a star of the eleventh magnitude has been observed at the Greenwich Observatory, and fluctuations of light were suspected in this small star by Hind and others.

A very remarkable star, sometimes called the “Blaze Star,” suddenly appeared in Corona Borealis, in May, 1866. It was first seen by the late Mr. Birmingham, at Tuam, Ireland, about midnight on the evening of May 12, when it was of the second magnitude, and equal to Alphecca, “the gem of the coronet.” Its appearance must have been very sudden, for Schmidt, the Director of the Athens Observatory, stated that he was observing the constellation on the same evening, about two and one-half hours previous to Birmingham’s discovery, and observed nothing unusual. He was certain that no star, of even the fifth magnitude, could possibly have escaped his notice.

On the following night it was seen by several observers in different parts of the world.

A remarkable and very interesting temporary star was discovered in 1892 in the constellation Auriga.

It is a remarkable fact that the great majority of the temporary stars appeared in or near the Milky Way. The chief exceptions to this rule are: the star of 76 B. C., in the Plow, the star recorded by Hepidannus in Aries, 1012 A. D., and the “Blaze Star” of 1866 in Corona Borealis.

A WORLD ON FIRE—NOVA PERSEI.—

ALEXANDER W. ROBERTS

In the small hours of the morning of 22d February, 1901, Dr. Anderson of Bonnington, Edinburgh, saw a bright star shining in the constellation of Perseus, where he knew no such star was ever seen before. The circumstances connected with this discovery afford another striking instance of how Nature keeps her secrets for her true amateur, using the word in its highest sense.

The evening of 21st February was cloudy, and nine out of ten astronomers would have gone to bed when there seemed little prospect of the night clearing; but Dr. Anderson was the tenth man. At twenty minutes to three in the morning the clouds rolled away from over the old gray Scottish capital, and the trained eye of the patient observer saw right in the heart of Perseus a new star. Never before had its light, blue-white, like an unpolished diamond, shone down on this strange earth of ours.

Next day the news of the wonderful discovery was flashed to all the great observatories of the world, and telescopes and spectroscopes, cameras and photometers, were directed toward the strange phenomenon, and by testing, measuring, examining, sought to wrest its secrets from it.

Much is still a mystery; but what has been ascertained during the period that the rhythm of its light-waves beat upon our shores is of great interest and importance as bearing directly on the life-history of each individual star in the heavens, and of our own sun and planet among them.

The first and simplest question that arises for settlement is the date when the new star blazed forth in our terrestrial sky. The curious reader will notice the reservation: in *our* terrestrial sky. When the star *actually* burst forth into resplendent light is another matter, as we shall discover later on. It was certainly before Dr. Anderson was born, and probably before another Scotsman—Ferguson by name—combined, like many another sage, counting and watching sheep with counting and watching stars.

With regard to the date of the appearance in our sky of the new star, Nova Persei, as it is called in astronomical literature, when Dr. Anderson discovered it at twenty minutes to three o'clock on the morning of 22d February, it was bright enough to be straightway evident to a trained astronomer. In these later days of strenuous scientific activities every portion of the sky is constantly being examined and charted, and no sooner was the discovery of Nova Persei announced than a searching of records began, in order to ascertain if, at any time, the star had ever been seen before.

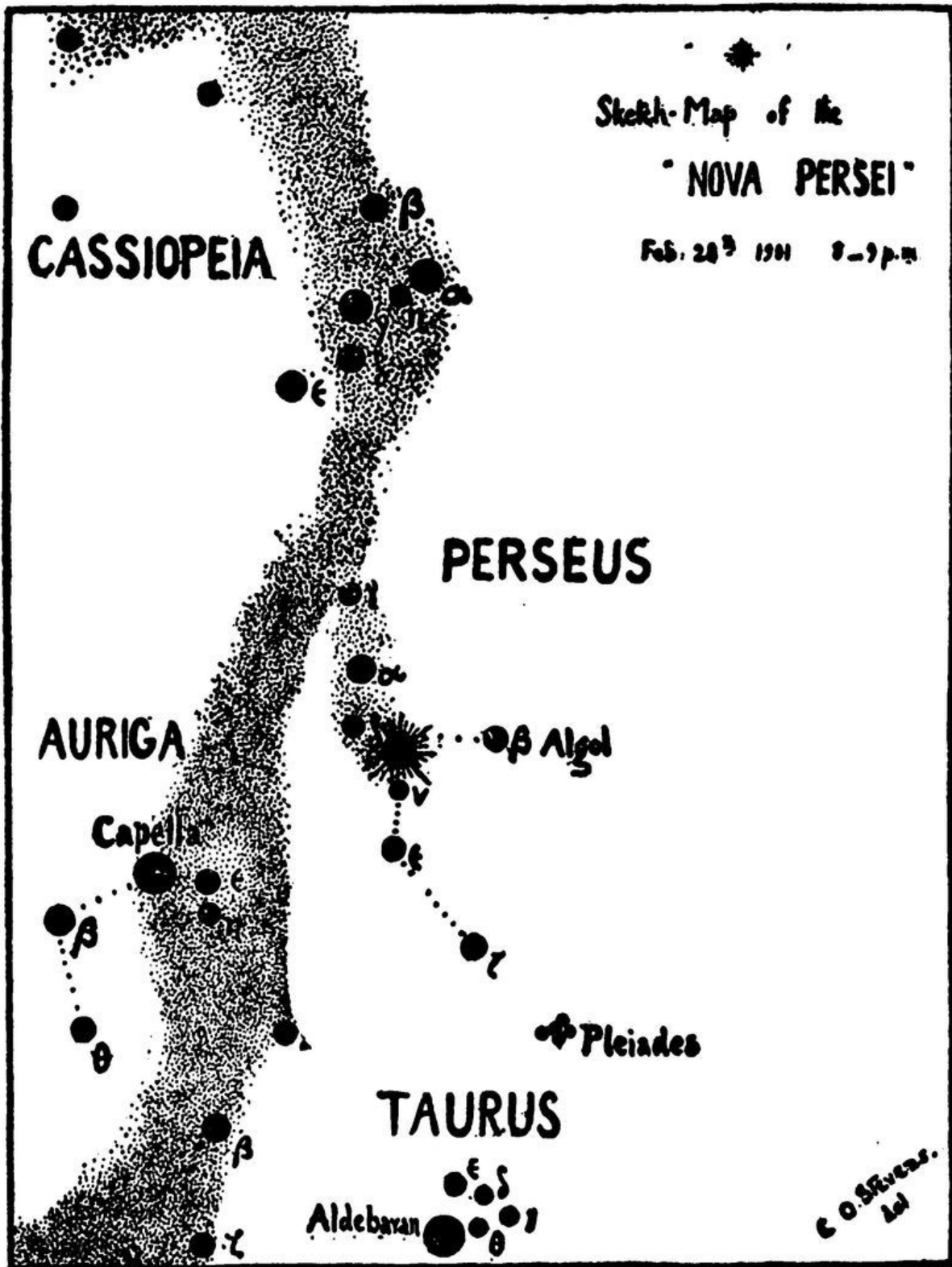


Fig. 19.—Chart Showing Position of Nova Persei

It so chanced that on the evenings of 18th and 19th February two photographs of the very spot where three days later the new star appeared were taken at Harvard Observatory. On neither of these photographs is there the slightest evidence of the star's existence. It was, therefore, on these dates non-existent so far as our earth was concerned. On the evening of 20th February a well-known English observer, Mr. Stanley Williams, had also taken a photograph of the same portion of the sky; and again there was no trace of the star. Mr. Williams's photograph was taken twenty-eight hours before Dr. Anderson saw it. Still more strange is the fact that on the evening of 21st February three observers on the Continent testify that they had the constellation Perseus under observation from seven o'clock to eleven, and had the new star then been visible they could not have failed to see it. The star, therefore, blazed out some time between eleven o'clock and three on the night of its discovery.

Now, what does this mean? It means this: that by some cause a star, quite dark before, or so faint that it could not be seen even by means of a powerful telescope, in a few hours, or perhaps in a few minutes, blazed forth as a star of conspicuous brightness. In this brief space of time a dark and probably chill globe became a seething mass of fire, a million times hotter than it was before. Fierce, fervent heat lit up the orb with a glow that reached from rim to rim of the stellar universe. We have here a catastrophe that goes beyond our wildest conceptions: the conflagration of a world, the ruin of a star. What guarantee have we for an assumption of this kind? What of certitude is there in our vision of such a Day of Doom for any part of our universe? Let us consider the salient facts regarding the recent changes in the appearance and structure of this star. We shall relate only those facts that are beyond controversy, as far as our present knowledge goes.

Nova Persei did not reach its maximum brightness till the evening of 25th February, when it was probably the most conspicuous object in the midnight sky. It was then at least six times brighter than at the time of its discovery. After this date it began to wane slowly. At intervals there were spurts of brightness lasting for two or three days, as if the fires had not exhausted themselves. On the whole, however, the light of the star waned, and by the end of the year its enfeebled light was just bright enough to be evident to the naked eye; twelve months after its appearance it could only be seen with the aid of a telescope.

Now, one of the most powerful instruments of research in the new astronomy is the spectroscope. It takes hold of the rays of light that come to us from a star, and makes these rays reveal the condition of things in the world they come from. One of the spectroscopes turned on the new star in Perseus was Professor Copeland's magnificent instrument at Blackford Hill Observatory, Edinburgh. Professor Copeland described the new star as "a feebly developed" sun. As the star, however, increased in brightness the spectroscope chronicled the fact that great physical changes were taking place in its composition and structure. The star soon ceased to be a feebly developed sun, for development had gone on apace with the increase of light. Round the solid or semi-molten mass there was rapidly aggregating an ocean of fiery gases, probably thrown up from the nucleus.

Put simply, Nova Persei, for long ages a cold, dark, solid globe, was in the brief space of a few days transformed from circumference to core into a luminous, heated gaseous sphere. By what chance or circumstance this vast change came about may be inquired into later on. We only note here that this was the story spelled out by those skilled in deciphering the observations recorded by the spectroscope. In July, 1901, Professor Pickering of Harvard Observatory announced that the star had become a nebula; that, indeed, its once solid globe had practically dissolved into thinnest air. Not only had its elements become molten with fervent heat, but they had become transformed into shimmering wisps of matter more diaphanous than a gossamer web.

Everything connected with the history of this star is of exceptional interest; but all that had already been ascertained was completely overshadowed by the astonishing discovery made in November, 1902, that nebulous prominences were observed darting out from the star with a velocity of at least 100,000 miles every second of time. These astonishing changes have been confirmed at the two great American observatories, the Yerkes and the Lick.

Whence and how had destruction come upon this particular star? At one hour the star is dark, cold, solid. A few hours later this dark, solid, cold body is a blazing world, its solid mass blown apparently into countless fragments; from every fragment, big or little, there pour streams of fiery vapor; for millions of miles round the star there is a whirlpool of fire, a

tempest of flame; and from end to end of this great universe of ours the brightness of the burning star pulsates. Three explanations have been given.

The one that naturally arises in our mind is that it was struck by another star. Two worlds, each moving at the rate of twenty miles a second, come into collision, and the result is the annihilation of both. The force of their impact, changed into heat, drives their elements into vapor. Such a catastrophe is quite possible in a universe like ours, where stars and worlds, millions and millions in number, sweep down the great avenues of space with a velocity far beyond our comprehension.

We take it that when the crack of doom comes to this earth of ours it will be in this fashion. Some great dark star will strike our sun fair and square, and then in the twinkling of an eye, before the inhabitants of earth know what has taken place, sun and moon and planet will be wrapped up and dissolved in an atmosphere of fire.

We can in a certain rough way compute the increase in temperature that would arise from the collision of two great orbs. Thus, let us suppose that Nova Persei was moving onward through space with a velocity of ten miles a second—a moderate velocity, be it noted, for a star—when it collided with the body that wrought its destruction. The impact would be terrific, and the result of it would be not only the complete disintegration of both stars, but a sudden rise in temperature of about five hundred thousand degrees, an increase sufficient to vaporize the hardest adamant.

The second theory which has been suggested as explanatory not only of Nova Persei, but of all new stars, is a modification of the foregoing. This theory is that the new star in its flight through space suddenly plunged into a nebula, or into some portion of space denser than that through which it had already passed. This explanation is not only intelligible but reasonable. If the new star plunged into a region filled with matter even as rare as air, the friction would immediately set the star on fire. We see the same phenomenon every night when a meteor hustles through our atmosphere. The meteoric rocks, with the chill of empty space in and around them, dash into our upper air. A few seconds are ample for the practical annihilation of most of them: in that brief space of time they have been subjected to a heat many times greater than that of a Bessemer furnace.

We can imagine Nova Persei as some monster meteor, a meteor larger than the sun, plunging into a gaseous mass somewhat like our air. In a few hours its temperature would be increased a million-fold. This increase would fill the surrounding space with fire, and there would be an immense and ever-increasing area at fervent heat.

To the mind of the writer this explanation has most to commend it. It is the one that is most in harmony with the information which has been gathered by hundreds of observers aided by the finest of modern scientific equipment. But there are other explanations. There will always be other explanations so long as the world lasts.

One of these explanations is of more interest than the rest, inasmuch as it makes a link of connection between the recent terrible volcanic eruption in the West Indies and the sudden appearance of a new star like Nova Persei. It is suggested that Nova Persei is, or rather was, a world somewhat like our own, only vastly larger—that is, there was an inner core of molten matter and an outer shell of solid material. One day, according to the explosion theory, this outer shell burst, and the interior fires rushed hither and thither like a devouring flood all over the stellar globe. Vast chemical changes went on as the lambent flames turned everything solid into streams of lava. Great electrical disturbances took place all round the star. The whole phenomenon of Nova Persei, according to this theory, is just the destruction of St. Pierre on a sidereal scale.

Such a doom, of course, is possible in any star or planet whose interior is still molten. At any moment the imprisoned fires might break their barriers and change a cold, fruitful, life-bearing earth into a furnace; but it is far from probable that any such fate will ever be meted out to our planet or to any other, and, at any rate, destruction did not come to Nova Persei in this manner. No explosion could account for an access of heat and light any way comparable to that which was observed. Neither could any interior disruption be violent enough to hurl the star into fragments. The gravitational hold of the star would prevent this dismemberment. Yet during the ages the mind of man has been irresistibly drawn to this conception of the world's end, so much so that perhaps, after all, our instinct is right and our science wrong, and the vision of the Minorite Celano of the

Dies iræ, dies illa
Solvat sæculum in favilla,

is a vision of those things that will be in the later days.

We have already touched on one strange circumstance connected with the appearance of Nova Persei. Dr. Anderson saw it for the first time at a few minutes to three o'clock on the morning of 22d February—that is, the news of the strange occurrence reached our planet then; but when did the event actually take place?

At Greenwich and at some of the other foremost observatories attempts have been made directly and indirectly to determine the distance of Nova Persei. And yet this distance defies measurement. The star is so far away that we have no instruments refined enough to deal with the problem. But we know that the sudden blazing up of Nova Persei was over and done with before our great-grandfathers were born. It happened more than two hundred years ago—perhaps two thousand years ago. All this time the news was swiftly traveling earthward, traveling on and on and on, two hundred thousand miles every second of the clock, past star and nebula and system, never halting, never faltering—yet it took hundreds of years to come to us; and beyond us lie countless worlds that will not see the new star for centuries to come. Hundreds of years hence in *their* sky will appear suddenly in the constellation of Perseus a strange star; it will increase in brightness for a few days just as it did in ours; it will fade away intermittently just as it did in ours. There is no imagination here; only sober facts.

We may be allowed, in closing our narrative of this wonderful star, to make one excursion into the region of imagination. As the news of the star passes on through space, are there any beings beyond ourselves who will take record of its appearance? It has taken centuries to come to us. Did any other creatures in some far-off world lift their eyes to the stars and wonder, as we do, what all this meant? Will some mortal, like ourselves, in some remoter world, in a day yet to come, see the sight, and have the intelligence to say, “Lo! a new star?” We have room enough here for the most extravagant fancy. Perhaps there is so much room that we shall lose ourselves if we venture to stray in such directions.

TELESCOPES.—A. FOWLER

THE REFRACTING TELESCOPE.—The function of a telescope is twofold. First, to magnify the heavenly bodies, or, what comes to the same thing, to make them look as if they were nearer to us, so that we can see them better. Second, to collect a much greater number of rays of light than the unassisted eye alone can grasp, so that objects too dim to be otherwise perceptible are brought within our range of vision.

There are two forms of telescope, distinguished as *Refractors* and *Reflectors*. The simplest form of refracting telescope is exemplified by the common opera-glass, and large refractors are not essentially different. Such instruments depend for their action upon the formation of an image by a lens. One can easily illustrate this by producing upon the wall of a room an inverted image of a candle or gas flame with a spectacle lens (one adapted for a long-sighted person), or with one of the larger lenses from an opera-glass. Having such an image, it may be magnified by means of another lens, just as one may magnify a photograph with an ordinary reading glass. Technically, the lens which forms the primary image is called the *object-glass* of the telescope, and that which is used to magnify this image is called the *eye-piece*. The object-glass is usually a large lens, which is placed at one end of a tube, while the eye-piece is a much smaller lens, placed at the other end. Means are provided for adjusting the distance between the two lenses so as to admit of distinct vision.

Matters are, however, not quite so simple as has been stated. There is a very great difficulty introduced by the fact that a lens made out of a single piece of glass gives an image which is surrounded by fringes of color, so that some device has to be adopted in order to destroy, as far as possible, this enemy of good definition. In the early history of the telescope, this so-called *chromatic aberration* was considerably reduced by making small object-glasses of very great focal length.^[22]

Lenses of 100-foot focus, however, are not easy to employ as object-glasses, and astronomy was, therefore, greatly benefited by Dollond's invention of the *achromatic lens* in 1760. This is a compound lens, usually

consisting of a double convex crown-glass lens and a concavo-convex, or double concave, lens of flint glass. The curvatures of the lenses, and the optical properties of the two kinds of glass composing them, are such that the color due to one of them is practically neutralized by that due to the other acting in opposition. A section of such an object-glass, with the “cell” in which it rests, is shown in Fig. 20.

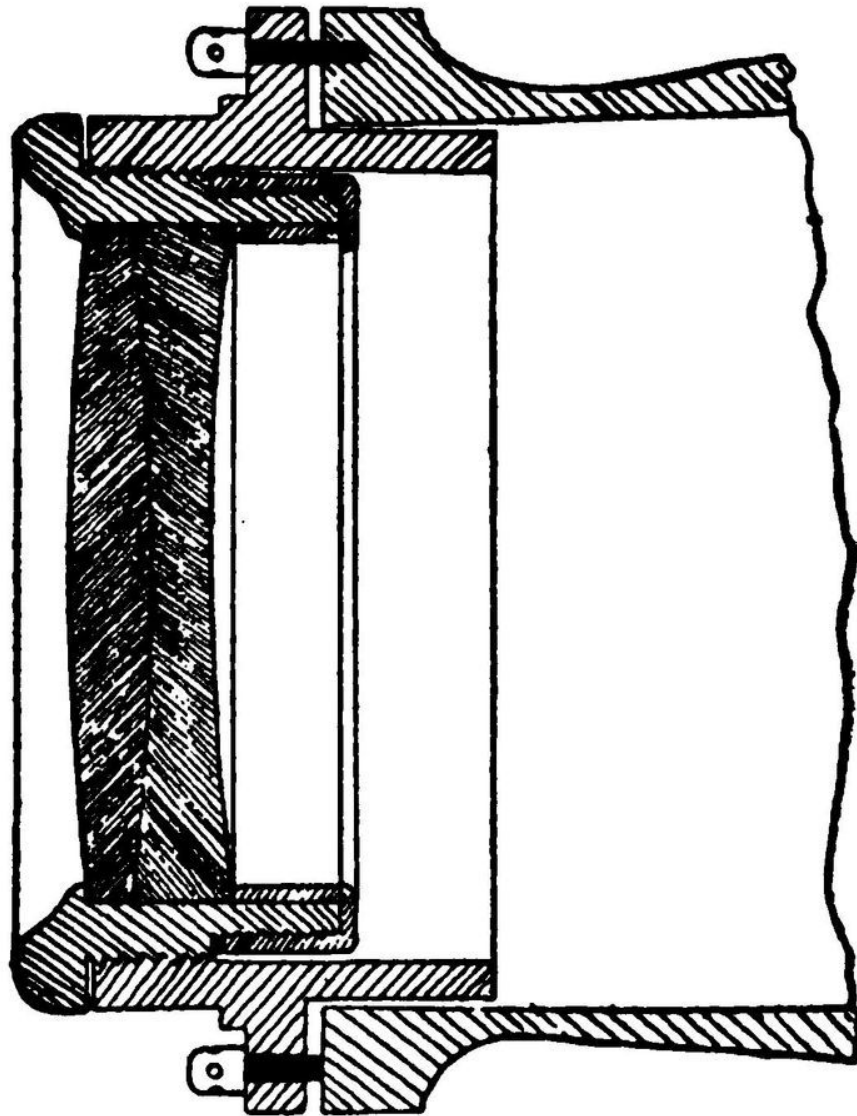


Fig. 20.—The Achromatic Object-Glass

In this way the focal length of the lens, and, therefore, the length of the telescope tube, can be kept within reasonable dimensions, while the definition is improved. There is, however, usually a little outstanding color,

due to the imperfect matching of the two lenses, and if one looks through a large refractor, even of a good quality, a purple fringe will be noticed round all very bright objects. This only affects a few of the brighter objects, while millions of others which are dimmer may be seen free from spurious color.

It may be remarked that the curved surfaces of the lenses forming telescopic object-glasses must not be parts of spheres. If they are, the images will be rendered indistinct by *spherical aberration*, and the optician has to design his curves to get rid of this defect at the same time as chromatic aberration.

A new form of telescopic objective, consisting of three lenses, which has many important advantages, has been invented by Mr. Dennis Taylor, of the well-known firm of T. Cooke & Sons, York, England.

Such a lens as this illustrates the perfection which the optician's art has now attained. Six surfaces of glass have to be so accurately figured that every ray of light falling upon the surface of the lens shall pass through the finest pin-hole at a distance of eighteen times the diameter of the lens.

THE REFLECTOR.—In a reflecting telescope, the object-glass of the refractor is replaced by a concave mirror. In order that such a mirror may reflect all the rays from a star to a single point, its concave surface must be part of a paraboloid of revolution, that is, a surface produced by the revolution of a parabola on its axis. If a spherical surface be employed, all the rays will not be reflected to a single point and the images which it gives will be ill-defined. Yet it is astonishing to find that the difference between a parabolic and spherical surface, even in the case of a large mirror, is exceedingly small. Sir John Herschel states that in the case of a mirror four feet in diameter, and forming an image at a distance of forty feet, the parabolic only departs from the spherical form at the edges by less than a twenty-one thousandth part of an inch.

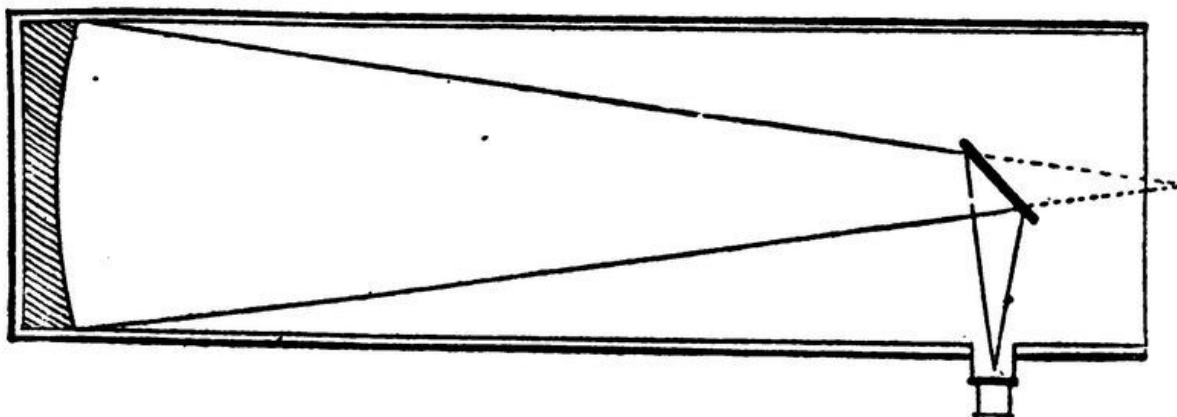


Fig. 21.—The Newtonian Reflector

An image being formed by a mirror, it is next to be viewed with an eye-piece just as in the case of a refracting telescope. Here there is a little difficulty, for if the eye-piece be applied in the direct line of the mirror, the interposition of the observer's head will block out the light. Several ways of overcoming this have been devised, but the plan most generally followed is that which Newton adopted in the first reflecting telescope which was ever constructed. With his own hands Newton made a small reflector, $6\frac{1}{4}$ inches long and having an aperture of $1\frac{1}{3}$ inches, with which he was able to study the phases of Venus and the phenomena of Jupiter's satellites. This precious little instrument is now one of the greatest treasures in the collection of the Royal Society of London. The general design of this telescope is shown in Fig. 21. The concave mirror is at the bottom of the telescope tube, and normally it would form an image of a star near the end of the tube. A plane mirror, however, of small size intercepts the rays and reflects them to the side, where they converge to a focus. This image is observed and magnified by an eye-piece, as in the refractor. It is true that in this arrangement the plane mirror, or *flat*, renders the central part of the principal mirror ineffective, but the loss of light is very much less than would be the case if the eye-piece were placed in position to view the image centrally.

In the hands of Sir William Herschel the reflecting telescope was greatly developed. The great telescope with which he enriched astronomical science had a mirror four feet in diameter, and its tube was forty feet in length. With the view of utilizing the whole surface of the mirror and dispensing with a second reflecting surface, the 4-foot mirror was placed at

a small angle to the bottom of the tube, so that its principal focal point was no longer at the centre, but at the side of the tube.

In practice, however, it is found that the Herschelian form of reflector does not give the best definition, and it is now very seldom seen.

Among other forms, the "Cassegrain" is perhaps the most important. During the last years this form has received a great deal of attention, more especially in regard to its special adaptability for photographic purposes.

In the Cassegrain telescope, the plane mirror of the Newtonian form is replaced by a small convex mirror which is part of a hyperboloid of revolution, its axis and focal point being coincident with those of the primary mirror. The rays are in this way reflected back to the mirror at the bottom of the tube, and in order that the image may be seen, it is necessary to cut out the middle part of the mirror to admit the eye-piece.

Although the small mirror must theoretically be hyperbolic, tolerable definition is obtained even if it be spherical or ellipsoidal, and its actual departure from these forms is so slight as to be beyond detection by measurement, so that the figuring of such mirrors can only be tested in the telescope. For photographic purposes this telescope has the very important advantage that a short telescope is equivalent to a very long one of the Newtonian form, or refracting telescope, so that the image of sun, moon, or planets formed at the focus is very large in comparison with the size of the telescope. A modification of this form of telescope, in which the small mirror is out of the path of the rays falling upon the larger one, and no longer obstructing the central part, has been revived by Dr. Common, and has become generally known as the "Skew Cassegrain."

In reflecting telescopes the mirrors were formerly made of speculum metal (an alloy of copper and tin), and the word *speculum* is even now commonly employed to signify a telescopic mirror, although it is usual to make the mirror of glass, with the concave surface silvered and highly polished.

One is frequently asked for an opinion as to which is the better form of telescope, the reflector or refractor, and it is a question that one finds some little difficulty in answering. On one point, however, all are agreed, namely, that the reflector has the advantage in regard to its achromatism; it is indeed

perfectly achromatic, while the so-called “achromatic” refractor is at best only a compromise. For the rest, one can not do better than quote the evidence of Dr. Isaac Roberts before the International Astrophotographic Congress: “The reflector requires the exercise of great care and patience, and a thorough personal interest on the part of the observer using it. In the hands of such a person it yields excellent results, but in other hands it might be a bad instrument. The reflector gives results at least equal, if not superior, to those obtained with the refractor, if the observer be careful of the centring, and of the polish of the mirror, and keeps the instrument in the highest state of efficiency; but when intrusted to an ordinary assistant the conditions necessary for its best performance can not be so well fulfilled as the same could be in the case of the refractor.” One great practical advantage of the reflector is that there are fewer optical surfaces, so that a large reflector may be obtained for the price of a much smaller refractor.

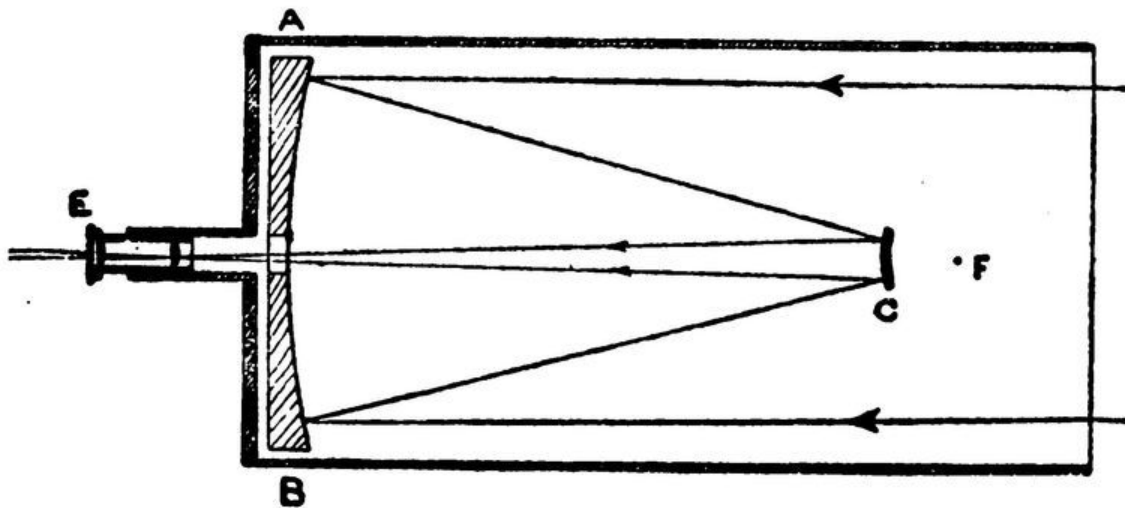


Fig. 22.—The Cassegrain Reflector

EYE-PIECES.—So far we have regarded the eye-piece of a telescope as a simple lens, but it is evident that the spherical and chromatic aberration of such a lens will interfere with its performance. For occasional use, however, even a simple lens is very serviceable if the object observed is brought to the centre of the field of view.

Compound eye-pieces are of various forms, each having certain advantages, the desiderata being freedom from color and “flatness of field”—that is, stars in different parts of the field are to be equally well in focus. Those most commonly employed are the Ramsden and Huyghenian

eye-pieces. The former consists of two plano-convex lenses of equal focal lengths, having their curved faces toward each other, and being placed at a distance apart equal to two-thirds of the focal length of either lens. Such an eye-piece can be used as a magnifying-glass, and it is therefore placed outside the focal image formed by the telescope with which it is used; on this account it is called a *positive eye-piece*. This kind of eye-piece is not quite achromatic, but its flat field of view gives it a special value for many purposes.

In the Huyghenian eye-piece there are again two lenses, made of the same kind of glass. That which comes nearest to the eye has a focal length of only one-third that of the *field* lens, and the distance between the two lenses is half the sum of the focal lengths. This form of eye-piece can not be used as a magnifying-glass in the ordinary sense, and as the field lens must be placed on the object-glass or mirror side of the focus, it is called a *negative eye-piece*. The Huyghenian eye-piece is more achromatic than the Ramsden, and is more widely used when it is only required to view the heavenly bodies. In instruments employed for purposes of measurement, a positive eye-piece is essential in order that the spider threads may be placed at the focus of the telescope. The images formed by an astronomical telescope are upside down, and neither of the eye-pieces described reinverts them.

A special form of eye-piece is therefore used when a telescope is employed for terrestrial sight-seeing. The desired result is obtained by the introduction of additional lenses, but there is a corresponding reduction of brightness.

For viewing the sun some device is necessary to reduce the quantity of light entering the eye. To look at the sun directly, even with a small instrument, is very dangerous. The arrangement usually adopted is a *solar diagonal*, in which the light is reflected from a piece of plane glass before entering the eye-piece; the piece of glass is wedge-shaped, so that the reflection from one surface only is effective; if the glass had parallel sides, the solar image would be double.

MAGNIFYING POWER.—The magnifying power of a telescope depends upon the focal length of the object-glass, or speculum, and that of the eye-piece. Optically, it is equal to the former divided by the latter, so that the

greater the focal length of an object-glass, or the smaller the focal length of the eye-piece, the greater will be the magnifying power. In a given telescope, the object-glass, or speculum, is a constant factor and the magnifying power can only be varied by changing the eye-piece. The focal length of the Lick telescope, for example, is about 600 inches; with an eye-piece which is equivalent to a lens of one-inch focus, the magnifying power would be 600; with a lens of half an inch focus, it would be 1,200, and so on.

The magnifying power which can be effectively employed, however, depends upon a great variety of circumstances. First, the clearness and steadiness of the air; then there is the quality of the object-glass, or speculum, to be considered; and also the brightness of the object to be observed, for when the object is very dim, its light will be spread out into invisibility if too high a power be used.

In practice, good refractors perform well with powers ranging up to 80 or 100 for each inch in the diameter of the object-glass. Thus, on sufficiently bright objects, a six-inch telescope will work well with a power of about 500, while a 30-inch may be effectively employed with powers between 2,000 and 3,000.

ILLUMINATING POWER.—It has already been pointed out that magnification is not the only function of a telescope. As a matter of fact, the most powerful telescopes in the world fail to produce the slightest increase in the apparent size of a star, for even if these objects be brought to apparently a 3,000th part of their real distances, they are still too far away to have any visible size. But although a star can not be magnified, it can be rendered more visible by the telescope, for the reason that the object-glass collects a greater number of rays than the naked eye. The pupil of the eye may be taken to have a diameter of one-fifth of an inch; a lens one-inch in diameter will have twenty-five times the *area* of the pupil, and will therefore collect twenty-five times the amount of light from a star; a two-inch lens will grasp one hundred times, and a 36-inch 32,400 times as much light as the pupil alone. Practically all these rays collected by the object-glass, or speculum, of a telescope can not be brought into the eye; some are lost through the imperfect transparency of the glass, or the imperfect reflecting power of the speculum. Still, allowing a considerable percentage for loss, there is an enormous concentration of light when a large telescope is employed.

THE ALTAZIMUTH MOUNTING.—Having got a telescope, we have next to see how it can be best supported, for unless it be a very small instrument indeed, it will be impossible to hold it in the hand like a spy-glass. However a telescope be mounted, provision must be made for turning it to any part of the sky whatsoever. Very frequently one of the axes on which the instrument turns is vertical, while the other is horizontal. Such a stand for a telescope is called an *altazimuth mounting*, for the reason that it permits the instrument to be moved in altitude and in azimuth.

As a rule, one finds only small telescopes mounted in this manner. The objection to it is that, as one continues to observe a heavenly body, two independent movements must be given to the telescope in order to follow the body in its diurnal movement across the heavens. If we commence observing a star newly risen, for example, the telescope must trace a star-like path in order to follow it as it ascends into the heavens.

THE EQUATORIAL TELESCOPE.—A much more convenient method of setting up a telescope is to mount it as an *equatorial*. The essential feature of this instrument is that one of the axes of movement, instead of being vertical, is placed parallel to the axis of the earth. This is called the *polar axis*, and, when the telescope is turned around such an axis, it traces out curves in the sky which are identical with those described by the stars in their diurnal motions. If, then, the telescope be directed to a star or other heavenly body, it can be made to follow the object and keep it in view by a single movement. The axis at right angles to the polar axis is called the declination axis, and is necessary in order that the telescope may be moved toward and from the poles so that all the heavenly bodies above the horizon may be included in its sweep.

One very important advantage of the equatorial is that, as only one motion is required to keep a star in view, so long as it is above the horizon, the necessary movement may be furnished by clockwork. A good equatorial is accordingly provided with a driving-clock, which is regulated so that it would drive the telescope through a whole revolution once a day. Unlike an ordinary clock, the driving-clock of a telescope is regulated by a governor, in order that the instrument may have a continuous and not a jerky movement.

The telescope is also provided with clamps and fine adjustments, one each in R. A. and declination, in order that it may be under the control of the observer. It is evident that the telescope must be capable of moving independently of the driving-gear, so that it may first be placed in the desired direction; when this is accomplished, the R. A. clamp is used to put the telescope in gear with the clock. The declination clamp is then made to fix the telescope firmly to the declination axis. Fine adjustments in both directions are necessary, because it is impossible to sight a large instrument with such precision as to bring an object exactly to the centre of the field of view.

Some of the driving-clocks fitted to equatorials are very elaborate. As clocks regulated by governors are not such reliable timekeepers as those regulated by pendulums, arrangements are made by which the accuracy of a pendulum can be electrically communicated to a governor clock. One of the best forms of electrically controlled clocks is that devised by Sir Howard Grubb.

Another important feature of an equatorial is that it can be provided with circles which enable the telescope to be pointed to any desired object of known right ascension and declination. One of these is the declination circle, attached to the declination axis and read by a vernier fixed to the sleeve in which the axis turns; this is adjusted so as to read 0° when the telescope points to any part of the celestial equator, and 90° when it is directed to the pole. The other circle is attached to the polar axis, and determines the position of the telescope with regard to the meridian; this is called the *hour circle*, and is divided into twenty-four hours. When the telescope is on the meridian, the hour circle reads zero, so that its reading in any other position gives the hour angle of the telescope. Having given the right ascension and declination of a heavenly body which it is desired to observe, the telescope is turned until the declination circle reads the proper angle, and the hour circle indicates the hour angle which is calculated for the particular moment of pointing the telescope. [The hour angle is the difference between the right ascension of the object and the sidereal time of observation.] In this way it is easy to find objects of known position which are invisible to the naked eye, and one can even pick up the planets and brighter stars in full sunshine. Conversely, one can determine from the circles the right ascension and declination of any object under observation,

but for various reasons only approximate results can be obtained in this way. The chief use of the circles on an equatorial is therefore to provide a means of pointing the telescope.

Telescopes of four inches aperture and upward are usually provided with a smaller companion called a *finder*. This has a larger field of view than the main telescope, so that objects which are of sufficient brightness can readily be picked up and brought to the centre of the finder, the adjustments being such that the object is then also at the centre of the field of the large telescope.

There are, of course, many practical details connected with the working of an equatorial with which space does not permit us to deal. It may be remarked, however, that the adjustment of the polar axis is very simply performed by first inclining it at an angle approximately equal to the latitude of the place where it is set up, and setting it as nearly as possible in the meridian by means of a compass or by observations of the sun at noon. The final adjustment is then made by a series of observations of stars of known position.

SOME OF THE WORLD'S GREAT TELESCOPES.—Thanks to the wide public interest taken in astronomical matters, a large number of powerful telescopes have been set up in various parts of the world. To the British Islands belongs the honor of possessing the largest telescope in the world. This is the giant reflector erected by Lord Rosse, in 1842, at Parsonstown, the mirror being six feet in diameter, and the focal length sixty feet. Many very valuable observations were made with this instrument in its early days, but of late years it seems to have fallen into disuse. One reason may be that the mounting is not of the most convenient form, and makes the telescope unsuitable for photographic work.

Coming next in point of size to the Rosse telescope is the reflector erected at Ealing by Dr. A. A. Common. The glass mirror of this telescope is five feet in diameter, five inches thick, and weighs more than half a ton. Dr. Common aimed specially at constructing the largest possible telescope which could be equatorially mounted and provided with a driving-clock, and he was only limited to an aperture of five feet by the impossibility of obtaining a glass disk of larger size. He has attained such great skill in this work that he was able to produce a perfect mirror five feet in diameter in

three months' time, although no less than 410,000 strokes of the polishing machine were required.

The telescope is of the Newtonian form, and the mounting is quite unique. The polar axis consists of an iron cylinder, made up of boiler plate, seven feet eight inches in diameter, and about fifteen feet long. From the top of the cylinder, near its outer edge, two horns, each six feet long, project outward, and the tube of the telescope swings on trunnions attached to the ends of the horns. The main part of the telescope tube is square, built up of steel angle iron, and carries the mirror at its lower end; the upper part of the tube, which carries the "flat" and eye-piece, is round, and of tinned steel strengthened by a skeleton framework.

It is evident that such an enormous instrument as this can not be made to travel by clockwork with the necessary uniformity without some very efficient arrangement for reducing friction. Dr. Common's plan—and it is here that his instrument is unlike others—is to make the hollow polar axis watertight, and to fix it in a tank of water. At the bottom of the polar axis is a ball and socket joint to keep it in position, and at the top is another bearing, which can be adjusted so that the polar axis lies truly in the meridian. It was found necessary to introduce nine tons of iron into the bottom of the hollow polar axis in order to sink it to the proper angle, and to put sufficient weight on the bearings to give stability to the instrument. In this way the great mass is brought into the region of manageability, and the driving-clock, which is driven by a weight of one and a half tons, is able to do its work efficiently. Such, in general outline, is this wonderful telescope, which, although not so large as Lord Rosse's famous instrument, is undoubtedly its superior in light-grasping power and general utility, and more especially in its adaptability for photographing the heavens.

Among other large reflecting telescopes now in use are the 4-foot reflectors at Melbourne and Paris, and the 3-foot reflectors at South Kensington and the Lick Observatory, California.

The largest refracting telescope yet constructed is one of forty inches aperture for the Yerkes Observatory of the University of Chicago. It is interesting to note here that Professor Keeler, in his report as an expert upon the performance of the object-glass, considers that there is "evidence for the first time that we are approaching the limit of size in the construction of

great objectives.” Unlike a mirror, a lens can be supported only upon its circumference, and it is the bending by its own weight that proves detrimental to its defining power. If the lens be made thicker with a view of overcoming this defect, the absorption of light by the glass increases, so that there is in the end no special gain by increasing the size.

The length of the Yerkes telescope is 62 feet, and is provided with all accessories pertaining to astrophysical research. The world-renowned Lick Telescope is of thirty-six inches aperture. The story of the foundation of this monster instrument is not much less wonderful than the telescope itself. Brought up in poor circumstances, with few opportunities for intellectual development, James Lick, nevertheless, amassed a fortune in business, and having few relations, he was anxious to dispose of his wealth in such a way as to bring him that fame which he had failed to achieve in other directions. Although it is very probable that he had never looked through a telescope in his life, the idea of a large telescope had taken a very firm hold upon his mind, and, thanks to the influence of his advisers, it was definitely announced in 1873 that Mr. Lick’s bid for immortality was to take this form. Several sites were examined by experts, and finally Mount Hamilton, California, 4,200 feet above sea-level, was selected. An excellent road, twenty-six miles in length, made at the cost of the county authorities, connects the observatory with the nearest town, San José, thirteen miles distant.

Owing to various delays, operations were not commenced until 1880, and five years were consumed in clearing away 72,000 tons of rocks and in erecting the buildings.

Mr. Lick had stipulated for the erection of “a telescope superior to and more powerful than any telescope yet made,” and Messrs. Alvan Clark & Co. contracted to supply a lens of thirty-six inches aperture for the sum of \$50,000. It turned out, however, that it was much easier to make such a contract than to fulfil it. To produce large disks of optically perfect glass, even in the rough, requires the greatest possible skill and patience, and this part of the work was undertaken by Feil & Co. of Paris. The flint glass disk was safely delivered in America in 1882, but the crown disk was cracked in packing. The elder Feil having retired from business, the duty of providing a new block of crown glass devolved upon his sons, who, after two years spent in vain attempts, ended in bankruptcy, and it was only through the

elder Feil again resuming business that the much-required disk was finally completed in 1885. After the lapse of another year, the rough disks were fashioned, in the workshops of the Clarks, into the most marvelous of telescopic lenses.

The mounting of the object-glass is worthy of the occasion. The tube is no less than thirty-seven feet long, and four feet in diameter in the middle part. An iron pier, thirty-eight feet high, beneath which lie the remains of Mr. Lick, supports the equatorial head, and a winding staircase enables the observer to reach the setting circles. Inside the hollow pier is the powerful driving-clock which turns the telescope to follow the heavenly bodies in their apparent movements. Finders of six, four, and three inches diameter, rods for the manipulation of the instrument, and all necessary accessories, complete what must long remain one of the most perfect instruments at the service of astronomical science. The \$200,000 expended upon it have already been amply justified by the work accomplished, while Mr. Lick's dream of immortality has become a reality.

The following list indicates some of the large refractors now doing active service:

Aperture	Observatory
36 inch	[Lick] California.
30 "	Pulkowa, Russia.
30 "	[Bischoffeim] Nice.
28 "	Greenwich.
27 "	Vienna.
26 "	Washington.
25 "	[Newall] Cambridge.
24 "	[Lowell] Mexico.
23 "	Princeton, New Jersey.

It is right to add, however, that opinion is still greatly divided as to whether these telescopes of large aperture really repay the expense and labor involved in their erection and use. On the very rare occasion when the "seeing" is practically perfect—which occurs perhaps only a few hours in a year—it is probable that the superiority of a large telescope is very marked, but under average conditions there seems to be little advantage over instruments of moderate size for many classes of observations.

Certain it is that a great deal of valuable work is done with comparatively small telescopes, ranging from six to fifteen inches aperture, and this in all departments of astronomical research. Hence, some of the most active observatories do not figure in the above list; among them may be mentioned the observatories of Harvard College, Potsdam, Paris, Heidelberg, Cape of Good Hope, Edinburgh, South Kensington, Stonyhurst College, and the observatory of Dr. Isaac Roberts at Crowborough, England.

HOUSING OF EQUATORIALS.—The building which accommodates an equatorial telescope must evidently be designed to admit of giving a clear opening to any part of the sky. Usually this is accomplished by making the roof, or *dome*, with a circular base, provided with wheels, which run on rails. It is then only necessary to open a narrow portion of the dome, extending from top to base, and to turn the dome until this aperture is in the required direction. One of the most elaborate domes now in existence is that built by M. Eiffel for the great refractor of the Nice Observatory. The lower part of the building is in the form of a square, having a side of about eighty-seven feet and a height of about thirty feet. The dome itself is seventy-four feet in diameter, and the moving parts alone weigh ninety-five tons.

There are two shutters, each a little wider than half the possible opening; these run on short rails, and are moved simultaneously by means of an endless rope. The whole of the dome is built up of steel angle iron, covered with very thin sheet steel. In order to facilitate the manipulation of the dome, its great weight is buoyed up by means of a float attached to its base and immersed in a circular tank of water of a little greater size than the base of the dome. If any mishap occurs with this gigantic tank, the dome rests on wheels which run on a circular rail, so that the work need not be interrupted. The whole arrangement is very easily turned with the aid of a winch by one man when the dome is floating, but when resting on the wheels several men are required at the winch.

This brief description will serve to illustrate some of the problems which confront the possessor of a very large telescope. For smaller instruments, the observatories follow pretty nearly the same plan, except that it is unnecessary to provide an arrangement for floating the dome.

The observatory which shelters a reflecting telescope need not differ very greatly from one which contains a refractor. If the instrument be a

Newtonian, it is generally convenient to sink the polar axis below the level of the floor in order that the observer may not be at too great a height from the ground, and in that case, the dome, or its equivalent, is all that is necessary. For his five-foot reflector, Dr. Common designed an observatory which is not of the ordinary form, but gives the necessary opening partly by means of large shutters and partly by a revolution of the whole house. It is not every one who is able to lay out \$40,000 on such a dome as that erected at Nice by M. Bischoffeim.

The varying position of the eye end of a telescope, when it is turned to different parts of the sky, makes it necessary to provide comfortable and safe seating accommodation for the observer, more especially when the telescope is a very large one. In the case of the Yerkes telescope, the eyepiece is thirty feet higher when observing near the horizon than when observing near the zenith, and the observer must necessarily follow the telescope. The most convenient arrangement in such a case is to raise or lower the floor of the observatory as occasion demands. The floor of the Yerkes Observatory is seventy-five feet in diameter, and by means of electric motors it can be given a vertical motion of twenty-two feet. A similar arrangement was provided for the Lick telescope from the designs of Sir Howard Grubb. With smaller instruments, observing ladders and adjustable chairs of various forms are employed.

THE EQUATORIAL COUDÉ.—A form of equatorial telescope which has possibly a great future before it is one introduced at Paris under the name of the *equatorial coudé*, or elbowed telescope. Its practical advantage is that the observer remains in a constant and comfortable position, so that revolving domes and elevating floors, or other arrangements serving similar purposes, are no longer necessary. The telescope tube is of two parts of nearly equal length, and what is ordinarily the lower half of the tube forms part of the polar axis, while the other half is attached to it at right angles. At the point of intersection of the two halves of the tube is a plane mirror, and there is another mirror in front of the object-glass. If the latter mirror were removed, such a telescope would only enable the observer to see objects lying along the celestial equator, but by its means objects in all parts of the heavens can be brought within range to an observer gazing down the hollow polar axis. The largest instrument is that at the Paris Observatory, which has

an object-glass $23\frac{1}{2}$ inches in diameter for visual observations, and another of the same size for photographic purposes.

FIXED TELESCOPES.—There is still another method of using a telescope. The telescope itself may be fixed, and the light of the heavenly bodies may be reflected into it by means of a mirror which is made to revolve so as to keep pace with their movements. Foucault devised an instrument called the *siderostat* for this purpose, and although it is not largely employed for telescopic observations, it is very widely utilized for spectroscopic work, where the spectroscope is of a kind not readily attached to a telescope.

Another instrument used for the same purpose has recently been brought forward under the name of the *cælost*. This is simply a mirror which is made to turn on a polar axis in its own plane, and since a reflected ray of light moves through twice the angle that the reflecting surface turns through, the mirror is made to revolve at the rate of one revolution in two days. As the name indicates, the whole heavens appear stationary in such an instrument, whereas in a *siderostat* only one star at a time appears at rest, while its neighbors slowly revolve round it.

PHOTOGRAPHIC TELESCOPES.—The application of photography to the study of the heavenly bodies marks one of the greatest advances of the present century. The instruments which are employed for this purpose range from the ordinary tourist camera to the largest telescope. Unlike a person sitting for a portrait, the heavenly bodies can not be made to stand still for the purpose, and as instantaneous photographs can only be obtained in the case of the sun and moon, it is usually necessary to make the camera follow the stars very exactly during the time of exposure in order that the images may fall on precisely the same parts of the photographic plate.

Some guiding arrangement is, therefore, essential, and generally the photographic camera or telescope is attached to an ordinary equatorial which is driven by clockwork, or very carefully by hand if the camera be a small one. In the guiding telescope are two spider-threads at right angles to each other, and it is by constantly keeping the image of a star at the intersection of these “wires” that the operator ensures the images remaining in a constant position upon the sensitive plate.

An ordinary portrait camera, in the hands of a skilled observer, yields very beautiful pictures, but they are naturally on a small scale. The field of

view of such an instrument is so large that a whole constellation may be photographed with a single exposure.

Portrait lenses of six inches aperture in the hands of Dr. Max Wolf and Professor Barnard have given magnificent delineations of the Milky Way, and of the extremely faint nebulosities which are to be found in many parts of the heavens.

For many purposes, however, telescopes of greater power are required, and here it may be remarked that the distance between the images of any two adjacent stars will vary in direct proportion to the focal length of the telescope. In the same way the size of the image of a planet, the moon, or a comet, increases as the focal length of the objective is increased.

Refracting telescopes which are employed for photography require object-glasses which are specially “corrected” for the photographic rays. White light is compounded of light of all colors, but it is the blue and violet constituents which are effective in producing photographic action on an ordinary sensitive plate. Now, an object-glass which is intended for visual purposes is made to focus at the same point as many as possible of the rays which are most effective to the human eye, that is the green, yellow, and red, and usually there is a blue or purple halo round the images of the brighter objects, which is, however, too feeble as a rule to interfere with visual observations. This blue halo will evidently result in defective definition if the lens be employed for photography. By putting the plate at the point where the blue rays are most nearly focused, a better image is obtained; but for really good work a photographic object-glass must be so designed that all the blue and violet rays are brought to one and the same focus. Such a lens will consequently be a very poor one for visual observations.

The new “photo telescopic” object-glass now manufactured by Messrs. Cooke appears to be full of promise. In this lens all the colors of the spectrum are brought to almost exactly the same focal point, so that it serves equally well for photographic or visual purposes.

This difficulty in regard to achromatism does not exist in the case of the reflecting telescope, since rays of light of every color are reflected at precisely the same angles. For this reason reflectors, when properly managed, give the best photographic results. Dr. Isaac Roberts and Dr.

Common are especially identified with the application of the reflecting telescope for celestial photography. The instrument employed by the former consists of a 20-inch reflector and a 7-inch guiding telescope of the refracting form. The two telescopes are mounted on the extreme ends of the declination axis of an equatorial.

Dr. Common does not employ a guiding telescope at all. The photographic plate which he places at the focus of the reflector is smaller than the field of view, so that by means of an eye-piece fitted with a cross wire at the side of the dark slide, he is able to watch a star near the edge of the field. Both eye-piece and dark slide are attached to a frame which can be controlled by two screws at right angles to each other. If the guiding star leaves the cross wire through errors in driving, or other causes, the eye-piece and dark slide are bodily moved after it by means of the adjusting screws. This method not only has the advantage of saving the cost of a guiding telescope, but reduces the effects of vibration consequent upon the correction of errors by moving the whole telescope.

For photographing the sun a special instrument called a *photoheliograph* is usually employed. This differs only from an ordinary photographic telescope in being provided with a secondary magnifier, by which means the focal image formed by the object-glass is amplified before falling upon the photographic plate. On a bright, clear day pictures of the sun eight inches in diameter can be taken with an exposure of about $\frac{1}{500}$ th of a second, and such a photograph will frequently record more facts as to the state of the solar surface than a whole day's observation. Lenses or mirrors of very long focus are also occasionally employed in solar photography, and in this way a large image is obtained without the use of a secondary magnifier.

Photographs of the moon and planets may be taken either with or without a secondary magnifier, but in either case the exposures are longer than for the sun.

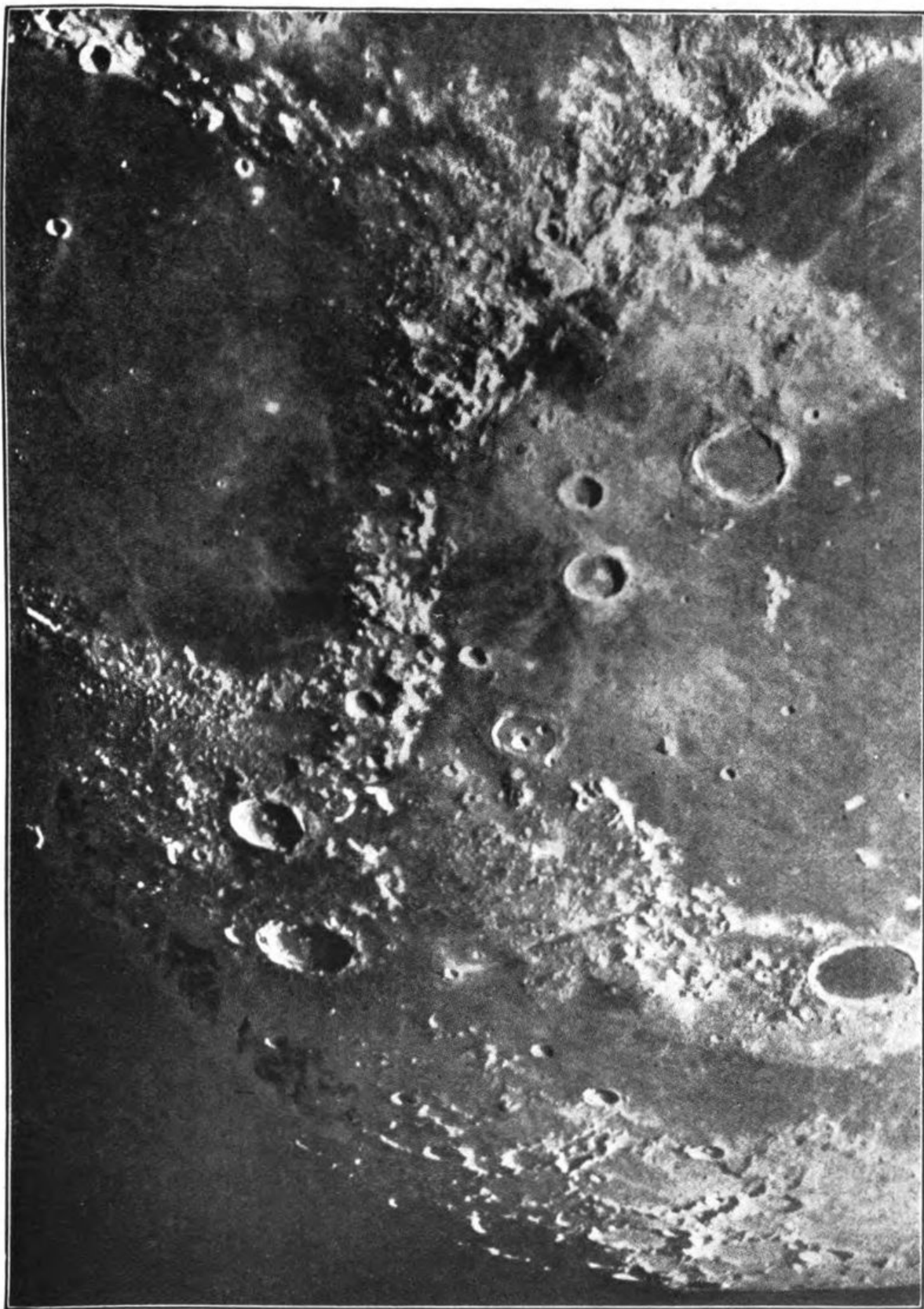
Finally, it may be added that the sensitive plates and processes used in astronomical photography do not differ from those employed by ordinary photographers.

FOOTNOTES:

[\[22\]](#) The focal length of a lens is the distance from its centre at which an image of a very distant object, such as the sun, is formed.

METEORS.—SIR ROBERT S. BALL

Our present knowledge as to the natural history of the shooting stars has been mainly acquired during the last hundred years. The first important step in the comprehension of these bodies was to recognize that the brilliant flash of light was caused by some object which came from without and plunged into our air. This was known at the end of the Eighteenth Century, largely by the labors of the philosopher Chladni in 1794.



A Portion of the Moon's Disk
Where Four Mountain Ranges Meet

Could an ordinary shooting star tell us its actual history, the narrative would run somewhat as follows:

“I was a small bit of material, chiefly, if not entirely, composed of substances which are formed from the same chemical elements as those you find on the earth. Not improbably I may have had some iron in my constitution, and also sodium and carbon, to mention only a few of the most familiar elements. I only weighed an ounce or two, perhaps more, perhaps less—but you could probably have held me in your closed hand, or put me into your waistcoat pocket. You would have described me as a sort of small stone, yet I think you would have added that I was very unlike the ordinary stones with which you were familiar. I have led a life of the most extraordinary activity; I have never known what it was to stay still; I have been ever on the move. Through the solitudes of space I have dashed along with a speed which you can hardly conceive. Compare my ordinary motion with your most rapid railway trains; my journey will be done ere the best locomotive ever built could have drawn the train out of the station. Pit me against your rifle bullets, against the shots from your one-hundred-ton guns; before the missile from the mightiest piece of ordnance ever fired shall have gone ten yards I have gone 1,000 yards. I do not assert that my speed has been invariable—sometimes it has been faster, sometimes it has been slower; but I have generally done my million miles a day at the very least. Such has been my career, not for hours or days, but for years and for centuries, probably for untold ages. And the grand catastrophe in which I vanished has been befitting to a life of such transcendent excitement and activity; I have perished instantly, and in a streak of splendor. In the course of my immemorial wanderings I have occasionally passed near some of the great bodies in the heavens; I have also not improbably in former years hurried by that globe on which you live. On those occasions you never saw me, you never could have seen me, not even if you had used the mightiest telescope that has ever been directed to the heavens. But too close an approach to your globe was at last the occasion of my fall. You must remember that you live on the earth buried beneath a great ocean of air. Viewed from outside space, your earth is seen to be a great ball, everywhere swathed with this thick coating of air. Beyond the appreciable limits of the

air stretches the open space, and there it is that my prodigious journeys have been performed. Out there we have a freedom to move of which you who live in a dense atmosphere have no conception. Whenever you attempt to produce rapid motion on the earth, the resistance of your air largely detracts from the velocity that would be otherwise attainable. Your quick trains are impeded by air, your artillery ranges are shortened by it. Movements like mine would be impossible in air like yours.

“And this air it is which has ultimately compassed my destruction. So long as I merely passed near your earth, but kept clear of that deadly net which you have spread, in the shape of your atmosphere, to entrap the shooting stars, all went well with me. I felt the ponderous mass of the earth, and I swerved a little in compliance with its attraction; but my supreme velocity preserved me, and I hurried past unscathed. I had many narrow escapes from capture during the lapse of those countless ages in which I have been wandering through space. But at last I approached once too often to the earth. On this fatal occasion my course led me to graze your globe so closely that I could not get by without traversing the higher parts of the atmosphere. Accordingly, a frightful catastrophe immediately occurred. Not to you; it did you no harm; indeed, quite the contrary. My dissolution gave you a pleasing and instructive exhibition. It was then, for the first time, that you were permitted to see me, and you called me a shooting star or a meteor.

“When from the freedom of open space I darted into the atmosphere, I rubbed past every particle of air which I touched in my impetuous flight, and in doing so I experienced the usual consequence of friction—I was warmed by the operation.

“You can readily comprehend the immense quantity of heat that will have been produced ere friction could deprive me of a speed of twenty miles a second. That heat not merely warmed me, but I rapidly became red-hot, white-hot, then I melted, even though composed of materials of a most refractory kind. Still friction had much more to do, and it actually drove me off into vapor, and I vanished. You, standing on your earth many miles below, never saw me—never could have seen me—until this supreme moment, when, glowing with an instantaneous fervor, I for a brief second became visible.

“Nature knows no annihilation, and though I had been driven off into vapor and the trial by fire had scattered and dispersed me, yet in the lofty heights of the atmosphere those vapors cooled and condensed. They did not, they never could again reunite and reproduce my pristine structure. Here and there in wide diffusion I repassed from the vaporous to the solid form, and in this state I wore the appearance of a streak of minute granules distributed all along the highway I had followed. These granules gradually subsided through the air to the earth. On Alpine snows, far removed from the haunts of men and from contamination of chimneys, minute particles have been gathered, many of which have unquestionably been derived from the scattered remains of shooting stars. Into the sea similar particles are forever falling, and they have been subsequently dredged up from profound depths, having subsided through an ocean of water after sinking through an ocean of air.”

Those splendid shooting stars which are often called fire-balls move in every direction. They come from the east, and from the west, from the north, and from the south. There is no hour of the night at which they have not occasionally been seen. Even in daylight it has happened not once or twice, but on several occasions, that a brilliant meteor has forced itself upon our astonished notice. They generally first make their appearance at a height which is between fifty and one hundred miles above the ground. They hurry down their inclined path, but generally become extinguished while still at least twenty miles aloft. In their more ambitious flights meteors have been known to span a kingdom. Nor are even greater strides unrecorded. The length of a continent may be compared with the track of that terrific meteor of 5th September, 1868, which broke into visibility at an appalling height above the Black Sea, and had not expended its stupendous energy until it passed over the smiling vineyards of France.

Great fire-balls are much more numerous than any one would suppose who had not paid attention to the subject. Nor need this be a matter for surprise if it be remembered that when a fire-ball does arrive it is only by a favorable combination of circumstances that any particular individual is privileged to witness the exhibition. As a random example of the yearly crop of fire-balls, I take from the middle of 1877 to the middle of 1878. A list of the fire-balls noticed during this period will be found in that store-house of valuable information, the Reports of the British Association. In the

year referred to I see that eighty-six great fire-balls have been recorded. They have appeared in various localities, both in the old hemisphere and in the new. The most arduous observer may think himself fortunate if he has even seen one of them.

As to the brilliant light from some of these great fire-balls, there are numerous statements. We are not infrequently told that even the beams of the full moon are ineffectual in comparison with the blaze of the meteor; and we find a high authority asserting that one of these bodies displayed a flash as “blinding as the sun.” On the 29th July, 1878, a fire-ball was seen which created so splendid an illumination that “the smallest objects were visible at Manchester.”

Fortunate, indeed, would the astronomer have been who, guided by some miraculous prescience, had gone to the ancient city of York on the evening of the 23d of February, 1879, and on the tower of the glorious minster spent the night in observation of the heavens. It would have been his privilege to witness a majestic meteor under circumstances of almost unique magnificence.

It was at seven minutes before three that such few stragglers as the streets of York still contained saw a pear-shaped ball of fire traveling across the sky. It drenched the ancient city with a flood of light. The superb front of the minster never before glowed with a more romantic illumination. The unwonted brilliancy streamed through every aperture in every window in the city; every wakeful eye was instantly on the alert; every light sleeper started up suddenly to know what was the matter. Even those whom the blaze of midnight light had failed to awaken were only permitted to protract their slumbers for another minute and a half—only until an awful crash, like a mighty peal of thunder, burst over the town, shaking the doors, the windows, and even the houses themselves. The whole city was thus alarmed. Every one started at the noise. But that noise was not a clap of thunder. Nor was it produced by an earthquake. It was merely the explosion of the fire-ball which flung itself against the atmosphere after its immeasurable voyage through space.

Perhaps the most remarkable instance of the *explosion* of a meteor is recorded in the case of the great fire-ball so widely observed in America on the 21st of December, 1876. The movements of this superb object have

been carefully studied by Professor H. A. Newton and Professor D. Kirkwood. For the prodigious span of a thousand miles this meteor tore over the American continent with a speed of some ten or fifteen miles a second. It first appeared over Kansas at a height of seventy-five miles. Thence it glided over the Mississippi, over the Missouri; it passed to the south of Lake Michigan; it made a short voyage over Lake Erie, and it can not have been very far from the Falls of Niagara, when by becoming invisible all further traces of its movements were lost. While passing a point midway between Chicago and St. Louis a frightful explosion shattered the meteor into a cluster of brilliant balls of fire, which seemed to chase each other across the sky. This cluster must have been about forty miles long and five miles wide. The detonation by which the explosion was accompanied was a specially notable incident of this meteor. It was not only heard with terrific intensity in the neighborhood, but the volume of sound was borne to great distances.

The glory of a meteor is often so evanescent that we just get a glimpse and it is gone. The sky resumes its ordinary aspect; the familiar stars are there, and even the very situation of the brilliant streak has become unrecognizable. But this is not always so; it sometimes happens that the brief career of the meteor leaves a notable trace behind it, so that for seconds and for minutes the sky is diversified by an unwonted spectacle. The path of the meteor leaves a stain of pearly light on the sky to mark the highway pursued by our celestial visitor.

In its fearful career the meteor is often rent to fragments, reduced to dust, dissolved into vapor. The glowing atoms of the wreck lie strewn along the path, just as the ghastly remnants of Napoleon's mighty army limned out the awful retreat from Moscow.

A pencil-shaped cloud of meteoric débris, perhaps eighty or a hundred miles in length, and four or five miles in diameter, thus hangs poised in air. It is at night. The sun has sunk so far below the horizon that there is no trace of the feeblest twilight glow. An ordinary cloud would, of course, be invisible except as concealing the stars; no beams of light fall upon it; there is nothing to render it luminous. So, too, the meteoric streak will often pass instantly into invisibility, but, as I have said, this is not always the case. There is a well-authenticated instance in which the trail of a superb meteor remained visible for nearly an hour. I have endeavored up to the present to

explain the various phenomena presented to us in the fall of a meteor, but here, for the first time, we have to note a circumstance for which it is not easy to account. We can explain why it is that the long meteoric cloud should be there, but we can not so easily explain why we should be able to see it. Whence comes this beautiful pearly luminosity? It seems that the meteoric dust must glow with some intrinsic luminosity.

We have spoken of dazzling fire-balls which generate for a brief moment a light which eye-witnesses, with possibly a pardonable exaggeration, have ventured to compare with the beams of the sun himself. Other meteors are described as being as bright as the full moon. Descending still lower in the scale of splendor, we read of fire-balls as bright as Venus or Jupiter, as bright as Sirius, or as a star of the first magnitude. With each step downward in brilliancy we find the meteors to increase in numerical abundance. Shooting stars as bright as the stars of the second or third magnitude are comparatively frequent; they are still more numerous of the fourth and fifth magnitudes. Every night brings its tale of shooting stars whose brightness is just sufficient to impress the unaided eye. Nor do the shooting stars which even the most attentive eye can detect represent a fraction of their entire number. As there are telescopic stars which the unaided eye can not see, so it might fairly be conjectured that, as we can trace meteors of successive stages of brightness down to the limit of unaided eye visibility, so there may be meteors still and still smaller which would be detected could we only direct a telescope toward them.

If we reflect that for every one that is seen there must be thousands which dart in unseen, we obtain an imposing idea of the myriads of shooting stars that daily rain in upon our globe.

The world is thus pelted on all sides day and night, year after year, century after century, by troops and battalions of shooting stars of every size, from objects not much larger than grains of sand up to mighty masses which can only be expressed in tons. In the lapse of ages our globe must thus be gradually growing by the everlasting deposit of meteoric débris. Looking back through the vistas of time past, it becomes impossible to estimate how much of the solid earth may not owe its origin to this celestial source.

The first and most important truth with regard to the recurrence of the meteors is their occasional appearance in what are known as “meteoric showers.” During such displays it sometimes happens that shooting stars in shoals break forth simultaneously, so as to produce a spectacle which we now regard as of the utmost beauty and interest, but which in earlier times has often been the source of the direst terror and dismay.

Let me, for the sake of illustration, give some account of one of these great showers of shooting stars.

In the year 1866 I occupied the position of astronomer to the late Earl of Rosse. The memorable night between November 13th and 14th, 1866, was a very fine one; the moon was absent—a very important consideration in regard to the effectiveness of the display. The stars shone out clearly, and I was diligently examining some faint nebulae in the eye-piece of the great telescope, when a sudden exclamation from the attendant caused me to look up from the eye-piece just in time to catch a glimpse of a fine shooting star, which, like a great sky-rocket, but without its accompanying noise, shot across the sky over our heads. The great shooting star which had already appeared was merely the herald announcing the advent of a mighty host. At first the meteors came singly, and then, as the hours wore on, they arrived in twos and in threes, in dozens, in scores, in hundreds. Our work at the telescope was forsaken; we went to the top of the castellated walls of the great telescope and abandoned ourselves to the enjoyment of the gorgeous spectacle.

To number the meteors baffled all our arithmetic; while we strove to count on the one side many of them hurried by on the other. The vivid brilliance of the meteors was sharply contrasted with the silence of their flight. We heard on that marvelous night no sounds save those with which we were familiar. The flights of the celestial rockets were attended with no noises that we could hear. The meteors were no doubt somewhat various as to size, but the characteristic feature of this shower, as contrasted with another great shower I have also seen, was the remarkable brilliance of the shooting stars. It was their exceptional splendor even more than their innumerable profusion that gave to the shower its peculiarity. As to the actual brilliancy of the meteors, I am enabled to give the accurate estimate made by Mr. Baxendell at Manchester, where the shower was well seen. Out of every hundred of these meteors ten were brighter than a first

magnitude star, and two or three of them were brighter than Sirius. Fifteen out of each hundred were between the first and second magnitudes, and twenty-five were between the second and third magnitudes, while the remainder were smaller.

Some important facts with regard to ancient shooting-star showers have survived the thousand and one casualties to which historical records are exposed. A careful discussion of those which are sufficiently accurate to be intelligible discloses to us the startling fact that in general every thirty-three years a grand shooting-star shower has rained down on our earth. It sometimes happens that two consecutive years are rendered memorable by great showers. At present the day of the year on which this particular shower is wont to appear is about the 13th November; but in earlier ages we find the date to shift slowly toward the commencement of the year. Thus the display which took place in A. D. 1698 was on the 9th of November; while, looking back still further to one of the very earliest records, viz., that of the year 934, we find the date has receded to October 14th. This change of the day on which the shower occurs is of profound theoretical importance in connection with the discovery of the orbit which these meteors pursue. The advance of the date is, however, so slow that for the past few generations, as well as for the next, we may sufficiently define this particular shower by the meteors which enliven the skies between the 12th and the 14th of November. In fact, the poetaster has parodied the well-known lines for the days of the month by a similar effort, which will serve to remind us also of another periodic shower of shooting stars which occurs in August. He writes:

“If you November’s stars would see,
From twelfth to fourteenth watching be.
In August too stars shine from heaven,
On nights between nine and eleven.”

These lines are intended to imply that the days named will usually bring, in every November, a few meteors at all events belonging to the grand shower. These are stragglers, as it were, from the mighty host which visits us three times in the century.

Astronomers have a special name for this group of November meteors. They are called the “Leonids.” To explain why this name has been given,

This is a detailed star chart of the Northern Hemisphere, showing constellations like Virgo, Coma Berenices, Ursa Major, Leo Minor, Sextans, and Hydrus. It includes a grid of celestial coordinates and labels for various stars and the ecliptic.

Fig. 23.—Position of Leo, Source of the Leonids

Among the constellations there is a fine sickle-shaped group, forming a part of Leo, one of the signs of the Zodiac. That part of the sky defined by Leo is curiously related to the meteors of the 12th to the 14th of November. Every shooting star truly belonging to that great shower pursued a track across the heavens, the direction of which, if carried back far enough, was always found to pierce through the sickle of Leo. Indeed, the paths of all the meteors formed a set of rays spreading away from that one point in the constellation. An invariable characteristic of this particular shower is its connection with the constellation of Leo, hence the appropriateness of the name of Leonids.

It must be borne in mind that we can never see the meteors until the fatal moment when they dive into our atmosphere. We could, indeed, at any time point our telescope to the spot in the heavens where we know the great shoal must certainly be located. But the mightiest telescope in the world does not disclose the shoal to us. In fact, we would never have seen these Leonids at all, we would never have become conscious that such a shoal of meteors existed, had it not been for a certain circumstance, which, for want of a better expression, I must speak of as accidental.

Our globe pursues a certain definite track around the sun. Year after year, with undeviating regularity, the earth performs the stages of its journey. If it reaches certain points on the 1st of January and the 12th of October in one year, then it reaches the same points on the 1st of January and the 12th of October respectively on next year, or any other year.

The Leonids and the earth have each a certain track. It might of course have happened that one of these tracks lay quite outside or quite inside the other. In the case of the Leonids, it has chanced that their orbit does intersect the orbit of the earth, and to this circumstance we are indebted for the glorious displays every thirty-three years.

There are many other periodic showers of shooting stars besides those notable Leonids. None of the other showers, however, possess the same importance as the Leonids, nor do they ever manifest celestial splendor comparable with that of those of the 13th of November. The Perseids, for example, which appear from the 9th to the 11th of August, are tolerably constant in their appearance, but have little spectacular interest. There is also another shower called the Andromedes, which occurs on the 27th of

November. It has produced certain displays, one of the most remarkable of which took place in 1872. The meteors were excessively numerous on that occasion, but they were so short in their paths, and so insignificant as to brilliance, that the spectacle, though of great scientific interest, could not be compared as to splendor with that of the Leonids in 1866.

There are also several other showers which appear with greater or less regularity. Each of these possesses two distinct characteristics by which its meteors can be identified. One of these characters is the date on which the shower appears. The other is the constellation or point on the heavens from which all the meteors appear to radiate. Thus when we speak of the Andromedes on the 27th of November, we express that the shower on the 27th November comes from the part of the heavens marked by the constellation of Andromeda.

A striking discovery has been made which points to a curious connection between comets and shooting stars. It has been found that the track followed by a great shower of meteors is often identical with the track pursued by a comet. It is wholly beyond the province of mere chance that an orbit such as that of the Leonids should, both as to its size and its position in space, be likewise that of a comet, unless the comet and the meteor swarm were objects related to each other.

The great sun guides our world through its long annual journey. The mighty mass of the earth yields compliance to the potent sway of the ruler of our system. But the sun does not merely exercise a control over the vast planets which circulate around him. The supreme law of gravitation constrains the veriest mote that ever floated in a sunbeam, with the same unremitting care that it does the mightiest of planets. Thus it is that each little meteor is guided in its journeys for untold ages. Each of these little objects hurries along, deflected at every moment, to follow its beautifully curved path by the incessant attraction of the sun. At last, however, the fatal plunge is taken. The long wanderings of the meteor have come to an end and it vanishes in a streak of splendor.

COMETS.—SIR JOHN HERSCHEL

The extraordinary aspect of comets, their rapid and seemingly irregular motions, the unexpected manner in which they often burst upon us, and the imposing magnitudes which they occasionally assume, have in all ages rendered them objects of astonishment, not unmixed with superstitious dread to the uninstructed, and an enigma to those most conversant with the wonders of creation and the operations of natural causes. Even now, that we have ceased to regard their movements as irregular, or as governed by other laws than those which retain the planets in their orbits, their intimate nature, and the offices they perform in the economy of our system, are as much unknown as ever. No distinct and satisfactory account has yet been rendered of those immensely voluminous appendages which they bear about with them, and which are known by the name of their tails (though improperly, since they often precede them in their motions), any more than of several other singularities which they present.

The number of comets which have been astronomically observed, or of which notices have been recorded in history, is very great, amounting to several hundreds, and when we consider that in the earlier ages of astronomy, and indeed in more recent times, before the invention of the telescope, only large and conspicuous ones were noticed; and that, since due attention has been paid to the subject, scarcely a year has passed without the observation of one or two of these bodies, and that sometimes two and even three have appeared at once; it will be easily supposed that their actual number must be at least many thousands. Multitudes, indeed, must escape all observation, by reason of their paths traversing only that part of the heavens which is above the horizon in the daytime. Comets so circumstanced can only become visible by the rare coincidence of a total eclipse of the sun—a coincidence which happened, as related by Seneca, sixty-two years before Christ, when a large comet was actually observed very near the sun. Several, however, stand on record as having been bright enough to be seen with the naked eye in the daytime, even at noon and in bright sunshine. Such were the comets of 1402, 1532 and 1843, and that of 43 B. C. which appeared during the games celebrated by Augustus in honor

of Venus shortly after the death of Cæsar, and which the flattery of poets declared to be the soul of that hero taking its place among the divinities.

Comets consist for the most part of a large and more or less splendid but ill-defined nebulous mass of light called the head, which is usually much brighter toward its centre, and offers the appearance of a vivid *nucleus*, like a star or planet. From the head, and in a direction *opposite to that in which the sun is situated* from the comet, appear to diverge two streams of light, which grow broader and more diffused at a distance from the head, and which most commonly close in and unite at a little distance behind it, but sometimes continue distinct for a great part of their course; producing an effect like that of the trains left by some bright meteors, or like the diverging fire of a sky-rocket (only without sparks or perceptible motion). This is the tail. This magnificent appendage attains occasionally an immense apparent length. Aristotle relates of the tail of the comet of 371 B. C., that it occupied a third of the hemisphere, or 60° ; that of A. D. 1618 is stated to have been attended by a train no less than 104° in length. The comet of 1680, the most celebrated of modern times, and on many accounts the most remarkable of all, with a head not exceeding in brightness a star of the second magnitude, covered with its tail an extent of more than 70° of the heavens, or, as some accounts state, 90° ; that of the comet of 1769 extended 97° , and that of the comet of 1843 was estimated at about 65° when longest.

The tail is, however, by no means an invariable appendage of comets. Many of the brightest have been observed to have short and feeble tails, and a few great comets have been entirely without them. Those of 1585 and 1763 offered no vestige of a tail; and Cassini describes the comets of 1665 and 1682 as being as round and as well defined as Jupiter. On the other hand, instances are not wanting of comets furnished with many tails or streams of diverging light. That of 1744 had no less than six, spread out like an immense fan, extending to a distance of nearly 30° in length. The small comet of 1823 had two, making an angle of about 160° , the brighter turned as usual from the sun, the fainter toward it, or nearly so. The tails of comets, too, are often somewhat curved, bending, in general, toward the region which the comet has left, as if moving somewhat more slowly, or as if resisted in their course.

The smaller comets, such as are visible only in telescopes, or with difficulty by the naked eye, and which are by far the most numerous, offer very frequently no appearance of a tail, and appear only as round or somewhat oval vaporous masses, more dense toward the centre, where, however, they appear to have no distinct nucleus, or anything which seems entitled to be considered as a solid body. This was shown in a very remarkable manner in the case of the comet discovered by Miss Mitchell in 1847, which on the 5th of October in that year passed *centrally* over a star of the fifth magnitude: so centrally that with a magnifying power of 100 it was impossible to determine in which direction the extent of the nebulosity was greatest. The star's light seemed in no degree enfeebled; yet such a star would be completely obliterated by a moderate fog, extending only a few yards from the surface of the earth. And since it is an observed fact that even those larger comets which have presented the appearance of a nucleus have yet exhibited *no phases*, though we can not doubt that they shine by the reflected solar light, it follows that even these can only be regarded as great masses of thin vapor, susceptible of being penetrated through their whole substance by the sun-beams, and reflecting them alike from their interior parts and from their surfaces. Nor will any one regard this explanation as forced, or feel disposed to resort to a phosphorescent quality in the comet itself, to account for the phenomena in question, when we consider the enormous magnitude of the space thus illuminated, and the extremely small *mass* which there is ground to attribute to these bodies. It will then be evident that the most unsubstantial clouds which float in the highest regions of our atmosphere, and seem at sunset to be drenched in light, and to glow throughout their whole depth as if in actual ignition, without any shadow or dark side, must be looked upon as dense and massive bodies compared with the filmy and all but spiritual texture of a comet. Accordingly, whenever powerful telescopes have been turned on these bodies, they have not failed to dispel the illusion which attributes *solidity* to that more condensed part of the head which appears to the naked eye as a nucleus; though it is true that in some a very minute stellar point *has* been seen, indicating the existence of something more substantial.

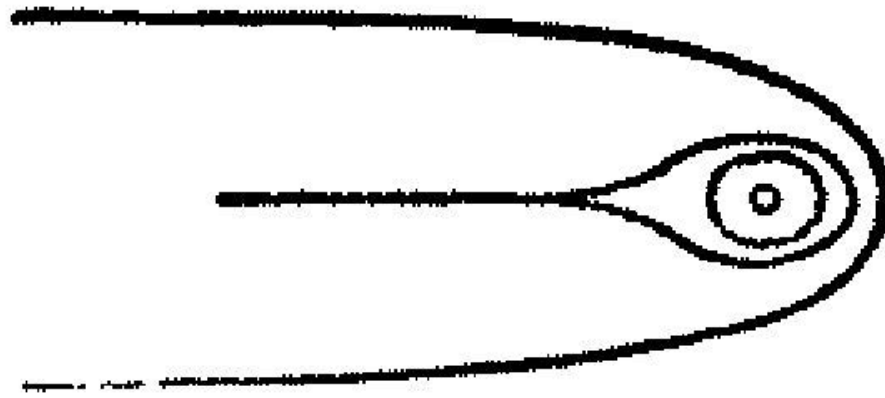


Fig. 24.—Head of Comet

That the luminous part of a comet is something in the nature of a smoke, fog, or cloud, suspended in a transparent atmosphere, is evident from a fact which has been often noticed, viz., that the portion of the tail where it comes closest to and surrounds the head is yet separated from it by an interval less luminous, as if sustained and kept off from contact by a transparent stratum, as we often see one layer of clouds over another with a considerable clear space between. These, and most of the other facts observed in the history of comets, appear to indicate that the structure of a comet, as seen in section in the direction of its length, must be that of a hollow envelope, of a parabolic form, inclosing near its vertex the nucleus and head, something as represented in the preceding figure. This would account for the apparent division of the tail into two principal lateral branches, the envelope being oblique to the line of sight at its borders, and therefore a greater depth of illuminated matter being there exposed to the eye. In all probability, however, they admit great varieties of structure, and among them may very possibly be bodies of widely different physical constitution, and there is no doubt that one and the same comet at different epochs undergoes great changes, both in the disposition of its materials and in their physical state.

We come now to speak of the motions of comets. These are apparently most irregular and capricious. Sometimes they remain in sight only for a few days, at others for many months; some move with extreme slowness, others with extraordinary velocity; while not infrequently the two extremes of apparent speed are exhibited by the same comet in different parts of its course. The comet of 1472 described an arc of the heavens of 40° of a great

circle in a single day. Some pursue a direct, some a retrograde, and others a tortuous and very irregular course; nor do they confine themselves, like the planets, within any certain region of the heavens, but traverse indifferently every part. Their variations in apparent size, during the time they continue visible, are no less remarkable than those of their velocity; sometimes they make their first appearance as faint and slow-moving objects, with little or no tail; but by degrees accelerate, enlarge, and throw out from them this appendage, which increases in length and brightness till (as always happens in such cases) they approach the sun, and are lost in his beams. After a time they again emerge on the other side, receding from the sun with a velocity at first rapid, but gradually decaying. It is for the most part after thus passing the sun that they shine forth in all their splendor, and that their tails acquire their greatest length and development; thus indicating plainly the action of the sun's rays as the exciting cause of that extraordinary emanation. As they continue to recede from the sun, their motion diminishes and the tail dies away, or is absorbed into the head, which itself grows continually feebler, and is at length altogether lost sight of, in by far the greater number of cases never to be seen more.

Without the clew furnished by the theory of gravitation, the enigma of these seemingly irregular and capricious movements might have remained forever unresolved. But Newton, having demonstrated the possibility of any conic section whatever being described about the sun, by a body revolving under the dominion of that law, immediately perceived the applicability of the general proposition to the case of cometary orbits; and the great comet of 1680, one of the most remarkable on record, both for the immense length of its tail and for the excessive closeness of its approach to the sun (within one-sixth of the diameter of that luminary), afforded him an excellent opportunity for the trial of his theory. The success of the attempt was complete. From that time it became a received truth, that the motions of comets are regulated by the same general laws as those of the planets.

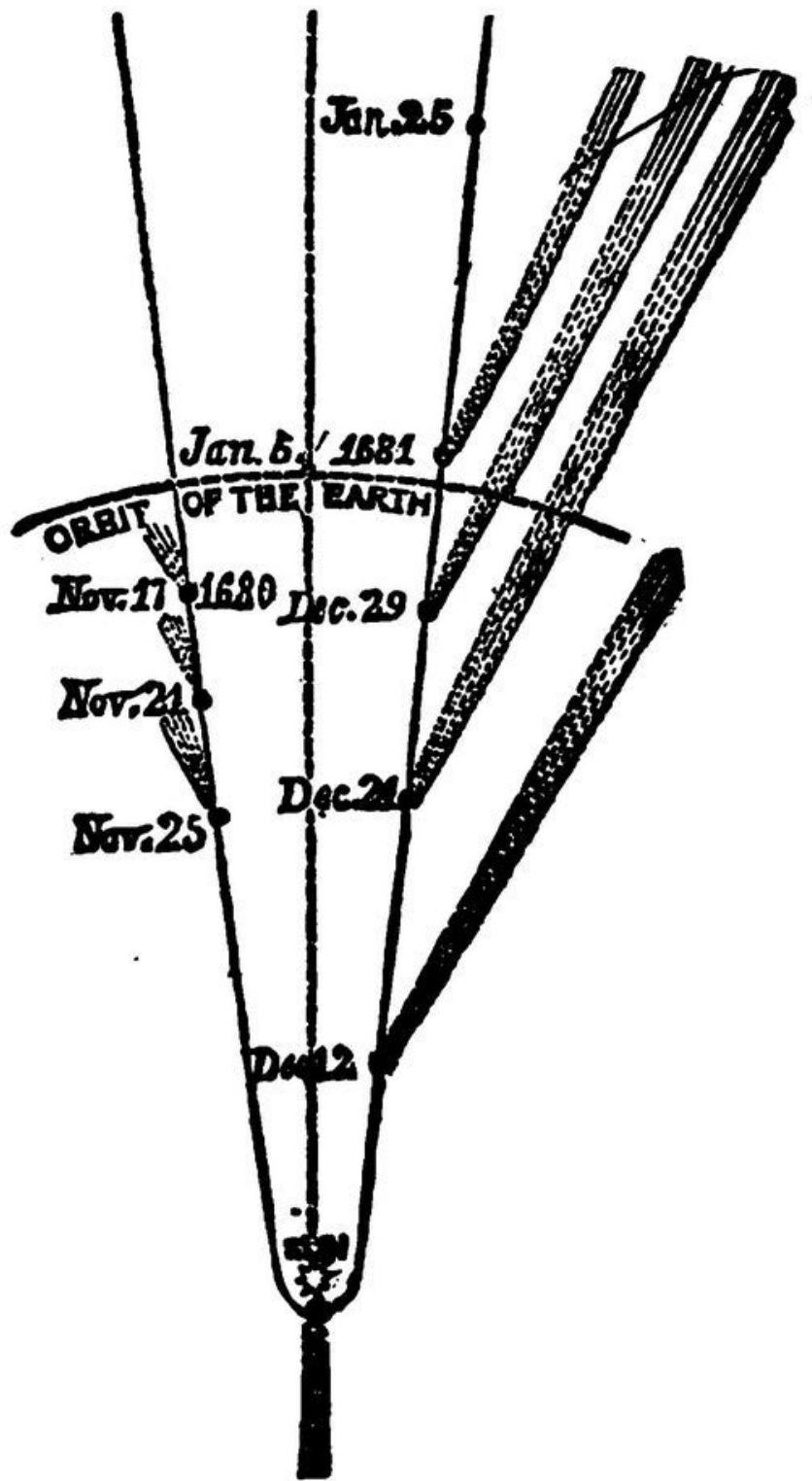


Fig. 25.—Orbit of Newton's Comet (1680)

Now calculations lead to the surprising fact, that the comets are by far the most voluminous bodies in our system. The following are the dimensions of some of those which have been made the subjects of such inquiry.

The tail of the great comet of 1680, immediately after its perihelion passage, was found by Newton to have been no less than 20,000,000 of leagues in length, and to have occupied only two days in its emission from the comet's body! a decisive proof this of its being darted forth by some active force, the origin of which, to judge from the direction of the tail, must be sought in the sun itself. Its greatest length amounted to 41,000,000 leagues, a length much exceeding the whole interval between the sun and earth. The tail of the comet of 1769 extended 16,000,000 leagues, and that of the great comet of 1811, 36,000,000. The portion of the head of this last, comprised within the transparent atmospheric envelope which separated it from the tail, was 180,000 leagues in diameter. It is hardly conceivable that matter once projected to such enormous distances should ever be collected again by the feeble attraction of such a body as a comet—a consideration which accounts for the surmised progressive diminution of the tails of such as have been frequently observed.

The most remarkable of those comets which have been ascertained to move in elliptic orbits is that of Halley, so called from the celebrated Edmund Halley, who, on calculating its elements from its perihelion passage in 1682, when it appeared in great splendor, with a tail 30° in length, was led to conclude its identity with the great comets of 1531 and 1607, whose elements he had also ascertained. The intervals of these successive apparitions being 75 and 76 years, Halley was encouraged to *predict* its reappearance about the year 1759. So remarkable a prediction could not fail to attract the attention of all astronomers, and, as the time approached, it became extremely interesting to know whether the attractions of the larger planets might not materially interfere with its orbital motion. The computation of their influence from the Newtonian law of gravity, a most difficult and intricate piece of calculation, was undertaken and accomplished by Clairaut, who found that the action of Saturn would retard its return by 100 days, and that of Jupiter by no less than 518, making in all 618 days, by which the expected return would happen later than on the supposition of its retaining an unaltered period—and that, in short, the time of the expected perihelion passage would take place within a month,

one way or other, of the middle of April, 1759. It actually happened on the 12th of March in that year. Its next return was calculated by several eminent geometers, and fixed successively for the 4th, the 7th, the 11th, and the 26th of November, 1835; the two latter determinations appearing entitled to the higher degree of confidence, owing partly to the more complete discussion bestowed on the observations of 1682 and 1759, and partly to the continually improving state of our knowledge of the methods of estimating the disturbing effect of the several planets. The last of these predictions, that of M. Lehmann, was published on the 25th of July. On the 5th of August the comet first became visible in the clear atmosphere of Rome as an exceedingly faint telescopic nebula, within a degree of its place as predicted by M. Rosenberger for that day. On or about the 20th of August it became generally visible, and, pursuing very nearly its calculated path among the stars, passed its perihelion on the 16th of November; after which, its course carrying it south, it ceased to be visible in Europe, though it continued to be conspicuously so in the Southern Hemisphere throughout February, March, and April, 1836, disappearing finally on the 5th of May.

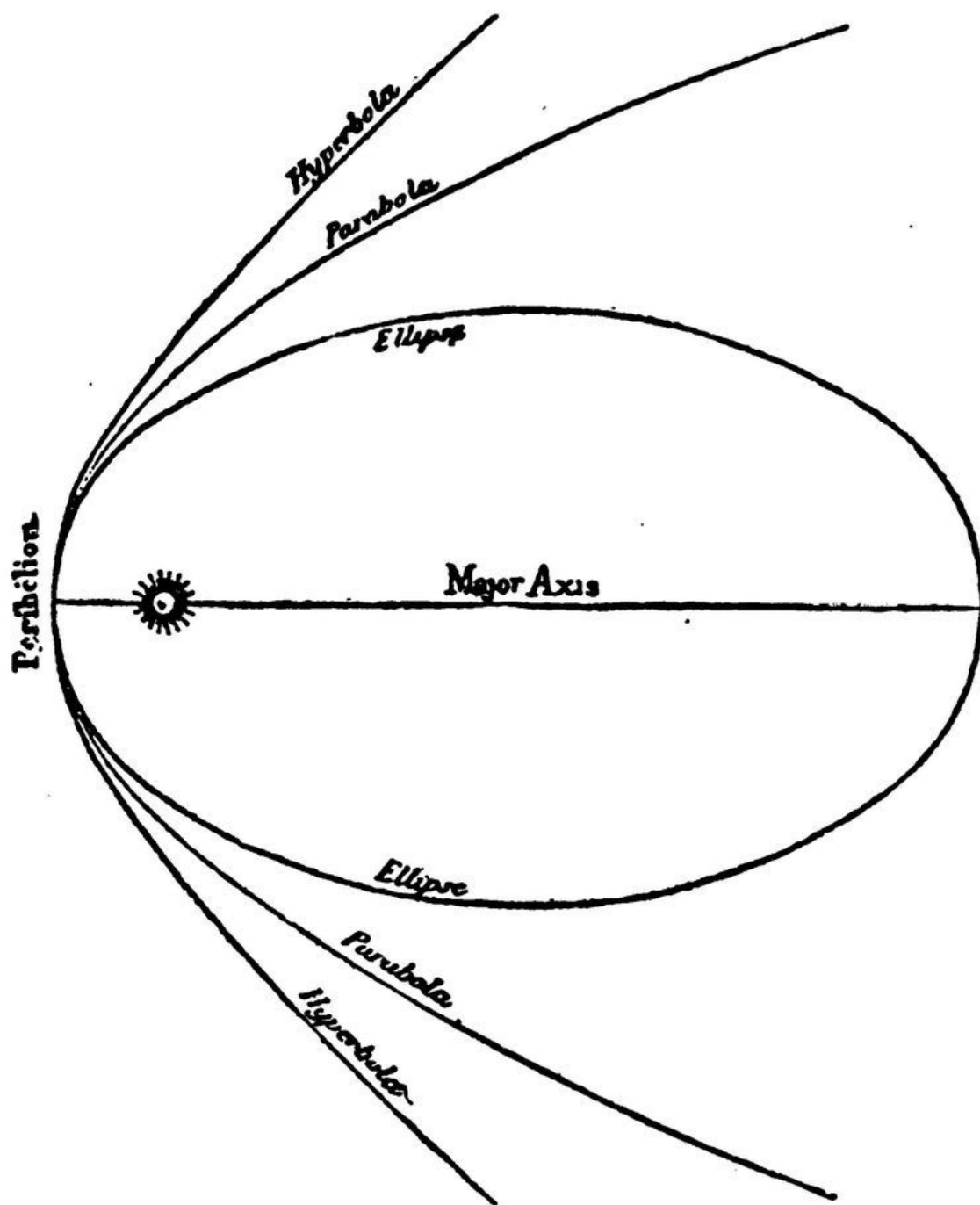


Fig. 26.—Forms of Cometary Orbits

Its first appearance, while yet very remote from the sun, was that of a small round or somewhat oval nebula, quite destitute of tail, and having a minute point of more concentrated light eccentrically situated within it. It was not before the 2d of October that the tail began to be developed, and

thenceforward increased pretty rapidly, being already 4° or 5° long on the 5th. It attained its greatest apparent length (about 20°) on the 15th of October. From that time, though not yet arrived at its perihelion, it decreased with such rapidity that already on the 29th it was only 3° , and on November the 5th $2\frac{1}{2}^{\circ}$ in length. There is every reason to believe that before the perihelion, the tail had altogether disappeared, as, though it continued to be observed at Pulkowa up to the very day of its perihelion passage, no mention whatever is made of any tail being then seen.

Reflecting on these phenomena, and carefully considering the evidence afforded by the numerous and elaborately executed drawings which have been placed on record by observers, it seems impossible to avoid the following conclusions: 1st. That the matter of the nucleus of a comet is powerfully excited and dilated into a vaporous state by the action of the sun's rays, escaping in streams and jets at those points of its surface which oppose the least resistance, and in all probability throwing that surface or the nucleus itself into irregular motions by its reaction in the act of so escaping, and thus altering its direction.

2. That this process chiefly takes place in that portion of the nucleus which is turned toward the sun; the vapor escaping chiefly in that direction.

3. That when so emitted, it is prevented from proceeding in the direction originally impressed upon it by some force directed *from* the sun, drifting it back and carrying it out to vast distances behind the nucleus, forming the tail or so much of the tail as can be considered as consisting of material substance.

4th. That this force, whatever its nature, acts unequally on the materials of the comet, the greater portion remaining unvaporized, and a considerable part of the vapor actually produced remaining in its neighborhood, forming the head and coma.

5th. That the force thus acting on the materials of the tail can not possibly be identical with the ordinary gravitation of matter, being centrifugal or repulsive, as respects the sun, and of an energy very far exceeding the gravitating force toward that luminary. This will be evident if we consider the enormous velocity with which the matter of the tail is carried backward, in opposition both to the motion which it had as part of the nucleus and to that which it acquired in the act of its emission, both which motions have to

be destroyed in the first instance, before any movement in the contrary direction can be impressed.

6th. That unless the matter of the tail thus repelled from the sun be retained by a peculiar and highly energetic attraction to the nucleus, differing from and exceptional to the ordinary power of gravitation, it must leave the nucleus altogether; being in effect carried far beyond the coercive power of so feeble a gravitating force as would correspond to the minute mass of the nucleus; and it is therefore very conceivable that a comet may lose, at every approach to the sun, a portion of that peculiar matter, whatever it be, on which the production of its tail depends, the remainder being of course less excitable by the solar action, and more impassive to his rays, and therefore, *pro tanto*, more nearly approximating to the nature of the planetary bodies.

7th. That, considering the immense distances to which at least some portion of the matter of the tail is carried from the comet, and the way in which it is dispersed through the system, it is quite inconceivable that the whole of that matter should be reabsorbed—that therefore it must lose during its perihelion passage some portion of its matter, and if, as would seem far from improbable, that matter should be of a nature to be repelled from, not attracted by, the sun, the remainder will, by consequence, be, *pro quantitate inertiae*, more energetically attracted to the sun than the mean of both. If then the orbit be elliptic, it will perform each successive revolution in a shorter time than the preceding, until, at length, the whole of the repulsive matter is got rid of.

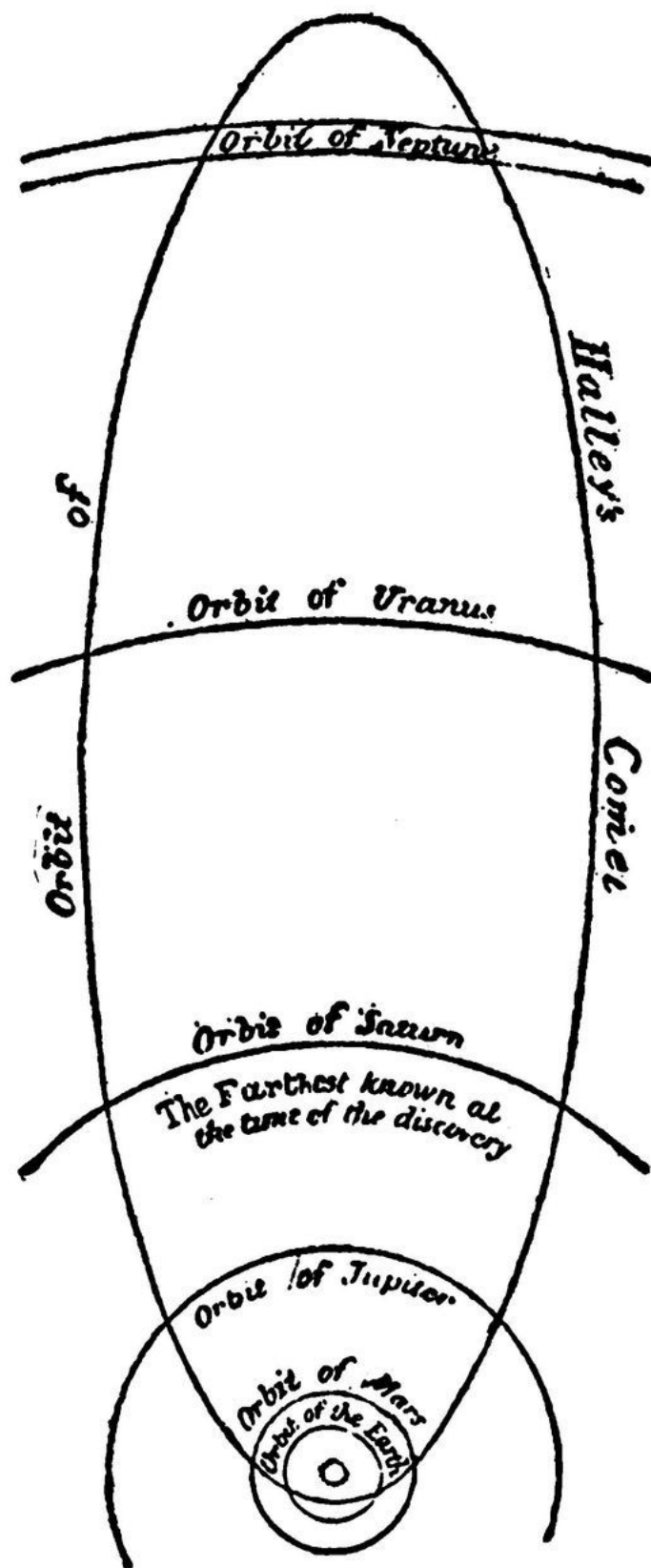


Fig. 27.—Halley's Comet

Besides the comet of Halley, several other of the great comets recorded in history have been surmised with more or less probability to return periodically, and therefore to move in elongated ellipses around the sun. Such is the great comet of 1680, whose period is estimated at 575 years, and which has been considered, with at least a high *prima facie* probability, to be identical with a magnificent comet observed at Constantinople and in Palestine, and referred by contemporary historians, both European and Chinese, to the year A. D. 1106; with that of A. D. 531, which was seen at noonday close to the sun; with the comet of 43 B. C., already spoken of as having appeared after the death of Cæsar, and which was also observed in the daytime; and finally with two other comets, mention of which occurs in the Sibylline Oracles, and in a passage of Homer, and which are referred, as well as the obscurity of chronology and the indications themselves will allow, to the years 618 and 1194 B. C. It is to the assumed near approach of this comet to the earth, about the time of the Deluge, that Whiston ascribed that overwhelming tide-wave to whose agency his wild fancy ascribed that great catastrophe—a speculation, it is needless to remark, purely visionary. These coincidences of time are certainly remarkable, especially when it is considered how very rare are the appearances of comets of this class. Professor Encke, however, has discussed, with all possible care, the observations recorded of the comet of 1680, taking into consideration the perturbations of the planets (which are of trifling importance, by reason of the great inclination of its orbit to the ecliptic), and his calculations show that no elliptic orbit, with such a period as 575 years, is competent to represent them within any probable or even possible limits of error, the most probable period assigned by them being 8814 Julian years. Independent of this consideration, there are circumstances recorded of the comet of A. D. 1106 incompatible with its motion in any orbit identical with that of the comet of 1680, so that the idea of referring all these phenomena to one and the same comet, however seducing, must be relinquished.

Another great comet, whose return about the year 1848 had been considered by more than one eminent authority in this department of astronomy highly probable, is that of 1556, to the terror of whose aspect some historians have attributed the abdication of the Emperor Charles V.

This comet is supposed to be identical with that of 1264, mentioned by many historians as a great comet, and observed also in China.

In 1661, 1532, 1402, 1145, 891, and 243 great comets appeared—that of 1402 being bright enough to be seen at noonday. A period of 129 years would conciliate all these appearances, and should have brought back the comet in 1789 or 1790 (other circumstances agreeing). That no such comet was observed about that time is no proof that it did not return, since, owing to the situation of its orbit, had the perihelion passage taken place in July it might have escaped observation.

We come now, however, to a class of comets of short period, respecting whose return there is no doubt, inasmuch as two at least of them have been identified as having performed successive revolutions round the sun; have had their return predicted already several times; and have on each occasion scrupulously kept to their appointments. The first of these is the comet of Encke, so called from Professor Encke of Berlin, who first ascertained its periodical return. It revolves in an ellipse of great eccentricity (though not comparable to that of Halley's), the plane of which is inclined at an angle of about $13^{\circ} 22'$ to the plane of the ecliptic, and in the short period of 1,211 days, or about $3\frac{1}{3}$ years. This remarkable discovery was made on the occasion of its fourth recorded appearance, in 1819. From the ellipse then calculated by Encke, its return in 1822 was predicted by him, and observed at Paramata, in New South Wales, by M. Rümker, being invisible in Europe: since which it has been repredicted and reobserved in all the principal observatories, both in the Northern and Southern Hemispheres, as a phenomenon of regular occurrence.

Another comet of short period is that of *Biela*, so called from M. Biela of Josephstadt, who first arrived at this interesting conclusion on the occasion of its appearance in 1826. It is considered to be identical with comets which appeared in 1772, 1805, etc., and describes its very eccentric ellipse about the sun in 2,410 days, or about $6\frac{3}{4}$ years; and in a plane inclined $12^{\circ} 34'$ to the ecliptic. It appeared again, according to the prediction, in 1832 and in 1846.

This comet is small and hardly visible to the naked eye, even when brightest. Nevertheless, as if to make up for its seeming insignificance by the interest attaching to it in a physical point of view, it exhibited, at its

appearance in 1846, a phenomenon which struck every astronomer with amazement, as a thing without previous example in the history of our system. It was actually seen to separate itself into two distinct comets, which, after thus parting company, continued to journey along amicably through an arc of upward of 70° of their apparent orbit, keeping all the while within the same field of view of the telescope pointed toward them. The first indication of something unusual being about to take place might be, perhaps, referred to the 19th of December, 1845, when the comet appeared to Mr. Hind pear-shaped, the nebulosity being unduly elongated in a direction inclining northward. But on the 13th of January, at Washington, in America, and on the 15th and subsequently in every part of Europe, it was distinctly seen to have become double; a very small and faint cometic body, having a nucleus of its own, being observed appended to it, at a distance of about $2'$ (in arc) from its centre, and in a direction forming an angle of about 328° with the meridian, running northward from the principal or original comet. From this time the separation of the two comets went on progressively, though slowly. On the 30th of January the apparent distance of the nucleus had increased to $3'$, on the 7th of February to $4'$, and on the 13th to $5'$, and so on, until on the 5th of March the two comets were separated by an interval of $9' 19''$, the apparent direction of the line of junction all the while varying but little with respect to the parallel.

During this separation, very remarkable changes were observed to be going on, both in the original comet and its companion. Both had nuclei, both had short tails, parallel in direction and nearly perpendicular to the line of junction; but whereas at its first observation, on January 13th, the new comet was extremely small and faint in comparison with the old, the difference both in point of light and apparent magnitude diminished. On the 10th of February they were nearly equal, although the day before the moonlight had effaced the new one, leaving the other bright enough to be well observed. On the 14th and 16th, however, the new comet had gained a decided superiority of light over the old, presenting at the same time a sharp and star-like nucleus, compared by Lieutenant Maury to a diamond spark. But this state of things was not to continue. Already, on the 18th, the old comet had regained its superiority, being nearly twice as bright as its companion, and offering an unusually bright and star-like nucleus. From this period the new companion began to fade away, but continued visible up

to the 15th of March. On the 24th the comet appeared again single, and on the 22d of April both had disappeared.

While this singular interchange of light was going forward, indications of some sort of communication between the comets were exhibited. The new or companion comet, besides its tail, extending in a direction parallel to that of the other, threw out a faint arc light which extended as a kind of bridge from the one to the other; and after the restoration of the original comet to its former pre-eminence, it, on its part, threw forth additional rays, so as to present the appearance of a comet with three faint tails forming angles of about 120° with each other, one of which extended toward its companion.

On the 22d of August, 1844, Signor de Vico, director of the observatory of the Collegio Romano, discovered a comet, the motions of which, a very few observations sufficed to show, deviated remarkably from a parabolic orbit. It passed its perihelion on the 2d of September, and continued to be observed until the 7th of December. Elliptic elements of this comet, agreeing remarkably well with each other, were accordingly calculated by several astronomers, from which it appears that the period of revolution is about 1,990 days, or $5\frac{1}{2}$ (5.4357) years, which (supposing its orbit undisturbed in the interim) would bring it back to the perihelion on or about the 13th of January, 1850, on which occasion, however, by reason of its unfavorable situation with respect to the sun and earth, it could not be observed.

This comet, when brightest, was visible to the naked eye, and had a small tail. It is especially interesting to astronomers from the circumstance of its having been rendered exceedingly probable by the researches of M. Leverrier, that it is identical with one which appeared in 1678, with some of its elements considerably changed by perturbation. This comet is further remarkable from having been concluded, by Messrs. Laugier and Mauvais, to be identical with the comet of 1585 observed by Tycho Brahe, and possibly also with those of 1743, 1766, and 1819.

By far the most remarkable comet, however, which has been seen during the present century, is that which appeared in the spring of 1843, and whose tail became visible in the twilight of the 17th of March in England as a great beam of nebulous light, extending from a point above the western horizon, through the stars of Eridanus and Lepus, under the belt of Orion. This

situation was low and unfavorable; and it was not till the 19th that the head was seen, and then only as a faint and ill-defined nebula, very rapidly fading on subsequent nights. In more southern latitudes, however, not only the tail was seen, as a magnificent train of light extending 50° or 60° in length; but the head and nucleus appeared with extraordinary splendor, exciting in every country where it was seen the greatest astonishment and admiration. Indeed, all descriptions agree in representing it as a stupendous spectacle, such as in superstitious ages would not fail to have carried terror into every bosom. In tropical latitudes in the Northern Hemisphere, the tail appeared on the 3d of March, and in Van Diemen's Land so early as the 1st, the comet having passed its perihelion on the 27th of February.

There is abundant evidence of the comet in question having been seen in full daylight, and in the sun's immediate vicinity. It was so seen on the 28th of February, the day after its perihelion passage, by every person on board the H.E.I.C.S. "Owen Glenndower," then off the Cape, as a short dagger-like object close to the sun a little before sunset. On the same day at 3^h 6^m p. m., and consequently in full sunshine, the distance of the nucleus from the sun was actually measured with a sextant by Mr. Clarke of Portland, United States, the distance centre from centre being then only $3^{\circ} 50' 43''$.

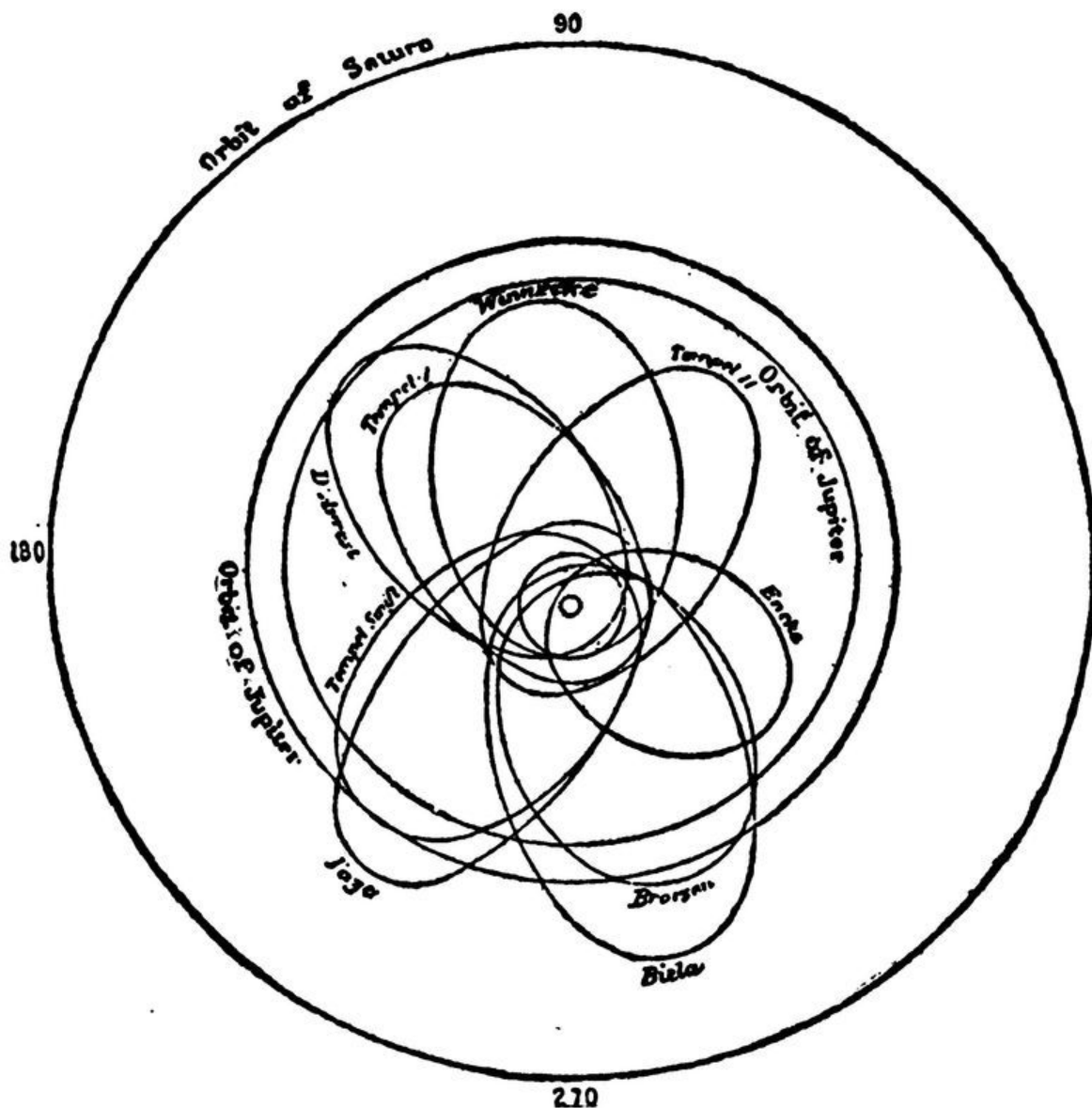


Fig. 28.—Orbits of the Nine Comets Captured by Jupiter
 Scale: 5 millimetres = 1 radius of the Earth's orbit

It is by no means merely as a subject of antiquarian interest, or on account of the brilliant spectacle which comets occasionally afford, that astronomers attach a high degree of importance to all that regards them. Apart even from the singularity and mystery which appertains to their physical constitution, they have become, through the medium of exact calculation, unexpected instruments of inquiry into points connected with the planetary system itself, of no small importance. We have seen that the movements of the comet Encke, thus minutely and perseveringly traced by the eminent astronomer whose name is used to distinguish it, have afforded

ground for believing in the presence of a resisting medium filling the whole of our system. Similar inquiries, prosecuted in the cases of other periodical comets, will extend, confirm, or modify our conclusions on this head. The perturbations, too, which comets experience in passing near any of the planets, may afford, and have afforded, information as to the magnitude of the disturbing masses, which could not well be otherwise obtained. Thus the approach of this comet to the planet Mercury in 1838 afforded an estimation of the mass of that planet the more precious, by reason of the great uncertainty under which all previous determinations of that element labored. Its approach to the same planet in the year 1848 was still nearer. On the 22d of November their mutual distance was only fifteen times the moon's distance from the earth.

It is, however, in a physical point of view that these bodies offer the greatest stimulus to our curiosity. There is, beyond question, some profound secret and mystery of nature concerned in the phenomenon of their tails. Perhaps it is not too much to hope that future observation, borrowing every aid from rational speculation, grounded on the progress of physical science generally (especially those branches of it which relate to the ethereal or imponderable elements), may ere long enable us to penetrate this mystery, and to declare whether it is really *matter* in the ordinary acceptation of the term which is projected from their heads with such extravagant velocity, and if not impelled, at least *directed*, in its course by a reference to the sun, as its point of avoidance. In no respect is the question as to the materiality of the tail more forcibly pressed on us for consideration than in that of the enormous sweep which it makes round the sun in perihelio, in the manner of a straight and rigid rod, in defiance of the law of gravitation, nay, even of the received laws of motion, extending (as we have seen in the comets of 1680 and 1843) from near the sun's surface to the earth's orbit, yet whirled round unbroken: in the latter case through an angle of 180° in little more than two hours. It seems utterly incredible that in such a case it is one and the same material object which is thus brandished. If there could be conceived such a thing as a *negative shadow*, a momentary impression made upon the luminiferous ether behind the comet, this would represent in some degree the conception such a phenomenon irresistibly calls up. But this is not all. Even such an extraordinary excitement of the ether, conceive it as we will, will afford no account of the projection of lateral streamers; of the effusion of light from the nucleus of a comet toward the sun; and its

subsequent *rejection*; of the irregular and capricious mode in which that effusion has been seen to take place; none of the clear indications of alternate evaporation and condensation going on in the immense regions of space occupied by the tail and coma—none, in short, of innumerable other facts which link themselves with almost equally irresistible cogency to our ordinary notions of matter and force.

LIFE IN OTHER WORLDS.—J. E. GORE

The question is often asked, Are the stars inhabited? To this we can confidently answer, No. The stars themselves are certainly not habitable by any forms of life with which we are familiar. That the stars are luminous incandescent bodies, similar to the sun, seems almost self-evident. That they shine by their own inherent light, and not by light reflected from another body, like the planets of the Solar System, is a fact which scarcely needs demonstration. There are no bright objects near them from which they could derive their light, and they are too far from the sun to obtain any illumination from that source. But if any proofs were necessary, we have the evidence of the spectroscope, which shows unmistakably that their light emanates from incandescent bodies. Many of the stars show spectra very similar to that of the sun. The light of others, although differing somewhat in quality when analyzed by the prism, indicates clearly that they are at a very high temperature—in many cases, indeed, suggesting that they are actually hotter than the sun. It may be objected, however, that in the case of binary or revolving double stars, the smaller component may possibly shine by light reflected from the brighter star. Indeed, this has been suggested in the case of Sirius and its faint companion. But, if the companion of Sirius shone merely by reflected light from its primary, it would be much fainter than it is, and, indeed, would be utterly invisible in our largest telescopes. Further, in some double stars, spectroscopic observations suggest that the component stars have different spectra. This is, of course, conclusive evidence against the hypothesis of borrowed light; for were the smaller star to shine by reflected light from the larger, the spectra of both would be identical, as in the case of the sun and moon. We may therefore conclude that all the visible stars are suns, and totally unfit for the habitation of living creatures.

But may not the stars have planets revolving round them, forming solar systems similar to our own? As they are evidently suns shining by inherent light, may they not form centres of planetary systems? In the case of those stars having spectra differing from the solar spectrum, we can not speak with any confidence; but for those which show spectra similar to that of our

sun, and having, therefore, probably a similar chemical constitution, the existence of planets revolving round them seems from analogy very probable. I refer to *single* stars, that is stars which have no telescopic close companion; for the double stars may, perhaps, form systems differently constituted. In any case these binary systems would not be strictly comparable with ours, for the sun is certainly a single star.

Whether systems of planets really revolve round the stars referred to, is a question which, unfortunately, can not be decided by observation. If a planet equal in size to the “giant planet,” Jupiter, were revolving round the nearest star—Alpha Centauri—at the same distance from that star that Jupiter is from the sun, it would be utterly invisible in our largest telescopes. The invisibility of planets circling round the stars is therefore no proof whatever of their non-existence. Each star of the solar type may possibly be attended by a retinue of planets which may, perhaps, remain forever invisible in the largest telescopes which man can construct. We can, therefore, draw our conclusions only from analogy. If other suns exist resembling our own sun in chemical constitution, which we know to be a fact, is it not reasonable to suppose that they also form centres of planetary systems similar to the Solar System?

“Consult with reason, reason will reply,
Each lucid point which glows in yonder sky,
Informs a system in the boundless space,
And fills with glory its appointed place;
With beams unborrowed brighten other skies,
And worlds to the unknown with heat and light supplies.”

The suns, which we call stars, were clearly not created for our benefit. They are of very little practical use to the earth’s inhabitants. They give us very little light; an additional small satellite—one considerably smaller than the moon—would have been much more useful in this respect than the millions of suns revealed by the telescope. They must, therefore, have been formed for some other purpose.

On Laplace’s Nebular Hypothesis, the condensation of an original nebulous mass endowed with a motion of rotation would result not only in the formation of a sun, similar to ours, but also in a system of planets revolving round the central body. If, indeed, the primitive nebula had no rotation or motions of any kind, the result would be a sun without planets or satellites; but the motions with which all the stars seem to be animated lead

us to suppose that this would be a case of very rare occurrence. We may therefore conclude, with a high degree of probability, that the stars—at least those with spectra of the solar type—form centres of planetary systems somewhat similar to our own.

This being surmised, let us consider the conditions necessary for the existence of life on these planets. There are various conditions which must be complied with before we can imagine life, as we know it, to be possible on any planet. Perhaps the most important of these is the question of temperature. We know that in the universe a great range of temperature exists, from the cold of interstellar space—estimated at about 460° below the freezing-point of water—to the intense heat which rages in the solar photosphere. In this long thermal scale life is, at least on the earth, restricted within rather narrow limits. Below a certain low temperature life can not exist. The point is, however, far above the temperature of space. On the other hand, above a certain high temperature—a low one, however, compared with the intense heat of the solar surface—life is also impossible, at least for highly organized beings like man and the larger animals. For minute microscopic organisms the scale may, perhaps, be somewhat extended; but even in its widest limits, the range of temperature within which life is possible is, so far as we know, certainly a narrow one.

For the support of life and vegetation, light is also necessary, for without it no flowers would bloom, nor corn grow and ripen to maturity. To obtain this supply of light and heat it is necessary that a life-bearing planet should revolve at a suitable distance from, and in a nearly circular orbit round, a central sun. These conditions, it is hardly necessary to say, are fulfilled in the case of the earth. Were we much nearer to the sun than we are, we should suffer from excessive heat, and were we much further away, we should probably perish from the cold. For this reason the existence of life on the other planets of the Solar System seems very doubtful. Mercury is probably too hot, and the other planets are certainly too cold, so far as heat from the sun is concerned, unless, indeed, their internal heat is sufficient to raise the temperature of their surface to a point sufficient for the maintenance of life. Indeed, there is good reason to suppose that in the planets Jupiter, Saturn, Uranus, and Neptune, this internal heat is still so great that life would be quite impossible on their surface. Venus, inside the earth's orbit, and Mars, outside, are the two planets which seem to approach

nearest to the required conditions. We know that both these planets possess atmospheres somewhat similar to ours, and, in Mars at least, land and water most probably exist on its surface. Venus is, of course, much hotter than the earth, and Mars much colder, but possibly the polar regions of Venus and the equatorial regions of Mars may form suitable abodes for some forms, at least, of animal and vegetable life.

Let us proceed, however, to consider some other conditions necessary for the existence of life on a planet. A suitable temperature is, of course, indispensable, but this is not all. There are other conditions which must be complied with. The planet must have a rotation on its axis, so that every portion shall in turn receive its due share of light and heat. Each point on its surface must have its day and night, the day for work and the night for rest. The axis of rotation must not lie in the plane of the planet's orbit, but must have a suitable inclination, so that each hemisphere may enjoy its seasons, summer and winter, "seed-time and harvest," in due course. Further the velocity of rotation on its axis must not be too rapid. If the earth rotated in a period of one and a quarter hours, bodies at the equator would have no weight, and life would be impossible in those regions.

The planet must also possess a mass sufficient to retain bodies on its surface by the force of gravity. In the case of very small bodies, such as the moons of Mars and some of the minor planets between Mars and Jupiter, objects thrown into the air would pass away into space never to return.

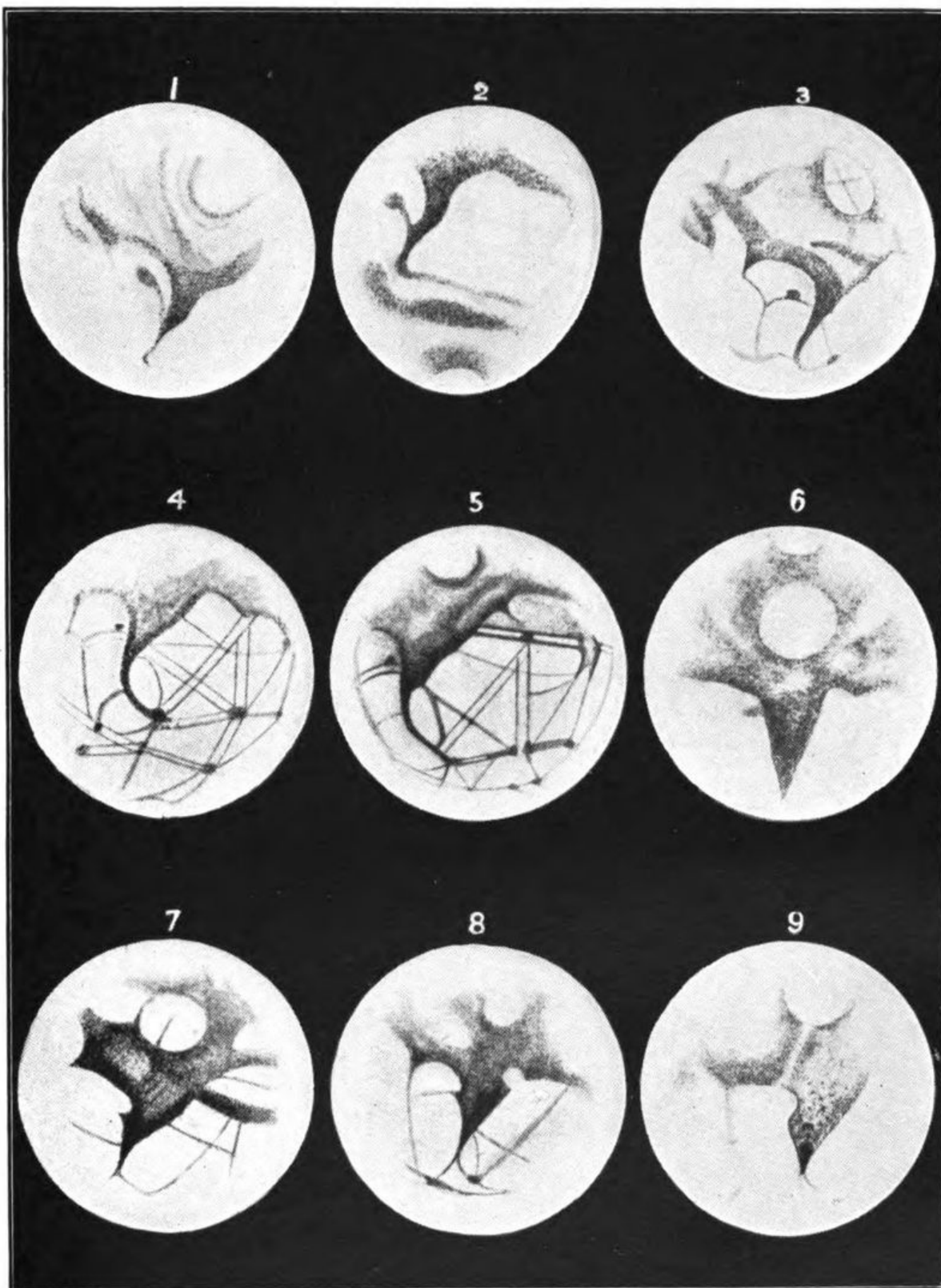
The planet should also have a mean density greater than that of water, otherwise the seas would possess no stability, and destructive waves would quickly destroy all life on its surface. All these conditions are fulfilled in the case of Mars as well as on the earth. In the planet Saturn, however, the density is less than that of water, and in Uranus and Neptune only slightly greater.

The planet must also possess a suitable atmosphere. This is an all-important condition for the support of animal life—at least for the existence of man and the higher orders of animals. This atmosphere must consist—so far as we know—of oxygen and nitrogen gases mechanically mixed in proper proportions, and with a small quantity of carbonic acid gas. Were the oxygen in smaller quantity than it exists in the earth's atmosphere, life could not be supported. On the other hand, were it much in excess of its

present amount, a fever would be produced in the blood which would very soon put an end to animal life. The presence of other gases in excessive quantities would also render the air unfit for breathing. We see, therefore, that a comparatively slight change in the composition of a planet's atmosphere would—so far as our experience goes—render the planet uninhabitable by any of the higher forms of life with which we are familiar.

For the support of life on a planet, water is also absolutely necessary. Without this useful fluid the world would soon become a desert, and life and vegetation would speedily vanish from its surface.

Geological conditions must also be considered. It is clearly necessary for the welfare of human beings at least that the surface soil and rocks should contain coal, iron, lime, and other minerals, substances almost indispensable for the ordinary wants of civilized existence.



Nine Views of the Hour-Glass Sea on Mars

1, Nov. 26, 1864; 2, June 29, 1873; 3, Oct. 28, 1879; 4, June 2, 1888; 5, June 20, 1890; 6, Aug. 6,

That all or any of the conditions considered would be complied with in the case of a planet revolving round a star it is, of course, impossible to say. But when we find stars showing by their spectra that they contain chemical elements identical with those which exist in the sun and the earth, analogy would lead us to suppose that very possibly a planet resembling our earth may revolve round each of these distant suns. I say *a* planet, for evidently there would be only *one* distance from the central luminary—a distance depending on its size—at which the temperature necessary for the support of life would exist, as in the case of the earth, over the whole of the planet's surface. For other planets of the stellar system, life would be, if it existed at all, most probably confined to restricted regions of the planet's surface. There would, therefore, be in each system one planet, and only one, *especially* suitable for the support of animal life as we know it. This is with reference to light and heat. If the other conditions were not complied with, then life would probably not exist even on this one planet. In the case of a star larger than the sun, the planet should be placed at a greater distance than the earth is from the sun, but in this case the length of the year and the seasons would be longer than ours.

The star which more nearly resembles the sun in the character of the light which it emits is the bright star Capella. Arcturus has a somewhat similar spectrum. But these are probably suns of enormous size, if any reliance can be placed on the measures of their distance from the earth. Other bright stars with spectra of the solar type are Pollux, Aldebaran, Beta Andromedæ, Alpha Arietis, Alpha Cassiopeiæ, Alpha Cygni, and Alpha Ursæ Majoris. Another star is Eta Herculis. The magnitude of this star as measured with the photometer is about $3\frac{1}{2}$. A parallax found by Bélyopolsky and Wagner places it at a distance of 515,660 times the sun's distance from the earth. If the sun were placed at this distance, I find that it would be reduced to a star of the third magnitude. This result would imply that Eta Herculis is a slightly smaller sun than ours; and a planet placed a little nearer to the star than the earth is to the sun might, perhaps, fulfil the conditions of a life-bearing world.

The number of stars visible in our largest telescopes is usually estimated at 100,000,000. Of these we may perhaps assume that 10,000,000 have a spectrum of the solar type, and therefore closely resemble our sun in their

chemical constitution. If we suppose that only one in ten of these is similar in size to the sun, and has a habitable planet revolving round it, we have a total of 1,000,000 worlds in the visible universe fitted for the support of animal life.

We may therefore conclude, with a high degree of probability, that among the “multitudinous” stellar hosts there are probably many stars having life-bearing planets revolving round them.

THE SUN—WHAT WE LEARN FROM IT.—

RICHARD A. PROCTOR

The Sun, the central and ruling body of the planetary system, and the source of light and heat to our earth and all the members of that system, is a globe about 852,900 miles in diameter. So far as observation extends, his figure is perfectly spherical, no difference having been observed between his polar and spherical diameters. It has been well remarked, indeed, by Sir G. Airy, that if any observer could by ordinary modes of measurement satisfy himself that a real difference existed between the diameters, that observer would have proved the inexactness of his own work; for the absence of any measurable compression comes out as the result of comparisons between thousands of observations of the sun's limbs made at Greenwich and other leading observatories. The volume of the sun exceeds the earth's 1,252,700 times. His mean density is almost exactly one-fourth of the earth's, and his mass exceeds hers about 316,000 times. Gravity at the surface of the sun exceeds terrestrial gravity about 27.1 times, so that a body dropped from rest near the sun's surface would fall through 436 feet in the first second, and have acquired a velocity of 872 feet per second.

Let the reader consider a terrestrial globe three inches in diameter, and search out on that globe the tiny triangular speck which represents Great Britain. Then let him endeavor to picture the town in which he lives as represented by the minutest pin-mark that could possibly be made upon this speck. He will then have formed some conception, though but an inadequate one, of the enormous dimensions of the earth's globe, compared with the scene in which his daily life is cast. Now, on the same scale, the sun would be represented by a globe about twice the height of an ordinary sitting-room. A room about twenty-six feet in length, and height, and breadth, would be required to contain the representation of the sun's globe on this scale, while the globe representing the earth could be placed in a moderately large goblet.

Such is the body which sways the motions of the Solar System. The largest of his family, the giant Jupiter, though of dimensions which dwarf

those of the earth or Venus almost to nothingness, would yet only be represented by a thirty-two inch globe, on the scale which gives to the sun the enormous volume I have spoken of. Saturn would have a diameter of about twenty-eight inches, his ring measuring about five feet in its extreme span. Uranus and Neptune would be little more than a foot in diameter, and all the minor planets would be less than the three-inch earth. It will thus be seen that the sun is a worthy centre of the great scheme he sways, even when we merely regard his dimensions.

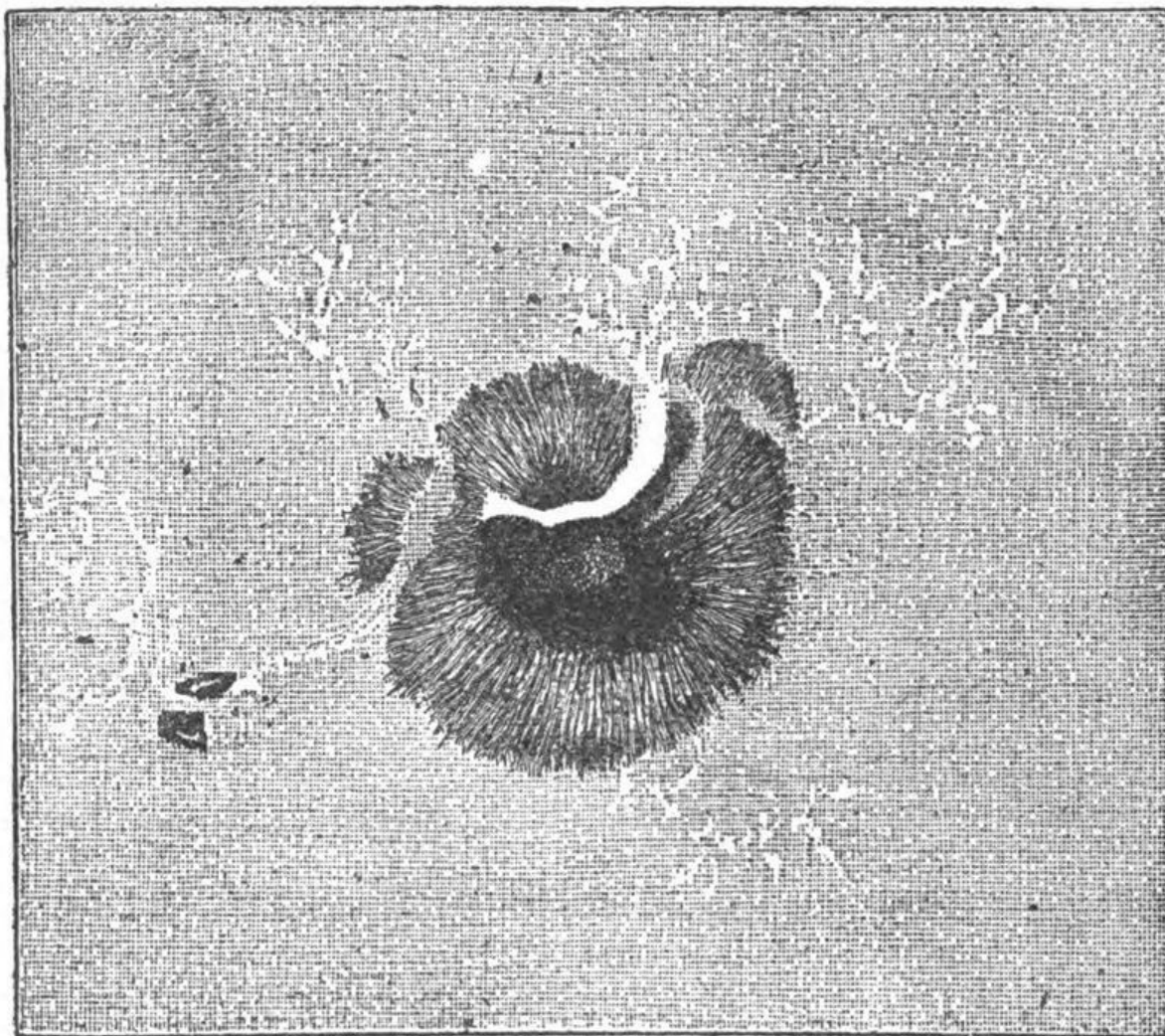


Fig. 29.—Sun Spot seen in 1870

The sun outweighs fully seven hundred and forty times the combined mass of all the planets which circle around him, so that, when we regard the energy of his attraction, we still find him a worthy ruler of the planetary scheme.

Viewed with the naked eye, the sun appears only as a luminous mass of intense and uniform brightness; but when examined with the telescope, his surface is frequently observed to be mottled over with a number of dark spots, of irregular and ill-defined forms, constantly varying in appearance, situation, and magnitude. These spots are occasionally of immense size, so as to be visible even without the aid of the telescope; and their number is frequently so great that they occupy a considerable portion of the sun's surface. Sir W. Herschel observed one in 1779 the diameter of which exceeded 50,000 miles, more than six times the diameter of the earth; and Scheiner affirms that he has seen no less than fifty on the sun's disk at once. Most of them have a deep black nucleus, surrounded by a fainter shade, or *umbra*, of which the inner part, nearest to the nucleus, is brighter than the exterior portion. The boundary between the nucleus and umbra is in general tolerably well defined; and beyond the umbra a stripe of light appears more vivid than the rest of the sun.

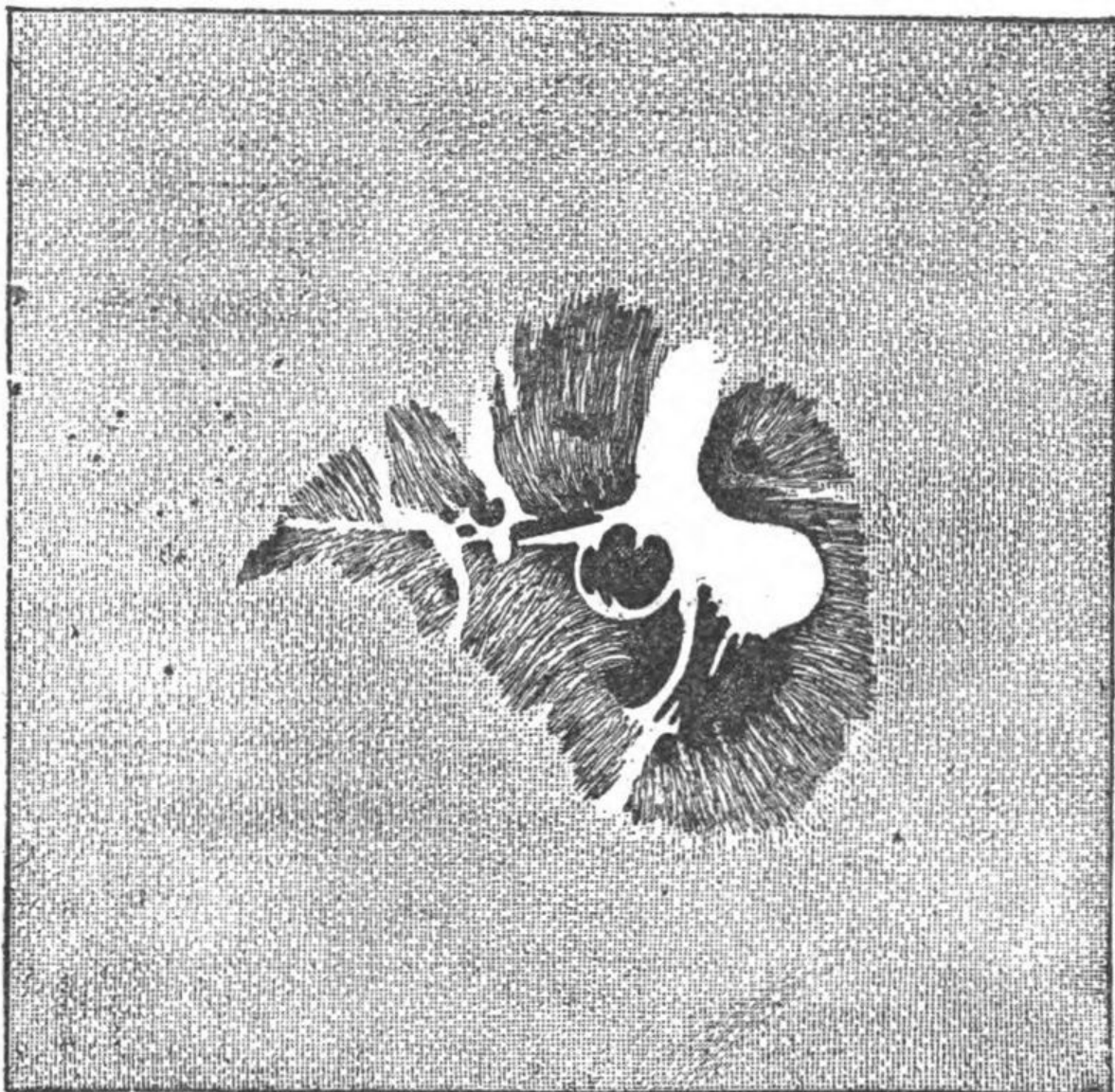


Fig. 30.—Phase of Spot

The discovery of the sun's spots has been attributed to Fabricius, Galileo, and Scheiner, and has been claimed for the English astronomer Harriot. Amid these conflicting pretensions it is perhaps impossible to arrive at the truth; but the matter is of little importance; the discovery is one which followed inevitably that of the telescope, and an accidental priority of observation can hardly be considered as establishing any claim to merit.

The study of solar physics may be said to have commenced with the discovery of the sun spots, about two hundred and sixty years ago. These spots were presently found to traverse the solar disk in such a way as to indicate that the sun turns upon an axis once in about twenty-six days. Nor

will this rotation appear slow, when we remember that it implies a motion of the equatorial parts of the sun's surface at a rate exceeding some seventy times the motion of our swiftest express train.

Next came the discovery that the solar spots are not surface stains, but deep cavities in the solar substance. The changes of appearance presented by the spots as they traverse the solar disk led Dr. Wilson to form this theory so far back as 1779; but, strangely enough, it is only in comparatively recent times that the hypothesis has been finally established, since even within the last ten years a theory was put forward which accounted satisfactorily for most of the changes of appearance observed in the spots, by supposing them to be due to solar clouds hanging suspended at a considerable elevation above the true photosphere.

Sir William Herschel, reasoning from terrestrial analogies, was led to look on the spot-cavities as apertures through a double layer of clouds. He argued that, were the solar photosphere of any other nature, it would be past comprehension that vast openings should form in it, to remain open for months before they close up again. Whether we consider the enormous rapidity with which the spots form and with which their figure changes, or the length of time that many of them remain visible, we find ourselves alike perplexed, unless we assume that the solar photosphere resembles a bed of clouds. Through a stratum of terrestrial clouds openings may be formed by atmospheric disturbances, but while undisturbed the clouds will retain any form once impressed upon them, for a length of time corresponding to the weeks and months during which the solar spots endure.

And because the solar spots present two distinct varieties of light, the faint penumbra and the dark umbra or nucleus, Herschel saw the necessity of assuming that there are two beds of clouds, the outer self-luminous and constituting the true solar photosphere, the inner reflecting the light received from the outer layer, and so shielding the real surface of the sun from the intense light and heat which it would otherwise receive.

But while recent discoveries have confirmed Sir William Herschel's theory about the solar cloud-envelopes, they have by no means given countenance to his view that the body of the sun may possibly be cool. The darkness of the nucleus of a spot is found, on the contrary, to give proof that in that neighborhood the sun is hotter, because it parts less readily with its

heat. We shall see presently how this is. Meantime let it be noticed, in passing, that a close scrutiny of large solar spots has revealed the existence of an intensely black spot in the midst of the umbra. This black spot must be regarded as the true nucleus.

The circumstance that the spots appear only on two bands of the sun's globe, corresponding to the sub-tropical zones on our own earth, led the younger Herschel to conclusions as important as those which his father had formed. He reasoned, like his father, from terrestrial analogies. On our own earth the sub-tropical zones are the regions where the great cyclonic storms have their birth, and rage with their chief fury. Here, therefore, we have the analogue of the solar spots, if only we can show reason for believing that any causes resembling those which generate the terrestrial cyclone operate upon those regions of the sun where the solar spots make their appearance.

We know that the cyclone is due to the excess of heat at the earth's equator. It is true that this excess of heat is always in operation, whereas cyclones are not perpetually raging in sub-tropical climates. Ordinarily, therefore, the excess of heat does not cause tornadoes. Certain aerial currents are generated whose uniform motion suffices, as a rule, to adjust the conditions which the excess of heat at the equator would otherwise tend to disturb. But when through any cause the uniform action of the aerial currents is either interfered with or is insufficient to maintain equilibrium, then cyclonic or whirling motions are generated in the disturbed atmosphere, and propagated over a wide area of the earth's surface.

Now we recognize the reason of the excess of heat at the earth's equator in the fact that the sun shines more directly upon that part of the earth than on the zones which lie in higher latitudes. Can we find any reason for suspecting that the sun, which is not heated from without as the earth is, should exhibit a similar peculiarity? Sir John Herschel considers that we can. If the sun has an atmosphere extending to a considerable distance from his surface, then there can be little doubt that, owing to his rotation upon his axis, this atmosphere would assume the figure of an oblate spheroid, and would be deepest over the solar equator. Here, then, more of the sun's heat would be retained than at the poles, where the atmosphere is shallowest. Thus, that excess of heat at the solar equator which is necessary to complete the analogy between the sun spots and terrestrial cyclones seems satisfactorily established.

It must be remarked, however, that this reasoning, so far as the excess of heat at the sun's equator is concerned, only removes the difficulty a step. If there were indeed an increased depth of atmosphere over the sun's equator sufficing to retain the requisite excess of heat, then the amount of heat we receive from the sun's equatorial regions ought to be appreciably less than the amount emitted from the remaining portions of the solar surface. This is not found to be the case, so that either there is no such excess of absorption, or else the solar equator gives out more heat, in other words, is essentially hotter, than the rest of the sun. But this is just the peculiarity of which we want the interpretation.

It may be taken for granted, however, that there is an analogy between the sun spots and terrestrial cyclonic storms, though as yet we are not very well able to understand its nature.

Then next we come to one of the most interesting discoveries ever made respecting the sun—the discovery that the spots increase and diminish in frequency in a periodic manner. We owe this discovery to the laborious and systematic observations made by Herr Schwabe of Dessau.

Schwabe found, in the course of about ten and a half years, the solar spots pass through a complete cycle of changes. They become gradually more and more numerous up to a certain maximum, and then as gradually diminish. At length the sun's face becomes not only clear of spots, but a certain well-marked darkening around the border of his disk disappears altogether for a brief season. At this time the sun presents a perfectly uniform disk. Then gradually the spots return, become more and more numerous, and so the cycle of changes is run through again.

The astronomers who have watched the sun from the Kew Observatory have found that the process of change by which the spots sweep in a sort of "wave of increase" over the solar disk is marked by several minor variations. As the surface of a great sea wave will be traversed by small ripples, so the gradual increase and diminution in the number of the solar spots are characterized by minor gradations of change, which are sufficiently well marked to be distinctly cognizable.

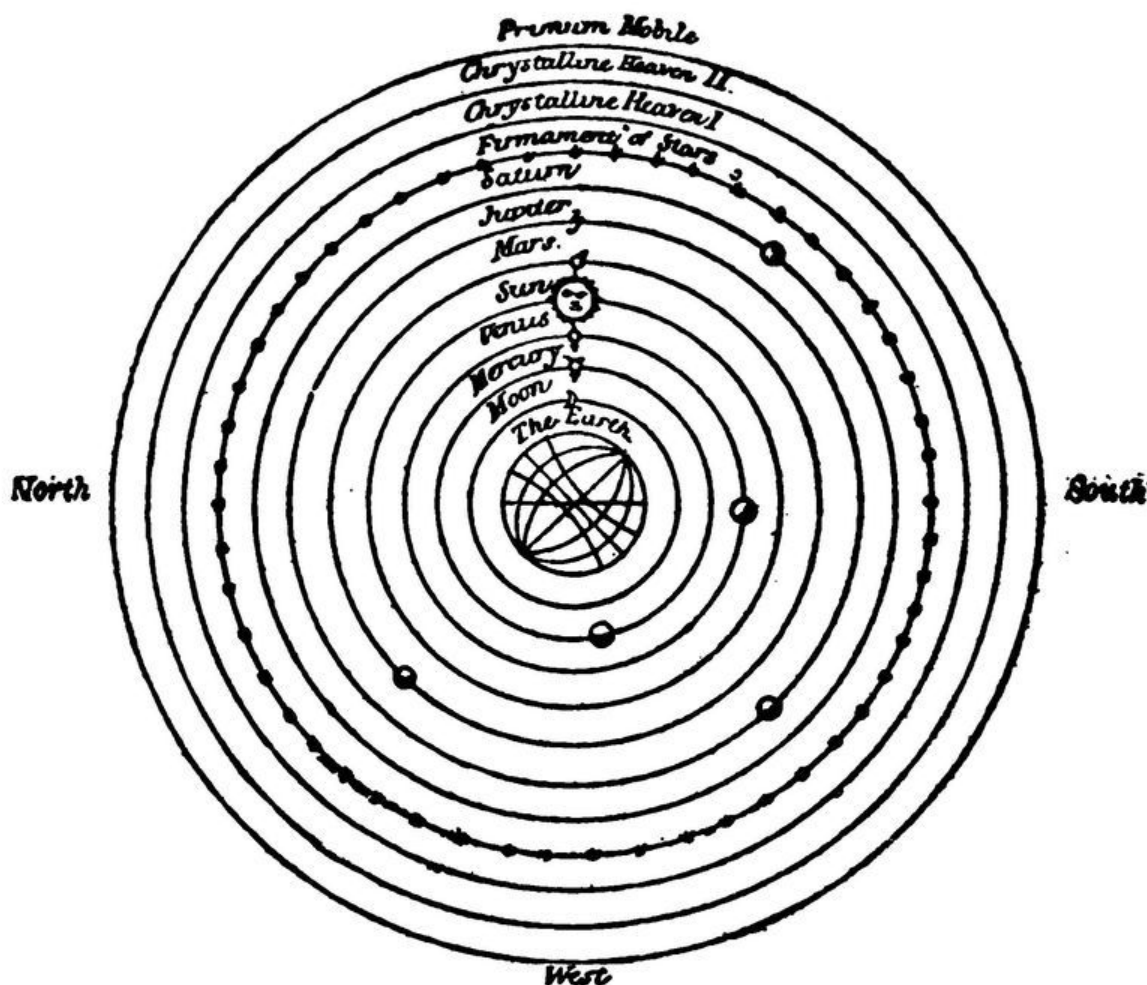


Fig. 31.—Ptolemaic System

There seems every reason for believing that the periodic changes thus noticed are due to the influence of the planets upon the solar photosphere, though in what way that influence is exerted is not at present perfectly clear. Some have thought that the mere attraction of the planets tends to produce tides of some sort in the solar envelopes. Then, since the height of a tide so produced varies as the cube or third power of the distance, it has been thought that a planet when in perihelion would generate a much larger solar tide than when in aphelion. So that, as Jupiter has a period nearly equal to the sun-spot period, it has been supposed that the attractions of this planet are sufficient to account for the great spot period. Venus, Mercury, the Earth, and Saturn have, in a similar manner, been rendered accountable for the shorter and less distinctly marked periods.

Without denying that the planets may be, and probably are, the bodies to whose influence the solar-spot periods are to be ascribed, I yet venture to express very strong doubts whether the attraction of Jupiter is so much greater in perihelion than in aphelion as to account for the fact that, whereas at one season the face of the sun shows many spots, at another it is wholly free from them.^[23]

However, we are not at present concerned so much with the explanation of facts as with the facts themselves. We have to consider rather what the sun is and what he does for the Solar System than why these things are so.

Let us note, before passing to other circumstances of interest connected with the sun, that the variable condition of his photosphere must cause him to change in brilliancy as seen from vast distances. If Herr Schwabe, for instance, instead of observing the sun's spots from his watch-tower at Dessau, could have removed himself to a distance so enormous that the sun's disk would have been reduced, even in the most powerful telescope, to a mere point of light, there can be no doubt that the only effect which he would have been able to perceive would have been a gradual increase and diminution of brightness, having a period of about ten and a half years.

Our sun, therefore, viewed from the neighborhood of any of the stars, whence undoubtedly he would simply appear as one among many fixed stars, would be a "variable," having a period of ten and a half years. And further, if an observer, viewing the sun from so enormous a distance, had the means of very accurately measuring its light, he would undoubtedly discover that, while the chief variation of the sun takes place in a period of ten and a half years, its light is subjected to minor variations having shorter periods.

The discovery that the periodic changes of the sun's appearance are associated with the periodic changes in the character of the earth's magnetism is the next that we have to consider.

It had long been noticed that, during the course of a single day, the magnetic needle exhibits a minute change of direction, taking place in an oscillatory manner. And, when the character of this vibration came to be carefully examined, it was found to correspond to a sort of effort on the needle's part to turn toward the sun. For example, when the sun is on the magnetic meridian, the needle has its mean position. This happens twice in

a day, once when the sun is above the horizon and once when he is below it. Again, when the sun is midway between these two positions—which also happens twice in the day—the needle has its mean position, because the northern and the southern ends make equal efforts (so to speak) to direct themselves toward the sun. Four times in the day, then, the needle has its mean position, or is directed toward the magnetic meridian. But, when the sun is not in one of the four positions considered, that end of the needle which is nearest to him is slightly turned away from its mean position toward him. The change of position is very minute, and only the exact modes of observation made use of in the present age would have sufficed to reveal it. There it is, however, and this minute and seemingly unimportant peculiarity has been found to be full of meaning.

The minute vibrations of the magnetic needle, thus carefully watched—day after day, month after month, year after year—were found to exhibit a yet more minute oscillatory change. They waxed and waned within narrow limits of variation, but yet in a manner there was no mistaking. The period of this oscillatory change was not to be determined, however, by the observations of a few years. Between the time when the diurnal vibration was least until it had reached its greatest extent, and thence returned to its first value, no less than ten and a half years elapsed, and a much longer time passed before the periodic character of the change was satisfactorily determined.

The reader will at once see what these observations tend to. The sun spots vary in frequency within a period of ten and a half years, and the magnetic diurnal vibrations vary within a period of the same duration. It might seem fanciful to associate the two periodic series of changes together, and doubtless when the idea first occurred to Lamont, it was not with any great expectation of finding it confirmed that he examined the evidence bearing on the point. Judging from known facts, we may see reasons for such an expectation in the correspondence of the needle's diurnal vibration with the sun's apparent motion, and the law which has been found to associate the annual variations of the magnet's power with the sun's distance. But undoubtedly when the idea occurred to Lamont it was an exceedingly bold one, and the ridicule with which the first announcement of the supposed law was received, even in scientific circles, suffices to show how unexpected that relation was which is now so thoroughly established. For a careful

comparison between the two periods has demonstrated that they agree most perfectly, not merely in length, but maximum for maximum, and minimum for minimum. When the sun spots are most numerous, then the daily vibration of the magnet is most extensive, while, when the sun's face is clear of spots, the needle vibrates over its smallest diurnal arc.

Then the intensity of the magnetic action has been found to depend upon solar influences. The vibrations by which the needle indicates the progress of those strange disturbances of the terrestrial magnetism which are known as magnetic storms have been found not merely to be most frequent when the sun's face is most spotted, but to occur simultaneously with the appearance of signs of disturbance in the solar photosphere. For instance, during the autumn of 1859, the eminent solar observer, Carrington, noticed the apparition of a bright spot upon the sun's surface. The light of this spot was so intense that he imagined the dark glass which protected his eye had been broken. By a fortunate coincidence, another observer, Mr. Hodgson, happened to be watching the sun at the same instant, and witnessed the same remarkable appearance. Now it was found that the self-registering magnetic instruments of the Kew Observatory had been sharply disturbed at the instant when the bright spot was seen. And afterward it was learned that the phenomena which indicate the progress of a magnetic storm had been observed in many places. Telegraphic communication was interrupted, and in some cases, telegraphic offices were set on fire; auroras appeared both in the Northern and Southern Hemisphere during the night which followed; and the whole frame of the earth seemed to thrill responsively to the disturbance which had affected the great central luminary of the Solar System.

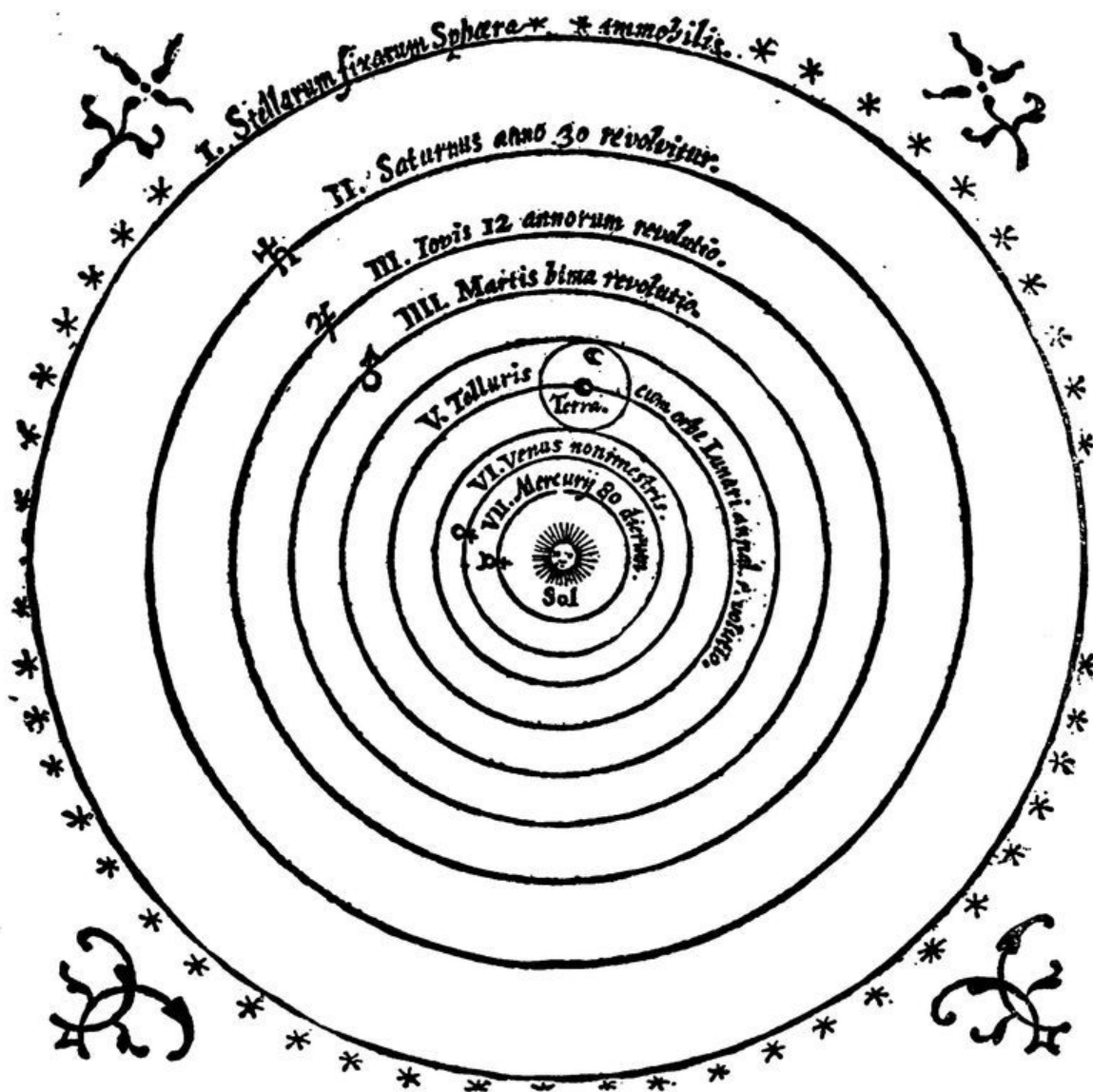


Fig. 32.—Copernican System: Facsimile of the Drawing in the Volume by Copernicus Published in 1543

The reader will now see why I have discussed relations which hitherto he may perhaps have thought very little connected with my subject. He sees that there is a bond of sympathy between our earth and the sun; that no disturbance can affect the solar photosphere without affecting our earth to a greater or less degree. But if our earth, then also the other planets. Mercury and Venus, so much nearer the sun than we are, surely respond even more swiftly and more distinctly to the solar magnetic influences. But beyond our earth, and beyond the orbit of moonless Mars, the magnetic impulses speed with the velocity of light. The vast globe of Jupiter is thrilled from pole to

pole as the magnetic wave rolls in upon it; then Saturn feels the shock, and then the vast distances beyond which lie Uranus and Neptune are swept by the ever-lessening yet ever-widening disturbance wave. Who shall say what outer planets it then seeks? or who, looking back upon the course over which it has traveled, shall say that planets alone have felt its effects? Meteoric and cometic systems have been visited by the great magnetic wave, and upon the dispersed members of the one and the subtle structure of the other effects even more important may have been produced than those striking phenomena which characterize the progress of the terrestrial or planetary magnetic storms.

When we remember that what is true of a relatively great solar disturbance, such as the one witnessed by Messrs. Carrington and Hodgson, is true also (however different in degree) of the magnetic influences which the sun is at every instant exerting, we see that a new and most important bond of union exists between the members of the solar family. The sun not only sways them by the vast attraction of his gravity, not only illumines them, not only warms them, but he pours forth on all his subtle yet powerful magnetic influences. A new analogy between the members of the Solar System is thus introduced to reinforce those other analogies which have been held so strikingly to indicate that the ends for which our earth has been created are not different from those which the Creator had in view when He planned the other members of the Solar System.

The real end and aim of the telescope, as applied by the astronomer to the examination of the celestial objects, is to gather together the light which streams from each luminous point throughout space. We may regard the space which surrounds us on every side as an ocean without bounds or limits, an ocean across which there are ever sweeping waves of light, either emitted directly from the various bodies subsisting throughout space, or else reflected from their surfaces. Other forms of waves also speed across those limitless depths in all directions, but the light-waves are those which at present concern us. Our earth is as a minute island placed within the ocean of space, and to the shores of this tiny isle the light-waves bear their message from the orbs which lie like other isles amid the fathomless depths around us. With the telescope the astronomer gathers together portions of light-waves which else would have traveled in diverging directions. By thus intensifying their action, he enables the eye to become cognizant of their

true nature. Precisely as the narrow channels around our shores cause the tidal wave, which sweeps across the open ocean in almost insensible undulations, to rise and fall through a wide range of variation, so the telescope renders sensible the existence of light-waves which would escape the notice of the unaided eye.

The telescope, then, is essentially a *light-gatherer*.

The spectroscope is used for another purpose. It might be called the *light-sifter*. It is applied by the astronomer to analyze the light which comes to him from beyond the ocean of space, and so to enable him to learn the character of the orbs from which that light proceeds.

The principle of the instrument is simple, though the appliances by which its full powers can alone be deduced are somewhat complicated.

A ray of sunlight falling on a prism of glass or crystal does not emerge unchanged in character. Different portions of the ray are differently bent, so that when they emerge from the prism they no longer travel side by side as before. The violet part of the light is bent most, the red least; the various colors from violet through blue, green, and yellow, to red being bent gradually less and less.

The prism then *sorts*, or *sifts*, the light-waves.

But we want the means of sifting the light-waves more thoroughly. The reader must bear with me while I describe, as exactly as possible in the brief space available to me, the way in which the first rough work of the prism has been modified into the delicate and significant work of the spectroscope. It is well worth while to form clear views on this point, because so many of the wonders of modern science are associated with spectroscopic analysis.

If, through a small round hole in a shutter, light is admitted into a darkened room, and a prism be placed with its refracting angle downward and horizontal, a vertical spectrum, having its violet end uppermost, will be formed on a screen suitably placed to receive it.

But now let us consider what this spectrum really is. If we take the light-waves corresponding to any particular color, we know, from optical considerations, that these waves emerge from the prism in a pencil exactly resembling in shape the pencil of white light which falls on the prism. They

therefore form a small circular or oval image on their own proper part of the spectrum. Hence the spectrum is in reality formed of a multitude of overlapping images, varying in color from violet to red. It thus appears as a rainbow-tinted streak, presenting every gradation of color between the utmost limits of visibility at the violet and red extremities.

If we had a square aperture to admit the light, we should get a similar result. If the aperture were oblong, there would still be overlapping images; but if the length of the oblong were horizontal, then, since each image would also be a horizontally placed oblong, the overlapping would be less than when the images were square. Suppose we diminish the overlapping as much as possible? in other words, suppose we make the oblong slit as narrow as possible? Then, unless there were in reality an infinite number of images distributed all along the spectrum from top to bottom, the images might be so narrowed as not to overlap; in which case, of course, there would be horizontal dark spaces or gaps in our spectrum. Or, again, if we failed in finding gaps of this sort by simply narrowing the aperture, we might lengthen the spectrum by increasing the refracting angle of the prism, or by using several prisms, and so on.

The first great discovery in solar physics, by means of the analysis of the prism (though the discovery had little meaning at the time), consisted in the recognition of the fact that, by means of such devices as the above, dark gaps or cross-lines *can* be seen in the solar spectrum. In other words, light-waves of the various gradations corresponding to all the tints of the spectrum from violet to red do *not* travel to us from the great central luminary of our system. Remembering that the effect we call color is due to the length of the light-waves, the effect of red corresponding to light-waves of the greatest length, while the effect of violet corresponds to the shortest light-waves, we see that in effect the sun sends forth to the worlds which circle around him light-waves of many different lengths, but not of all. Of so complex and interesting a nature is ordinary daylight.

But spectroscopists sought to interpret these dark lines in the solar spectrum, and it was in carrying out this inquiry—which even to themselves seemed almost hopeless, and to many would appear an utter waste of time—that they lighted upon the noblest method of research yet revealed to man.

They examined the spectra of the light from incandescent substances (white-hot metals and the like), and found that in these spectra there are no dark lines.

They examined the spectra of the light from the stars, and found that these spectra are crossed by dark lines resembling those in the solar spectrum, but differently arranged.

They tried the spectra of glowing vapors, and they obtained a perplexing result. Instead of a number of dark lines across a rainbow-tinted streak, they found bright lines of various colors. Some gases would give a few such lines, others many, some only one or two.

Then they tried the spectrum of the electric spark, and they found here also a series of bright lines, but not always the same series. The spectrum varied according to the substances between which the spark was taken and the medium through which it passed.

Lastly, they found that the light from an incandescent solid or liquid, when shining through various vapors, no longer gives a spectrum without dark lines, but that the dark lines which then appear vary in position, according to the nature of the vapor through which the light has passed.

Here were a number of strange facts, seemingly too discordant and too perplexing to admit of being interpreted. Yet one discovery only was wanting to bring them all into unison.

In 1859, Kirchhoff, while engaged in observing the solar spectrum, lighted on the discovery that a certain double dark line, which had already been found to correspond exactly in position with the double bright line forming the spectrum of the glowing vapor of sodium, was intensified when the light of the sun was allowed to pass through that vapor. This at once suggested the idea that the presence of this dark line (or, rather, pair of dark lines) in the spectrum of the sun is due to the existence of the vapor of sodium in the solar atmosphere, and that this vapor has the power of absorbing the same order of light-waves as it emits. It would of course follow from this that the other dark lines in the solar spectrum are due to the presence of other absorbent vapors in its atmosphere, and that the identity of these would admit of being established in the same way, supposing this

general law to hold, that a vapor emits the same light-waves that it is capable of absorbing.

Kirchhoff was soon able to confirm his views by a variety of experiments. The general principles to which his researches led—in other words, the principles which form the basis of spectrum analysis—are as follows:

1. An incandescent solid or liquid gives a continuous spectrum.
2. A glowing vapor gives a spectrum of white lines, each vapor having its own set of bright lines, so that, from the appearance of a bright-line spectrum, one can tell the nature of the vapor or vapors whose light forms the spectrum.
3. An incandescent solid or liquid shining through absorbent vapors gives a rainbow-tinted spectrum crossed by dark lines, these dark lines having the same position as the bright lines belonging to the spectra of the vapors; so that, from the arrangement of the dark lines in such a spectrum, one can tell the nature of the vapor or vapors which surround the source of light.^[24]

The application of the new method of research to the study of the solar spectrum quickly led to a number of most interesting discoveries. It was found that, besides sodium, the sun's atmosphere contains the vapors of iron, calcium, magnesium, chromium, and other metals. The dark lines corresponding to these elements appear unmistakably in the solar spectrum. There are other metals, such as copper and zinc, which seem to exist in the sun, though some of the corresponding dark lines have not yet been recognized. As yet it has not been proved that gold, silver, mercury, tin, lead, arsenic, antimony, or aluminium exist in the sun—though we can by no means conclude, nor indeed is it at all probable, that they are absent from his substance. The dark lines belonging to hydrogen are very well marked indeed in solar spectrum, and, as we shall see presently, the study of these lines has afforded most interesting information respecting the physical constitution of the sun.

Now we notice at once how importantly these researches into the sun's structure bear upon the subject of this treatise. It would be indeed interesting to consider the actual condition of the central orb of the planetary scheme, to picture in imagination the metallic oceans which exist

upon his surface, the continual evaporation from those oceans, the formation of metallic clouds, and the downpour of metallic showers upon the surface of the sun. But apart from such considerations, and viewing Kirchhoff's discoveries simply in their relation to the subject of other worlds, we have enough to occupy our attention.

If it could have been shown that, in all probability, the substance of the sun consists of materials wholly different from those which exist in this earth, the conclusion obviously to be drawn from such a discovery would be that the other planets also are differently constituted. We could not find any just reason for believing that in Jupiter or Mars there exist the elements with which we are acquainted, when we found that even the central orb of the planetary system exhibits no such feature of resemblance to the earth. But now that we know, quite certainly, that the familiar elements, iron, sodium, and calcium, exist in the sun's substance, while we are led to believe, with almost perfect assurance, that all the elements we are acquainted with also exist there, we see at once that, in all probability, the other planets are constituted in the same way. There may of course be special differences: in one planet the proportionate distribution of the elements may differ, and even differ very markedly, from that which prevails in some other planet. But the general conclusion remains, that the planets are formed of the elements which have so long been known as terrestrial; for we can not recognize any reason for believing that our earth alone, of all the orbs which circle around the sun, resembles that great central orb in general constitution.

Now, we have in this general law a means of passing beyond the bounds of the Solar System, and forming no indistinct conceptions as to the existence and character of worlds circling around other suns. For these orbs, like our sun, contain in their substance many of the so-called terrestrial elements, while it may not unsafely be asserted that all, or nearly all, those elements, and few or no elements unknown to us, exist in the substance of every single star that shines upon us from the celestial concave. Hence we conclude that round those suns also there circle orbs constituted like themselves, and therefore containing the elements with which we are familiar. And the mind is immediately led to speculate on the uses which those elements are intended to subserve. If iron, for example, is present in some noble orb circling around Sirius, we speculate not unreasonably

respecting the existence on that orb—either now or in the past, or at some future time—of beings capable of applying that metal to the useful purposes which man makes it subserve. The imagination suggests immediately the existence of arts and sciences, trades and manufactures, on that distant world. We know how intimately the use of iron has been associated with the progress of human civilization, and though we must ever remain in ignorance of the actual condition of intelligent beings in other worlds, we are yet led, by the mere presence of an element which is so closely related to the wants of man, to believe, with a new confidence, that for such beings those worlds must in truth have been fashioned.

I would fain dwell longer on the thoughts suggested by the researches of Kirchhoff. Gladly too would I enter at length on an account of those interesting discoveries which have been made in connection with the total eclipses of the sun. One point, however, remains which is too intimately connected with my subject to be passed over.

I refer to the sun's corona.

It has been proved that the solar prominences consist of glowing vapors, hydrogen being their chief constituent. It has been found also, by comparing Mr. Lockyer's observations of the prominence-spectra with Dr. Frankland's elaborate researches into the peculiarities presented by the spectrum of hydrogen at different pressures, that even in the very neighborhood of the solar photosphere these vapors probably exist at a pressure so moderate as to indicate that the limits of the sun's vaporous envelope can not lie very far (relatively) from the outer solar cloud-layer.

Now, the solar corona has been seen, during total eclipses of the sun, to extend to a distance at least equal to the sun's diameter from the eclipsed orb. So that, assuming the corona to be a solar atmosphere, it would have a depth of about eight hundred and fifty thousand miles, and being also drawn toward the sun by his enormous attractive energy (exceeding more than twenty-seven times that of the earth), it could not fail to exert a pressure on his surface exceeding many thousand-fold that of our air upon the earth. In fact, such an atmosphere, let its outermost layers be as rare as we can conceive, would yet have its lower layers absolutely liquefied, if not solidified, by the enormous pressure to which they would be subjected. We can not, then, believe this corona to be a solar atmosphere.

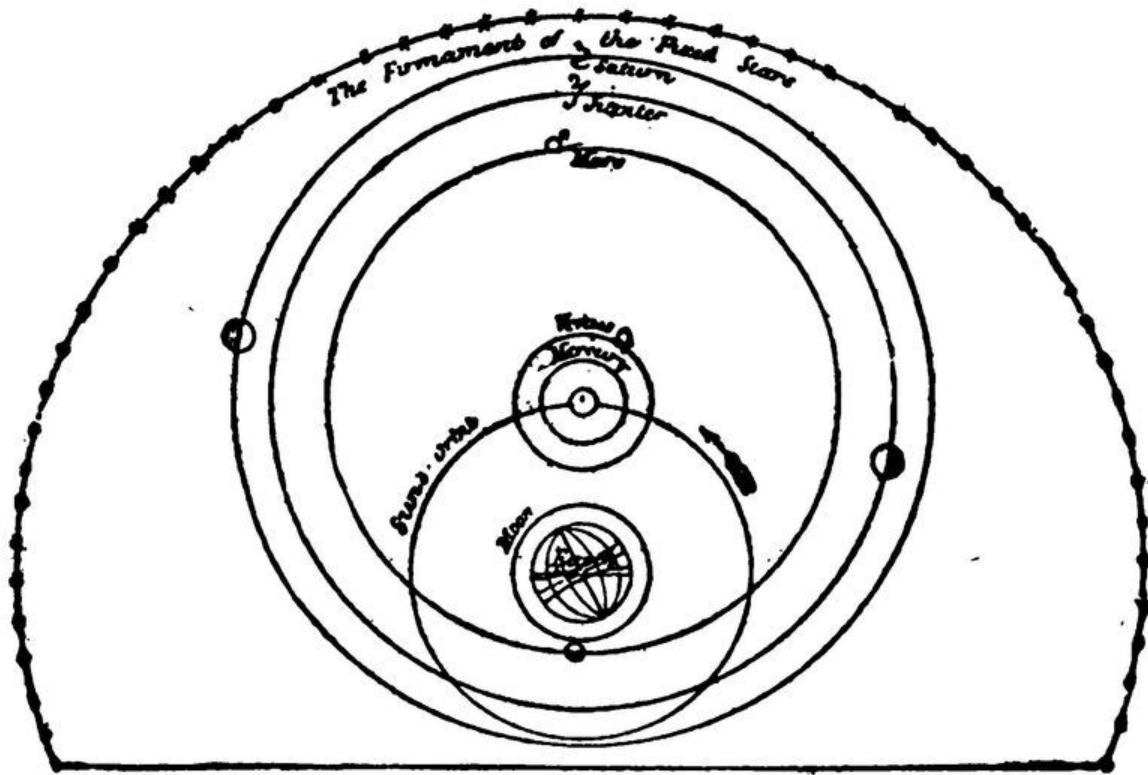


Fig. 33.—Tychonic System

Yet it is quite impossible to dissociate the corona, either wholly or in part, from the sun. I am aware that physicists of eminence have attempted to do this, and not only so, but to make of the zodiacal light a terrestrial phenomenon. But they have overlooked considerations which oppose themselves irresistibly to such a conclusion.

In the first place, the mere fact that, during a total eclipse, the moon looks black, in the very heart of the corona, affords, when properly understood, the most conclusive evidence that the light of the corona comes from behind the moon. If the glare of our atmosphere could by any possibility account for the corona (which is not the case), then that glare should appear over the moon's disk also. That this is so is proved by the fact that, when the glare really does cover the moon, as while the sun is but slightly eclipsed, the moon is not projected as a black disk on the background of the sky, though, where her outline crosses the sun, it appears black, by contrast with the intensity of his light.^[25] The point seems, however, too obvious to need discussion.

And, secondly, as Mr. Baxendell has pointed out, during totality the part of the earth's atmosphere between the eye and the corona is not illuminated by the sun. Over a wide space all round the sun we are looking through an atmosphere which is completely dark. In fact, if the earth's atmosphere alone were in question, we ought to see a dark or negative corona around the sun, the illuminated atmosphere only beginning to be faintly visible at a considerable angular distance from the sun. This argument, rightly understood, is altogether decisive of the question.^[26]

But the spectroscope has given certain very perplexing evidence respecting the light of the corona, and it remains that we should endeavor to see how that evidence bears on the interesting problem which the corona presents to our consideration.

During the total eclipse of 1868 the American observers found that the spectrum of the corona is continuous, but crossed by certain bright lines. If we accept the absence of dark lines as established by the evidence (which is doubtful), this result seems at first sight very difficult to explain. Referring to the principles of spectroscopic analysis stated on [pp. 338-339](#), it will be seen that we should be led to infer that the corona consists of incandescent matter surrounded by certain glowing gases. It is difficult to suppose that this is the real explanation of the phenomenon.

Mr. Lockyer suggests that, if the corona shone by reflecting the solar light, the continuous spectrum might be accounted for by supposing the light from the glowing vapors around the sun to supply the part wanting where the solar dark lines are, and that some of these vapors shining yet more brightly would exhibit their bright lines upon the continuous background of the spectrum. This view, as applied by Mr. Lockyer to the theory that the corona is a terrestrial phenomenon, is untenable, for the reasons already adduced. But, independently of those reasons, there are others which render such a solution of the difficulty unavailable.

Now, remembering that we have two established facts for our guidance—(1) the fact that the corona can not be a solar atmosphere, and (2) the fact that it must be a solar appendage—I think a way may be found toward a satisfactory explanation.

Let it be premised that the bright lines of the coronal spectrum correspond in position to those seen in the spectrum of the aurora, and that

the same lines are seen in the spectrum of the Zodiacal Light, and in that of the phosphorescent light occasionally seen over the heavens at night.

Since we have every reason to believe that the light of the aurora is due to electrical discharges taking place in the upper regions of the air, we are invited to the belief that the coronal light may be due to similar discharges taking place between the particles (of whatever nature) constituting the corona.

Now, though the appearance of an aurora is due to some special terrestrial action (however excited), yet the material substances between which the discharges take place must be assumed to be at all times present in the upper regions of air. In all probability, they are the particles of those meteors which the earth is continually encountering. And since we know that meteor-systems must be aggregated in far greater numbers near the sun than near the earth, we may regard the coronal light as due to electrical discharges excited by the sun's action, and taking place between the members of such systems. Besides this light, however, there must necessarily be a large proportion of light reflected from these meteoric bodies. In this way the peculiar character of the coronal spectrum may be readily accounted for. We know, from the auroral spectrum, that the principal bright lines due to the electrical discharges would be precisely where we see bright lines in the coronal spectrum. But, besides these, there would be fainter bright lines corresponding to the various elements which exist in the meteoric masses. These elements, we know, are the same as those in the substance of the sun. Thus the bright lines would correspond in position with the dark lines of the solar spectrum. Hence, as light reflected by the meteors would give the ordinary solar spectrum, there would result from the combination a continuous spectrum, on which the bright lines first mentioned would be seen, as during the American eclipse.

What the polariscope has told us respecting the corona is in accordance with this view.

In the same way the quality of the Zodiacal Light admits of being perfectly accounted for, without resorting to the hypothesis that this phenomenon is a terrestrial one.

The explanation thus put forward has at least the advantage of being founded on well-established relations. We know that the auroral light is

associated with the earth's magnetism, and that meteoric bodies are continually falling upon the earth's atmosphere. We know, also, that the sun exerts magnetic influences a thousand-fold more intense than those of the earth, and that in his neighborhood there must be many million times more meteoric systems.

But we have other and independent reasons, which must not be overlooked, for considering the corona to be of some such nature as I have suggested. Leverrier has shown that there probably exists in the neighborhood of the sun a family of bodies whose united mass suffices appreciably to affect the motions of the planet Mercury. It would not be safe to neglect considerations thus vouched for.

Mr. Baxendell also has shown that certain periodic variations in the earth's magnetism point to the existence of such a family of bodies; and he has been able to assign to them a position according well with that determined by Leverrier.

Now, whatever opinion we form as to the exact character of the system of bodies pointed to by the researches of Leverrier and Baxendell—whether we suppose that system to form a zone around the sun, or that (as I believe) the system is merely due to the aggregation of meteoric perihelia in the sun's neighborhood—we may be quite certain of this, that during a total solar eclipse the system could not fail to become visible. Hence there is a double objection to the view put forward by Mr. Lockyer and others. In the first place, it fails to account for the appearance presented by the corona; in the second place, it fails to render an account of the implied non-appearance of the system which, according to the researches of Leverrier and Baxendell, circles around the sun.

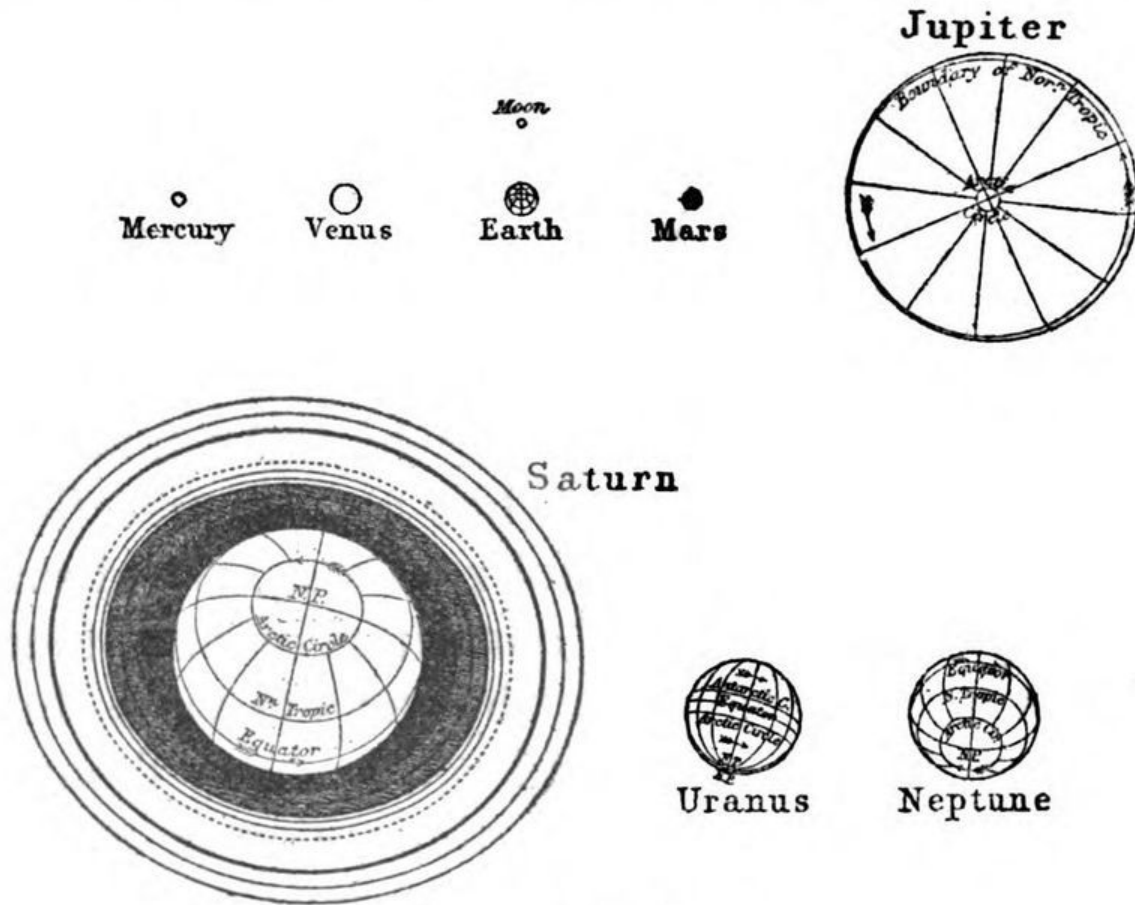


Fig. 34.—Scale of Planets
Jupiter and Saturn are shown in their true axial positions, Uranus and Neptune in the axial positions inferred from the motions of their satellites

We know that the sun is the sole source whence light and heat are plentifully supplied to the worlds which circle around him. The question immediately suggests itself—Whence does the sun derive those amazing stores of force from whence he is continually supplying his dependent worlds? We know that, were the sun a mass of burning matter, he would be consumed in a few thousand years. We know that, were he simply a heated body, radiating light and heat continually into space, he would in like manner have exhausted all his energies in a few thousand years—a mere day in the history of his system. Whence, then, comes the enormous supply of force which he has afforded for millions on millions of years, and which also our reason tells us he will continue to afford while the worlds which circle around him have need of it—in other words, for countless ages to come?

Now, there are two ways in which the solar energies might be maintained. The mere contraction of the solar substance, Helmholtz tells us, would suffice to supply such enormous quantities of heat that, if the heat actually given out by the sun were due to this cause alone, there would not, in many thousands of years, be any perceptible diminution of the sun's diameter. But, secondly, the continual downfall of meteors upon the sun would cause an emission of heat in quantities vast enough for the wants of all the worlds circling round him; while his increase of mass from this cause would not be rendered perceptible in thousands of years, either by any change in his apparent size or by changes in the motions of his family of worlds.

It seems far from unlikely that both these processes are in operation at the same time. Certainly the latter is, for we know, from the motions of the meteoric bodies which reach the earth, that myriads of these bodies must continually fall upon the sun. And if the corona and Zodiacal Light really be due to the existence of flights of meteoric systems circling around the sun, or to the existence in his neighborhood of the perihelia of many meteoric systems, then there must be a supply of light and heat from this source very nearly if not quite sufficient to account for the whole solar emission.

It is well worthy of notice, too, that the association between meteors and comets has an important bearing on this question. We know that the most remarkable characteristic of comets is the enormous diffusion of their substance. Now, in this diffusion there resides an enormous fund of force. The contraction of a large comet to dimensions corresponding to a very moderate mean density would be accompanied by the emission of a vast supply of heat. And the question is worth inquiring into, whether we can indeed assume that the meteors which reach our atmosphere are solid bodies, and not rather of cometic diffusion; since it is difficult otherwise to account for the light and heat which they emit. Friction through the rarer upper strata of our atmosphere will certainly not account for these phenomena; nor, I think, will the compression of the atmosphere in front of the meteors; on the other hand, the sudden contraction of a diffused vapor would be accompanied by precisely such results. But, be this as it may, it is certain that a large portion of the substance of every comet is in a singularly diffused state. And since the meteoric systems circling in countless millions round the sun are, in all probability, associated in the most intimate manner

with comets, we may recognize in this diffusion, as well as in the mere downfall of meteors, the source of an enormous supply of light and heat.

And lastly, turning from our sun to the other suns which shine in uncounted myriads throughout space, we see the same processes at work upon them all. Each star-sun has its coronal and its zodiacal disks, formed by meteoric and cometic systems; for otherwise each would quickly cease to be a sun. Each star-sun emits, no doubt, the same magnetic influences which give to the Zodiacal Light and to the solar corona their peculiar characteristics. And thus the worlds which circle round those orbs may resemble our own in all those relations which we refer to terrestrial magnetism, as well as in the circumstance that on them also there must be, as on our own earth, a continual downfall of minute meteors. In those worlds, perchance, the magnetic compass directs the traveler over desert wastes or trackless oceans; in their skies, the aurora displays its brilliant streamers; while, amid the constellations which deck their heavens, meteors sweep suddenly into view, and comets extend their vast length athwart the celestial vault, a terror to millions, but a subject of study and research to the thoughtful.

FOOTNOTES:

[23] Professor Kirkwood has published a most interesting series of inquiries, going far to prove that the real secret or the planetary influences lies in the fact that the sun's surface is not uniform, and that on a certain solar longitude the planetary influences are more effective than elsewhere.

[24] To these may be added the following law:

4. Light reflected from any opaque body gives the same spectrum as it would have given before reflection.

5. But if the opaque body be surrounded by vapors, the dark lines corresponding to these vapors make their appearance in the spectrum with a distinctness proportioned to the extent to which the light has penetrated those vapors before being reflected to us.

6. If the reflecting body be itself luminous, the spectrum belonging to it is superadded to the spectrum belonging to the reflected light.

7. Glowing vapors surrounding an incandescent source of light may cause bright lines or dark lines to appear in the spectrum, according as they are more or less heated; or, they may emit just so much light as to make up for what they absorb, in which case there will remain no trace of their presence.

8. The electric spark presents a bright-line spectrum, compounded of the spectra belonging to the vapors of those substances between which, and of those through which, the discharge takes place. According to the nature of these vapors and of the discharge itself, the relative intensity of the component parts of the spectrum will be variable.

Lastly, the appearance of the spectrum belonging to any element will vary according to the circumstances of pressure and temperature under which the element may emit light.

[25] It is also shown most conclusively, by a photograph of the eclipse of August, 1868, taken an instant before the totality. Here we see the glare trenching upon the moon's disk (elsewhere black), as it should theoretically. So soon as totality commenced, the glare had reached the moon's limb, whence it must immediately have passed quickly away.

[26] In fact, if we take the mode of reasoning by which Mr. Lockyer has endeavored to get over certain physical difficulties presently to be mentioned, we shall be able to point definitely to the place where his argument fails. He says, conceive a tiny moon placed so as to appear coincident with the centre of the sun's disk. There will be atmospheric glare as well as direct sunlight. Now, conceive this small moon to expand until it all but covers the sun. Still there will be glare and a certain small proportion of direct sunlight. So far his reasoning is most just. But when he allows his expanding moon to cover the sun, and to extend beyond the solar disk as in total eclipse, the atmospheric glare can no longer be assumed to exist all round the expanding moon: at the moment when the moon just hides the sun, the glare begins to leave the moon, a gradually expanding black ring being formed round that body. It is only necessary to consider where the glare comes from to see that this must be so.

I have taken no account of diffraction here, because it has been abundantly proved that no corona of appreciable width could be formed around the moon during total eclipse by the diffraction of the rays of light as they pass near the moon's limb.

MERCURY.—WILLIAM F. DENNING

Mercury is the nearest known planet to the sun. It is true that a body, provisionally named Vulcan, has been presumed to exist in the space inferior to the orbit of Mercury; but absolute proof is lacking, and every year the idea is losing strength in the absence of any confirmation of a reliable kind. Not one of the regular and best observers of the sun has recently detected any such body during its transits (which would be likely to occur pretty frequently), and there is other evidence of a negative character; so that the ghost of Vulcan may be said to have been laid, and we may regard it as proven that no major planet revolves in the interval of 36,000,000 miles separating Mercury from the sun.

Copernicus, amid the fogs of the Vistula, looked for Mercury in vain, and complained in his last hours that he had never seen it. Tycho Brahe, in the Island of Huen, appears to have been far more successful. The planet is extremely fugitive in his appearances, but is not nearly so difficult to find as many suppose. Whenever the horizon is very clear, and the planet well placed, a small sparkling object, looking more like a scintillating star than a planetary body, will be detected at a low altitude and may be followed to the horizon.

Mercury revolves round the sun in 87 days, 23 hours, 15 minutes, and 44 seconds in an eccentric orbit, so that his distance from that luminary varies from 43,350,000 to 28,570,000 miles. When in superior conjunction the apparent diameter of the planet is 4".5; at inferior conjunction it is 12".9, and at elongation 7". His real diameter is 3,000 miles.

Being situated so near to the sun, it is obvious that to an observer on the earth he must always remain in the same general region of the firmament as that body. His orbital motion enables him to successively assume positions to the east and west of the sun, and these are known as his elongations, which vary in distance from 18° to 28° . He becomes visible at these periods either in the morning or evening twilight, and under the best circumstances may remain above the horizon two hours in the absence of the sun. The best times to observe the planet are at his E. elongations during the first half of

the year, or at his W. elongations in the last half; for his position at such times being N. of the sun's place, he remains a long while in view.

Occasionally he presents quite a conspicuous aspect on the horizon, as in February, 1868, when I thought his lustre vied with that of Jupiter, and in November, 1882, when he shone brighter than Sirius. The planet is generally most conspicuous *a few mornings after his W. elongations and a few evenings before his E. elongations.*

In the course of his orbital round, Mercury exhibits all the phases of the moon. Near his elongations the disk is about half illuminated, and similar in form to that of our satellite when in the first or third quarter. But the phase is not to be distinctly made out unless circumstances are propitious. Galileo's telescope failed to reveal it, and Hevelius, many years afterward, found it difficult. This is explained by the small diameter of the planet and the rarity with which his disk appears sharply defined. The phase is sometimes noted to be less than theory indicates; for the planet has been seen crescented when he should have presented the form of a semicircle. Several observers have also remarked that his surface displays a rosy tint, and that the terminator is more deeply shaded and indefinite than that of Venus.

The atmosphere of Mercury is probably far less dense than that of Venus. The latter being furthest from the sun might be expected to shine relatively more faintly than the former, but the reverse is the case. Mercury has a dingy aspect in comparison with the bright white lustre of Venus. On May 12, 1890, when the two planets were visible as evening stars, and separated from each other by a distance of only 2° , I examined them in a 10-inch reflector, power 145. The disk of Venus looked like newly polished silver, while that of Mercury appeared of a dull leaden hue. A similar observation was made by Mr. Nasmyth on September 28, 1878. The explanation appears to be that the atmosphere of Mercury is of great rarity, and incapable of reflection in the same high degree as the dense atmosphere of Venus.

As a naked-eye object, Mercury must necessarily be looked for when near the horizon; but there is no such need in regard to telescopic observation, which ought to be only attempted when the planet surmounts the dense lower vapors and is placed at a sufficient elevation to give the instrument a fair chance of producing a steady image. The presence of

sunshine need not seriously impair the definition, or make the disk too faint for detail.

I have occasionally seen Mercury, about two or three hours after his rising, with outlines of extreme sharpness and quite comparable with the excellent views obtained of Venus at the time of sunrise or sunset. Those who possess equatorials should pick up the planet in the afternoon and follow him until after sunset, when the horizontal vapors will interfere. Others who work with ordinary altazimuth stands will find it best to examine the planet at his western elongations during the last half of the year, when he may be found soon after rising by the naked eye or with an opera-glass, and retained in the telescope for several hours after sunrise if necessary.

Mercury was displayed under several advantages in the morning twilight of November, 1882, and I made a series of observations with a 10-inch reflector, power 212. Several dark markings were perceived, and a conspicuous white spot. The general appearance of the disk was similar to that of Mars, and I forwarded a summary of my results to Professor Schiaparelli of Milan, who favored me with the following interesting reply:

“I have myself been occupied with this planet during the past year (1882). You are right in saying that Mercury is much easier to observe than Venus, and that his aspect resembles Mars more than any other of the planets of the Solar System. It has some spots which become partially obscured and sometimes completely so; it has also some brilliant white spots in a variable position.”

Professor Schiaparelli used an 8½-inch refractor in this work, and was able under some favorable conditions to apply a power of 400. The outcome of his researches, encouraged since 1882 by the addition of an 18-inch refractor to the appliances of his observatory, was announced in the curious fact that the rotation of Mercury is performed in the same time that the planet revolves round the sun! If this conclusion is just, Mercury constantly presents one and the same hemisphere to the sun, and the behavior of the moon relatively to the earth has found an analogy.

Spots or markings of any kind have rarely been distinguished on Mercury. On June 11, 1867, Prince recorded a bright spot, with faint lines diverging from it northeast and south. The spot was a little south of the

centre. Birmingham on March 13, 1870, glimpsed a large white spot near the planet's east limb, and Vögel, at Bothkamp, observed spots on April 14 and 22, 1871. These instances are quoted by Webb, and they, in combination with the markings seen by Schiaparelli at Milan and by the author at Bristol in 1882, sufficiently attest that this object deserves more attentive study.

One of the most interesting phenomena, albeit a somewhat rare event, in connection with Mercury, is that of a transit across the sun. The planet then appears as a black circular spot. Observers have noticed one or two very small luminous points on the black disk, and an annulus has been visible round it. These features are probably optical effects.

THE PLANET VENUS.—CAMILLE FLAMMARION

Revolving round the sun in 224 days, Venus has its motion combined with ours in such a manner that it passes its inferior conjunction, between the sun and us, every 584 days; but the plane in which it revolves is inclined $3^{\circ} 23'$ to that in which the earth itself moves. When Venus attains its greatest elongations from the sun it shines in the west in the evening, then in the morning in the east, with a splendid brightness which eclipses that of all the stars. It is, without comparison, the most magnificent star of our sky. Its light is so vivid that it casts a shadow. Sometimes, even, it pierces the azure of the sky, in spite of the presence of the sun above the horizon, and *shines in full daylight*.

The maximum visibility of Venus is produced by its greatest phase, by its greatest elongation from the sun, and by the clearness of our atmosphere.

The brilliant Venus was certainly the first planet noticed by the ancients, as much on account of its brightness as its rapid motion. Hardly is the sun set than it sparkles in the twilight; from evening to evening it removes further from the west and increases in brightness; during several months it reigns sovereign of the skies, then plunges into the solar fires and disappears. It was pre-eminently the star of the evening, the shepherd's star, the star of sweet confidences. It was the first of celestial beauties, and the names conferred upon it correspond to the direct impression which it produced on contemplative minds. Homer called it "Callistos," the *Beautiful*; Cicero named it *Vesper*, the evening star, and *Lucifer*, the morning star, a name likewise given in the Bible and the ancient mythologies to the chief of the celestial army.

The most ancient astronomical *observation* we have of Venus is a Babylonian record of the year 685 B. C. It is written on a brick and preserved in the British Museum.

The best hours for examining Venus in a telescope are those of daylight. In the night the irradiation produced by the brilliant light of this beautiful planet prevents us from distinguishing clearly the outlines of its phases.

When Venus occupies the region of its orbit behind the sun, with reference to us—which is called the point of superior conjunction—it is at its greatest distance, and is reduced to a disk of $9\frac{1}{2}$ seconds in diameter. It comes imperceptibly toward us, and when it passes its quadrature, at its mean distance, it presents the aspect of a half-moon. It soon attains its most brilliant light, at the epoch when it shines at a distance of 39° from the sun, and shows the third phase 69 days before its inferior conjunction. Its apparent diameter is then 40 seconds, and the width of its illuminated part is scarcely 10 seconds. In this position we see the fourth of the disk illuminated; but this quarter emits more light than the more complete phases. Finally, when it reaches the region of its orbit nearest to the earth, it shows us nothing more than an excessively thin crescent, since it is then between the sun and us, and presents to us, so to say, its dark hemisphere. This is the position where its apparent size is greatest, and it then measures 62 seconds in diameter. After passing its inferior conjunction the phases are reproduced, in inverse order, as a morning star.

Venus is constantly visible in full daylight in astronomical instruments, even at the moment of its superior conjunction. It is then round and quite small. At the epochs of its inferior conjunction it presents itself under the form of a very thin crescent.

We sometimes notice that the interior of the crescent of Venus, the remainder of the disk, is less black than the background of the sky. This has been called the ashy light (*lumière cendrée*) of Venus, although it has no satellite to produce it. It seems to me that this visibility, rather subjective than objective, arises from clouds on the planet, which whiten its disk and vaguely reflect the stellar light scattered through space. The eye instinctively continues the outline of the crescent, and imagines, rather than sees, the rest.

The revolution of Venus round the sun is performed in an orbit almost exactly circular, and without perceptible eccentricity (0.0068), in a period of 224 days, 16 hours, 49 minutes, 8 seconds.

The days of Venus, also, are a little more rapid than ours, but not much. Since the year 1666 attentive observation of the planet led Cassini to conclude that it turns on itself in 23 hours, 15 minutes. This observation is

extremely difficult, on account of the brightness of the planet and the faintness of the irregularities visible on its disk.

The year of Venus, composed of 224 terrestrial days, consequently contains 231 of its own, since the day is a little shorter there than here.

These same observations show that the axis of rotation of this planet is much more inclined than ours, and that this inclination is 55 degrees. It follows that the seasons, although each lasting but 56 terrestrial days, or 58 Venusian days, are much more intense on this world than on ours. They pass, without transition, from summer to winter.

The inclination of the world of Venus being more than twice as great as ours, we have only to take a terrestrial globe and incline it by the same quantity to understand the climates and seasons which will result. We may easily see that the torrid zone extends, in this case, up to the frigid zone, and even beyond it; and, reciprocally, the frigid zone extends to the torrid zone, and even encroaches on it; so that no place remains for a temperate zone. There is not, then, on Venus any temperate climate, but all latitudes are both tropical and arctic.

It follows, then, from all these circumstances, that the seasons and climates are much more violent and more varied than ours. This neighboring world shows nearly the same dimensions as ours. Thus this planet is truly the twin sister of ours.

The resemblance will be still more complete if we add that this world is certainly surrounded by an atmosphere.

When we examine with the spectroscope the light reflected by this planet we first find the lines of the solar spectrum (and this is natural, since the planets have no light of their own, and merely reflect that of the sun); but we notice besides several absorption lines similar to those which the terrestrial atmosphere gives, and particularly those of clouds and water vapor.

We may also add that attentive observation of the indentations visible on the crescent of Venus has shown that the surface of this planet is quite as uneven as that of the earth, and even more so; that there are there Andes, Cordilleras, Alps, and Pyrenees, and that the most elevated summits attain a

height of 44,000 metres (27 miles). It has even been ascertained that the Northern Hemisphere is more mountainous than the Southern.

Even the study of the geography of Venus has already been commenced. But it is extremely difficult to draw, and the hours of sufficiently pure atmosphere and possible observation are very rare. This difficulty will be easily understood if we reflect that it is exactly when Venus arrives at its nearest to us that it is least visible, since, its illuminated hemisphere being always turned toward the sun, it is its dark hemisphere which is presented to us. The nearer it approaches us, the narrower the crescent becomes. Add to this its vivid light and its clouds, and you may imagine what difficulty astronomers have in dealing with it.



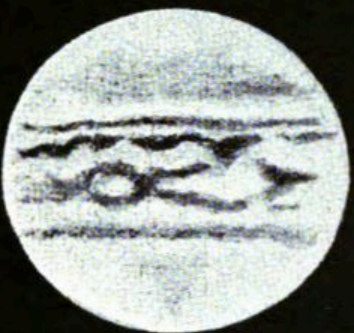
1882 Dec. 8. 11^h29^m



1882 Dec. 22. 10^h44^m



1882 Dec. 22. 11^h54^m



1883 Jan. 15. 10^h57^m



1883 Jan. 15. 12^h9^m



1883 Jan. 18. 9^h53^m



1883 Jan. 18. 10^h25^m



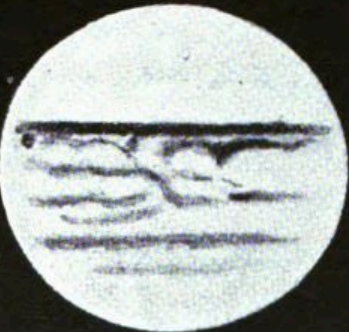
1883 Jan. 18. 11^h26^m



1883 Jan. 19. 9^h58^m



1883 Jan. 19. 10^h35^m



1883 Jan. 19. 11^h18^m



1883 Jan. 19. 11^h59^m

Twelve Views of Jupiter
Taken at Intervals within Six Consecutive Weeks

However, by observing it in the daytime to avoid the glare, and not waiting till the crescent becomes too thin, by choosing the quadratures, and making use of moments of great atmospherical purity, observers succeed, from time to time, in perceiving grayish spots, which may indicate the place of its seas.

Of what nature are the inhabitants of Venus? Do they resemble us in physical form? Are they endowed with an intelligence analogous to ours? Do they pass their life in pleasure, as Bernardin de St. Pierre said, or, rather, are they so tormented by the inclemency of their seasons that they have no delicate perception, and are incapable of any scientific or artistic attention? These are interesting questions, to which we have no reply. All that we can say is, that organized life on Venus must be little different from terrestrial life, and that this world is one of those which resembles ours most. The imaginary travelers to these worlds of the sky have always carried with them their terrestrial ideas. The only scientific conclusion which we can draw from astronomical observation is that this world differs little from ours in volume, in weight, in density, and in the duration of its days and nights; that it differs a little more in the rapidity of its years, the intensity of its climates and seasons, the extent of its atmosphere, and its greater proximity to the sun. It should, then, be inhabited by vegetable, animal, and human races but little different from those which people our planet. As to imagining it desert or sterile, this is a hypothesis which could not arise in the brain of any naturalist. The action of the divine sun must be there, as in Mercury, still more fertile than his terrestrial work, already so wonderful. We may add that Venus and Mercury, having been formed after the earth, are relatively younger than our planet.

The inhabitants of Venus see us shining in their sky like a magnificent star of the first magnitude, soaring in the zodiac, and showing motions similar to those which the planet Mars presents to us; but instead of showing a reddish brightness, the earth shines in the sky as a bluish light. It is from Venus that we are most luminous. The inhabitants of Venus with the naked eye see our moon shining beside the earth and revolving round it in twenty-seven days. They form a magnificent couple. Our planet seen from there measures 65", and the moon nearly 18"; the moon seen from Venus

shows the same diameter as the earth seen from the sun. Mercury is brilliant, and comes immediately after the earth in brightness. Mars, Jupiter, and Saturn are also visible as from here, but a little less luminous. The constellations of the whole sky show exactly the same aspect as seen from the earth.

THE EARTH AS A PLANET.—ÉLISÉE RECLUS

The earth on which we dwell is one of the lowest in rank among the heavenly bodies. If an astronomer in some other planet were exploring the immensity of space, our earth, owing to its small size, might readily elude his intelligent view. A mere satellite of the sun, the volume of which is 1,255,000 times greater, the earth is but a point as compared with the immense tract of ether traversed by the planets in their courses round their central globe. The sun itself is only a spark, which seems lost amid the eighteen millions of stars which Herschel's telescope discerned in the Milky Way; the latter, an immense agglomeration of suns and planets, which looks to us like a broad streak of light round the whole universe, is in reality nothing but a nebula. Beyond our own sky, other skies stretch far away into infinity, and others beyond these, so that light notwithstanding its prodigious rapidity, takes eternities to cross them. How small the earth seems in this fathomless abyss of stars!

In the form of its orbit, in its movements round the sun and on its own axis, in the succession of days and seasons, and in all the phenomena governed by the great law of attraction, the earth becomes the representative of all the other planets; in studying it, we study all the heavenly bodies.

Our planet is a spheroid; that is, a sphere flattened at the two poles and enlarged at the equator, so that all the circles passing through the extremity of the polar axis form ellipses. The presumed depression of each pole is about thirteen miles, nearly a three-hundredth part of the radius of the earth; but it is not altogether certain that the two poles are equally flattened. Perhaps a contrast exists between the two hemispheres, not only in the features of their continents and the distribution of seas, but also in their geometrical shape. Be this as it may, it appears to be proved that the curvature is not exactly the same at all points of the earth at an equal distance from the poles; the meridians appear without exception to be irregular ellipses.

The dimensions of the earth, as we have already seen, are almost as nothing compared with the larger celestial bodies, and especially with the extent of space which can be explored by the telescope. If light, the speed of which has been adopted in astronomy as a term of comparison, could be diffused in a curved line, it would travel seven times round the globe in a second of time; this standard of measurement, therefore, the only one suited to the stellary field, is completely inapplicable to the surface of our globe.

The isolated globule in the immensity of space which we call the earth is not motionless, as the ancients necessarily supposed, looking upon it, as they did, as the immovable base of the firmament of heaven. Hurried on in the vortex of universal vitality, our globe is ever actuated by ceaseless motion, describing in ether a series of elliptic spirals so complicated that astronomers have not yet been able to calculate their various curves. Besides rotating on its own axis, the earth describes an ellipse round the sun, and, under the influence of this body, is drawn along from one heaven to another toward distant constellations. It also oscillates and rocks on its axis, and deviates more or less from its path, to salute, as it were, every heavenly body which meets it. It is probable that it never passes a second time through the same regions of the air; yet, if it has again to traverse the spiral line of ellipses it has already described, it would be after a cycle of so many thousands of millions of years, that the earth itself, completely transformed, would be no longer the same planet.

The motion of the earth, the immediate effects of which are the most obvious to the notice of men, is the daily rotation which takes place round an ideal axis passing through the two poles. The globe turns from right to left, or from west to east—that is, in a contrary direction to the apparent motion of the sun and stars, which seem to rise in the east and to set in the west. As the earth's axis terminates at each pole, there is least surface-motion at those points, and the motion is the more rapid in any part of the surface of the globe the further it is from the central axis. At St. Petersburg, in 60° latitude, the speed of rotation is about nine miles a minute; in Paris, it exceeds eleven and a half miles during the same brief time; on the equatorial line, which may be looked upon as the ring of an immense wheel, the speed of the earth is twice as great as it is at 60° of latitude—that is, about eighteen miles a minute, or 528 yards a second—a rapidity equal to the flight of a 26-pound cannon-ball impelled by thirteen pounds of powder.

By means of this rotatory motion, the earth presents toward the sun each of its faces alternately, and each also in turn toward the comparatively darker regions of space; the succession of day and night is thus constituted. In addition to this, the rotation of the earth is an important fact which must always be taken into account in determining the direction of fluids in motion on the surface of the globe, such as streams and rivers, also marine and atmospheric currents.

The annual revolution which the earth performs round the sun follows the line of an ellipse, one of the *foci* of which is occupied by the central star; the eccentricity of the ellipse is nearly equal to $\frac{17}{1000}$ th of the great axis. The distance between the sun and the earth always varies according to the particular point of its orbit which the latter is traveling over. At its *aphelion*, that is, at its greatest remoteness, this distance is about $93\frac{3}{4}$ millions of miles; at the period of its *perihelion*, when the two heavenly bodies are nearest to each other, it is approximately 90,259,000 miles. The mean distance, as estimated by astronomers since the corrections of Encke, Hansen, Foucault, and Hind, is 91,839,000 miles. This extent of space is traversed by the solar rays in 8 minutes, 16 seconds; sound would take fifteen years in passing through the same distance.

As Kepler has laid down in his celebrated laws, our planet moves with an increased rapidity as it approaches nearer to the sun and travels more slowly in proportion to its distance from that luminary; but its mean speed may be estimated at nearly nineteen miles a second, or sixty times the rapidity of a ball from the cannon's mouth. This speed, which makes one dizzy to think of, is to be added, as regards each point in the surface of the earth, to the rotatory motion which impels it round the polar axis.

After having turned round 366 times on its axis, our planet has terminated its orbicular course, and is in the same position relatively to the sun as at its starting-point; it has then accomplished its *year*.

This daily rotation of the earth round its axis produces the succession of days and nights, and, in the same way, its annual revolution round the sun causes the alternations of the seasons. If the axis of the earth, that is the ideal line which passes through its two poles, were perpendicular to the plane of its annual orbit, it is evident that the portion of the globe lighted by the sun would invariably extend from one pole to the other, and that in both

hemispheres the days and nights would always consist of twelve hours each. But this is not the case. The earth performs its revolutionary movements in an inclined position; its ideal polar axis is sloped about $23^{\circ} 28'$ from a perpendicular to its plane, and this position is so far maintained that as regards the comparatively rapid succession of days and seasons it may be looked upon as invariable. This obliquity of axis causes continued changes in the phase presented to the sun. The portion of the earth illumined by the rays of the sun varies every day; for, although the planetary axis may appear to maintain its extremity in a fixed position as regards some point in infinite space, in respect to the sun it presents a constantly varying degree of inclination, in consequence of the continual motion of the earth. Twice during the course of the year it so happens that the solar rays fall perpendicularly upon the equator of the earth; at every other period in the annual revolution, sometimes the Northern and sometimes the Southern Hemisphere receives the greatest amount of light.

The astronomical year commences on the 20th of March, at the exact moment when the sun illumines the equator in a vertical direction, and the line of separation between light and shade passes through the two poles. The period of darkness is then equal to that of light, and admits of exactly twelve hours at all points of the earth. Hence the name of “equinox” (equality of nights). But after this day, which in the Northern Hemisphere serves as the starting-point of spring, the earth continues its translatory movement. In consequence of the inclination of its axis, the Northern Hemisphere, being turned toward the sun, receives a greater quantity of light, while the southern half of the globe is less vividly lighted. The vertical rays of the sun now fall more and more to the north of the equator, and the circle of light, far from arresting its progress at the poles, where the day of six months’ duration is commencing to dawn, extends far beyond it over the regions of the north. On the 21st of June, the day of the first solstice, the axis of the earth being deeply inclined toward the sun, this luminary shines on the zenith of the tropic of Cancer at $23\frac{1}{2}^{\circ}$ north of the equator, and its light illumines the whole of the arctic zone, that is, the portion of the earth’s surface extending to $23\frac{1}{2}^{\circ}$ round the North Pole. Then spring ceases and summer begins as regards the Northern Hemisphere. In the Southern Hemisphere, on the contrary, autumn is giving place to winter. Above the equator long days are prevailing, interrupted by short nights; while in the south it is the nights which last the longest. In the arctic zone

the sun performs its apparent course of diurnal rotation entirely above the horizon. The six months' day, which spring inaugurated at the North Pole, attains its high noon on the first day of summer. At the same moment midnight arrives in the darkness which is oppressing its antipodes.

Immediately after the 21st of June all the phenomena which took place during the preceding season are directly reversed. The sun appears to retrograde toward the southern horizon; its vertical rays cease to fall on the line of the northern tropic, and constantly approach the equator. The zone of light in the northern pole and of shade in the southern equally diminish, and the days shorten in the Northern Hemisphere in the same proportion as they lengthen in the Southern; an equilibrium is gradually being re-established between the two halves of the earth. On the 22d of September the position of the sun is again exactly above the equator, and its light just reaches both poles. The equinox, or the absolute equality of day and night in every part of the globe, occurs for the second time in the year; but this moment of equilibrium is, so to speak, but a mathematical point between the two seasons. The axis of the earth which, during the six months past, turned the North Pole toward the sun, now presents to him the South Pole; the vertical rays of the central luminary fall to the south of the earth's equator, and the Southern Hemisphere, in its turn, is the best endowed of the two halves of the globe in the amount of light it receives and in the length of its days. In the Southern Hemisphere spring is commencing; in the Northern, autumn. Three months afterward, on the 21st of December, the sun comes directly over the southern tropic, or the tropic of Capricorn, $23\frac{1}{2}^{\circ}$ south of the equator, and the whole of the antarctic zone is presented to the solar rays. Summer has begun in the Southern Hemisphere, and at the same time winter commences in that of the north. Then, as the globe moves on, these two seasons follow each other in their course, until at length the earth attains a position similar to that from which it started; the March equinox, the first day of spring in Europe, and the first day of autumn in Australia, commences anew the astronomical year.

The elliptical form of the earth's orbit and the unequal pace of the globe in the various points of its course cause some considerable variations in the duration of the seasons. In fact, from the 20th of March to the 22d of September, that is, during the spring and summer of the Northern Hemisphere, the earth takes 186 days to travel over the first and largest half

of its orbit, while during the winter period, from the 22d of September to the 20th of March, only 179 days are required to accomplish the second half of its journey. The summer period of the Northern Hemisphere actually exceeds by seven or eight days, or about 187 hours, the corresponding period in the southern half of the globe; added to this, in consequence of the longer space of time during which the Arctic Pole remains inclined toward the sun in the regions north of the equator, the hours of daylight exceed the hours of night, while in the south the hours of darkness predominate. This is, however, to some extent compensated for; as, although in the southern regions of the earth the summer lasts a shorter time, our planet is then closer to the sun; it is at its perihelion, and consequently receives a larger proportion of heat. There is, however, no doubt about the fact—as it is proved by a direct observation, both of the winds and currents, and also of their various temperatures—that, taking an equal distance from the equator, the southern regions are colder than those of the north.

If an equality of seasons between the two halves of the world does not at present exist, it will not fail to be established after a long series of centuries by means of a slow terrestrial movement, which has been known by the name of the *precession of the equinoxes*. Just as a top (if we may be allowed to avail ourselves of so old an illustration) turns round on the ground and bends over successively in every direction, thus describing with its axis an ideal cone, so the earth revolves in space, and slowly sways the line of its poles. This line, which is always sloped at an angle of $66^{\circ} 32'$ to the plane of the terrestrial orbit, turns round with a slight lateral motion, so as always to point to a new region of the sky; if it were prolonged indefinitely it would describe a circle amid the distant stars. As the axis of the earth is constantly changing its direction in this way, the plane of the equator must vary exactly to the same extent in its position as regards the sun. In fact, every year the exact moment of the March equinox anticipates by about twenty minutes the time at which the corresponding equinox fell in the year preceding. Each revolution of the earth round the sun brings a fresh advance of twenty minutes in the determination of the equinox; and as, during the long course of ages, the axis of the earth does not intermit in this swaying motion, the time must come, after a period of 12,900 years, that the conditions of the seasons will be altogether changed. The hemisphere which hitherto received the larger proportion of heat will receive the lesser share, and that half of the globe which has endured the larger number of wintry

days will now, in its turn, enjoy the more lengthened period of summer. Then, after a second period of 12,900 years, during which the relation between the seasons of the two hemispheres is being gradually modified, the axis of the earth completes its round of swaying, which has lasted for 258 centuries, and the position of the globe in respect to the sun being nearly the same as at its starting-point, a second cycle of seasons will then commence.

We might call this period *the earth's great year*, if, at the end of it, the earth were in an identical position to that which it occupied at the commencement; but this is not the case. The attraction of the moon, and the disturbances caused by the vicinity of certain planets, are incessantly modifying the curve described in the starry fields of space by the earth's axis, and complicate it with a multitude of spirals, the various periods of which do not coincide with the great period of the swaying of the axis. The successive undulations form a continuous system of interwoven spirals. "It is a manifestation of the infinite."

But even this is not all. In addition to all the motions of the globe which we have already pointed out—its diurnal rotation, its annual revolution round the sun, the rhythmical swaying of its axis, proved by the precession of the equinoxes, the nutation or more rapid swaying which is caused by the attraction of the moon—we must now notice the enormous translatory movement which is dragging it through endless tracks of space in the train of the sun. Not many years ago, this motion was entirely unknown to astronomers, and yet it is going on with inconceivable rapidity—a rapidity more than double that of the course of the planet round its central luminary. In one second of time the earth moves about forty-four miles toward the point of the heavens where we find the constellation of Hercules. During one year only she travels 1,382 millions of miles in this direction. Our own little earth itself is carried on from space to space, and never closes the cycle of its revolutions. Ever since the time when its particles were first grouped together, it has been describing in space the infinite spiral of its ellipses, and thus will it go on turning and oscillating in ether until the moment when it will exist no longer as an independent planet. For the earth, too, must have an end; like every other body in the universe, it comes into existence, and lives only to die when its turn comes. Already its annual motion of rotation is diminishing in speed; certainly this slackening of pace

is not very observable, since no astronomer from Hipparchus to Laplace has yet exactly defined it. But, unless some cosmical force acting in a contrary direction compensates for the loss of speed caused by the friction of the tides against the bed and the shores of the ocean, the impetus of our planet will every century diminish. After various catastrophes which it is impossible to foresee, the earth will eventually completely change its course of action, and lose its independent existence, either uniting itself with other planetary bodies or breaking up into fragments; or it will perhaps terminate its course by falling like a mere aerolite upon the surface of the sun.

THE MOON.—THOMAS GWYN ELGER

We know, both by tradition and published records, that from the earliest times the faint gray and light spots which diversify the face of our satellite excited the wonder and stimulated the curiosity of mankind, giving rise to superstitions more or less crude and erroneous as to their actual nature and significance. It is true that Anaxagoras, five centuries before our era, and probably other philosophers preceding him—certainly Plutarch at a much later date—taught that these delicate markings and differences of tint, obvious to every one with normal vision, point to the existence of hills and valleys on her surface; the latter maintaining that the irregularities of outline presented by the “terminator,” or line of demarcation between the illumined and unillumined portion of her spherical superficies, are due to mountains and their shadows; but more than fifteen centuries elapsed before the truth of this sagacious conjecture was unquestionably demonstrated. Selenography, as a branch of observational astronomy, dates from the spring of 1609, when Galileo directed his “optic tube” to the moon, and in the following year, in the *Sidereus Nuncius*, or the “Intelligencer of the Stars,” gave to an astonished and incredulous world an account of the unsuspected marvels it revealed.

The bright and dusky areas, so obvious to the unaided sight, were found by Galileo to be due to a very manifest difference in the character of the lunar surface, a large portion of the Northern Hemisphere, and no inconsiderable part of the southeastern quadrant, being seen to consist of large gray monotonous tracts, often bordered by lofty mountains, while the remainder of the superficies was much more conspicuously brilliant, and, moreover, included by far the greater number of those curious ring-mountains and other extraordinary features whose remarkable aspect and peculiar arrangement first attracted his attention.

Before the close of the century when selenography first became possible, Hevel of Dantzic, Scheiner, Langrenus (cosmographer to the King of Spain), Riccioli, the Jesuit astronomer of Bologna, and Dominic Cassini, the celebrated French astronomer, greatly extended the knowledge of the

moon's surface, and published drawings of various phases and charts, which, though very rude and incomplete, were a clear advance upon what Galileo, with his inferior optical means, had been able to accomplish. Langrenus, and after him Hevel, gave distinctive names to the various formations, mainly derived from terrestrial physical features, for which Riccioli subsequently substituted those of philosophers, mathematicians, and other celebrities; and Cassini determined by actual measurement the relative position of many of the principal objects on the disk, thus laying the foundation of an accurate system of lunar topography; while the labors of T. Mayer and Schröter in the Eighteenth Century, and of Lohrmann, Mädler, Neison (Nevill), Schmidt, and other observers in the Nineteenth, have been mainly devoted to the study of the minuter detail of the moon and its physical characteristics.

As was manifest to the earliest telescopic observers, its visible surface is clearly divisible into strongly contrasted areas, differing both in color and structural character. Somewhat less than half of what we see of it consists of comparatively level dark tracts, some of them many thousands of square miles in extent, the monotony of whose dusky superficies is often unrelieved for great distances by any prominent object; while the remainder, everywhere manifestly brighter, is not only more rugged and uneven, but is covered to a much greater extent with numbers of quasi-circular formations differing widely in size, classed as walled-plains, ring-plains, craters, craterlets, crater-cones, etc. (the latter bearing a great outward resemblance to some terrestrial volcanoes), and mountain ranges of vast proportions, isolated hills and other features.

Though nothing resembling sheets of water, either of small or large extent, has ever been detected on the surface of the moon, the superficial resemblance, in small telescopes, of the large gray tracts to the appearance which we may suppose our terrestrial lakes and oceans would present to an observer on the moon, naturally induced the early selenographers to term them *Maria*, or "seas"—a convenient name, which is still maintained, without, however, implying that these areas, as we now see them, are, or ever were, covered with water.

There are twenty-three of these dusky areas which have received distinctive names; seventeen of them are wholly, or in great part, confined to the northern and to the southeastern quarter of the Southern Hemisphere

—the southwestern quadrant being to a great extent devoid of them. By far the largest is the vast Oceanus Procellarum, extending from a high northern latitude to beyond latitude 10° in the southeastern quadrant, and, according to Schmidt, with its bays and inflections, occupying an area of nearly two million square miles, or more than that of all the remaining Maria put together. Next in order of size come the Mare Nubium, or about one-fifth the superficies, covering a large portion of the southeastern quadrant, and extending considerably north of the equator, and the Mare Imbrium, wholly confined to the northeastern quadrant, and including an area of about 340,000 square miles. These are by far the largest lunar “seas”. The Mare Fœcunditatis, in the Western Hemisphere, the greater part of it lying in the southwestern quadrant, is scarcely half so big as the Mare Imbrium; while the Maria Serenitatis and Tranquilitatis, about equal in area (the former situated wholly north of the equator and the latter only partially extending south of it), are still smaller. The arctic Mare Frigoris, some 100,000 square miles in extent, is the only remaining large sea; the rest, such as the Mare Vaporum, the Sinus Medii, the Mare Crisium, the Mare Humorum, and the Mare Humboldtianum, are of comparatively small dimensions, the Mare Crisium not greatly exceeding 70,000 square miles, the Mare Humorum (about the size of England) 50,000 square miles, while the Mare Humboldtianum, according to Schmidt, includes only about 42,000 square miles, an area which is approached by some formations not classed with the Maria.

Among the Maria which exhibit the most remarkable arrangement of ridges is the Mare Humorum, in the southeastern quadrant. Here, if it be observed under a rising sun, a number of these objects will be seen extending from the region north of the ring-mountain Vitello in long undulating lines, roughly concentric with the western border of the “sea,” and gradually diminishing in altitude as they spread out, with many ramifications, to a distance of 200 miles or more toward the north. At this stage of illumination they are strikingly beautiful in a good telescope, reminding one of the ripple-marks left by the tide on a soft, sandy beach. Like most other objects of their class, they are very evanescent, gradually disappearing as the sun rises higher in the lunar firmament, and ultimately leaving nothing to indicate their presence beyond here and there a ghostly streak or vein of a somewhat lighter hue than that of the neighboring surface.

The Maria, like almost every other part of the visible surface, abound in craters of a minute type, which are scattered here and there without any apparent law or ascertained principle of arrangement.

Walled-plains, approximating more or less to the circular form, though frequently deviating considerably from it, are among the largest inclosures on the moon. They vary from upward of 150 to 160 miles or under in diameter, and are often encircled by a complex rampart of considerable breadth, rising in some instances to a height of 12,000 feet or more above the inclosed plain. This rampart is rarely continuous, but is generally interrupted by gaps, crossed by transverse valleys and passes and broken by more recent craters and depressions. As a rule, the area within the circumvallation (usually termed “the floor”) is only slightly, if at all, lower than the region outside: it is very generally of a dusky hue, similar to that of the gray plains of Maria, and, like them, is usually variegated by the presence of hills, ridges, and craters, and is sometimes traversed by delicate furrows, termed clefts or rills.

Ptolemæus, in the third quadrant and not far removed from the centre of the disk, may be taken as a typical example of the class. Here we have a vast plain, 115 miles from side to side, encircled by a massive but much broken wall, which at one peak towers more than 9,000 feet above a level floor, which includes details of a very remarkable character. The adjoining Alphonsus is another, but somewhat smaller object of the same type, as are also Albategnius and Arzachel; and Plato, in a high northern latitude, with its noble, many-peaked rampart and its variable steel-gray interior, Grimaldi, near the eastern limb (perhaps the darkest area on the moon), Schickard, nearly as big on the southeastern limb, and Bailly, larger than either (still further south in the same quadrant), although they approach some of the smaller “seas” in size, are placed in the same category. The conspicuous central mountain, so frequently associated with other types of ringed inclosures, is by no means invariably found within the walled-plains; though, as in the case of Petavius, Langrenus, Gassendi, and several other noteworthy examples, it is very prominently displayed. The progress of sunrise on all these objects affords a magnificent spectacle. Very often when the rays infringe on their apparently level floor at an angle of from 1° to 2° , it is seen to be coarse, rough grained, and covered with minute elevations, although an hour or so afterward it appears as smooth as glass.

The more massive and extended mountain ranges of the moon are found in the Northern Hemisphere, and (what is significant) in that portion of it which exhibits few indications of other superficial disturbances. The most prominently developed systems, the Alps, the Caucasus, and the Apennines, forming a mighty western rampart to the Mare Imbrium and giving it all the appearance of a vast walled-plain, present few points of resemblance to any terrestrial chain. The former include many hundred peaks, among which Mont Blanc rises to a height of 12,000 feet, and a second, some distance west of Plato, to nearly as great an altitude; while others ranging from 5,000 to 8,000 feet are common. They extend in a southwest direction from Plato to the Caucasus, terminating somewhat abruptly, a little west of the central meridian in about N. lat. 42° . One of the most interesting features associated with this range is the so-called great Alpine valley, which cuts through it west of Plato.

The Caucasus consist of a massive wedge-shaped mountain land, projecting southward, and partially dividing the Mare Imbrium from the Mare Serenitatis, both of which they flank. Though without peaks so lofty as those pertaining to the Alps, there is one, immediately east of the ring-plain Calippus, which, towering to 19,000 feet, surpasses any of which the latter system can boast. The Apennines, however, are by far the most magnificent range on the visible surface, including as they do some 3,000 peaks, and extending in an almost continuous curve of more than 400 miles in length from Mount Hadley, on the north, to the fine ring-plain Eratosthenes, which forms a fitting termination, on the south. The great headland Mount Hadley rises more than 15,000 feet, while a neighboring promontory on the southeast of it is fully 14,000 feet, and another, close by, is still higher above the Mare. Mount Huyghens, again in N. lat. 20° , and the square-shaped mass Mount Wolf, near the southern end of the chain, include peaks standing 18,000 and 12,000 feet respectively above the plain to which their flanks descend with a steep declivity. The counterscarp of the Apennines, in places 160 miles in width from east to west, runs down to the Mare Vaporum, with a comparatively gentle inclination. It is everywhere traversed by winding valleys of a very intricate type, all trending toward the southwest, and includes some very bright craters and mountain-rings.

Whether variations in the visibility of lunar details, when observed under apparently similar conditions, actually occur from time to time from some

unknown cause, is one of those vexed questions which will only be determined when the moon is systematically studied by experienced observers using the finest instruments at exceptionally good stations; but no one who examines existing records of rills by Gruithuisen, Lohrmann, Mädler, Schmidt, and other observers, can well avoid the conclusion that the anomalies brought to light therein point strongly to the probability of the existence of some agency which occasionally modifies their appearance or entirely conceals them from view. In short, the more direct telescopic observations accumulate, and the more the study of minute detail is extended, the stronger becomes the conviction that, in spite of the absence of an appreciable atmosphere, there may be something resembling low-lying exhalations from some parts of the surface which from time to time are sufficiently dense to obscure, or even obliterate, the region beneath them.

Sir John Herschel maintained that “the actual illumination of the lunar surface is not much superior to that of weathered sandstone rock in full sunshine. I have,” he says, “frequently compared the moon setting behind the gray perpendicular façade of the Table Mountain, illumined by the sun just risen in the opposite quarter of the horizon when it has been scarcely distinguishable in brightness from the rock in contact with it. The sun and moon being at nearly equal altitudes, and the atmosphere perfectly free from cloud or vapor, its effect is alike on both luminaries.” Zöllner’s elaborate researches on this question are closely in accord with the above observational result. Though he considers that the brightest parts of the surface are as white as the whitest objects with which we are acquainted, yet, taking the reflected light as a whole, he finds that the moon is more nearly black than white. The most brilliant object on the surface is the central peak of the ring-plain Aristarchus, the darkest the floor of Grimaldi, or perhaps a portion of that of the neighboring Riccioli. Between these extremes there is every gradation of tone. Proctor, discussing this question on the basis of Zöllner’s experiments respecting the light reflected by various substances, concludes that the dark area just mentioned must be notably darker than the dark gray syenite which figures in his tables, while the floor of Aristarchus is as white as newly fallen snow.

MARS.—AGNES M. CLERKE

The furthest terrestrial planet from the sun is Mars, the “star of strength.” No other heavenly body, except the moon, is so well placed for observation from our position in space.

The diameter of Mars is 4,200 miles; its surface is equal to two-sevenths, its volume to one-seventh those of the earth. But, in consequence of its inferior mean density, nine such spheres would go to make up the mass of our world. The superficial force of gravity on Mars, compared with its terrestrial value, is as thirty-eight to a hundred. A man could leap there a wall eight feet four inches in height with no more effort than it would cost him here to spring over a two-foot fence.

The planet’s rotation is performed in 24 hours, 37 minutes, on an axis deviating from the vertical by $24^{\circ} 50'$. Hence its seasons resemble our own, except in being nearly twice as long, for the Martian year is of 687 days.

The disk of Mars is diversified with three shades of color—reddish, or dull orange, dark grayish-green, and pure white. The last shows mainly in two diametrically opposite patches. Each pole is surrounded by a brilliant cap, suggesting the deposition of ice or snow over the chilly spaces corresponding to our arctic and antarctic regions. Nor is this all. Each of the polar hoods shrinks to a mere remnant as the local summer advances, but regains its original size when wintry influences are again in the ascendant. Here, and nowhere else in the planetary system, we meet evidence of seasonal change; and seasonal change is associated with vital possibilities. Again, a globe upon which snow visibly melts must contain water; hence the green markings can not but image to our minds seas and inlets subdividing continents, the blond complexion of which may be caused by some native peculiarity of the soil. It is in no way connected with vegetation, since it neither fades nor flushes with the advent of spring; and an atmospheric origin is excluded by the circumstance that it becomes effaced by a whitish haze near the limb, just where the densest atmospheric strata are traversed by the line of sight.

The spots on Mars are by no means so sharply defined as lunar craters and *maria*; yet they are fundamentally permanent. Some can be recognized from drawings made over two hundred years ago; and these antique records have served modern astronomers to determine with minute accuracy the rotation-period of the planet. Continents are somewhat vaguely outlined. Great tracts of them are of an uncertain and variable hue, as if subject to inundations. This peculiarity, thoroughly certified during the favorable opposition of 1892, makes a strong distinction between Mars and the Earth. Terrestrial oceans keep within the limits assigned to them. On the neighboring planet—as M. Faye observed in 1892—“water seems to march about at its ease,” flooding from time to time regions as wide as France. The imperfect separation of the two elements recalls the conditions prevailing during the terrestrial carboniferous era.

The main part of the land of Mars is situated in the Northern Hemisphere. It covers two-thirds of the entire globular surface. Rather than land, indeed, it should be called a network of land and water. The great continental block—so its orange tint declares it to be—is cut up in all possible directions by an intricate system of what appear to be waterways, running in perfectly straight lines—that is, along great circles of the globe—for distances varying from 350 to upward of 4,000 miles. They are frequently seen in duplicate, strictly parallel companions developing thirty to three hundred miles apart from the original formations. This mysterious phenomenon is evanescent, or rather periodical.

The canals invariably connect two bodies of water; hence they need no locks or hydraulic machinery; their course is on a dead level. The broadest of them are comparable with the Adriatic; those at the limit of visibility, stretching like the finest spider-threads across the disk, have a width of eighteen miles. “The canals,” Schiaparelli says, “may intersect among themselves at all possible angles, but by preference they converge toward the small spots to which we have given the name of lakes. For example, seven are seen to converge in Lacus Phœnicis, eight in Trivium Charontis, six in Lunæ Lacus, and six in Ismenius Lacus.”

These “lakes” evidently form an integral part of the canal system. They resemble huge railway junctions; and the largest of them—the “Eye of Mars” (Schiaparelli’s Lacus Solis)—seems, in Mr. Lowell’s phrase, like the

hub of a five-spoked wheel. Mr. W. H. Pickering in 1892, and Mr. Percival Lowell in 1894, were amazed at their extraordinary abundance.

“Scattered over the orange-ochre groundwork of the continental regions of the planet,” the latter wrote, “are any number of dark, round spots. How many there may be it is not possible to state, as the better the seeing, the more of them there seem to be. In spite, however, of their great number, there is no instance of one occurring unconnected with a canal. What is more, there is apparently none which does not lie at the junction of several canals. Reversely, all the junctions appear to be provided with spots.”

Most of these foci are about 120 miles in diameter, and appear most precisely circular when most clearly seen. “Plotted upon a globe,” Mr. Lowell continues, “they and their connecting canals make a most curious network over all the orange-ochre equatorial parts of the planet, a mass of lines and knots, the one marking being as omnipresent as the other. Indeed, the spots are as peculiar and distinctive a feature of Mars as the canals themselves.”

Like the canals, too, they emerge periodically, and in the same but a retarded succession. They “are, therefore, in the first place, seasonal phenomena, and, in the second place, phenomena that depend for their existence upon the prior existence of the canals.”

Mr. Lowell terms them “oases,” and does not shrink from the full implication of the term.

The most important result of the numerous observations of Mars, made during the oppositions of 1892 and 1894, was the recognition of a regular course of change dependent upon the succession of its seasons. Schiaparelli had long anticipated this result; he is commonly in advance of his time. These changes, moreover, when closely watched, are really self-explanatory. The alternate melting of the northern and southern snow-caps initiates and to some extent determines them. As summer advances in either hemisphere, the wasting of the corresponding white calotte can be followed in every minute particular. “The snowy regions are then seen to be successively notched at their edges; black holes and huge fissures are formed in their interiors; great isolated fragments many miles in extent stand out from the principal mass, dissolve, and disappear a little later. In short, the same divisions and movements of these icy fields present

themselves to us at a glance that occur during the summer of our own arctic regions.”

Indeed, glaciation on Mars is much less durable than on the earth. In 1894 the southern snow-cap vanished to the last speck 59 days after the solstice and the remnant usually left looks scarcely enough to make a comfortable cap for Ben Nevis. An immense quantity of water is thus set free. The polar seas overflow; gigantic inundations reinforced, doubtless, from other sources, spread to the tropics; Syrtis regions of marsh or bog deepen in hue, and become distinctly aqueous; canals dawn on the sight, and grow into undeniable realities. We seem driven to believe that they discharge the function of flood-emissaries.

Mr. Lowell does not hesitate to pronounce them of artificial formation, and, on that large assumption, the purpose of their connection with his “oases” becomes transparently clear. They bring to these Tadmors in the wilderness the water supply by which they are made to “blossom as the rose.” The junction-spots, we are told, do not enlarge when the vernal freshet reaches them; they only darken through the sudden development of vegetation. These circular “districts, artificially fertilized by the canal system,” are strewn broadcast over vast desert areas, the orange-ochreous sections of Mars, covering the greater part of its surface, but deep buried in the millennial dust of disintegrated red sandstone strata.

“Here, then,” Mr. Lowell remarks, “we have an end and reason for the existence of canals, and the most natural conceivable—namely, that the canals are constructed for the express purpose of fertilizing the oases. When we consider the amazing system of the canal lines, we are carried to this conclusion as forthright as is the water itself; what we see being not the canal itself, indeed, but the vegetation along its banks.”

The proportion of water to land is much smaller on Mars than on the earth. Only two-sevenths of the disk are covered by the dusky areas, and of late the aqueous nature of some, if not all, of these has been seriously called in question. Professor Pickering was convinced by his observations, in 1892 and 1894, “that the permanent water area upon Mars, if it exist at all, is extremely limited in its dimensions.” He estimated it at about half the size of the Mediterranean. Professor Schaeberle is similarly incredulous. If the dark markings are seas, he asks, how explain the irregular gradations of

shade in them? How, above all, explain their apparent intersection by well-marked canals? Professor Barnard, observing with the Lick thirty-six inch in 1894, discerned on the Martian surface an astonishing wealth of detail, "so intricate, small, and abundant, that it baffled all attempts to properly delineate it." It was embarrassing to find these minute features belonging more characteristically to the "seas" than to the "continents." Under the best conditions, the dark regions lost all trace of uniformity. Their appearance resembled that of a mountainous country, broken by cañon, rift, and ridge, seen from a great elevation. These effects were especially marked in the "ocean" area of the Hour-Glass Sea.

Evidently the relations of solid and liquid in that remote orb are abnormal; they can not be completely explained by terrestrial analogies. Yet a series of well-attested phenomena are intelligible only on the supposition that Mars is, in some real sense, a terraqueous globe. Where snows melt there must be water; and the origin of the Rhone from a great glacier is scarcely more evident to our senses than the dissolution of Martian ice-caps into pools and streams.

The testimony of the spectroscope is to the same effect. Dr. Huggins found, in 1867, the spectrum of Mars impressed with the distinct traces of aqueous absorption, and the fact, although called in question by Professor Campbell of Lick, in 1894, has been reaffirmed both at Tulse Hill and at Potsdam. That clouds form and mists rise in the thin Martian air, admits of doubt. During the latter half of October, 1894, an area much larger than Europe remained densely obscured. Whether or no actual rain was at that time falling over the Maraldi Sea and the adjacent continent it would be useless to conjecture. We only know that with the low barometric pressure at the surface of Mars, the boiling point of water must be proportionately depressed (Flammarion puts it at 115° Fahrenheit), which implies that it evaporates rapidly, and can be transported easily.

If the Martian atmosphere be of the same proportionate mass as that of our earth, it can possess no more than one-seventh its superficial density. That is to say, it is more than twice as tenuous as the air at the summits of the Himalayas. The corresponding height of a terrestrial barometer would be four and a half inches. Owing, however, to the reduced strength of gravity on Mars, this slender envelope is exceedingly extensive. In the pure sky scarcely veiled by it, the sun, diminished to less than half his size at our

horizons, probably exhibits his coronal streamers and prominences as a regular part of his noontide glory; atmospheric circulation proceeds so tranquilly as not to trouble the repose of a land “in which it seemeth always afternoon”; no cyclones traverse its surface, only mild trade-winds flow toward the equator, to supply for the volumes of air gently lifted by the power of the sun, to carry reinforcements of water-vapor north and south. Aerial movements are, in fact, by a very strong presumption, of the terrestrial type, but executed with greatly abated vigor.

Brilliant projections above the terminator of Mars were first distinctly perceived at the Lick Observatory in 1890. They have been reobserved at Nice, Arequipa, and Flagstaff (Mr. Lowell’s observatory), coming into view, as a rule, when circumstances concur to favor their visibility. They strictly resemble lunar peaks and craters, catching the first rays of the sun, while the ground about them is still immersed in darkness; and Professor Campbell connects them with “mountain chains lying *across* the terminator of the planet,” and in some cases possibly snow-covered. He calculates their height at about ten thousand feet. Their presence was unlooked for, since a flat expanse is a condition *sine quâ non* for the minute intersection of land by water, which seems to prevail on Mars.

Although the sun is less than half as powerful on Mars as it is here, the Martian climate, to outward appearance, compares favorably with our own. Polar glaciation is less extensive and more evanescent, and little snow falls outside the arctic and antarctic regions. Yet the theoretical mean temperature is minus 4° C., or 61° of Fahrenheit below freezing. This means a tremendous ice-grip. The coldest spot on the earth’s surface is considerably warmer than this cruel average. Fortunately, it exists only on paper. Some compensatory store of warmth must then be possessed by Mars, and it can scarcely be provided by its attenuated air. Possibly, internal heat may still be effective, and we see exemplified in Mars the geological period when vines and magnolias flourished in Greenland, and date-palms ripened their fruit on the coast of Hampshire.

The climate of Mars, according to Schiaparelli, “must resemble that of a clear day upon a high mountain. By day a very strong solar radiation hardly at all mitigated by mist or vapor; by night a copious radiation from the soil toward celestial space, and hence a very marked refrigeration; consequently, a climate of extremes, and great changes of temperature from day to night,

and from one season to another. And as on the earth, at altitudes of from 17,000 to 20,000 feet, the vapor of the atmosphere is condensed only into the solid form, producing those whitish masses of suspended crystals which we call cirrus-clouds, so in the atmosphere of Mars it would be rarely possible to find collections of cloud capable of producing rain of any consequence. The variation of temperature from one season to another would be notably increased by their long duration, and thus we can understand the great freezing and melting of the snow, renewed in turn at the poles at each complete revolution of the planet round the sun.”

The German astronomer Mädler searched in 1830 for a Martian satellite, and although his telescope was of less than four inches aperture, he satisfied himself that none with a diameter of as much as twenty-three miles could be in existence. As it happened, he was right. The pair of moons detected by Professor Asaph Hall with the Washington twenty-six refractor, August 11 and 17, 1877, are unquestionably below that limit of size. Neither of them can well be more than ten miles across. Their names, “Deimos” and “Phobos,” are taken from the *Iliad*, where Fear and Panic are introduced as attendants upon the God of War. Deimos revolves in 30 hours and 18 minutes at a distance of 14,600 miles from the centre of Mars. And since the planet rotates in 24 hours, 37 minutes, the diurnal motion of the sphere from east to west is so nearly neutralized by the orbital circulation of the satellite from west to east that nearly 132 hours elapse between its rising and its setting. During the interval, it changes four times from new to full, and *vice versa*.

Phobos is more effective in illumination, both because it is larger and because it is less distant. At the Martian equator, its brightness is equal to $\frac{1}{60}$ th that of our moon, but beyond 69° of latitude it is permanently shut out from view by the curvature of the globe.

THE PLANETOIDS.—CAMILLE FLAMMARION

On the first day of the last century (January 1, 1801), Piazzi, an astronomer devoted to the sky, was observing at Palermo the small stars of the constellation Taurus, and noting their exact positions, when he remarked one which he had never seen before. The following evening (January 2) he directed his telescope again toward the same region of the sky, and remarked that the star was no longer at the point where he had seen it the day before, and that it had retrograded by 4'. It continued to retrograde up to the 12th, stopped, and then moved in the direct way—that is to say, from west to east. What was this moving star? The idea that it might be a planet did not immediately occur to the mind of the observer, and he took it for a comet, as William Herschel had done in 1781, when he discovered Uranus.

However, the skilful Sicilian observer was a member of an association which had for its special object the search for an unknown planet between Mars and Jupiter. From the earliest times of modern astronomy Kepler had described the disproportion, the void which exists between the orbit of Mars and that of Jupiter. If we omit, in fact, the orbit of the small planets or asteroids, we notice that the four planets, Mercury, Venus, the earth, and Mars, are in some measure crowded quite close to the sun, while Jupiter, Saturn, Uranus, and Neptune extend far into immensity. The law of Titius indicates a number, the number 28, as not being represented by any planet. It was in 1772 that this *savant* published this relation in a German translation which he had made of the *Contemplation de la Nature* of Charles Bonnet. Bode, Director of the Berlin Observatory, was so astonished at the coincidence that he announced this arithmetical relation as being a real law of nature, and spoke of it in such a way that it is generally known only by his name. He even organized an association of twenty-four astronomers to explore each hour of the Zodiac and search for the unknown. This systematic exploration had not yet produced any result when, by the merest chance, Piazzi saw his moving star, and at first believed it to be a comet. But on receipt of the news, Bode was convinced that this was the looked-for planet.

The new planet was found to be at the distance 2.77, and to revolve within a few days of the predicted period. Piazzi gave to the new body the name of *Ceres*, the protecting divinity of Sicily in the “good old times” of mythology.

The gap being thus filled up at the distance 28 by the discovery of Ceres, no one thought that other planets might exist there; and if Piazzi had supposed so, he might have at once discovered a dozen of the small bodies which revolve in this region. An astronomer of Bremen, Olbers, observed this planet on the evening of March 28, 1802, when he perceived in the constellation of the Virgin a star of the seventh magnitude which was not marked on Bode’s chart, which he used. The following day he found it had changed its place, and recognized by this fact that it was a second planet. But it was much more difficult to give citizenship to it than to its elder sister, because, the gap being filled up, it was not required, and it was more inconvenient than agreeable. They looked upon it, then, as a comet until its motion proved that it revolved in the same region as Ceres at the distance 2.77, and in 1,685 days (the period of Ceres is 1,681 days). They gave it the name of *Pallas*.

The unexpected discoveries of Ceres and Pallas led astronomers to revise the catalogues of stars and celestial charts. Harding was of the number of the zealous revisers. He was soon rewarded for his trouble. On September 1, 1804, at ten o’clock in the evening, he saw in the constellation of Pisces a star of the eighth magnitude which was not noted in the *Histoire Céleste* of Lalande. On September 4, he found it had perceptibly changed its place: it was a new planet. It received the name of *Juno*. Its distance from the sun is expressed by the number 2.67, and its revolution is performed in 1,592 days.

After these three discoveries, Olbers, noticing that the orbits of these planets crossed each other in the constellation of the Virgin, advanced the hypothesis that they might be nothing else but fragments of a large shattered planet. Mechanics show that, in this case, the fragments would again pass every year—that is to say, at each of their revolutions—through the spot where the catastrophe took place. Olbers then set himself to explore the constellation Virgo carefully, and found on March 29, 1807, a fourth small planet, to which he gave the name of *Vesta*. Its distance is but 2.36, and its revolution only 1.326 days. This is the brightest of the small planets, and it

is sometimes seen with the naked eye (when we know where it is), like a star of the sixth magnitude.

It seems surprising that after these brilliant beginnings thirty-eight years should then have passed without the discovery of a single planet, for it was only in 1845 that the fifth, *Astræa*, was discovered by Hencke (who should not be confused with the astronomer Encke), a simple amateur astronomer, postmaster at Berlin, who amused himself by constructing charts of the stars. The principal reason for this must be attributed to the want of good star-charts, for to find these little moving points the first thing necessary is to have a very precise chart of the region of the Zodiac which we observe, in order to see whether one of the stars observed is in motion. The earliest good Zodiacal charts are those which the Academy of Berlin commenced to publish in 1830, taking as a basis the zones of Bessel continued by Argelander. Those of the Paris Observatory, which are more perfect, were only begun in 1854.

These small planets are all telescopic, invisible to the naked eye, with the exception of Vesta, and sometimes Ceres, which good sight can occasionally succeed in distinguishing; they are of the seventh, eighth, ninth, tenth, and eleventh magnitudes, and even still smaller, and it was for this reason also that so long an interval of time elapsed between the fourth and fifth discoveries. It is probable that all the small planets of any importance are now known, but that a great number—several hundreds, perhaps—still remain to be discovered of which the average brightness does not exceed that of stars of the twelfth magnitude, and of which the diameter is but a few miles. The diameter of the largest, Vesta, may be estimated at 400 kilometres (248 miles).

Hencke found successively the 5th and the 6th in 1845 and 1847; Hind, the English astronomer, the 7th and 8th in 1847; Graham, an English observer, the 9th in 1848; Gasparis, an Italian astronomer, the 10th and 11th in 1849 and 1850, and afterward seven others. Hind has further discovered eight others; Goldschmidt, a German painter (a naturalized Frenchman), discovered fourteen between 1852 and 1861.^[27] They are now discovered by swarms; Paliser alone has found sixty-eight since 1874.

The names given to these small bodies commenced with the mythological army of divinities of the earth and ancient heaven; but even

before the list had been exhausted certain scientific, or even national or political, circumstances caused the preference to be given to more modern names. It was thus that the 11th, discovered at Naples, received the name of Parthenope; the 12th, discovered in England, that of Victoria; the 20th, that of Massilia; the 21st, that of Lutetia; the 25th, that of Phocæa, before even Urania had been restored to the skies; the 45th was named in honor of the Empress of the French; the 54th, in honor of the illustrious Alexander von Humboldt; etc. The 87th, 107th, 141st, 154th, and 169th have been named in honor of a young astronomer who has devoted his best years to the culture of astronomy.

A rather curious fact is that they have put Wisdom (*Sapientia*) in the sky only at the 275th, discovered in 1888; Bellona has been placed there since the 28th (1854).

Of all this number of planets, the nearest to the sun is No. 149, Medusa, of which the distance is 2.17—that is to say, about twice as far from the sun as the earth; and the most distant is No. 279, Thule, of which the distance is 4.26, about $4\frac{1}{4}$ times our distance.

A large number of these small bodies are remarkable for their great eccentricity and for their high inclination to the ecliptic, an inclination so great that some of them leave the Zodiac; thus, Pallas (2) goes 34 degrees from the ecliptic; Euphrosyne (31) and Anna (265) and Istria (183), to 26 degrees. They are sometimes northern circumpolar stars, always above the horizon, sometimes southern stars, not arising above the horizon of Paris. All these orbits are so interlaced with each other that, if they were material hoops, we could by means of one or two taken by chance raise all the others.

Are they globes? Yes, doubtless, for the most part. But several among the smaller ones may be polyhedral, and may have proceeded from subsequent explosions; the variations of brightness which have been sometimes observed seem to imply surfaces irregularly broken.

Are they *worlds*? Why not? Is not a drop of water shown in the microscope peopled with a multitude of various beings? Does not a stone in a meadow hide a world of swarming insects? Is not the leaf of a plant a world for the species which inhabit and prey upon it? Doubtless among the multitude of small planets there are those which must remain desert and

sterile because the conditions of life (of any kind) are not found united. But we can not doubt that on the majority the ever-active forces of nature have produced, as in our world, creations appropriate to these minute planets. Let us repeat, moreover, that for nature there is neither great nor little. And there is no necessity to flatter ourselves with a supreme disdain for these little worlds, for in reality the inhabitants of Jupiter would have more right to despise us than we have to despise Vesta, Ceres, Pallas, or Juno: the disparity is greater between Jupiter and the earth than between the earth and these planets.

FOOTNOTES:

[27] Goldschmidt passionately loved astronomy, and I have found among his papers, which his family left me, numerous observations and remarks which show how he loved the study of the sky. His greatest ambition had been, at first, to possess a small telescope, in order to make some observations, and the best day of his life was that on which he found one in the possession of a dealer in old stores. He hastened to direct it to the sky from his modest studio, situated in one of the most frequented streets of Paris (Rue de l'Ancienne-Comédie), above the Café Procope, formerly used as a rendezvous by the stars of literature. There, *from his window*, he discovered, in 1852, the 21st small planet, which received from Arago the name of Lutetia; then, in 1854, the 32d (Pomona); then, in 1855, the 36th (Atlanta); and afterward eleven others, all from his window. Having often removed in search of a pure atmosphere, he finally retired to Fontainebleau, where the forest offered him on all sides admirable subjects for painting; and here he died in 1866.

JUPITER.—AGNES M. CLERKE

Jupiter is by far the most important member of the solar family. The aggregate mass of all the other planets is only two-fifths of his, which 316 earths would be needed to counterbalance. His size is on a still more colossal scale than his weight, since in volume he exceeds our globe 1,380 times. His polar and equatorial diameters measure respectively 84,570 and 90,190 miles, giving a mean diameter of 88,250 miles, and a polar compression of $\frac{1}{16}$ th. The corresponding equatorial protuberance rises to 2,000 miles, so that the elliptical figure of the planet strikes an observer at the first glance. This at once indicates rapid axial movement; and Jupiter's rotation is accordingly performed in nine hours and fifty-five minutes, with an uncertainty of a couple of minutes.

The numbers just given imply that this great planet is of somewhat slight consistence, and its mean density is, in fact, a little less than that of the sun. The sun is heavier than an equal bulk of water in the proportion of 1.4 to 1, Jupiter in the proportion of 1.33 to 1. The earth is thus more than four times specifically heavier than the latter globe. Three Jupiters would keep in equipoise four equal globes of water, while the earth would turn the scale against five and a half aqueous models of itself. This low density, an unfailing characteristic of all the giant planets, is charged with meaning. It at once gives us to understand that, in crossing the zone of asteroids, we enter upon a different planetary region from that left behind. The bodies revolving there are on an immensely larger scale of magnitude than those on the hither side; they are of solar, rather than terrestrial, density; they rotate much more rapidly, and are in consequence of a more elliptical shape; they display, and most likely possess, no solid surface; they are attended by retinues of satellites.

Jupiter circulates round the sun in 11.86 years, in an orbit deviating by less than one and a half degrees from the plane of the ecliptic, but of thrice the eccentricity of the ellipse traced out by the earth. With a mean distance from the sun of 483 millions of miles, it accordingly approaches within 462 at perihelion, and withdraws to 504 millions of miles at aphelion. Seasons it

has none worth mentioning; nor could they be of much effect even if they were better marked.

Under propitious circumstances Jupiter comes within 369 million miles of the earth. These occur when he is in opposition nearly at the epoch of his perihelion passage. His maximum opposition distance, on the other hand, is 411 million miles. He is then at aphelion. Thus, at the most favorable opposition, he is 42 million miles nearer to us than at the least favorable. The effect on his brightness is evident to the eye. When his midnight culmination takes place in October, he in fact sends us one and a half times more light than when the event comes round to April. We need only recall the unusual splendor of his appearance in September and October, 1892, when his lustre was double that of Sirius. His opposition period, as we may call it, is 399 days.

The intrinsic brilliancy of his surfaces is surprising, especially when we consider that it is somewhat deeply tinged with color.

The minimum diameter of the visible disk considerably exceeds the maximum of that of Mars. Even with a low power it thus makes a beautiful and interesting telescopic object. Its distinctive aspect is that of a belted planet, the belts varying greatly in number and arrangement. As many as thirty have, on occasions, been counted, delicately ruling the disk from pole to pole. They are always parallel to the equator, but are otherwise highly changeable, and can not be too closely studied as an index to the planet's physical constitution. Two in particular are remarkable. They are called the north and south equatorial belts, and inclose a lustrous equatorial zone. The poles are shaded by dusky hoods.

This general scheme of markings, however, when viewed with one of the great telescopes of the world, is so overlaid with minor particulars as sometimes to be scarcely recognizable. One can not see the wood for the trees. Lovely color-effects, too, come out under the best circumstances of definition and aerial transparency. The tropical belts may be summarily described as red; but they are of complex structure, and their subordinate features and formations are marked out, under the sway of alternating and tumultuous activities, by strips and patches of vermilion, pink, purple, drab, and brown. The intermediate space is divided into two bands by a line, or narrow ribbon, pretty nearly coinciding with the equator, and rosy or vivid

scarlet in hue. The polar caps are sometimes of a delicate wine-color, sometimes pale gray.

Professor Keeler made an elaborate study of the planet with the Lick 36-inch in 1889, and executed a series of valuable drawings. With a power of 320, the disk, he tells us, “was a most beautiful object, covered with a wealth of detail which could not possibly be accurately represented in a drawing.” Most of the surface was then “mottled with flocculent and irregular cloud-masses. The edges of the equatorial zone were brilliantly white, and were formed of rounded, cloud-like masses, which, at certain places, extended into the red belt as long streamers. These formed the most remarkable and curious feature of the equatorial regions. They are the cause of the double or triple aspect which the red belts present in small telescopes.”

Near their starting-points the streamers were white and sharply defined, but became gradually diffused over the ruddy surface of the belts. When at all elongated, they invariably flowed backward *against* the rotational drift, and were inferred to be cloud-like masses expelled from the equatorial region, and progressively left behind by its advance. This hypothesis was confirmed by the motion of some bright points, or knots, on the streamers. “The portions of the equatorial zone surrounding the roots of well-marked streamers were somewhat brighter,” Professor Keeler continues, “than at other places, and it is a curious circumstance that they were almost invariably suffused with a pale olive-green color, which seemed to be associated with great disturbance, and was rarely seen elsewhere.”

Now, if the material of the streamers had been simply a superficial overflow, it should have carried with it into higher latitudes an excess of linear rotational speed, and should hence have pushed its way onward as it proceeded north and south. But, instead, it fell behind; its velocity was less, not greater, than that of the belts with which it eventually became incorporated. What are we to gather from this fact? Evidently that the currents issuing north and south were of eruptive origin. Their motion, in miles per second, was slow, because they belonged to profound strata of the planet’s interior. Their backward drift measured the depth from which they had been flung upward.

The spots, red, white, and black, constantly visible on the Jovian surface, excite the highest curiosity. They are of all kinds and qualities, and their histories and adventures are as diverse as they are in themselves. Some are quite evanescent; others last for years. At times they come in undistinguished crowds, like flocks of sheep, then a solitary spot will acquire notoriety on its own account. White spots appear in both ways; black spots more often in communities; and it is remarkable that the former frequent distinctively, though not exclusively, the Southern, the latter the Northern Hemisphere. Red spots, too, develop pretty freely; but the attention due to them has been mainly observed by one striking specimen.

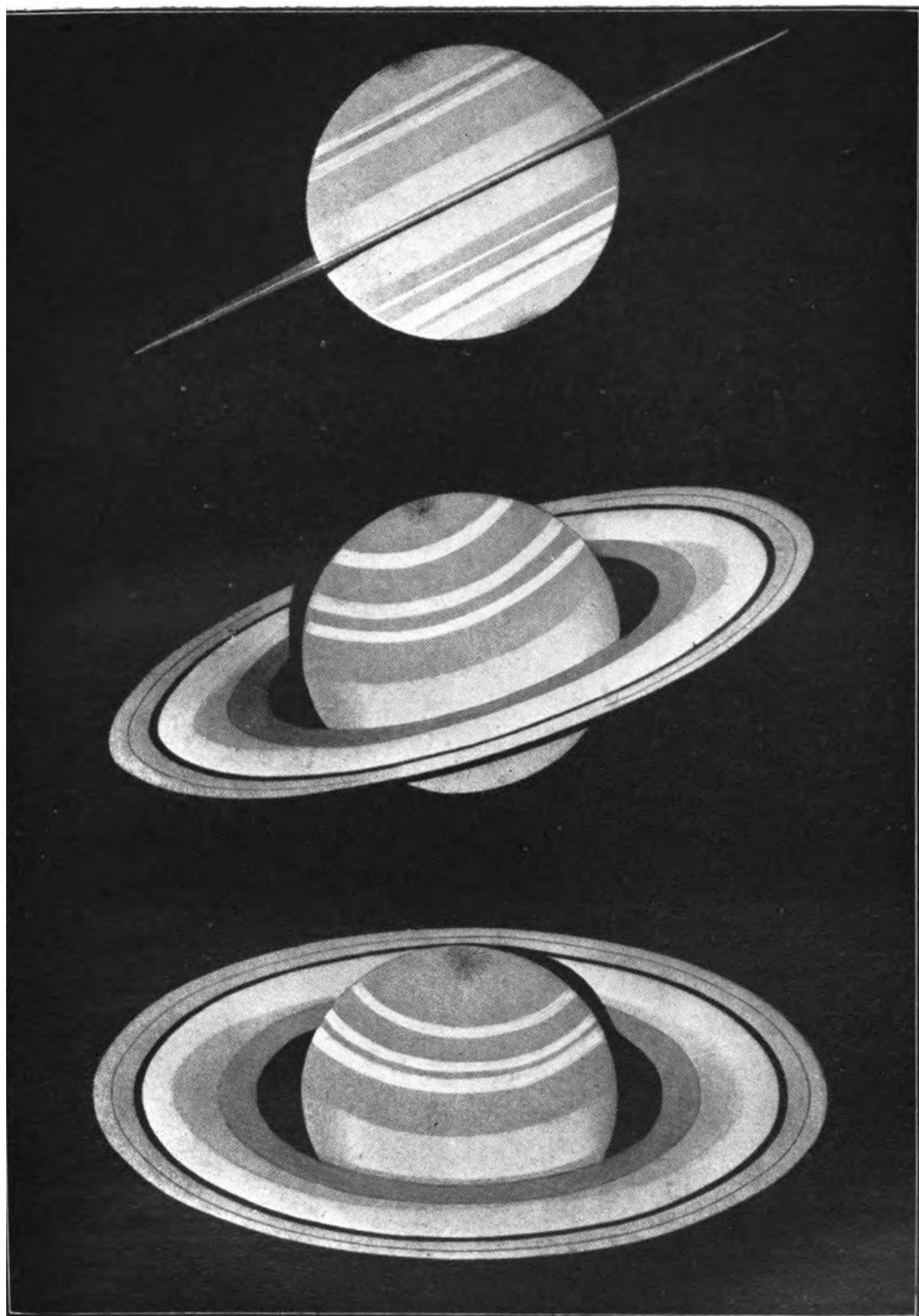
The Great Red Spot has been present with us for at least nineteen years; and it is a moot point whether its beginnings were not watched by Cassini more than two centuries ago. Its modern conspicuousness, however, dates from 1878. Then of a full brick-red hue, and strongly marked contour, it measured 30,000 by nearly 7,000 miles, and might easily have inclosed three such bodies as the earth. It has since faded several times to the verge of extinction, and partially recovered; but there has never been a time when it ceased to dominate the planet's surface-configuration. More than once it has been replaced by a bare elliptical outline, as if through an effusion of white matter into a mold previously filled with red matter; and just such a sketch was observed by Gledhill in 1870. The red spot is attached, on the polar side, to the southern equatorial belt. It might almost be described as jammed down upon it; for a huge gulf, bounded at one end by a jutting promontory, appears as if scooped out of the chocolate-colored material of the belt to make room for it. Absolute contact, nevertheless, seems impossible. The spot is surrounded by a shining aureola, which seemingly defends it against encroachments, and acts as a *chevaux-de-frise* to preserve its integrity. The formation thus constituted behaves like an irremovable obstacle in a strong current. The belt-stuff encounters its resistance, and rears itself up into a promontory or "shoulder," testifying to the solid presence of the spot, even though it be temporarily submerged. The great red spot, the white aureola, and the brownish shoulder are indissolubly connected.

The spot is then no mere cloudy condensation. Yet it has no real fixity. Its period of rotation is inconstant. In 1870-80, it was of 9 hours, 55 minutes, 34 seconds; in 1885-86, it was longer by 7 seconds. The object had

retrograded at a rate corresponding to one complete circuit of Jupiter in six years, or of the earth in seven months. It is not then fast moored, but floats at the mercy of the currents and breezes predominant in the strange region it navigates. A quiescent condition is implied by the approximate constancy of its rotation-period during the last ten years. With the paling of its color, its “proper motion” slackens or ceases. This must mean that, at its maxima of agitation, it is the scene of uprushes from great depths, which, bringing with them a slower linear velocity, occasion the observed laggings. It is not self-luminous, and shows no symptom of being depressed below the general level of the Jovian surface.

Jupiter has no certain and single period of rotation. Nearly all the spots that from time to time come into view on its disk are in relative motion, and thus give only individual results. The great red spot has the slowest drift of all (with the rarest exceptions), while the black cohorts of the Northern Hemisphere outmarch all competitors. Mr. Stanley Williams, as the upshot of long study, has delimited nine atmospheric surfaces with definite periods. They are well marked, and evidently have some degree of permanence, yet the velocities severally belonging to them are distributed with extreme irregularity. Thus, two narrow, adjacent zones differ in movement by 400 miles an hour. This state of things must obviously be maintained by some constantly acting force, since friction, if unchecked, would very quickly abolish such enormous discrepancies. The rotational zones are unsymmetrically placed; there is no correspondence between those north and south of the Jovian equator; and, although the equatorial drift is quicker than that of either tropic, it is outdone in 20° to 24° north latitude.

Jupiter’s equatorial rotation, as indicated by observations of spots, is accomplished in 9 hours 50 minutes; but Bélopolsky and Deslandres’s spectrographic determinations gave rates of approach and recession falling somewhat short of the corresponding velocity.



Three Views of Saturn

Showing Varying Aspects of the Ring taken at Different Intervals: 1, Feb. 2, 1862; 2, Nov. 3, 1858;
3, March 23, 1856

However this be, the rotation of the great planet, albeit ill-regulated (if the expression be permissible), is distinctly of the solar type. It is itself a “semi-sun,” showing no trace of a solid surface, but a continual succession of cloud-like masses belched forth from within. Jupiter’s low mean density, considered apart from every other circumstance, suffices to demonstrate the primitive nature of his state. In a sun-like body, the circulation is bodily and vertical. That the processes going on in Jupiter are of this kind is beyond question. Exchanges of hot and colder substances are effected, not by surface-flows, but by up and down rushes. The parallelism of his belts to his equator makes this visible to the eye. An occasional oblique streak betokens a current in latitude, but it is exceptional, and might be called out of character.

Jupiter’s true atmosphere encompasses the disturbed shell of vapors observed telescopically. Its general absorptive action upon light is betrayed by the darkening of the planet’s limb—another point of resemblance to the sun; while its special, or selective, absorption can only be detected with the spectroscope.

The actinic power of Jupiter’s light is very remarkable. It surpasses that of moonlight nine times, and that of Mars twenty-four times. Dr. Lohse further ascertained that the Southern Hemisphere is twice as chemically effective as the Northern. This superiority is doubtless connected with the greater physical agitation of the same region. A series of photographs of Jupiter, taken in 1891 with the great Lick refractor, were the first of any value for purposes of investigation.

Jupiter’s satellites were the first trophies of telescopic observation. They are, indeed, bright enough for naked-eye perception, could they be removed from the disk which obscures them with its excessive splendor; and the first and third have actually been seen, in despite of the glare, by a few persons with phenomenally good eyesight. The mythological titles of the Galilean group—Io, Europa, Ganymede, and Calypso (proceeding from within outward)—have been superseded by prosaic numbers.

The Jovian family presents an animated and attractive spectacle. The smallest of its original members (No. II) is almost exactly the size of our moon; the largest (No. III), with its diameter of 3,550 miles, considerably exceeds the modest proportions of Mercury. Satellite I revolves in $42\frac{1}{2}$ hours at the same average distance from Jupiter's surface that our moon does from that of the earth. No. II has a period of 3 days 13 hours, and its distance from Jupiter's centre is 415,000 miles. Both these orbits are sensibly circular; and Nos. III and IV travel in ellipses of very small eccentricity, the one at a mean distance of 664,000, the other at 1,167,000 miles, in periods respectively of 7 days 4 hours, and 16 days $16\frac{1}{2}$ hours. All four revolve strictly in the plane of Jupiter's equator.

They constitute a system bound together by peculiar dynamical relations, in consequence of which they can never be all either eclipsed or seen aligned at one side of their primary at the same time. They can all, however, be simultaneously hidden behind it, or in its shadow; although this moonless condition is looked out for as a telescopic rarity.

The transits of the satellites across the Jovian disk present many curious appearances, due to complicated and changeable effects of light and shade both upon the planetary background and upon the little circular objects self-compared with it. These, in the ordinary course, show bright while near the dusky limb, then vanish during the central passage, and re-emerge again bright at the opposite side. But instead of duly vanishing, they now and then darken even to the point of becoming indistinguishable from their own shadows, by which they are preceded or followed. This difference of behavior can not be attributed wholly to varieties of lustre in the sections of the disk transited; otherwise it could be predicted. But this has never been attempted; "black transits" come when least expected. The third and fourth satellites are those chiefly subject to these phases; the second has never been known to exhibit them; and they but slightly affect the first. Indeed, all the satellites, except, perhaps, No. II, are striped or spotted; and this leads to seeming deformations in their shape, as well as fluctuations in their brightness, the markings being evidently of atmospheric origin, and hence changeable. Their distinct and accurate perception has been made possible by the excellence of the Lick 36-inch refractor.

Jupiter's moons seem to resemble him in constitution. The first three possess the same high reflective power. No. II is as bright as the planet's

brightest parts, so that its albedo can not fall short of 0.70. And even No. IV (formerly designated “Calypso” in reference to its frequent obscurations) exactly matches, during its darkest phases, the blue-gray polar hoods of its primary. On an average, too, the satellites seem to be of about the same mean density as Jupiter, No. I being considerably the lightest for its bulk; and their spectra, according to Vogel’s observations in 1873, are composed of solar rays modified in precisely the same way as those reflected by the planet.

The discovery, September 9, 1892, of Jupiter’s “fifth satellite” was one of the keenest astronomical surprises on record. Professor Barnard seized the opportunity, lent by the specially favorable opposition of 1892, to rummage the system for novelties. Keeping the telescopic field dark by means of a metallic bar placed so as to occult the gorgeous planetary round, he sought, night after night, for what might appear. At length, on September 9, he caught the glimmer he wanted, and made sure, September 10, that it truly intimated the presence of a new satellite.

This small body revolves in a period of 11 hours, 57 minutes, 23 seconds at a mean distance of 112,160 miles from Jupiter’s centre, or 67,000 from his bulged equatorial surface. Hence, it should by right be called “No. I” instead of “No. V.” The major axis of the ellipse in which it circulates advances so rapidly, owing to the disturbance caused by Jupiter’s spheroidal figure, as to complete a revolution in five months. The implied eccentricity of its orbit, as M. Tisserand has shown, very slightly exceeds that of the orbit of Venus, yet it has been made obvious by Barnard’s observations of the differences between its east and west elongations. Its orbital velocity of $16\frac{1}{2}$ miles a second far surpasses that of any other satellite in the solar system. Close vicinity to a mass so vast as Jupiter’s demands counterbalancing swiftness. Its period of revolution being, however, longer by one hour than Jupiter’s period of rotation, it so far conducts itself normally as to rise in the east and set in the west. On the other hand, since its progress over the sphere is measured by the difference between the two periods, it spends five Jovian days in journeying from one horizon to the other, running, in the meantime, four times through all its phases. Yet it never appears full. Jupiter’s voluminous shadow cuts off sunlight from it during nearly one-fifth of each circuit.

SATURN.—AGNES M. CLERKE

Nearly twice as far from the sun as Jupiter revolves a planet, the spacious orbit of which was, until 1781, supposed to mark the uttermost boundary of the Solar System. The mean radius of that orbit is 886 millions of miles; but in consequence of its eccentricity, the sun is displaced from its middle point to the extent of 50 million miles, and Saturn is accordingly 100 million miles nearer to him at perihelion than at aphelion. The immense round assigned to the “saturnine” planet is traversed in $29\frac{1}{2}$ years, at the tardy pace of six miles a second. His seasons are thus twenty-nine times more protracted than ours, and are nominally more accentuated, since his axis of rotation deviates from the vertical by 27° . But solar heat, however distributed, plays an insignificant part in his internal economy. In the first place, its amount is only $\frac{1}{91}$ st its amount on the earth; in the second, Saturn, like Jupiter—even more than Jupiter—is thermally self-supporting. The bulk of his globe comparatively to its mass suffices in itself to make this certain. The mean diameter of Saturn is 71,000 miles, or nine times (very nearly) that of the earth; if of equal density, its mass should then be nine cubed, or 729 times the same unit. The actual proportion, however, is 95; hence the planet has a mean density of only $\frac{95}{729}$ th, or between $\frac{1}{7}$ th and $\frac{1}{8}$ th the terrestrial, and being thus composed of matter as light as cork, would float in water. Professor G. H. Darwin has, moreover, demonstrated, from the movements of its largest satellite, that its density gains markedly with descent into the interior, so that its surface-materials must be lighter than any known solid or liquid.

When at its nearest to the earth, Saturn is as large as a sixpence held up at a distance of 210 yards. But instead of being round like a sixpence, it is strongly compressed—more compressed even than Jupiter. The spectra of the two planets are almost identical. Both are impressed with traces of aqueous absorption, and include the “red star line.”

Saturn resembles to the eye a large, dull star; its rays are entirely devoid of the sparkling quality which distinguishes those of Jupiter. But it shows telescopically an analogous surface-structure. Its most conspicuous

markings are tropical dark belts of a grayish or greenish hue; the equatorial region is light yellow, diversified by vague white spots; while the poles carry extensive pale blue canopies. The apparent tranquillity of the disk may be attributed in part to the vast distance from which it is viewed; yet not wholly.

From measures executed by Barnard in 1895, it appears that the equatorial diameter of Saturn is 76,470, its polar diameter 69,770 miles, giving a mean diameter of 74,240, and a compression of about $\frac{1}{12}$ th. Gravity, at its surface, is only $\frac{1}{5}$ th more powerful than on the earth.

Thus, Saturn not only belongs to the same celestial species as Jupiter, but is a closely related individual of that species. There is no probability that either is to any extent solid. Both exhibit the same type of markings; both betray internal tumults by eruptions of spots which, by their varying movements, supply a measure for the profundity of their origin; both possess identically constituted atmospheres, and are darkened marginally by atmospheric absorption.

Saturn is, however, distinguished by the possession of a unique set of appendages. Nothing like them is to be seen elsewhere in the heavens; and when well opened they form, with the globe they inclose, and the retinue of satellites in waiting outside, a strange and wonderful telescopic object. The rings, since they lie in the plane of Saturn's equator, are inclined 27° to the Saturnian orbit, and 28° to the ecliptic. The earth is, however, comparatively to Saturn, so near the sun, that their variations in aspect, as viewed from it, may in a rough way be considered the same as if seen from the sun. They correspond exactly with the Saturnian seasons. At the Saturnian equinoxes, the rings are illuminated edgewise, and disappear, totally or approximately; at the Saturnian solstices, sunlight strikes them nearly at the full angle of 27° , first from *below*, then from *above*. At these epochs, we perceive the appendage expanded into an ellipse about half as wide as it is long. Two concentric rings (generally called A and B) are then very plainly distinguishable, the inner being the brighter. The black fissure which separates them is called "Cassini's division," because that eminent observer was, in 1675, the first to perceive it. A chasm known as "Encke's division," in the outer ring (A), is a thinning-out rather than an empty space; and temporary gaps frequently appear in A, while B is entirely exempt from

them. There are then two definite and permanent bright rings, and no more; but with them is associated the dusky formation discovered by W. C. Bond, November 15, 1850, and described by Lassell as “something like a crape veil covering a part of the sky within the inner ring.” It is semi-transparent, the limb of Saturn showing distinctly through it.

The exterior diameter of the ring-system is 172,800, while its breadth is 42,300 miles. The rings A and C are each 11,000 miles wide; while B measures 18,000, Cassini’s division 2,270, and the clear interval between C and the planetary surface somewhat less than 6,000 miles. Each ring, C included, is brightest at its outer edge; but there is no gap between the shining and the dusky structures, B shading by insensible gradations up to C, yet maintaining distinctness from it. The earliest exact determinations of the former were made by Bradley in 1719, since when they have been affected by no appreciable change. The theoretically inevitable subversion of the system is progressing with extreme slowness.

The thickness of the rings is quite inconsiderable. They are flat sheets, without (so to speak) a third dimension. For this reason, they disappear utterly in most telescopes, when their plane passes through the earth, as it does twice in each Saturnian year. Only under exceptional conditions, a narrow, knotted, often nebulous, streak survives as an index to their whereabouts. On October 26, 1891, Professor Barnard, armed with the Lick refractor, found it impossible to see them projected upon the sky, notwithstanding that their shadow lay heavily on the planet. It was not until three days later that “slender threads of light” came into view. The corresponding thickness of the formation was estimated at less than fifty miles. The phenomenon of ring disappearance will not recur until July 29, 1907.

The constitution of this marvelous structure is no longer doubtful. It represents what might be called the fixed form of a revolving multitude of diminutive bodies. This was demonstrated by Clerk Maxwell in the Adams Prize Essay of 1857. His conclusion proved irreversible. The pulverulent composition of Saturn’s rings is one of the acquired truths of science. An incalculable number of tiny satellites revolving independently in distinct orbits, in the precise periods prescribed by their several distances from the planet, are aggregated into the unmatched appendages of Galileo’s *tergeminus planeta*. The local differences in their brightness depend upon

the distribution of the component satelloids. Where they are closely packed, as in the outer margins of rings A and B, sunlight is copiously reflected; where the interspaces are wide, the blackness of the sky is barely veiled by the scanty rays thrown back from the thinly scattered cosmic dust. The appearance of the crape ring as a *dark* stripe on the planet results—as M. Seeliger has pointed out—not from the transits of the objects themselves, but from the flitting of their shadows in continual procession across the disk.

The albedo of these particles is so high as to render it improbable that they are of an earthy or rocky nature, such as the meteorites which penetrate our atmosphere. The rings they form are, on the whole, more lustrous than Saturn's globe; but this superiority is held to be due to the absence of atmospheric absorption. Their spectrum is that of unmodified sunlight.

An eclipse of Japetus, the eighth Saturnian moon, by the globe and rings, November 1, 1889, was highly instructive as to the nature of the dusky appendage. The satellite was never lost sight of during its passage behind it; but became more and more deeply obscured as it traveled outward; then, at the moment of ingress into the shadow of ring B, suddenly disappeared. Certainty was thus acquired that the particles forming the crape ring are most sparsely strewn at its inner edge—which is, nevertheless, perfectly definite—and gradually reach a maximum of density at its outer edge. Yet, while there is not the smallest clear interval, a sharp line of demarcation separates it from the contiguous bright ring. Professor Barnard was the only observer of these curious appearances. The distribution of the ring-constituents, like that of the asteroids, was governed by the law of commensurable periods, Saturn's moons replacing Jupiter as the perturbing and regulating power.

The “satellite-theory” of Saturn's rings has received confirmation from apparently the least promising quarters. Professor Seeliger of Munich showed, from photometric experiments in 1888, that their constant lustre under angles of illumination ranging from 0° to 30° was proof positive of their composition out of discrete small bodies. And Professor Keeler of Alleghany, by a beautiful and refined application of the spectroscopic method, arrived at the same result in April, 1895. “Under the two different hypotheses,” he remarked, “that the ring is a rigid body, and that it is a swarm of satellites, the relative motion of its parts would be essentially

different.” The former would necessarily involve increasing velocity *outward*, the latter, increase of velocity *inward*, just for the same reason that Mercury moves more swiftly than the earth, and the earth than Saturn; while the sections of a solid body, which could have but one period of rotation, should move faster, *in miles per second*, the further they were from the centre of attraction. The line of sight test is then theoretically available; but it was an arduous task to render it practically so. The difficulties were, however, one by one overcome; and a successful photograph of the spectra of Saturn and its rings gave the required information in unmistakable shape. From measurements of the inclinations of five dusky rays contained in it with reference to a standard horizontal line, rates of movement were derived of $12\frac{1}{2}$ miles per second for the inner edge of ring B, and of 10 miles for the outer edge of ring A. The agreement with theory was, as nearly as possible, exact; the components of the rings were experimentally demonstrated to be moving, each independently of every other, under the dominion of Kepler’s laws.

For the globe of Saturn, Professor Keeler obtained, by the same exquisite method, a rotational period of 10 hours, 14 minutes, 24 seconds, in precise accordance with that indicated by the white spot of 1876, which thus seems to have had no proper motion, but to have floated on the ochreous equatorial surface as tranquilly as a water-lily upon a stagnant pool. The result, so far as it goes, hints that Saturn may be really, as well as apparently, less ebullient than Jupiter.

Seers into the future of the heavenly bodies consider that the rings of Saturn, like the gills of a tadpole, are symptomatic of an early stage of development; and will be disposed of before he arrives at maturity. They can not be regarded otherwise than as abnormal excrescences. No other planet retains matter circulating round it in such close relative vicinity. It was proved by Roche of Montpellier that no secondary body of importance can exist within less than 2.44 mean radii of its primary; inside of that limit it would be rent asunder by tidal strain. But the entire ring-system lies within the assigned boundary; hence, being *where* it is, it can only exist *as* it is—in flights of discrete particles. Will it, however, always remain where it is?

“Clerk Maxwell,” wrote Mr. Cowper Ranyard, “used to describe the matter of the rings as a shower of brickbats, among which there would

inevitably be continual collisions. The theoretical results of such impacts would be a spreading of the ring both inward and outward. The outward spreading will in time carry the meteorites beyond Roche's limit, where, in all probability, they will, as Professor Darwin suggests, slowly aggregate, and a minute satellite will be formed. The inward spreading will in time carry the meteorites at the inner edge of the ring into the atmosphere of the planet, where they will become incandescent, and disappear as meteorites do in our atmosphere."

Yet it may be that collisions are infrequent in this conglomeration of "brickbats." There is the strongest presumption that they all circulate in the same direction, in orbits nearly circular, and scarcely deviating from the plane of the Saturnian equator. Those pursuing markedly eccentric tracks must long ago have been eliminated. Thus, encounters can only occur through gravitational disturbances by Saturn's moons, and they must be of a mild character, depending upon very small differences of velocity. The first sign of a "spreading outward" should be the formation of an exterior "crape ring," of which no faintest trace has yet been perceived.

Saturn's rings are entirely invisible from its polar regions, but occasion prolonged and complex eclipse-effects in its temperate and equatorial zones. They have been fully treated of from the geometrical point of view by Mr. Proctor in *Saturn and its System*.

Of this planet's eight satellites,^[28] the largest, Titan (No. VI), was discovered first (by Huygens in 1655), and the smallest, Hyperion (No. VII), last (by Lassell and Bond in 1848). The five others were detected by J. D. Cassini and William Herschel. Titan, alone of the entire group, equals our moon in size. It measures, according to Professor Barnard, 2,720 miles across. Its period of revolution is nearly sixteen days, its distance from Saturn's centre, 771,000 miles. The orbit of Japetus (No. VIII) is the largest, and its period the longest of any secondary body in the Solar System. It circulates in $79\frac{1}{2}$ days at a distance of 2,225,000 miles, equal to $59\frac{1}{2}$ of Saturn's equatorial radii. Hence its path is of about the same *proportional* dimensions as that of our moon. Japetus is remarkable for its variability in light. It is capable of tripling or quadrupling its minimum lustre. Sir William Herschel noticed that these maxima coincided with a position on the western side of the planet, and inferred rotation of the lunar kind. "From the changes in this body," he argued in 1792, "we may conclude that some

part of its surface, and this by far the largest, reflects much less light than the rest; and that neither the darkest nor the brightest side is turned toward the planet, but partly one and partly the other, though probably less of the bright side.”

This explanation, however, he admitted to be incomplete. There was, and is, outstanding variability, which seems to intimate the presence of an atmosphere and the formation of clouds. But no positive knowledge has yet been gained regarding the physical state of Saturn’s moons. We may, nevertheless, conjecture that, since tidal friction has destroyed the rotation (as regards Saturn) of the remotest member of the family, it has not spared those more exposed to its grinding-down action. All presumably rotate in the same time that they revolve.

The five inner satellites move in approximately circular orbits; the three outer in ellipses about twice as eccentric as the terrestrial path. All, Japetus only excepted, keep strictly to the plane of the rings. And since this makes an angle of 27° with the planet’s orbit, eclipses are much less frequent here than in the Jovian system. They can only occur when Saturn is within a certain distance (different for each) from the node of the satellite-orbit. Even Mimas (No. I), although it wheels round the ring at an interval of only 34,000 miles, often slips outside the obliquely projected shadow-cone. Its distance from Saturn’s centre is 118,000 miles, and it completes a circuit in $22\frac{1}{2}$ hours. Perpetually wrapped in the glare of its magnificent primary, it is a very shy object, only to be caught sight of in its timid excursions by the very finest telescopes. Like all the Saturnian moons, except Titan, and, by a rare conjunction, Japetus, it is far too much contracted to be visible in transit across the disk.

The movements of these bodies have been carefully studied, and their mutual perturbations to some extent unraveled. They have proved exceedingly interesting to students of celestial mechanics. Titan has, in this department, chiefly to be reckoned with. He exercises in the Saturnian system a similar overpowering influence to that wielded by Jupiter in the Solar System.

FOOTNOTES:

[28] A ninth satellite, Phœbe, was discovered in 1904. Its existence had been suspected for many years, and it was discovered at the Arequipa Observatory, Peru, on March 14, 1899, by means of photography. Since that date, it has been several times lost and rediscovered.—E. S.

URANUS AND NEPTUNE.—WILLIAM F. DENNING

While Sir W. Herschel was a musician at Bath he formed the design of making a telescopic survey of the heavens. While engaged in this, he accidentally effected a discovery of great importance, for on the night of March 13, 1781, an object entered the field of his 6.3-inch reflector which ultimately proved to be a new major planet of our system.

The acute eye of Herschel, directly it alighted upon the strange body, recognized it as one of unusual character, for it had a perceptible disk, and could be neither fixed star nor nebula. He afterward found the object to be in motion, and its appearance being "hazy and ill-defined," with very high powers, he was led to regard it as a comet, and communicated his discovery to the Royal Society at its meeting on April 26, 1781.

The supposed comet soon came under the observation of others, including Maskelyne, the Astronomer Royal, and Messier, the "Comet Ferret," of Paris. The latter, in a letter to Herschel, said: "Nothing was more difficult than to catch it, and I can not conceive how you could have hit this star or comet several times, for it was absolutely necessary for me to observe it for several days in succession before I could perceive that it was in motion."

As observations began to accumulate, it was seen that a parabolic orbit failed to accommodate them. Ultimately the secret was revealed. The only orbit to represent the motion of the new body was found to be an approximately circular one situated far outside the path of Saturn, and the inference became irresistible that the supposed "comet" must in reality be a new primary planet revolving on the outskirts of the Solar System. This conclusion was justified by facts of a convincing nature, and its announcement created no small excitement in the scientific world. Every telescope was directed to that part of the firmament which contained the new orb, and its pale blue disk, wrapped in tiny proportions, was viewed again and again with all the delight that so great a novelty could inspire. From the earliest period of ancient history, no discovery of the same kind had been effected. The Chaldeans were acquainted with five major planets,

in addition to the earth, and the number had remained constant until the vigilant eye of Herschel enlarged our knowledge, and Saturn was relieved as the sentinel planet going his rounds on the distant frontiers of our system.

When the elements of the new body had been computed, a search was instituted among the records of previous observers, and it was found that Herschel's planet had been seen on many occasions, but it had invariably been mistaken for a fixed star. Flamsteed observed it on six occasions between 1690 and 1715, while Le Monnier saw it on twelve nights in the years 1750 to 1771, and it seems to have been pure carelessness on the part of the latter which prevented him from anticipating Herschel in one of the greatest discoveries of modern times.

The name Uranus was applied to the new planet, though the discoverer himself called it *Georgium Sidus*, and there were others who termed it *Herschel* in honor of the man through whose sagacity it had been revealed.

Uranus revolves around the sun in 30,687 days, which very slightly exceeds 84 terrestrial years. His mean distance from the sun is 1,782,000,000 miles, but the interval varies between 1,699 and 1,865 millions of miles. The apparent diameter of the planet undergoes little variation; the mean is 3".6, but observers differ. His real diameter is approximately 31,000 miles, and the polar compression about $\frac{1}{13}$, though this value is not that found by all authorities.

The planet near opposition shines like a star of the sixth magnitude, and is observable with the naked eye. He emits a bluish light. While engaged in meteoric observations, I have sometimes followed the planet with the naked eye during several months, and noted the changes in his position relatively to the stars near. It is clear from this that Uranus admitted of detection before the invention of the telescope.

A luminous ring, similar to that of Saturn, was at first supposed to surround Uranus, and Herschel suspected the existence of such a feature on several occasions; but it scarcely survived his later researches, and modern observations have finally disposed of it.

In May and June, 1883, Professor Young, having the advantage of the fine 23-inch refractor at the Princeton Observatory, observed two faint belts, one on each side of the equator, and much like the belts of Saturn. On

March 18, 1884, Messrs. Thollon and Perrotin, with the 14-inch equatorial at Nice, remarked dark spots similar to those on Mars, toward the centre of the disk, and a white spot was seen on the limb. Two different tints were perceived, the color of the Northwest Hemisphere being dark and that of the Southeast a bluish-white color. In April observations were continued, and the white spot was seen “rather as a luminous band than a simple spot,” but it was most conspicuous near the limb. The observers thought the appearances indicated a rotation-period of about ten hours. The brothers Henry at Paris, in 1884, invariably noticed two belts lying parallel to each other, and including between them the brighter equatorial zone of the planet. Their results apparently show that the angle between the plane of the Uranian equator and that of the satellite-orbits is about 41° .

M. Perrotin, with the great 30-inch equatorial at Nice, reobserved the belts in May and June, 1889. He wrote that dark parallel bands were noticed several times, and they were very similar to the belts of Jupiter. M. Perrotin notes that the bands of Uranus do not always present the same aspect. They vary in size and number in different parts of their circumference.

For many years it was supposed that Uranus possessed six satellites, all of which were discovered by Sir W. Herschel, but later observations proved that four of these had no existence. They were small stars near the planet. But two of Herschel’s satellites were fully corroborated, and two new ones were discovered by Lassell and Struve. The number of satellites attending Uranus is four, and it is probable that many others exist, though they are too minute to be distinguished in the most powerful instruments hitherto constructed. The following are the known satellites: 3d Ariel, discovered in 1847; 4th Umbriel, discovered in 1847; 1st Titania, discovered in 1787, and 2d Oberon, discovered in 1787.

Titania and Oberon are the two brightest satellites, but none of them can be seen except in large instruments. From observations with large modern instruments it appears highly probable that the four known satellites must be considerably larger than any others which may be revolving round the planet. A curious fact in connection with these satellites is that their motions are retrograde.

The leading incidents in the narrative of the discovery of Uranus and Neptune present a great dissimilarity—Uranus was discovered by accident,

Neptune by design. Telescopic power revealed the former, while theory disclosed the latter. In one case optical appliance afforded the direct means of success, while in the other the unerring precision of mathematical analysis attained it. The telescope played but a secondary part in the discovery of Neptune, for this instrument was employed simply to realize or confirm what theory had proven.

Certain irregularities in the motion of Uranus could not be explained but on the assumption of an undetected planet situated outside the known boundaries of the system. Two able geometers applied themselves to study the problem of these irregularities, and to deduce from them the place of the disturbing body. This was effected independently by Messrs. Le Verrier and Adams; and Dr. Galle of Berlin, having received from Le Verrier the leading results of his computations, and the intimation that the longitude of the suspected planet was then 326° , found it with his telescope on the night of September 23, 1846, in longitude $326^{\circ} 52'$. The calculated place by Professor Adams was $329^{\circ} 19'$ for the same date.

The name given to the new planet was Neptune. When the elements were computed it was found that they presented rather large differences with those theoretically computed by Le Verrier and Adams. It was also found that the planet had been previously observed by Lalande on May 8 and 10, 1795, but its true character escaped detection. This astronomer had observed a star of the eighth magnitude on May 8; but on May 10, not finding the same star in the exact place noted on the former evening, he rejected the first observation as inaccurate and adopted the second, marking it doubtful. Lalande, like Le Monnier, the unsuspecting discoverer of Uranus, let a valuable discovery slip through his hands.

Neptune revolves round the sun in 60,126 days, which is equal to rather more than $164\frac{1}{2}$ of our years. His mean distance from the sun is 2,792,000,000 miles, and his usual diameter $2''.7$. He exceeds Uranus in dimensions, his real diameter being 37,000 miles.

Our knowledge of this distant orb is extremely limited, owing to his apparently diminutive size and feebleness. No markings have ever been sighted on his miniature disk, and we can expect to learn nothing until one of the large telescopes is employed in the work. No doubt this planet

exhibits the same belted appearance as that of Uranus, and there is every probability that he possesses numerous satellites.

Directly the new planet was discovered, Mr. Lassell turned his large reflector upon it and sought to learn something of its appearance, and possibly detect one or more of its satellites. On October 3 and 10, 1846, he was struck with the appearance of the disk, which was obviously not spherical. He subsequently confirmed this impression, and concluded that a ring, inclined about 70° , surrounded the planet. Professor Challis supported this view, but later observations in a purer sky led Mr. Lassell to abandon the idea. Thus the ring of Neptune, like the ring of Uranus, though apparently obvious at first, vanished in the light of more modern researches.

But if Mr. Lassell quite failed to demonstrate the existence of a ring, he nevertheless succeeded in discovering a satellite belonging to the planet. This was on October 10, 1846. The new satellite was found to have a period of 5 days, 21 hours, and 3 minutes, and to be situated about 220,000 miles distant from the planet.

END OF VOLUME ONE

TRANSCRIBER'S NOTE

Footnote ^[13] is referenced twice from [page 102](#).

Obvious typographical errors and punctuation errors have been corrected after careful comparison with other occurrences within the text and consultation of external sources.

Some hyphens in words have been silently removed, some added, when a predominant preference was found in the original book.

Except for those changes noted below, all misspellings in the text, and inconsistent or archaic usage, have been retained.

[Pg 3](#): 'Caliph Al-Mamum' replaced by 'Caliph Al-Mamun'.

[Pg 20](#): 'Nineteenth Century' replaced by 'Nineteenth Century'.

[Pg 21](#): 'Map of the' replaced by 'Chart of the'.

[Pg 21](#): 'Hourglass Sea' replaced by 'Nine Views of the Hour-Glass Sea'.

[Pg 74](#): 'cose che redire' replaced by 'cose che ridire'.

[Pg 74](#): 'Nè sa, nè' replaced by 'Né sa, né'.

[Pg 100](#): 'Hesoid' replaced by 'Hesiod'.

[Pg 122](#): 'familiar to most' replaced by 'familiar to most'.

[Pg 150](#): 'formed of myraids' replaced by 'formed of myriads'.

[Pg 223](#): 'may be interred' replaced by 'may be inferred'.

[Pg 238](#): 'Will some motral' replaced by 'Will some mortal'.

[Pg 292](#): 'its orbital motion' replaced by 'its orbital motion'.

[Pg 380](#): 'the Mare Humorom' replaced by 'the Mare Humorum'.

[Pg 390](#): 'present themselves' replaced by 'present themselves'.

[Pg 391](#): 'Mr. Lowell remarks,*' replaced by 'Mr. Lowell remarks,;' (the * anchor had no footnote and has been removed).

[Pg 396](#): 'permanently shut' replaced by 'permanently shut'.

[Pg 418](#): 'is a thining-out' replaced by 'is a thinning-out'.

*** END OF THE PROJECT GUTENBERG EBOOK THE STORY OF
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